The Rhythm of Numbers

Continued Fractions, Primes, Music and Self-Similarity

Teddy Rook

Teddy Rook Book
https://teddyrookbook.com/connections/rhythm-of-numbers/

November 2026

Abstract

This monograph grows out of one guess: that π is close to $3 + k/(7k+1)$ for a suitable whole number k. The guess is right at $k = 16$, where it gives $355/113$, the fraction Zu Chongzhi recorded in the fifth century AD, and it is right for a reason: $k/(7k+1)$ is a continued fraction cut off after two steps. We follow that reason in three directions.

First, *rhythm*. Playing a number α as a beat (a hit wherever $\lfloor n\alpha \rfloor$ steps up) produces a Sturmian word, and we give a self-contained proof that its blocks nest exactly as the continued fraction says: the blocks at level n are q_n beats long and hold p_n hits. For $\pi - 3 = [0; 7, 15, 1, 292, \dots]$ this gives bars of 7 beats, phrases of 106 and 113, and a single short phrase that begins at beat 32,996 and closes a block of 33,102 beats. Reading the levels as bar, phrase, section and movement turns a continued fraction into musical form; Teddy Rook used it to compose the six-movement *Continued-Fraction Suite* for piano, which we describe as a method.

Second, *primes*. The same machine (the Euclidean algorithm) generates Euclidean rhythms, tunings from $\log_2(3/2)$, mode-locking staircases and the factoring method of Lehmer, Powers, Morrison and Brillhart. Coprimality ties these to $6/\pi^2$, and we derive the exact correlation of two pulse trains, $(\gcd(a, b) - 1)/\sqrt{(a-1)(b-1)}$. We then pull two open threads with real computations. (a) *Prime convergents.* Combining Lévy's theorem with the equidistribution of convergent denominators modulo m (Szüsz; Moeckel; Jager and Liardet), we derive the heuristic that a typical real number has about $2 \ln 2 \cdot \ln N \approx 1.3863 \ln N$ prime denominators among its first N convergents. A sample of 400 random reals fits the predicted growth; π has 9 prime denominators among its first 5,000 convergents against 10.9 expected, which is typical. (b) *Spacings.* We place a crystal, the rhythm of π (a one-dimensional quasicrystal with two gaps), 5,000 zeros of the Riemann zeta function, random GUE matrices, the primes and a Poisson process on one scale of rigidity, using nearest-neighbor spacings, pair correlation and number variance. The picture organizes the Montgomery–Dyson connection; it proves nothing about the Riemann hypothesis.

Third, *self-similarity*. A continued fraction contains a copy of itself after every term, which is the idea behind Ford circles, the Stern–Brocot tree, Minkowski's question-mark function, the devil's staircase, continued-fraction Cantor sets (whose dimension we recompute as 0.531280506277), the Farey bulbs of the Mandelbrot set and the Fibonacci word fractal. A chi-square test of π's first 10,000 terms against the Gauss–Kuzmin law gives $p = 0.33$ (12 degrees of freedom).

Returning to six open questions (Chapter 15), we expand π to twelve million certified terms, search its convergents for primes to $n = 20{,}000$, and prove that for a random real number the *expected* number of prime denominators among the first N convergents is $2 \ln 2 \cdot \ln N + O(1)$, with exactly the heuristic's constant; the almost-everywhere version remains open. [Added 2026-11-05]

A second look (Chapter 16) closes that gap: for almost every real number the count of prime convergent denominators among the first N convergents is $2 \ln 2 \cdot \ln N$ plus an error of order at most $(\ln N)^{1/2} (\ln \ln N)^{3/2+\varepsilon}$, so Heuristic 8.3 holds as a theorem for almost every

number, though not, as far as anyone knows, for π. It also proves that e's convergent denominators modulo a prime ℓ repeat with period dividing 6ℓ, shows that the top eigenvalue of the correlation matrix grows faster than any power of $\ln m$ and that its spectrum holds near-copies of small gcd matrices, explains the 171-note temperament by three small commas, and adds two tests of π's terms. [Added 2026-11-06] A third look (Chapter 17), from Teddy Rook's notes, settles the growth of the correlation matrix's extreme eigenvalues: the logarithms of the largest eigenvalue and of the reciprocal of the smallest are both $(2 + o(1))\sqrt{\ln m / \ln \ln m}$. It proves that the eigenvalue distribution has a limit, and finds the limiting mean of $1/\lambda$. For e it proves that the denominators change sign modulo an odd prime ℓ every 3ℓ steps, so the period is exactly 6ℓ, and that a convergent denominator of e can be prime only at $n = 2$ or $n \equiv 1 \pmod 3$. The search of π's convergents, extended to $n = 50{,}000$ with a sieve that runs the recurrence modulo small primes, finds four new probable-prime denominators, at $n = 25{,}332$, 27,651, 32,394 and 37,721, with 13,017 to 19,465 digits, and a new probable-prime numerator at $n = 42{,}037$, with 21,667 digits. The denominators come in a cluster, and the prime convergents lean toward one note; both look like chance. Primes are far more plentiful among the intermediate fractions that belong to π's largest terms. With the new primes, π's own count of prime convergents favors the heuristic's constant $2 \ln 2$ over the value without its correction factor by about 30 to 1, and every small prime divides its denominators at the typical rate. [Added 2026-11-10] Another chapter audits what is classical, what is known to specialists, what is a new presentation, and what is new. The mathematics is classical or known; what we did not find elsewhere is the reading of a number's continued fraction as a complete musical form, the Suite as an artwork, the prime-convergent experiment at this scale with an explicit constant, and the single spacing spectrum. Every number in the text was computed by the scripts listed in the appendices. A further chapter (Chapter 18), also from Teddy Rook's notes, adds physics and astronomy to the Connections diagram and asks where these structures appear in both the very small and the very large. The inverse-square law gives a planet and Bohr's electron the same law of periods, and its hidden symmetry gives the shell sizes $2n^2$. The continued fraction of a ratio of orbital periods detects resonance by a large term: the locked moons of Jupiter and two resonant chains of exoplanets stand out, while the planets of the solar system taken together do not ($p = 0.14$). The inverse of the correlation matrix has a sign pattern given by Liouville's function, so its two extreme eigenvalues are Perron eigenvalues of two non-negative matrices built from common multiples; their ratio is computed to $m = 3 \times 10^7$ and stays near 2.57 for the gcd matrix, and whether it has a limit is open. The energy bands of an electron on a lattice in a magnetic field split in the same counts as a number's rhythm. Kepler's figure of the conjunctions of Jupiter and Saturn is read as a continued fraction, and his planetary harmonies are tested against chance: those of single planets are not established, and those of pairs hold for the six planets he knew but not for the planets found later. [Added 2026-11-10]

Keywords: continued fractions; Sturmian words; Christoffel words; Euclidean rhythms; musical form; coprimality; prime numbers; Lévy's constant; pair correlation; random matrices; quasicrystals; fractals; Gauss–Kuzmin law.

How to read this work

Each chapter is written to stand alone. Chapters 1–3 build the main tool. Chapters 4–7 follow it into music, coprimality and the primes, mostly retelling classical results with proofs or precise references. Chapters 8 and 9 are the two research threads, with new computations. Chapter 10 is about self-similarity and fractals, Chapter 11 about the music, Chapter 12 about the wider Connections diagram. Chapter 13 says what is new and what is not; Chapter 14 discusses limits and open questions, Chapter 15 reports further work on those questions [Added 2026-11-05], Chapter 16 a second round of that work [Added 2026-11-06], and Chapter 17 a third, with the digits of a new probable prime in Appendix D [Added 2026-11-10], and Chapter 18 carries the same structures into physics and astronomy [Added 2026-11-10]. The revision history below lists every addition.

Results that are classical carry a citation in their heading or the sentence that introduces them. Short standard proofs are included so that the text can be read by a strong undergraduate with calculus and a first course in number theory; longer ones are cited. A "Heuristic" is an argument that predicts a number without proving it, and we always say so. Interactive versions of most figures are on the companion pages [114, 115].

Every number in this monograph was computed or checked by the scripts in Appendix C. Numbers that come from data, from the terms of π to the last decimal of a p-value, are inserted into the text automatically from the scripts' output; small numbers that appear in the prose (such as $355/113$ or $292 \times 113 = 32{,}996$) are asserted by the same scripts. All figures were drawn for this work from computed data.

Revision history

This work is updated as new results come in. Every addition carries the date it was published, in a small gray note such as [Added 2026-11-06]; text without a note belongs to the first version.

2026-11-04
First version: Chapters 1–14 and Appendices A–C.

2026-11-05
Chapter 15, "Working on the open questions"; the status notes of Section 14.3; the abstract's paragraph on the open questions; six rows of the audit in Chapter 13; the long expansions in Appendix B and six scripts in Appendix C.

2026-11-06
Chapter 16, "A second look at the open questions": Heuristic 8.3 proved for almost every real number; the period of e's denominators modulo a prime; growth bounds and near-copies in the spectrum of the correlation matrix; why the 171-note temperament wins; two further tests of π's terms. Dated update notes in Section 14.3, in Chapter 15 and in the audit; one script in Appendix C.

2026-11-10
Chapter 17, "A third look at the open questions", from Teddy Rook's notes: the growth of the largest and smallest eigenvalues of the correlation matrix; a limiting law for its spectrum; the sign flip and forced divisors of e's convergents; π's convergents searched to $n = 50{,}000$, with four new probable-prime denominators, the eleventh to fourteenth, and a sixteenth probable-prime numerator, whose digits are in Appendix D; the cluster they form and the notes of the prime convergents; the rhythm sieve, prime intermediate fractions and the family of the opening guess; what the new primes say about the questions (π's own constant, the sieve survivors and the voices test). The chapter now follows Teddy Rook's finished text. A reader's map of what is proved (Section 13.2). Dated update notes in Chapters 5, 7, 8, 13, 14, 15 and 16, in the abstract and in the bibliography; new scripts in Appendix C.

2026-11-10
Chapter 18, "Large and small: the same patterns in atoms and in orbits", from Teddy Rook's notes: Kepler's third law in the atom and the symmetry behind the shell sizes; orbital resonance read through continued fractions, with a test against random ratios; the sign mirror and the harmonic form of the correlation matrix (Propositions 18.2 and 18.4) and the ratio of its two extreme eigenvalues to $m = 3 \times 10^7$; the energy bands of Harper's equation at the convergents of the Suite's numbers; conjunction figures (Proposition 18.6) and Kepler's harmonies tested. Fourteen rows of the audit in Chapter 13, three rows of the reader's map, forward references in Chapters 3, 4, 5, 12 and 14, twenty references and the scripts in Appendix C.

Contents

Abstract i

How to read this work iii

Revision history iv

1 Introduction: a guess about pi 1
- 1.1 The guess . . . 1
- 1.2 Why the guess works . . . 1
- 1.3 Three directions . . . 2
- 1.4 What is new and what is not . . . 3
- 1.5 Conventions . . . 3

2 Continued fractions: the machine 4
- 2.1 The algorithm . . . 4
- 2.2 Convergents and the recurrence . . . 5
- 2.3 How close the convergents are . . . 5
- 2.4 Patterns in the terms . . . 6
- 2.5 The statistics of a typical number . . . 7

3 The rhythm of a number 8
- 3.1 Playing a number as a beat . . . 8
- 3.2 Sturmian and Christoffel words . . . 8
- 3.3 The nested-block theorem . . . 9
- 3.4 Pi's levels, and the beat where the surprise lands . . . 10
- 3.5 From levels to musical form . . . 11
- 3.6 Three gaps . . . 12

4 Music from the same ladder 14
- 4.1 Tuning: the fifth and its convergents . . . 14
- 4.2 Euclidean rhythms . . . 15
- 4.3 Polyrhythm and coprimality . . . 16
- 4.4 Mode locking and the circle map . . . 16

5 Coprimality and $6/\pi^2$ 18
- 5.1 The orchard . . . 18
- 5.2 Coprime means uncorrelated . . . 18
- 5.3 Hydrogen's $1/n^2$: a shared formula, not shared physics . . . 20

6 Primes in sound and space 22

6.1 Primes as notes, and the bias of neighbors . . . 22
6.2 A sunflower seeded with primes . . . 22
6.3 Diffraction: a quasicrystal and the primes . . . 24
6.4 Primes as a sum of waves . . . 24

7 Tools that hunt primes **27**
7.1 Factoring with the continued fraction of $\sqrt{N}$ . . . 27
7.2 Mersenne necklaces and Fermat's little theorem . . . 28
7.3 The prime hunt . . . 28

8 Thread (a): prime convergents **31**
8.1 The question . . . 31
8.2 A heuristic with an explicit constant . . . 31
8.3 The test on random reals . . . 33
8.4 Is pi typical? . . . 34
8.5 Do the site's tools help? . . . 36
8.6 What is known, and what is not . . . 36

9 Thread (b): one spectrum, from crystal to randomness **37**
9.1 The question . . . 37
9.2 The point sets . . . 37
9.3 Nearest-neighbor spacings . . . 38
9.4 Pair correlation . . . 39
9.5 Number variance: a scale of rigidity . . . 39
9.6 The spectrum, and Dyson's bridge . . . 40
9.7 What this does and does not show . . . 41

10 Self-similarity and fractals **42**
10.1 Teddy Rook's picture . . . 42
10.2 Ford circles . . . 43
10.3 The Stern–Brocot tree . . . 44
10.4 Minkowski's question-mark function . . . 44
10.5 The devil's staircase . . . 45
10.6 Cantor dust: numbers with small terms . . . 46
10.7 The Mandelbrot set's Farey bulbs . . . 47
10.8 The Fibonacci word fractal . . . 47
10.9 Six numbers, six pictures . . . 47
10.10 Is pi typical? . . . 49

11 Composition: the Continued-Fraction Suite as a method **51**
11.1 The work . . . 51
11.2 The method . . . 51
11.3 What the method gives a composer . . . 53

12 Connections: music, patterns, math and chemistry **54**
12.1 Music and math: ratios, clocks and rhythms . . . 54
12.2 Music and chemistry: octaves of elements . . . 54
12.3 Music and patterns: sand that draws sound . . . 54
12.4 Math and patterns, patterns and chemistry . . . 55

12.5 Math and chemistry: why 2, 8, 18, 32 . . . 55
12.6 All four: waves and symmetry . . . 55
12.7 Where the Rhythm of Numbers sits . . . 56

13 What is new here, and what is re-found 57
13.1 The audit . . . 57
13.2 A reader's map of what is proved . . . 63
13.3 The two questions the owner asked . . . 64

14 Discussion 66
14.1 What the work shows . . . 66
14.2 Limitations . . . 66
14.3 Open questions . . . 67
14.4 Future work . . . 68

15 Working on the open questions 69
15.1 Twelve million terms of pi . . . 69
15.2 Question 1: the prime convergents of pi . . . 70
15.3 Question 2: Fibonacci and Pell primes . . . 72
15.4 Question 3: a theorem in the model . . . 73
15.5 Question 4: is pi typical? . . . 77
15.6 Question 5: the spectrum of the correlation matrix . . . 78
15.7 Question 6: ranking the equal temperaments . . . 82
15.8 Status after this work . . . 84

16 A second look at the open questions 86
16.1 Question 3: the heuristic for almost every number . . . 86
16.2 Question 1: e modulo a prime . . . 89
16.3 Question 5: the correlation matrix again . . . 90
16.4 Question 6: why 171 wins . . . 93
16.5 Question 4: two more tests of pi . . . 93
16.6 Status after the second look . . . 94

17 A third look at the open questions 95
17.1 Question 5: the largest eigenvalue, settled to leading order . . . 96
17.2 Question 5: the smallest eigenvalue mirrors the largest . . . 98
17.3 Question 5: the bulk has a limiting law . . . 100
17.4 Question 1: the number e, completed . . . 102
17.5 Question 1: five new prime convergents for pi . . . 104
17.6 Finding prime convergents faster . . . 108
17.7 What the new primes say about the six questions . . . 112
17.8 Status after the third look . . . 114

18 Large and small: the same patterns in atoms and in orbits 117
18.1 The atom and the solar system are one problem . . . 117
18.2 Resonance as rhythm . . . 118
18.3 The mirror in the correlation matrix . . . 120
18.4 The butterfly of a number . . . 123
18.5 Kepler, continued . . . 124

18.6 What is known and what is ours 128

A Algorithms **130**

B Computational methods **133**

C Reproducibility **135**

D The digits of the probable primes **138**

Bibliography **156**

List of Figures

1.1 The starting guess $\pi \approx 3 + k/(7k+1)$ . . . 2

3.1 The nested blocks of π's rhythm . . . 11
3.2 The three-gap theorem for $\pi - 3$ . . . 13

4.1 The fifth in equal temperaments . . . 15
4.2 Euclidean rhythms as necklaces . . . 16
4.3 Arnold tongues of the circle map . . . 17

5.1 The orchard of visible lattice points . . . 19
5.2 The pulse-train correlation matrix . . . 20
5.3 Two roads to $\pi^2/6$ . . . 21

6.1 The last digits of consecutive primes . . . 23
6.2 Pi's sunflower, seeded with primes . . . 23
6.3 Diffraction of π's rhythm and of the primes . . . 25
6.4 The prime staircase rebuilt from zeta zeros . . . 26

7.1 The prime hunt . . . 30

8.1 Convergent denominators avoid small primes . . . 33
8.2 Prime denominators among convergents . . . 34
8.3 Where π falls among random reals . . . 35

9.1 Nearest-neighbor spacings . . . 38
9.2 Pair correlation of zeta zeros . . . 39
9.3 Number variance: one scale of rigidity . . . 40
9.4 The zeros diffract at logarithms of prime powers . . . 41

10.1 The nested tail of π . . . 43
10.2 Ford circles near $\pi - 3$ . . . 44
10.3 The Stern–Brocot tree . . . 45
10.4 Minkowski's question-mark function . . . 45
10.5 The devil's staircase of the circle map . . . 46
10.6 Cantor dust of continued fractions with terms 1 and 2 . . . 47
10.7 The Farey bulbs of the Mandelbrot set . . . 48
10.8 The Fibonacci word fractal and its relatives . . . 48
10.9 The Suite's numbers as square-cutting ladders . . . 49
10.10 Pi's terms against the Gauss–Kuzmin law . . . 50
10.11 Khinchin and Lévy for π . . . 50

11.1 The forms of the six movements 52

12.1 Chladni figures 55

15.1 Pi's prime convergents to $n = 20{,}000$ 71
15.2 A theorem checked by simulation 77
15.3 Twelve million terms against Gauss–Kuzmin 79
15.4 Khinchin, Lévy and the records of pi 79
15.5 The spectrum of the arithmetic correlation matrix 82
15.6 Prime sizes do not stand out 84

List of Tables

2.1 The first convergents of π . . . 6

3.1 The levels of the rhythm of $\pi - 3$ . . . 11

4.1 Euclidean rhythms . . . 16

7.1 Continued-fraction factoring of $2^{32} + 1$ . . . 28
7.2 The prime hunt in rhythm prefixes . . . 29

8.1 Divisibility of convergent denominators by small primes . . . 32
8.2 Prime convergents of five constants . . . 35

11.1 The six movements . . . 52

13.1 The prior-art audit . . . 57
13.2 A reader's map of what is proved . . . 63

15.1 Pi's prime convergents against the heuristic . . . 70
15.2 Fibonacci and Pell primes against the heuristic . . . 73
15.3 The exact expectation against simulation . . . 76
15.4 The largest term against Galambos's law . . . 78
15.5 The spectrum of the correlation matrix . . . 81
15.6 The largest eigenvalue for large m . . . 81
15.7 The best equal temperaments, 5 to 200 notes . . . 83
15.8 Prime against composite sizes . . . 84
15.9 Status of the six questions . . . 85

16.1 Overlaps of the sets E_q and E_r . . . 89
16.2 The top eigenvalue against an empirical fit . . . 91
16.3 Localized eigenvectors and the spacing ratio . . . 92
16.4 Status of the six questions after the second look . . . 94

17.1 The top eigenvalue of C_m and its bounds . . . 98
17.2 The two extreme eigenvalues of C_m . . . 100
17.3 The new prime denominators and numerator of π . . . 105
17.4 Table 15.1 continued . . . 107
17.5 Pi's prime denominators to $n = 50{,}000$ . . . 107
17.6 The notes of π's prime convergents . . . 108
17.7 Prime intermediate fractions of π . . . 112
17.8 Pi's prime convergents against the heuristic with and without $\zeta(2)$ . . . 113

17.9 The voices test on two million convergents of π 114
17.10 Status of the six questions after the third look 115

18.1 Neighbouring period ratios (outer over inner): the expansion, the nearest fraction with denominator at most 8, the normalized error $q^2|r - p/q|$ of Table 2.1, and the term that follows p/q. The last line of each block compares the mean normalized error with that of random ratios drawn log-uniformly from $[1.25, 3]$. 119
18.2 The two ends of the spectrum and their ratio $(1/\lambda_{\min})/\lambda_{\max}$, for the correlation matrix C_m and, in the last column, for the gcd matrix G_m. 122
18.3 The gcd matrix on the numbers up to 10^7 whose prime factors are all at most y. With a fixed set of primes both ends have the same ceiling, $\prod_{p \le y} M_p$. 123
18.4 Energy bands of Harper's equation at convergents p/q: the sizes of the groups into which the widest gaps divide the q bands, and the total width of the bands against Thouless's estimate $32G/(\pi q) = 9.33/q$ (G is Catalan's constant). 124
18.5 Conjunction figures from mean periods. Each line gives the step α between meetings, its expansion, and the cycles that the convergents produce. An opposition of an outer planet is a conjunction of the Earth with it, seen from the Sun. . . . 125
18.6 Kepler's intervals for the single planets, and by how much the ratio of fastest to slowest angular speed misses each one, in cents (a semitone of the piano is 100 cents). The orbits of 1600 and 2000 are mean orbits from one formula valid from 3000 BC to 3000 AD. 126

C.1 The scripts 135

CHAPTER 1

Introduction: a guess about pi

1.1 The guess

This work began with a guess written on a scrap of paper. Pi is a little more than 3. The leftover, $\pi - 3 = 0.14159\ldots$, is a little less than $1/7$. So try

$$\pi \approx 3 + \frac{k}{7k+1} \tag{1.1}$$

for some whole number k. When k is large, $k/(7k+1)$ is just under $1/7$; when k is small it is further below. Somewhere in between it might hit $\pi - 3$ well. Teddy Rook tried values of k one at a time, and Fig. 1.1 shows what happens. The error falls steadily, reaches a sharp minimum at $k = 16$, and then rises again. At $k = 16$ the guess gives

$$3 + \frac{16}{113} = \frac{355}{113} = 3.14159292\ldots,$$

which differs from π by 2.67e-7. That is six correct decimal places from a fraction with a three-digit denominator.

The fraction 355/113 is famous. The Chinese mathematician Zu Chongzhi recorded it in the fifth century AD; it is called *Milü*, "the close ratio" [91, p. 281]; see also Berggren et al. [13]. Two other values of k give classical fractions too: $k = 10$ gives 223/71, the lower bound that Archimedes proved, and the limit $k \to \infty$ gives 22/7, his upper bound. So the guess (1.1) quietly contains three classical fractions for π.

1.2 Why the guess works

The guess works because of a small piece of algebra:

$$\frac{k}{7k+1} = \frac{1}{7 + \dfrac{1}{k}}. \tag{1.2}$$

The right-hand side is a *continued fraction* stopped after two steps. Solving $1/(7+1/k) = \pi - 3$ for k gives

$$k^* = \frac{\pi - 3}{1 - 7(\pi - 3)} = 15.99659,$$

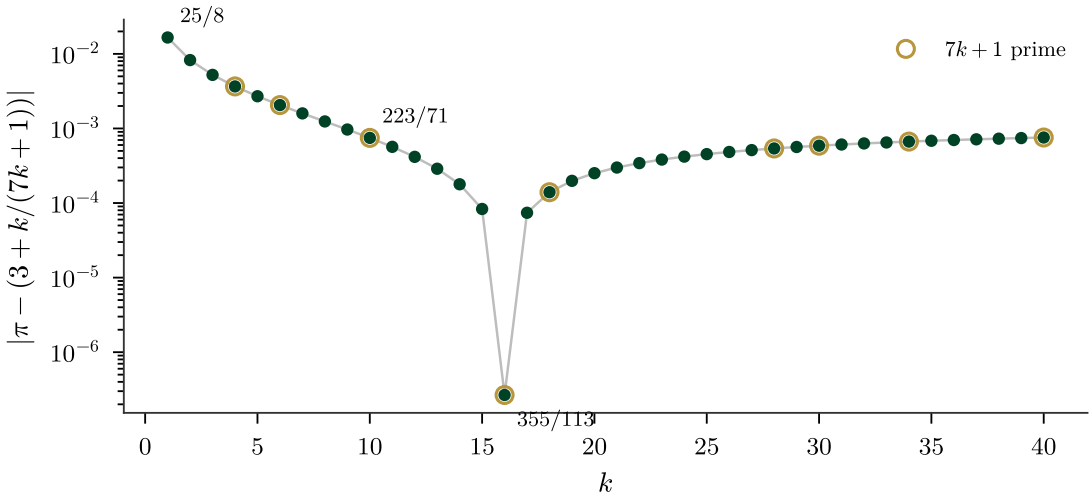


Figure 1.1. The starting guess. The error $|\pi - (3 + k/(7k + 1))|$ for $k = 1, \ldots, 40$ on a logarithmic scale. The sharp dip at $k = 16$ is 355/113; $k = 10$ gives Archimedes' lower bound 223/71 and $k = 1$ gives 25/8. Gold rings mark the k for which $7k + 1$ is prime.

not a whole number but very close to 16. Rounding k^* to the nearest integer is exactly the next step of the continued fraction of π, and the reason the rounding error is so small is that the step after that is huge. Writing out the first few steps,

$$\pi = 3 + \cfrac{1}{7 + \cfrac{1}{15 + \cfrac{1}{1 + \cfrac{1}{292 + \cdots}}}}, \tag{1.3}$$

and $k^* = 15 + 1/(1 + 1/(292 + \cdots)) \approx 16$. The terms of π's continued fraction begin

$$\pi = [3;\ 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, 1, 84, \ldots].$$

Chapter 2 develops this machinery from the beginning.

1.3 Three directions

The rest of this work follows the continued fraction of π, and of a few other numbers, in three directions.

Rhythm and music. Play a number α between 0 and 1 as a beat: on beat n, sound a hit if $\lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor$, and stay silent otherwise. For $\alpha = \pi - 3$ the hits come every 7 beats, with an occasional gap of 8. Listen longer and the gaps of 8 come in a pattern of their own, and that pattern has a pattern, and so on. Chapter 3 proves that the levels of this rhythm are exactly the levels of the continued fraction: the terms $7, 15, 1, 292, \ldots$ are repeat counts. Read as bar, phrase, section and movement, they give a musical form; Chapter 4 connects the same machinery to tuning, Euclidean rhythms and mode locking, and Chapter 11 describes the *Continued-Fraction Suite*, six piano pieces Teddy Rook composed this way.

Primes. The algorithm that builds a continued fraction is Euclid's algorithm, and Euclid's algorithm is the oldest test for whether two numbers share a prime factor. Chapters 5–7 follow that link through coprime pairs and $6/\pi^2$, the patterns of primes in residues, spirals and diffraction, the zeros of the zeta function, factoring with continued fractions, and a hunt for primes in rhythms. Two questions from the companion site get chapters of their own. Does π's continued fraction have infinitely many prime denominators (Chapter 8)? And what is the right way to picture the famous match between zeros of the zeta function and energy levels of heavy nuclei (Chapter 9)?

Self-similarity. Teddy Rook pictured π as "a number, then a smaller number after it, then a smaller one after that, and so on forever." Chapter 10 makes this precise: the tail of a continued fraction is again a continued fraction, built by the same rule. That one fact underlies a family of classical fractals, and we draw each of them for π.

1.4 What is new and what is not

Almost all of the mathematics here is classical, and we cite it as such. The rhythm of a number is a Sturmian word [88]; the even spreading of beats is the Euclidean rhythm of Bjorklund and Toussaint [132]; Ford circles, the Stern–Brocot tree and the question-mark function date from 1858 to 1938. Chapter 13 carries out a careful audit, idea by idea. In brief, the parts we did not find elsewhere are: the systematic reading of a number's whole continued fraction as nested musical form (bar, phrase, section, movement), together with its use as a compositional method; the Suite itself, which is new as an artwork whatever the prior art on its method; the explicit constant $2\ln 2$ in the heuristic count of prime convergents and its test on hundreds of random reals; and the placement of several point sets on a single spacing spectrum. Even for these we say "we did not find prior work," not "this is new," wherever the literature is too wide to search completely.

1.5 Conventions

Throughout, $\lfloor x \rfloor$ is the floor of x and $\{x\} = x - \lfloor x \rfloor$ its fractional part. A continued fraction is written $[a_0; a_1, a_2, \dots]$. For a number α between 0 and 1 we often write $\alpha = [0; a_1, a_2, \dots]$ and work with $\pi - 3$, $\varphi - 1$, $\sqrt{2} - 1$, $e - 2$ and $\log_2(3/2)$, the five numbers of the Suite. We count beats from 1 when we state the rhythm rule, but we index positions in a beat string from 0, as the companion site does; Section 3.4 says exactly what each count means. "Prime" for numbers with more than a few hundred digits means *probable prime* by the Baillie–PSW test (Appendix B).

CHAPTER 2

Continued fractions: the machine

This chapter collects the facts about continued fractions that the rest of the work uses. All of them are classical; good sources are Khinchin's short book [71] and Chapter X of Hardy and Wright [56]. We include the short proofs because the identities in them are the ones that turn into rhythms later.

2.1 The algorithm

Take a real number x. Write it as its whole part plus a leftover, and turn the leftover upside down:

$$x = a_0 + \frac{1}{x_1}, \qquad a_0 = \lfloor x \rfloor, \quad x_1 = \frac{1}{x - a_0} > 1. \tag{2.1}$$

Now do the same to x_1, then to x_2, and so on: $x_n = a_n + 1/x_{n+1}$ with $a_n = \lfloor x_n \rfloor$. The whole numbers $a_0, a_1, a_2, \ldots$ are the *terms* (or partial quotients) of x, and the real numbers $x_n > 1$ are its *complete quotients*. The process stops exactly when some leftover is zero, which happens exactly when x is rational. Writing the steps inside one another gives

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{x_n}}}} = [a_0; a_1, \ldots, a_{n-1}, x_n].$$

For π the first complete quotients are $x_1 = 7.06251$, $x_2 = 15.9966$, $x_3 = 1.00342$ and $x_4 = 292.6346$. The size of x_4 is the famous 292 at work.

When $x = a/b$ is rational, the algorithm is Euclid's algorithm in disguise: dividing a by b gives quotient a_0 and remainder r_1, then b by r_1 gives a_1, and so on. Running it on 355 and 113 gives the quotients $3, 7, 16$, so $355/113 = [3; 7, 16]$, and the last nonzero remainder is $\gcd(355, 113) = 1$.

For $0 < x < 1$ the step from x_n to x_{n+1} is one iteration of the *Gauss map*

$$T(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor, \qquad 0 < x < 1, \tag{2.2}$$

which reads off the term $\lfloor 1/x \rfloor$ and keeps the leftover. Most of the statistics in this work are statements about long runs of T.

2.2 Convergents and the recurrence

Stopping after the term a_n gives a fraction $p_n/q_n = [a_0; a_1, \dots, a_n]$, the nth *convergent*. Its numerator and denominator obey a two-term recurrence that is the engine of everything that follows.

Proposition 2.1 (the recurrence; see 71, Thm. 1)**.** *Set $p_{-1} = 1$, $q_{-1} = 0$, $p_0 = a_0$, $q_0 = 1$, and for $n \geq 1$*

$$p_n = a_n p_{n-1} + p_{n-2}, \qquad q_n = a_n q_{n-1} + q_{n-2}. \tag{2.3}$$

Then $[a_0; a_1, \dots, a_n] = p_n/q_n$ for every n, and more generally $[a_0; a_1, \dots, a_{n-1}, t] = (tp_{n-1} + p_{n-2})/(tq_{n-1} + q_{n-2})$ for every real $t > 0$.

Proof. In matrix form, (2.3) says

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix}. \tag{2.4}$$

A matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ acts on numbers by $t \mapsto (at + b)/(ct + d)$, and products of matrices act by composition. The matrix $\left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$ sends t to $a + 1/t$, which is one layer of a continued fraction. Composing the layers gives the general statement, and $t = a_n$ in the last layer gives the first. □

Taking determinants in (2.4), each factor has determinant -1, so

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}. \tag{2.5}$$

Two consequences will matter again and again. First, any common divisor of p_n and q_n divides ± 1, so every convergent is in lowest terms. Second, the same is true of neighboring denominators: $\gcd(q_n, q_{n-1}) = 1$. In the language of Chapter 3, the length of a rhythmic block and its number of hits never share a factor, and neither do the lengths of neighboring blocks.

2.3 How close the convergents are

Proposition 2.2 (the error of a convergent; 71, Thms. 9 and 13)**.** *If x is irrational, then*

$$x - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n \left(x_{n+1} q_n + q_{n-1}\right)}, \tag{2.6}$$

and therefore

$$\frac{1}{q_n(q_n + q_{n+1})} < \left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2}. \tag{2.7}$$

Proof. By Proposition 2.1 with $t = x_{n+1}$, $x = (x_{n+1}p_n + p_{n-1})/(x_{n+1}q_n + q_{n-1})$. Subtract p_n/q_n, put everything over a common denominator, and use (2.5). For the bounds, $a_{n+1} < x_{n+1} < a_{n+1} + 1$ gives $q_{n+1} < x_{n+1}q_n + q_{n-1} < q_{n+1} + q_n$. □

The formula (2.6) explains the 292. The error of $p_3/q_3 = 355/113$ is about $1/(113 \cdot x_4 \cdot 113)$, and $x_4 = 292.6346$ is large, so the error is tiny. A big term is a sign that the convergent just before it is unusually good. Table 2.1 lists the first convergents of π with their errors and the normalized error $q_n^2 |\pi - p_n/q_n|$, which by (2.6) is roughly $1/a_{n+1}$.

Convergents are not only close; they are the best possible for their size.

Table 2.1. The first convergents of π. The last column is $q_n^2\,|\pi - p_n/q_n|$, which is close to $1/a_{n+1}$; it is smallest just before the large term 292.

n	a_n	p_n	q_n	$\lvert\pi - p_n/q_n\rvert$	$q_n^2\,\lvert\pi - p_n/q_n\rvert$
0	3	3	1	0.142	0.1416
1	7	22	7	0.001,26	0.06196
2	15	333	106	8.32×10^{-5}	0.9351
3	1	355	113	2.67×10^{-7}	0.003406
4	292	103,993	33,102	5.78×10^{-10}	0.6332
5	1	104,348	33,215	3.32×10^{-10}	0.3659
6	1	208,341	66,317	1.22×10^{-10}	0.5381
7	1	312,689	99,532	2.91×10^{-11}	0.2887
8	2	833,719	265,381	8.72×10^{-12}	0.6138

Theorem 2.3 (best approximation; Lagrange, see 71, Thms. 16–17)**.** *If $1 \le q < q_{n+1}$, then $|qx - p| \ge |q_n x - p_n|$ for every integer p, with equality only for $p/q = p_n/q_n$. In particular no fraction with denominator smaller than q_{n+1} is closer to x than p_n/q_n.*

For π this means that no fraction with denominator below 33,102 beats 355/113; our script checks it directly by scanning every denominator. A partial converse is due to Legendre.

Theorem 2.4 (Legendre; 56, Thm. 184)**.** *If p/q is in lowest terms and $|x - p/q| < 1/(2q^2)$, then p/q is a convergent of x.*

We confirmed it for π by brute force: among all fractions with denominator up to 3,000, the only ones within $1/(2q^2)$ of π are convergents. Legendre's theorem reappears in Chapter 10 as a statement about circles.

2.4 Patterns in the terms

Some numbers have continued fractions with simple patterns:

$$\varphi = \tfrac{1+\sqrt{5}}{2} = [1;1,1,1,\dots], \qquad \sqrt{2} = [1;2,2,2,\dots],$$
$$e = [2;1,2,1,1,4,1,1,6,1,\dots], \quad \log_2 \tfrac{3}{2} = [0;1,1,2,2,3,1,5,2,23,2,2,1,1,55,\dots].$$

For φ the recurrence (2.3) with all terms equal to 1 is the Fibonacci recurrence, and the denominators are $1,1,2,3,5,8,13,21,34,55,89,144,\dots$. For $\sqrt{2}$ they are the Pell numbers $1,2,5,12,29,70,169,408,985,\dots$. The pattern of e, with $2,4,6,\dots$ in every third place, is due to Euler; our scripts verify it for the first 3,900 terms. The number $\log_2(3/2)$, which measures a musical fifth (Chapter 4), shows no pattern, and neither does π.

Theorem 2.5 (Euler and Lagrange; 56, Thms. 176–177)**.** *The continued fraction of x is eventually periodic if and only if x is a quadratic irrational, that is, an irrational root of a quadratic equation with integer coefficients.*

So φ and $\sqrt{2}$ have repeating terms, while π and e, which are not quadratic irrationals, do not. This dichotomy returns in every chapter: repeating terms give exactly self-similar rhythms, music and pictures; non-repeating terms give the same *rule* at every level but a new *shape*.

2.5 The statistics of a typical number

Pick x at random between 0 and 1. What do its terms look like? Gauss conjectured, and Kuzmin proved in 1928 [77], with a sharper version by Lévy [84], that for almost every x the terms follow one fixed law.

Theorem 2.6 (Gauss–Kuzmin law)**.** *For almost every $x \in (0,1)$, the proportion of indices $n \le N$ with $a_n = k$ tends, as $N \to \infty$, to*

$$P(k) = -\log_2\!\left(1 - \frac{1}{(k+1)^2}\right). \tag{2.8}$$

So about 41.5% of the terms are 1, 17.0% are 2 and 9.3% are 3. Behind the law is the *Gauss measure* $d\mu = dx/((1+x)\ln 2)$, which the Gauss map (2.2) leaves invariant, and the fact that T is ergodic for it; the frequency of $a_n = k$ is the Gauss measure of the interval $(1/(k+1), 1/k]$, which is (2.8). Two more laws follow from the same ergodic theory.

Theorem 2.7 (Khinchin 1935; Lévy 1936)**.** *For almost every x,*

$$(a_1 a_2 \cdots a_n)^{1/n} \to K_0 = 2.685452001\ldots \qquad \text{and} \qquad \frac{\ln q_n}{n} \to \frac{\pi^2}{12\ln 2} = 1.18657\ldots \tag{2.9}$$

The first limit is Khinchin's constant [72]; the second is Lévy's constant [85], and it says that the denominators grow like $e^{1.18657n} \approx 3.27582^n$. Lévy's theorem is the key input to the prime-convergent heuristic of Chapter 8. No one knows whether π obeys any of these laws; nobody has even proved that its terms are unbounded. What we can do is count. From π computed to 11,600 digits, our scripts extract its first 10,000 terms (two precisions agree on every one), and find 4,206 ones (42.1%), a geometric mean of 2.6632 and $\ln q_{10000}/10000 = 1.17941$. Chapter 10 turns these counts into a formal test.

CHAPTER 3

The rhythm of a number

3.1 Playing a number as a beat

Let $0 < \alpha < 1$. Imagine a drummer who keeps a running total, adding α on every beat, and strikes the drum whenever the total passes a whole number. On beat n the total goes from $n\alpha$ to $(n+1)\alpha$, so the rule is:

$$\text{beat } n \text{ is a hit} \iff \lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor , \qquad n = 1, 2, 3, \dots \tag{3.1}$$

We write a hit as `x` and a rest as `.`, or as 1 and 0. The resulting infinite word $c_\alpha = c_\alpha(1)\, c_\alpha(2)\, c_\alpha(3) \cdots$ is called the *characteristic word* of α. Since the total grows by less than 1 per beat, there is at most one hit per beat, and after L beats there have been exactly $\lfloor (L+1)\alpha \rfloor$ hits. The hits are spread as evenly as the whole numbers allow: the density of hits is α, and any two stretches of the same length differ by at most one hit. For $\alpha = \pi - 3$ the first 40 beats are

```
......x......x......x......x......x.....
```

a hit every 7 beats, because $1/\alpha = 7.06251$ lies between 7 and 8. The leftover $0.0625 \approx 1/16$ means that roughly every 16th gap is an 8 instead of a 7. That is already the start of a hierarchy, and this chapter shows that it continues forever, level by level, exactly as the continued fraction says.

3.2 Sturmian and Christoffel words

Words like c_α have a long history in combinatorics, where they are called *Sturmian words*; the standard reference is Chapter 2 of Lothaire's book, written by Berstel and Séébold [88], and the books of Allouche and Shallit [4, Ch. 9] and Berstel, Lauve, Reutenauer and Saliola [15].

Definition 3.1. A binary word is *balanced* if any two factors (blocks of consecutive letters) of the same length contain numbers of 1s that differ by at most 1. An infinite word is *Sturmian* if it is balanced and not eventually periodic. A *mechanical word* of slope α and intercept ρ has nth letter $\lfloor (n+1)\alpha + \rho \rfloor - \lfloor n\alpha + \rho \rfloor$; the characteristic word is the case $\rho = 0$. For a rational slope p/q in lowest terms, one period of the mechanical word with $\rho = 0$, read as $\lfloor (i+1)p/q \rfloor - \lfloor ip/q \rfloor$ for $i = 0, \dots, q-1$, is the *lower Christoffel word* of slope p/q.

For irrational α, the mechanical words of slope α are exactly the Sturmian words [88, Ch. 2]. A Christoffel word is the finite, rational version: it spreads p hits over q beats as evenly as possible. In music these are the maximally even rhythms of Clough and Douthett [29], and, up to rotation, the Euclidean rhythms of Chapter 4.

3.3 The nested-block theorem

The construction that the companion site uses to play a number builds the rhythm from its continued fraction rather than from the floor rule (3.1). It is the classical construction of *standard words*.

Definition 3.2 (standard words)**.** Let $\alpha = [0; a_1, a_2, \dots]$. Set $s_{-1} = \texttt{x}$, $s_0 = \texttt{.}$, and

$$s_1 = s_0^{a_1 - 1}\, s_{-1}, \qquad s_n = s_{n-1}^{a_n}\, s_{n-2} \quad (n \geq 2), \tag{3.2}$$

where w^a means a copies of w in a row.

In words: the level-1 block is $a_1 - 1$ rests and a hit; each later block is *(term) copies of the block just built, followed by the block before it.* For $\pi - 3$, s_1 is the 7-beat block `.....x`; s_2 is fifteen of those and one extra rest, 106 beats; $s_3 = s_2 s_1$ is 113 beats; $s_4 = s_3^{292} s_2$ is 33,102 beats. In pseudocode (Algorithm 3 in Appendix A):

```
this, prev = "."*(a1-1) + "x", "."; for a in a2, a3, ...: this, prev = this*a
                                    + prev, this
```

The theorem below says that this construction and the floor rule (3.1) produce the same rhythm, and that the block lengths are the convergent denominators.

Lemma 3.3 (floors near a convergent)**.** *Let $n \geq 1$ and $1 \leq m < q_n + q_{n+1}$. Then $\lfloor m\alpha \rfloor = \lfloor mp_n/q_n \rfloor - \varepsilon_n(m)$, where $\varepsilon_n(m) = 1$ if q_n divides m and n is odd, and $\varepsilon_n(m) = 0$ otherwise.*

Proof. By (2.5), p_n/q_n and p_{n+1}/q_{n+1} satisfy $|p_n q_{n+1} - p_{n+1} q_n| = 1$, and α lies strictly between them (Proposition 2.2). A fraction a/b strictly between two such fractions has $b \geq q_n + q_{n+1}$: indeed $|a/b - p_n/q_n| \geq 1/(bq_n)$ and $|p_{n+1}/q_{n+1} - a/b| \geq 1/(bq_{n+1})$, and the two distances add up to $1/(q_n q_{n+1})$, which forces $b \geq q_n + q_{n+1}$. Now take $m < q_n + q_{n+1}$. If an integer k lay strictly between $m\alpha$ and mp_n/q_n, then k/m would lie strictly between α and p_n/q_n, which is impossible. So if mp_n/q_n is not an integer, both numbers have the same floor. If it is an integer (that is, $q_n \mid m$), the floors agree when $\alpha > p_n/q_n$ and differ by one when $\alpha < p_n/q_n$; by (2.6) the convergent lies above α exactly when n is odd. □

Lemma 3.4 (near-periodicity)**.** *For $n \geq 1$ and $1 \leq j \leq q_{n+1} - 2$, $c_\alpha(j + q_n) = c_\alpha(j)$.*

Proof. Let $f(m) = \lfloor mp_n/q_n \rfloor - \varepsilon_n(m)$. Then $f(m + q_n) = f(m) + p_n$, because ε_n has period q_n. By Lemma 3.3, $c_\alpha(j) = f(j+1) - f(j)$ whenever $j+1 < q_n + q_{n+1}$. For $j \leq q_{n+1} - 2$ both $j+1$ and $j + q_n + 1$ are below $q_n + q_{n+1}$, so $c_\alpha(j + q_n) = f(j + q_n + 1) - f(j + q_n) = f(j+1) - f(j) = c_\alpha(j)$. □

Theorem 3.5 (nested blocks; Fraenkel, Mushkin and Tassa; Brown)**.** *Let $\alpha = [0; a_1, a_2, \dots]$ be irrational, with convergents p_n/q_n and standard words s_n. Then for every $n \geq 0$:*

(i) s_n is q_n beats long and contains p_n hits;

(ii) *the length and the number of hits of s_n are coprime, and so are the lengths of s_n and s_{n-1};*

(iii) *for $n \geq 1$, s_n is the beginning of the characteristic word c_α. Hence the first q_{n+1} beats of the rhythm consist of a_{n+1} copies of the level-n block followed by one level-$(n-1)$ block, and $c_\alpha = \lim_{n\to\infty} s_n$.*

Part (iii) is due to Fraenkel, Mushkin and Tassa [45] and Brown [22]; a proof through Sturmian morphisms is in Lothaire [88, Ch. 2]. We give a direct proof from the two lemmas, since the same counting reappears in every musical application.

Proof. (i) The lengths satisfy $|s_{-1}| = |s_0| = 1$, $|s_1| = a_1 = q_1$, and $|s_n| = a_n|s_{n-1}| + |s_{n-2}|$, which is the recurrence (2.3) for q_n with $q_0 = 1$. The hit counts satisfy $0, 1, a_n \cdot (\text{previous}) + (\text{one before})$, starting from s_0 with $0 = p_0$ hits and s_1 with $1 = p_1$ hit, which is the recurrence for p_n.

(ii) follows from (i) and the determinant identity (2.5).

(iii) For $n = 1$: α lies between $1/(a_1+1)$ and $1/a_1$, so $(j+1)\alpha < 1$ for $j < a_1$ and $(a_1+1)\alpha > 1$; the first a_1 beats are $a_1 - 1$ rests and a hit, which is s_1. For $n = 2$ when $a_1 = 1$ (so $q_1 = 1$ and the lemma below does not apply): then $1-\alpha$ lies between $1/(a_2+2)$ and $1/(a_2+1)$, hence $\lfloor j\alpha \rfloor = j - 1$ for $1 \leq j \leq a_2 + 1$ and $\lfloor (a_2+2)\alpha \rfloor = a_2$; the first $a_2 + 1$ beats are a_2 hits and a rest, which is $s_2 = s_1^{a_2} s_0$.

For the inductive step, suppose $n \geq 1$, $q_n \geq 2$, and s_n and s_{n-1} are beginnings of c_α (for $n = 1$ we use $s_0 = \,.\, = c_\alpha(1)$, which holds when $a_1 \geq 2$, the only case with $q_1 \geq 2$). Since $q_n \geq 2$, every position $j \leq q_{n+1} - q_n$ satisfies $j \leq q_{n+1} - 2$, so Lemma 3.4 lets us shift by q_n repeatedly: the first $q_{n+1} = a_{n+1}q_n + q_{n-1}$ beats are a_{n+1} copies of the first q_n beats followed by the first q_{n-1} beats. By hypothesis these are s_n and s_{n-1}, so the first q_{n+1} beats are $s_n^{a_{n+1}} s_{n-1} = s_{n+1}$. Since $q_n \geq 2$ for all $n \geq 2$, the induction runs forever. □

We also confirmed the theorem by computer: for 200 random term sequences (terms drawn from $\{1, 2, 3, 4, 5, 7, 12\}$, words up to several thousand beats) and for $\pi - 3$ over the first 140,000 beats, the standard words agree letter for letter with the floor rule (3.1), and every block has q_n beats and p_n hits.

Remark 3.6. The theorem is the precise form of the statement that "every term of the continued fraction is a repeat count." The term a_{n+1} says how many times the level-n block repeats before the level-$(n-1)$ block closes the level-$(n+1)$ block. Big terms mean long runs of identical blocks; a term equal to 1 means the level adds a single block. For the golden ratio every term is 1, so every level is "one long block, then one short block," and the levels are the Fibonacci numbers.

3.4 Pi's levels, and the beat where the surprise lands

Table 3.1 lists the first levels of the rhythm of $\pi - 3$, and Fig. 3.1 draws them.

The two artifacts on which the companion site is based disagreed about where the first short phrase of level 4 "lands": one said beat 32,996, the other 33,102. Both are right about different things. Number the beats from 0, as the site does. Level 4 is $s_4 = s_3^{292} s_2$: 292 phrases of 113 beats followed by one phrase of 106. The 292 long phrases fill beats 0 through $292 \times 113 - 1 = 32{,}995$, so the short phrase *begins* at beat 32,996; it fills beats 32,996 to 33,101, so the level-4 block *ends* after beat 33,101, and the next block begins at beat $33{,}102 = q_4$. Our

Table 3.1. The levels of the rhythm of $\pi - 3 = [0; 7, 15, 1, 292, 1, 1, \dots]$. Each block is (term) copies of the block above it in the table, then the block two rows up. The fraction is hits over beats, the nth convergent of $\pi - 3$; adding 3 gives the convergent of π.

level n	term a_n	block s_n	beats q_n	hits p_n	convergent of π
1	7	$.^{6}\mathtt{x}$	7	1	22/7
2	15	$s_1^{15}\, s_0$	106	15	333/106
3	1	$s_2\, s_1$	113	16	355/113
4	292	$s_3^{292}\, s_2$	33,102	4,687	103993/33102
5	1	$s_4\, s_3$	33,215	4,703	104348/33215

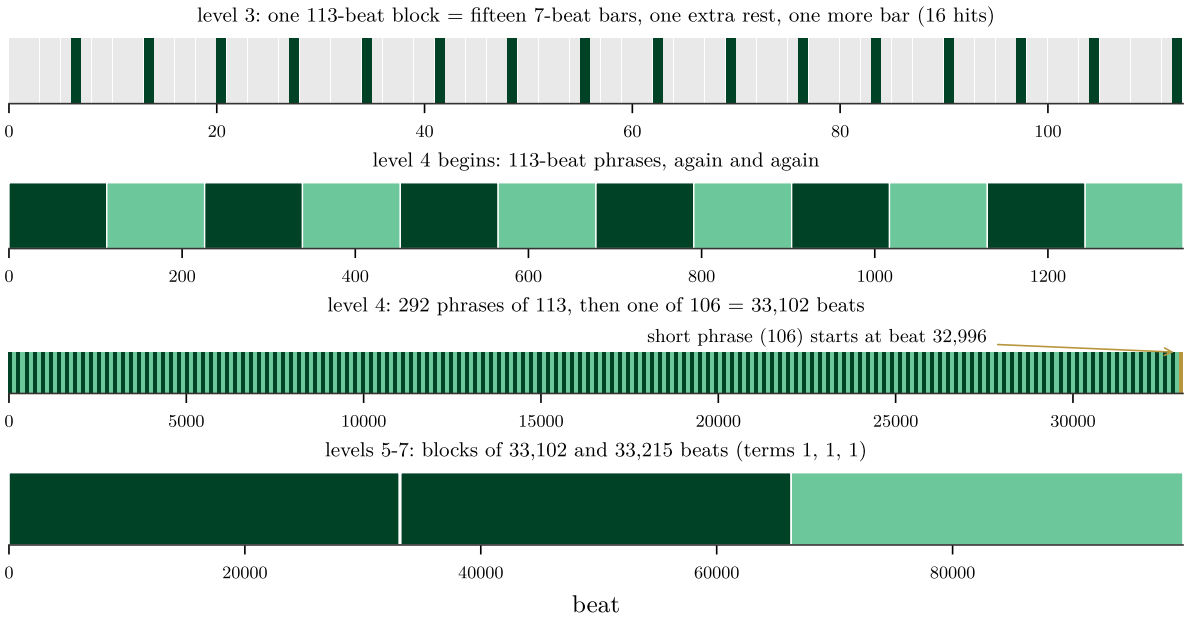


Figure 3.1. The nested blocks of π's rhythm, drawn to scale at four zoom levels. Top: one 113-beat block (hits in green). Second row: the start of level 4, a run of 113-beat phrases. Third row: all of level 4, 292 phrases of 113 beats and then the single 106-beat phrase (gold), which begins at beat 32,996 and closes the block at 33,102. Bottom: levels 5 to 7, where each term is 1.

script locates every short phrase in the first 140,000 beats: they begin at beats 32,996, 66,211, 99,313, 132,528. In the first 80,000 beats there are 706 complete phrases of 113 beats and 2 of 106.

When each beat is one bar of music, as in the fifth and sixth movements of the Suite (Chapter 11), bars are numbered from 1, and the same moment is the bar line after bar 32,996: the first bar of the short cycle is printed as bar 32,997, and the short cycle ends with bar 33,102. At a tempo of one bar every 2.5 seconds, reaching it takes 22.9 hours of continuous playing.

3.5 From levels to musical form

Theorem 3.5 turns any continued fraction into a hierarchy of blocks, and a hierarchy of blocks is what musicians call *form*: notes group into bars, bars into phrases, phrases into sections,

sections into movements. The reading proposed on the companion site, and used by Teddy Rook for the Suite, is simply to name the levels: the shortest block is a bar, the next a phrase, the next a section, and so on. Each term then says how many of one unit make the next, and the second, shorter block at each level is a contrasting unit, a "B" section that the number brings in on its own schedule.

Two small choices make the reading musical. First, a level that adds a single unit to the one below (a term equal to 1 at a low level, or the "extra rest" in $s_2 = s_1^{15} s_0$) can be absorbed: in the Composer on the site, π's bars are 7 beats, its phrases 113, and its sections 33,102. Second, the rhythm itself can be kept inside each bar, so that the accents of a piece follow the number at the smallest scale while its form follows the number at the largest. Chapter 11 shows both choices at work.

3.6 Three gaps

The rhythm of α has only two kinds of gaps between hits, $\lfloor 1/\alpha \rfloor$ and $\lfloor 1/\alpha \rfloor + 1$ (7 and 8 for $\pi - 3$), because it is balanced. A related classical fact concerns the points $\{\alpha\}, \{2\alpha\}, \ldots, \{N\alpha\}$ placed on a circle of circumference 1.

Theorem 3.7 (three-gap theorem; Sós, Świerczkowski)**.** *For any irrational α and any N, the points $0, \{\alpha\}, \ldots, \{N\alpha\}$ cut the circle into arcs of at most three different lengths, and when there are three, the largest is the sum of the other two.*

The statement was conjectured by Steinhaus. Proofs and the history are in Sós [121], Świerczkowski [126] and the survey of Alessandri and Berthé [3], which also explains the link with Sturmian words. The gap lengths are built from the distances $\|q_k \alpha\|$ of the multiples $q_k \alpha$ to the nearest integer, so the continued fraction controls them too. Figure 3.2 shows the distinct gap lengths for $\pi - 3$ and $N \le 120$; our script checks the theorem for every $N \le 1{,}200$. At $N = 100$ the three lengths are $0.0089, 0.0177, 0.0265$. In Chapter 9 this rigidity, two gaps in the rhythm and at most three on the circle, puts π's rhythm at the "quasicrystal" end of a scale whose other end is pure randomness.

Sections 18.2 and 18.5 apply the three-gap theorem to the meetings of two planets. [Added 2026-11-10]

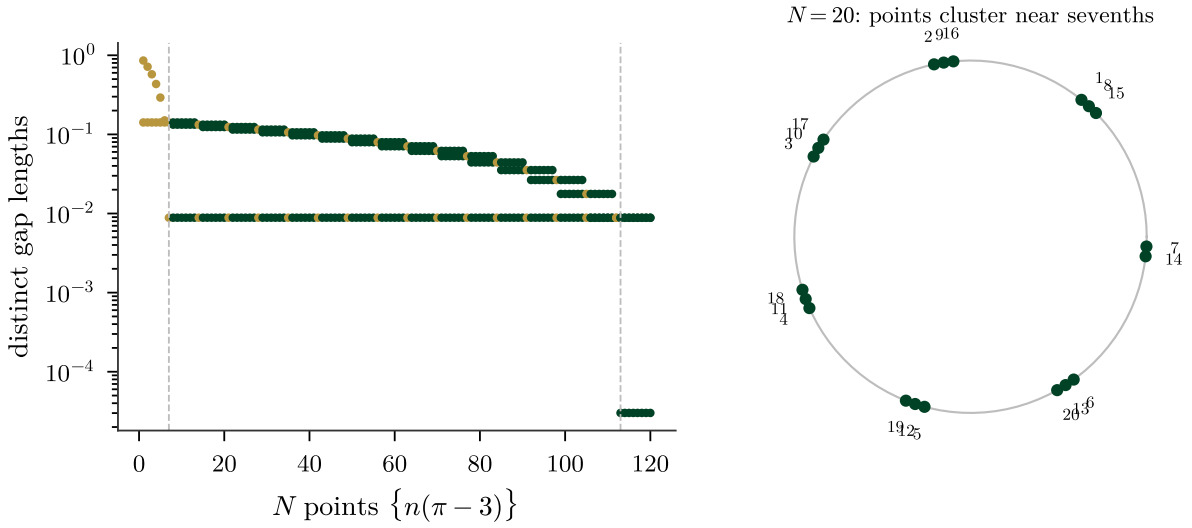


Figure 3.2. The three-gap theorem for $\alpha = \pi - 3$. Left: the distinct arc lengths among the points $0, \{\alpha\}, \ldots, \{N\alpha\}$ for $N = 1, \ldots, 120$ (green: three lengths; gold: two; dashed lines at the denominators $N = 7$ and $N = 113$). The count is two whenever the number of points, $N + 1$, is a denominator such as 7, 106 or 113. Right: the first 20 points, which cluster near multiples of $1/7$ because 7α is close to 1.

CHAPTER 4

Music from the same ladder

The continued fraction is not only a rhythm generator. The same ladder of fractions decides how musical scales are tuned, the same algorithm spreads drum hits around the world's traditional rhythms, and the same fractions are where coupled oscillators lock together. This chapter retells these classical connections in the language of Chapters 2 and 3.

4.1 Tuning: the fifth and its convergents

Two notes sound smooth together when their frequencies are in a simple ratio. The octave is $2:1$ and the fifth is $3:2$. Measured in octaves, a fifth is

$$\log_2 \tfrac{3}{2} = 0.584963\ldots = [0; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, \ldots].$$

A scale that divides the octave into n equal steps (an *n-note equal temperament*, or n-EDO) contains a good fifth when some k/n is close to $\log_2(3/2)$. The convergents of $\log_2(3/2)$ are

$$1/2, 3/5, 7/12, 24/41, 31/53, 179/306, \ldots$$

and they are the classical answers: $7/12$ is "a fifth is 7 of 12 half steps," and 41 and 53 are the next good stops, long known to theorists [12]. In cents (hundredths of an equal-tempered half step, so 1200 cents per octave), the 12-note fifth is 1.955 cents flat; the 41-note fifth misses by 0.484 cents and the 53-note fifth by 0.068.

Figure 4.1 shows the error of the best fifth in every n-note scale up to $n = 72$. The scales that set a new record for the fifth are $n = 1, 2, 3, 5, 7, 12, 29, 41, 53$. All are convergent denominators except 29, whose fifth $17/29$ is an *intermediate fraction* between the convergents $7/12$ and $24/41$. That is exactly what the theory of best approximations predicts: the record-setters for $|nx - k|$ are the convergents, and the record-setters for $|x - k/n|$ also include some intermediate fractions [71, Ch. II].

The Pythagorean comma. Twelve pure fifths overshoot seven octaves by a ratio, $3^{12}/2^{19} = 531441/524288$, called the Pythagorean comma, which is 23.46 cents. Equal temperament shares the comma out equally: each of its twelve fifths is $23.46/12 = 1.955$ cents flat, which our script confirms to nine decimals. The comma exists because $\log_2(3/2)$ is irrational: no number of pure fifths can ever equal a whole number of octaves.

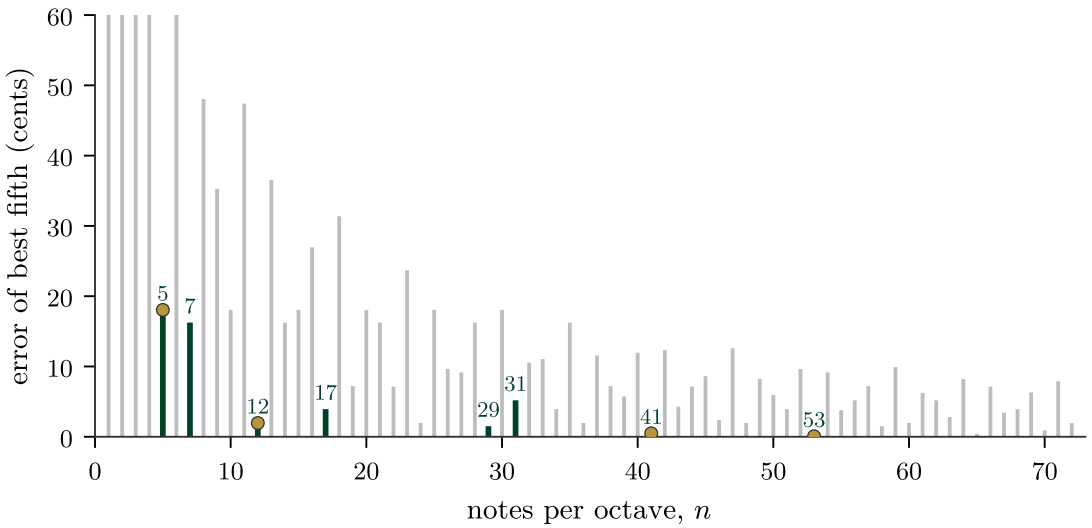


Figure 4.1. The error of the best fifth in every n-note equal temperament, $n = 1, \ldots, 72$, in cents. Green bars and labels mark well-known scales; gold dots mark the convergent denominators of $\log_2(3/2)$: 2, 5, 12, 41, 53.

Thirds, and a prime number of notes. The same comparison for the major third $5 : 4$ gives a 12-note third 13.69 cents sharp, while the 31-note scale, which Christiaan Huygens advocated [12], has a third within 0.78 cents of pure. Stepping around an n-note circle by s notes visits every note exactly when $\gcd(s, n) = 1$, so the number of step sizes that visit all notes is Euler's $\varphi(n)$. The circle of fifths visits all 12 notes because $\gcd(7, 12) = 1$. In a scale with a prime number of notes, such as 7, 31 or 53, every step size visits every note.

4.2 Euclidean rhythms

Suppose we want k drum hits in a cycle of n steps, spread as evenly as possible. In 2003 Eric Bjorklund described an algorithm for exactly this problem, for spacing pulses in the timing system of the Spallation Neutron Source accelerator; in 2005 Godfried Toussaint observed that its outputs include many traditional rhythms from around the world and named them *Euclidean rhythms*, because the algorithm has the structure of Euclid's algorithm [132]. Bjorklund's method starts with k groups `[x]` and $n - k$ groups `[.]`, and repeatedly appends one leftover group to each of the others until at most one leftover group remains. The group counts follow Euclid's algorithm on (n, k).

Proposition 4.1. *For $1 \le k < n$, the Euclidean rhythm $E(k, n)$ is a rotation of the lower Christoffel word of slope k/n, the period of the rhythm of k/n in the sense of (3.1).*

This is known; Gómez-Martín, Taslakian and Toussaint relate Euclidean rhythms to the maximally even sets of Clough and Douthett and to other even-spacing constructions [29, 49]. Our script confirms it for all 780 pairs with $n \le 40$. Two consequences connect the drum machine to π. Running Euclid's algorithm on 355 and 113 produces the quotients $3, 7, 16$, the beginning of π's continued fraction. And $E(16, 113)$ is a rotation of the level-3 block of π's rhythm (Table 3.1): the drum machine with 16 hits in 113 steps plays π at level 3. Figure 4.2 draws some of these rhythms as necklaces.

Table 4.1. Euclidean rhythms $E(k, n)$ from Bjorklund's algorithm, as computed by our script, with the gaps between hits. The names follow Toussaint [132]. The son clave (gaps $3, 3, 4, 2, 4$) is not a rotation of $E(5, 16)$, so it is not Euclidean.

name	rhythm	pattern	gaps
tresillo	$E(3, 8)$	`x..x..x.`	3, 3, 2
cinquillo	$E(5, 8)$	`x.xx.xx.`	2, 1, 2, 1, 2
West African bell	$E(7, 12)$	`x.xx.x.xx.x.`	2, 1, 2, 2, 1, 2, 2
ruchenitza	$E(3, 7)$	`x.x.x..`	2, 2, 3
khafif-e-ramal	$E(2, 5)$	`x.x..`	2, 3
Venda clapping	$E(5, 12)$	`x..x.x..x.x.`	3, 2, 3, 2, 2
samba	$E(7, 16)$	`x..x.x.x..x.x.x.`	3, 2, 2, 3, 2, 2, 2

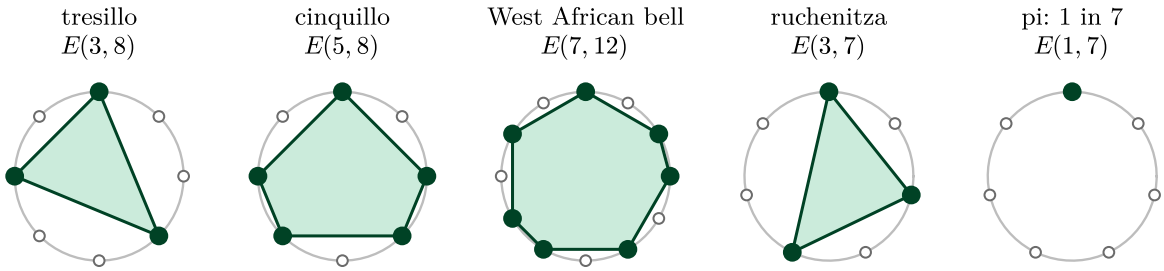


Figure 4.2. Euclidean rhythms drawn as necklaces: steps run clockwise from the top, hits are filled. The hits of each rhythm form a polygon that is as close to regular as the steps allow. The last necklace is the first level of π's rhythm, one hit in seven.

4.3 Polyrhythm and coprimality

Two pulse trains, one with a hit every a steps and one every b steps, coincide every $\operatorname{lcm}(a, b)$ steps. When a and b are coprime, they meet only once per cycle of ab steps: three against four meet only on the downbeat of a 12-step cycle. When they share a factor, they meet more often; four against six meet every 12 steps, twice per cycle of 24. The Polyrhythm layerer on the site stacks the rhythms of several convergents as concentric rings; for π, the rings $1/7$, $15/106$ and $16/113$ play three versions of one beat that drift apart and realign. Because $\gcd(7, 113) = 1$, the 7-ring and 113-ring meet only at the start of their common cycle. Chapter 5 turns "meeting less often" into an exact correlation.

4.4 Mode locking and the circle map

In 1665 Christiaan Huygens noticed that two pendulum clocks hanging from the same beam fell into step; a modern experimental study is Bennett, Schatz, Rockwood and Wiesenfeld [11]. Coupled oscillators tend to lock at simple frequency ratios, and the simplest model of this is the *sine circle map*

$$\theta_{t+1} = \theta_t + \Omega - \frac{K}{2\pi} \sin(2\pi\theta_t), \tag{4.1}$$

where Ω is the natural rotation per step and $K \geq 0$ the strength of the coupling. The *rotation number* $W = \lim(\theta_t - \theta_0)/t$ is the average rotation actually achieved. For $K = 0$, $W = \Omega$. For

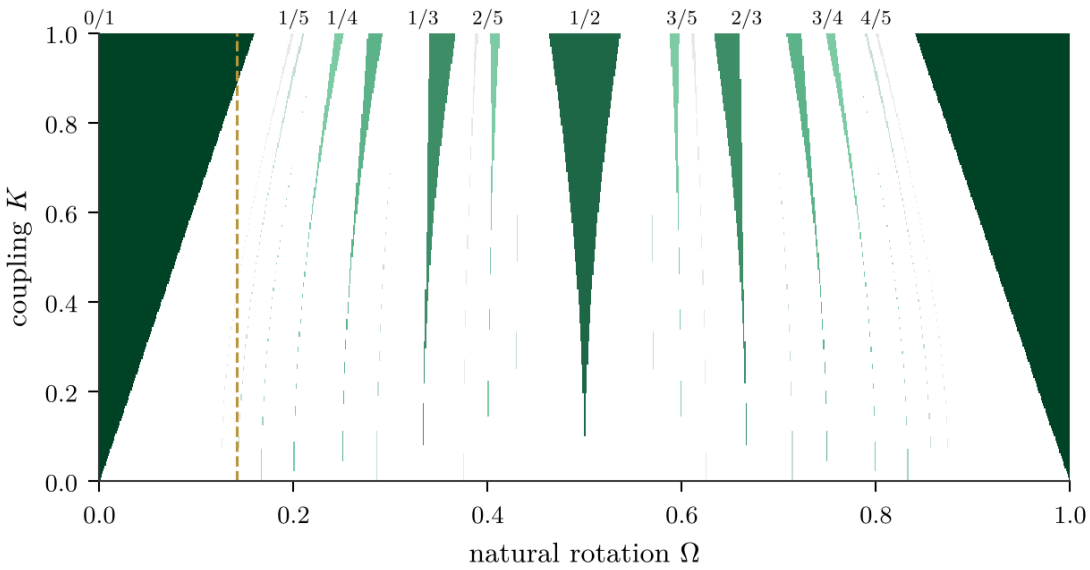


Figure 4.3. Arnold tongues of the sine circle map (4.1). Shaded points (Ω, K) have rotation number p/q with $q \le 8$ (darker: smaller q), computed from 400 iterations after a transient. The dashed gold line is $\Omega = \pi - 3$, which enters the 0/1 tongue at $K = 2\pi(\pi - 3) \approx 0.8897$.

$0 < K \le 1$ the map is a homeomorphism of the circle and W exists; it is locked to a fraction p/q on an interval of Ω around p/q, and these intervals widen with K into the *Arnold tongues* of Fig. 4.3. At $K = 1$ the locked intervals fill the whole line except a Cantor set of measure zero, a "complete devil's staircase" [67], which Chapter 10 explores.

The 0/1 tongue can be computed exactly. A fixed point of (4.1) (modulo 1) needs $\Omega = (K/2\pi)\sin(2\pi\theta)$, which has a solution if and only if $|\Omega| \le K/(2\pi)$. So at $K = 1$ the 0/1 step extends $1/(2\pi) = 0.1592$ on each side of 0; our numerical staircase gives 0.1592. The number $\pi - 3 = 0.14159\ldots$ is inside it, so a system whose natural rotation is $\pi - 3$ is captured into the 0/1 lock once $K \ge 2\pi(\pi - 3) = 0.8897$. Numerically, at $K = 1$ the 1/2 step is 0.074 wide and the 1/3 step 0.031, and the steps with denominators up to 10 already cover a fraction 0.598 of the interval $[0, 1]$.

Counting the tongues. Each tongue belongs to a fraction p/q in lowest terms, so the number of tongues with $0 \le p/q \le 1$ and $q \le N$ is $1 + \sum_{q \le N} \varphi(q)$, the length of the Farey sequence of order N. A classical theorem [56, Thm. 330] gives $\sum_{q \le N} \varphi(q) \sim 3N^2/\pi^2$. For $N = 50$, 100 and 200 our counts are 775, 3,045 and 12,233, against $3N^2/\pi^2 \approx 760$, 3040 and 12159. Here π appears through $6/\pi^2$, the subject of the next chapter.

Sections 18.2 and 18.5 apply this to orbits, where the locked ratios are resonances between periods. [Added 2026-11-10]

CHAPTER 5

Coprimality and $6/\pi^2$

Every level of a rhythm is a coprime pair: by Theorem 3.5(ii) a block of q_n beats holds p_n hits with $\gcd(p_n, q_n) = 1$. This chapter is about coprimality itself: how common it is, how it appears as statistical independence, and where π enters through $6/\pi^2$.

5.1 The orchard

Stand at the origin of the integer lattice and look out over an orchard with a tree at every point (a, b) with $a, b \geq 1$. The tree at (a, b) is visible exactly when no other tree stands on the segment from the origin to it, that is, when $\gcd(a, b) = 1$. Figure 5.1 shows the visible trees in a 40×40 corner, 0.6119 of the grid. For a 1000×1000 corner the share is 0.608383.

Theorem 5.1 (56, Thm. 332)**.** *The proportion of pairs* (a, b) *with* $1 \leq a, b \leq N$ *and* $\gcd(a, b) = 1$ *tends to* $6/\pi^2 = 0.607927\ldots$ *as* $N \to \infty$.

The heuristic behind it is short. A prime p divides both a and b with probability $1/p^2$, and divisibility by different primes behaves independently, so the probability that no prime divides both is

$$\prod_{p} \left(1 - \frac{1}{p^2}\right) = \frac{1}{\zeta(2)}, \qquad \zeta(2) = \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{5.1}$$

The first equality is Euler's product formula [41]: expanding each factor $(1 - p^{-2})^{-1} = 1 + p^{-2} + p^{-4} + \cdots$ and multiplying out, unique factorization produces each $1/n^2$ exactly once. The second is Euler's solution of the Basel problem [40]. The theorem makes the heuristic rigorous by counting with the Möbius function. Numerically, the product over the primes below one million is 0.60792714, against $6/\pi^2 = 0.60792710$.

So π, a number about circles, measures how often two whole numbers share no prime. The convergents of $\pi - 3$, read as pairs (beats, hits) such as $(7, 1)$, $(106, 15)$ and $(113, 16)$, are all visible trees.

5.2 Coprime means uncorrelated

The orchard looks like a matrix, and it is one. Treat "a hit every ath step" as a signal $x_t^{(a)} = 1$ if $a \mid t$ and 0 otherwise, and compare two such signals over one common period.

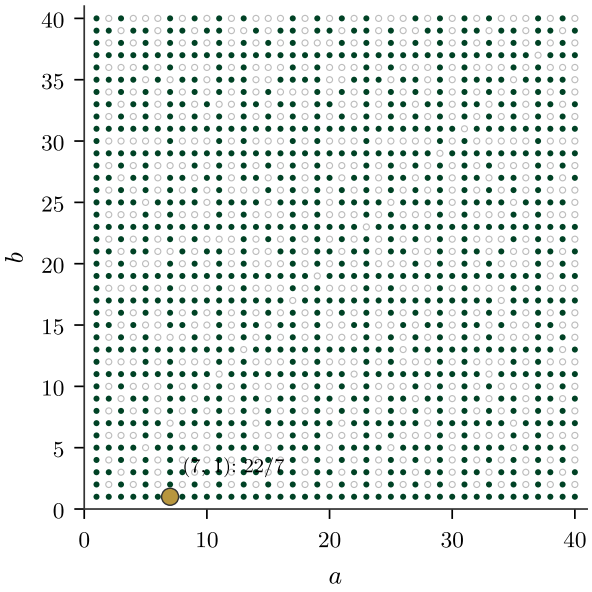


Figure 5.1. The orchard: lattice points (a, b) with $1 \le a, b \le 40$. Filled points are visible from the origin ($\gcd(a, b) = 1$); open circles are hidden. The gold point $(7, 1)$ is the first level of π's rhythm, read as (beats, hits).

Proposition 5.2 (the correlation of two pulse trains)**.** *For integers $a, b \ge 2$, the Pearson correlation of $x^{(a)}$ and $x^{(b)}$ over a full period of $\operatorname{lcm}(a, b)$ steps is*

$$\operatorname{corr}(a, b) = \frac{\gcd(a, b) - 1}{\sqrt{(a - 1)(b - 1)}}. \tag{5.2}$$

In particular the correlation is zero exactly when a and b are coprime, and it is 1 when $a = b$.

Proof. Over a period, $x^{(a)}$ has mean $1/a$ and, being 0 or 1, variance $\frac{1}{a}(1 - \frac{1}{a}) = (a - 1)/a^2$. The product $x_t^{(a)} x_t^{(b)}$ is 1 exactly when t is a multiple of $\operatorname{lcm}(a, b) = ab/\gcd(a, b)$, so its mean is $\gcd(a, b)/(ab)$. The covariance is therefore

$$\frac{\gcd(a, b)}{ab} - \frac{1}{a} \cdot \frac{1}{b} = \frac{\gcd(a, b) - 1}{ab},$$

and dividing by $\sqrt{(a - 1)(b - 1)}/(ab)$, the product of the standard deviations, gives (5.2). □

Our script checks (5.2) against a direct computation for every pair $2 \le a \ne b \le 30$. For example $\operatorname{corr}(4, 6) = 1/\sqrt{15} = 0.258$, which is why four against six "keeps colliding," while three against four, being coprime, is uncorrelated. In the 39×39 matrix of pulse trains $2, \ldots, 40$ (Fig. 5.2), the share of off-diagonal zeros is 0.607, close to $6/\pi^2$ for the same reason as the orchard. A prime's row is zero except at its multiples: primes are the "independent" pulse trains of arithmetic.

For periods 2 to 40 the matrix is positive definite: its eigenvalues run from 0.072 to 5.38. Random-matrix theory is used in finance to separate real correlations among stock returns from noise [78, 112]; how the spectrum of this arithmetic correlation matrix compares with such random benchmarks is a question we leave open (Chapter 14). [Update: see Sections 15.6 and 16.3, and Sections 17.1–17.3, where the extreme eigenvalues are found to leading order

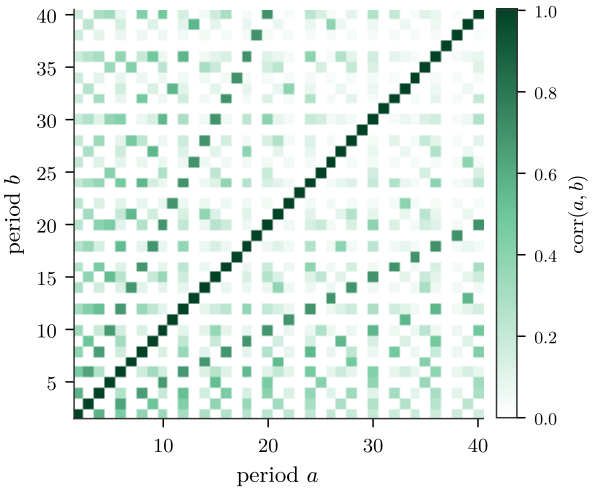


Figure 5.2. The correlation matrix (5.2) of pulse trains with periods 2 to 40. White cells are zero, exactly the coprime pairs; the pattern is the orchard of Fig. 5.1 seen as a matrix.

and the spectrum is shown to have a limiting law. [Added 2026-11-10]] Formula (5.2) itself is an elementary exercise; we did not find it stated in this form in the literature we searched, but we expect it is known, and we claim no novelty for it (Chapter 13).

5.3 Hydrogen's $1/n^2$: a shared formula, not shared physics

Hydrogen's energy levels are proportional to $-1/n^2$, and its visible spectral lines, the Balmer series, come from the differences $1/4 - 1/n^2$. In wavelengths,

$$\lambda_n = B\,\frac{n^2}{n^2-4}, \qquad B = 364.56 \text{ nm}, \tag{5.3}$$

which is the formula Johann Balmer found in 1885 [9]. It gives $\lambda_3 = 656.2\,\text{nm}$ (the red H-alpha line) and $\lambda_4 = 486.1\,\text{nm}$.

Add up the reciprocal squares that set the levels, $1 + \frac{1}{4} + \frac{1}{9} + \cdots$, and the answer is $\zeta(2) = \pi^2/6$, the reciprocal of the orchard's $6/\pi^2$, and also, by (5.1), a product over the primes (Fig. 5.3). It is tempting to say that "hydrogen knows about the primes." It does not. The match is a match of formulas: the sum of $1/n^2$ is the same series wherever it appears. No physical quantity of hydrogen is the sum $\sum 1/n^2$, and nothing in the atom depends on prime factorization. We include it because it is a good illustration of how one series can sit inside several subjects, which is the theme of Chapter 12.

Section 18.1 gives a neighbouring link that is shared physics. [Added 2026-11-10]

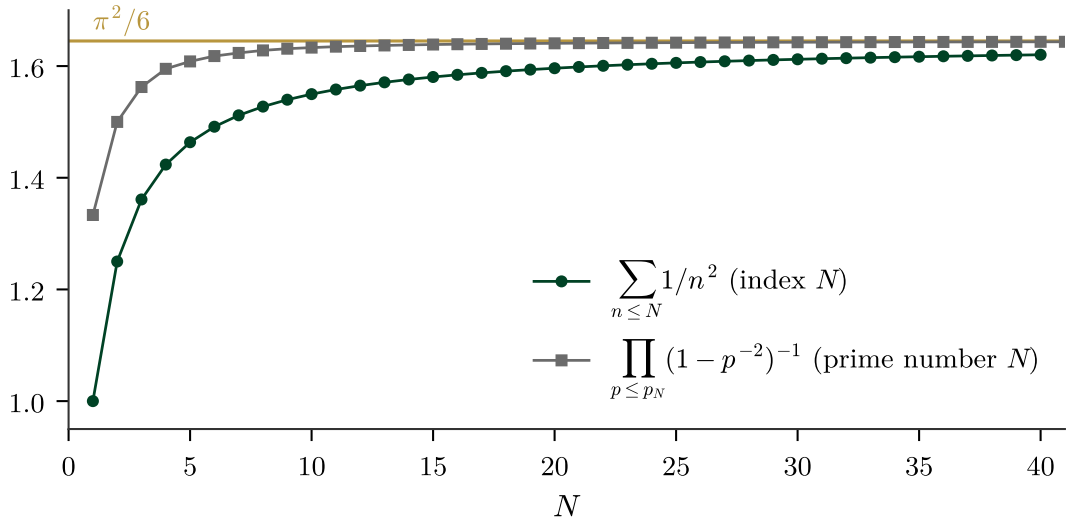


Figure 5.3. Two roads to $\pi^2/6$: the partial sums $\sum_{n \le N} 1/n^2$ (green, the reciprocal squares behind hydrogen's levels) and the partial Euler products $\prod (1-p^{-2})^{-1}$ over the first N primes (grey). The product converges much faster.

CHAPTER 6

Primes in sound and space

This chapter collects the places on the companion site where the primes are heard or seen: as notes, as arms of a spiral, as columns of a wheel, as peaks in a diffraction pattern and as a sum of waves. Each is a classical or recent published result; our contribution is to compute each one again and to tie it to the rhythm of π.

6.1 Primes as notes, and the bias of neighbors

Number the twelve notes of the octave $0, \dots, 11$. Every prime above 3 leaves remainder 1, 5, 7 or 11 on division by 12, because the other remainders share a factor with 12; these are the $\varphi(12) = 4$ notes C♯, F, G and B. Playing the primes in order as notes, with each gap to the next prime as a duration, gives the "prime music" instrument on the site.

If the remainders of consecutive primes were independent and equally likely, a prime would repeat the remainder of the one before it one time in four. They do not. Lemke Oliver and Soundararajan [82] showed numerically that consecutive primes avoid repeating their remainder, and explained the bias by a conjecture of Hardy–Littlewood type that predicts it fades only like $1/\ln x$. Our counts agree (Fig. 6.1). Modulo 10, among the primes from 7 up to 10^5, 10^6 and 10^7, the share of repeats is 13.9%, 15.4% and 16.3%, against 25% for independence; modulo 12 it is 12.8%, 15.0% and 16.4%. The share creeps up toward 25% as the range grows, slowly, as predicted.

6.2 A sunflower seeded with primes

Place seed number k at angle $2\pi k\alpha$ and distance $\sqrt{k}$ from the center. With $\alpha = \varphi - 1$ (the golden angle) this is the familiar model of a sunflower head; with $\alpha = \pi - 3$ the seeds fall into 7 spokes that curl into 113 arms (Fig. 6.2). The reason is the rhythm again: $113\alpha = 113\pi - 339$ differs from the integer 16 by only 3.0×10^{-5}, so seeds k and $k + 113$ sit at almost the same angle, and arm j of the 113 holds the seeds $k \equiv j \pmod{113}$.

Now light only the prime seeds. A prime larger than q can lie in arm j of q only if $\gcd(j, q) = 1$, each of those $\varphi(q)$ arms holds infinitely many primes by Dirichlet's theorem of 1837 [56, Thm. 15], and by the prime number theorem for arithmetic progressions they share the primes equally in the long run. Since 113 is prime, primes avoid only the arm $j = 0$; among

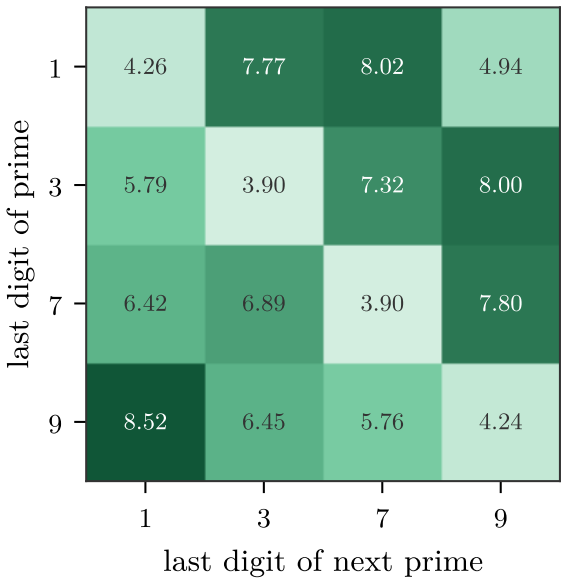


Figure 6.1. Last digits of consecutive primes up to 10^7 (664,576 primes from 7 on). Each cell is the percentage of pairs (prime, next prime) with the given last digits; independence would give 6.25% everywhere. The diagonal, where the digit repeats, is depleted.

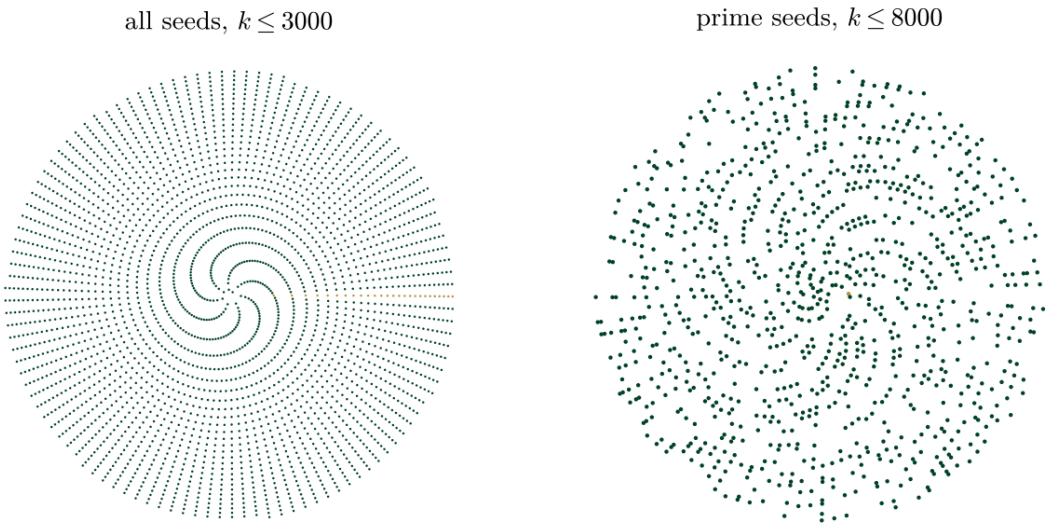


Figure 6.2. Pi's sunflower: seed k at angle $2\pi k(\pi - 3)$ and radius $\sqrt{k}$. Left, all seeds up to 3,000 form 7 spokes that curl into 113 arms (seeds divisible by 113 in gold). Right, only the prime seeds up to 8,000: they fill the arms evenly.

the primes up to 8,000 they occupy 112 of the 113 arms. With 106 arms, which is 2×53, primes occupy only $52 = \varphi(106)$ arms, and half the spiral goes dark.

Wheels. The same fact, straightened out, is *wheel factorization*: lay the integers in rows of width w, and the primes beyond w fall only in the $\varphi(w)$ columns coprime to w. With $w = 30$ a sieve can skip 73% of all numbers without testing them, and with $w = 210$, 77.1%. Pi's

rhythm, laid in rows of width 113, forms 16 nearly straight columns, one for each hit in a 113-beat block.

6.3 Diffraction: a quasicrystal and the primes

Crystallographers read the order of a material from its diffraction pattern, which is (up to constants) the squared modulus of the Fourier transform of the atom positions. For N points $x_1, \dots, x_N$ on a line, define the structure factor

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{2\pi i k x_j} \right|^2 . \tag{6.1}$$

A random set gives $S(k) \approx 1$ for all $k \neq 0$; a periodic crystal gives sharp peaks of height about N at the multiples of its reciprocal lattice. In 1982 Dan Shechtman observed sharp diffraction from a metal alloy whose symmetry no periodic crystal can have [118]; such ordered but aperiodic materials are *quasicrystals*, and he received the 2011 Nobel Prize in Chemistry.

Pi's rhythm is a one-dimensional quasicrystal. Put an atom at each hit of π's rhythm. Sturmian point sets are standard examples of one-dimensional quasicrystals: their diffraction is pure point, with Bragg peaks at the points $m\alpha + n$ of the "Fourier module" $\mathbb{Z} + \alpha\mathbb{Z}$ [8]. In Fig. 6.3 (top) the peaks sit at $\{m(\pi - 3)\}$, folded into $[0, \frac{1}{2}]$; the peak at $k = \pi - 3$ has height 397 for the hits among the first 3,000 beats.

The primes are "effectively limit-periodic." Torquato, Zhang and de Courcy-Ireland [131] computed the structure factor of the primes in long intervals and found dense Bragg-like peaks at rational wavenumbers $k = a/q$ with *square-free* q, and none at $1/4$ or $1/9$; they called this order "effectively limit-periodic." The rational positions and the square-free rule have a short explanation. If the primes in a window are equidistributed among the $\varphi(q)$ residue classes coprime to q, then for $\gcd(a, q) = 1$

$$\sum_{p} e^{2\pi i a p/q} \approx \frac{N}{\varphi(q)} \sum_{\substack{r \bmod q \\ \gcd(r,q)=1}} e^{2\pi i a r/q} = \frac{N\,\mu(q)}{\varphi(q)},$$

because the inner sum is a Ramanujan sum, which equals the Möbius function $\mu(q)$ [56, §16.6]. So $S(a/q) \approx N\mu(q)^2/\varphi(q)^2$, which vanishes unless q is square-free. For the 1,472 primes in $[10^6, 10^6 + 2 \cdot 10^4)$ our computation gives $S(1/2) = 1472$ (all are odd), $S(1/3) = 369$ (the prediction $N/4$ is 368), but only $S(1/4) = 1.20$ and $S(1/9) = 0.57$, at the level of the random background.

Freeman Dyson suggested that the zeros of the zeta function, if the Riemann hypothesis is true, form a one-dimensional quasicrystal in a suitable sense, and that classifying such quasicrystals might lead somewhere [35]. Chapter 9 takes up that remark.

6.4 Primes as a sum of waves

Let $\psi(x) = \sum_{p^m \le x} \ln p$, a staircase that climbs by $\ln p$ at every prime power p^m. Riemann's 1859 paper [113] introduced the idea, and von Mangoldt proved [134], that for $x > 1$ not a

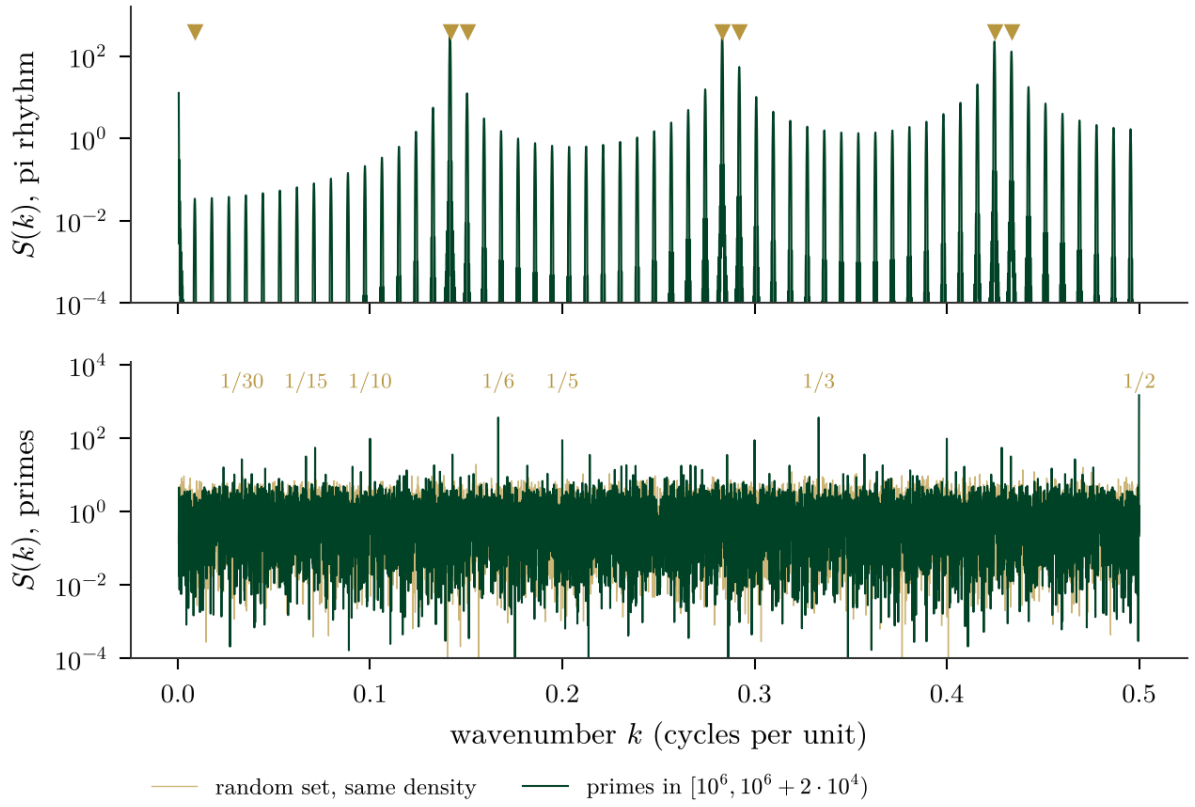


Figure 6.3. Structure factors (6.1) on a logarithmic scale (computed with a smooth taper to suppress side lobes). Top: atoms at the hits of π's rhythm in the first 3,000 beats; gold triangles mark $\{m(\pi-3)\}$ for $m = 1, \dots, 7$, folded into $[0, \frac{1}{2}]$. Bottom: atoms at the primes in $[10^6, 10^6 + 2 \cdot 10^4)$ (green) and at a random set of the same size (gold); the prime peaks sit at fractions with square-free denominators.

prime power

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \ln(2\pi) - \tfrac{1}{2} \ln\bigl(1 - x^{-2}\bigr)\,, \tag{6.2}$$

where ρ runs over the nontrivial zeros of the zeta function, summed symmetrically in pairs $\rho, \bar{\rho}$. Writing $\rho = \frac{1}{2} + i\gamma$ (as all computed zeros are), each pair contributes a wave $-2\,\mathrm{Re}(x^{\rho}/\rho)$ with amplitude about $\sqrt{x}/\gamma$ and frequency γ in the variable $\ln x$. The first zeros have $\gamma = 14.13473$, 21.02204 and 25.01086. Figure 6.4 rebuilds ψ from 3, 20 and 100 pairs of zeros: the waves add up to the staircase. The $\ln 2\pi$ in (6.2) is one more place where π sits next to the primes.

The Montgomery–Dyson observation, stated carefully. In 1972 Hugh Montgomery, studying the spacings between the γ's, derived a pair-correlation function $1 - (\sin \pi u/\pi u)^2$ for suitably normalized zeros; he proved it for a restricted class of test functions assuming the Riemann hypothesis, and conjectured it in general [96]. Freeman Dyson recognized the function as the pair correlation of eigenvalues of large random Hermitian matrices from the Gaussian Unitary Ensemble (GUE), a model of the energy levels of heavy nuclei [36, 92]. Odlyzko's large-scale computations showed excellent agreement [100]. What is known is the agreement of statistics; there is no known Hermitian operator whose eigenvalues are the zeros, and no proof that the statistics must agree. Chapter 9 computes these statistics for ourselves.

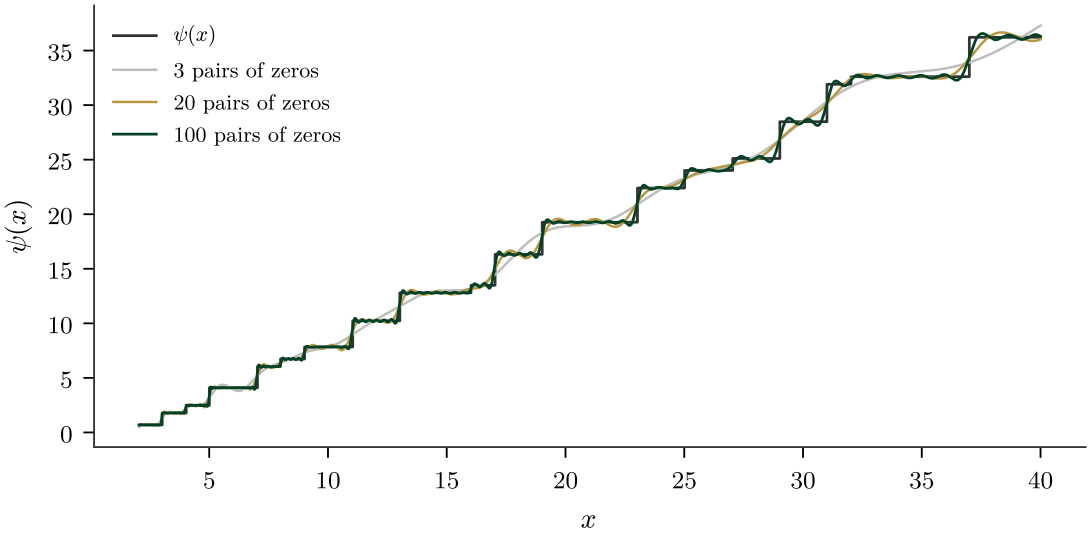


Figure 6.4. The explicit formula (6.2): the staircase $\psi(x)$ (black) and its reconstruction from the first 3, 20 and 100 pairs of zeta zeros.

CHAPTER 7

Tools that hunt primes

The continued fraction found 355/113. Can the same machine find primes? This chapter tries three tools from the companion site on the primes: a factoring method that is built from continued fractions, a picture of Fermat's little theorem made of rhythms, and an experiment that reads rhythms as binary numbers and tests them for primality. The first two are classical; the third is our experiment, and its result is a clear null.

7.1 Factoring with the continued fraction of $\sqrt{N}$

To factor an odd number N it is enough to find x and y with $x^2 \equiv y^2 \pmod N$ and $x \not\equiv \pm y$, because then $\gcd(x-y, N)$ is a proper factor. Continued fractions produce many congruences of the form $x^2 \equiv$ (small number) $\pmod N$. Expand $\sqrt{N}$ (or $\sqrt{kN}$ for a small multiplier k) with the integer recurrence

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = \frac{kN - P_{n+1}^2}{Q_n}, \qquad a_{n+1} = \left\lfloor \frac{\lfloor \sqrt{kN} \rfloor + P_{n+1}}{Q_{n+1}} \right\rfloor, \tag{7.1}$$

starting from $P_0 = 0$, $Q_0 = 1$, $a_0 = \lfloor \sqrt{kN} \rfloor$. If A_n are the numerators of the convergents, reduced modulo N, then

$$A_{n-1}^2 \equiv (-1)^n Q_n \pmod N, \qquad 0 < Q_n < 2\sqrt{kN}. \tag{7.2}$$

These are standard facts about the continued fraction of a square root. Using them to factor, by searching the Q_n for squares, was described by D. H. Lehmer and R. E. Powers in 1931 [81]. In 1975 Michael Morrison and John Brillhart made it practical by combining many Q_n that factor over a small set of primes, using linear algebra modulo 2 to find a product that is a square, and with it factored the seventh Fermat number $F_7 = 2^{128} + 1$ [98].

Our script implements the simple Lehmer–Powers search. It is instructive to run it on $F_5 = 2^{32} + 1$, which Euler showed is divisible by 641 [56, §2.5]. Because $F_5 = 65536^2 + 1$, the continued fraction of $\sqrt{F_5}$ is $[65536; 131072, 131072, \dots]$ with every $Q_n = 1$, a square that gives only the trivial factorization. With multiplier $k = 2$ the search succeeds: at step $n = 340$ it finds a square Q_n, and the gcd gives $F_5 = 641 \times 6{,}700{,}417$. Table 7.1 shows the first steps.

Table 7.1. The first steps of the continued fraction of $\sqrt{2(2^{32}+1)}$ in the recurrence (7.1). By (7.2) each Q_n is, up to sign, a square modulo N; the search waits for one that is a perfect square at an even step.

n	a_n	Q_n	square?
1	1	166,833	
2	9	18,530	
3	289	639	
4	1	181,815	
5	62	2,930	
6	3	49,383	
7	1	109,871	
8	2	52,014	

A product of two six-digit primes, $999{,}983 \times 1{,}000{,}003$, falls after 2 steps, because two factors this close make N nearly a square.

This is the clearest case of "pi's machine doing prime work": the algorithm that, run on π, finds 22/7 and 355/113 is the algorithm that, run on $\sqrt{N}$, finds factors. It does not use any property of π, of course.

7.2 Mersenne necklaces and Fermat's little theorem

Take $m = 2^p - 1$ points on a circle and join each point x to $2x \bmod m$. Write x in binary with p digits. Since $2^p \equiv 1 \pmod m$, doubling modulo m simply rotates the digits by one place. So every orbit of the doubling map is a binary word of length p together with all its rotations: a *necklace*, or in our language a p-beat rhythm and all its rotations.

Proposition 7.1 (Fermat's little theorem for base 2, by necklaces)**.** *If p is prime, then p divides $2^p - 2$.*

Proof. Exclude the two constant words (all rests and all hits). Every other binary word of prime length p has exactly p distinct rotations, since a word equal to a nontrivial rotation of itself has a period dividing p, hence period 1. So the $2^p - 2$ nonconstant words split into orbits of size exactly p. □

The same argument with an alphabet of a letters gives $p \mid a^p - a$; this combinatorial proof is due to Golomb [48]. For $p = 7$ there are $(2^7 - 2)/7 = 18$ necklaces, each a 7-beat rhythm. For the composite length $p = 6$ the orbits have mixed sizes: 1 of length 2, 2 of length 3, 9 of length 6. The picture does *not* make $2^p - 1$ prime: $2^{11} - 1 = 2047 = 23 \times 89$. The test that does is the Lucas–Lehmer test, and with it the Great Internet Mersenne Prime Search found the largest known prime, $2^{136{,}279{,}841} - 1$, with 41,024,320 digits, in October 2024 [54]; as of November 2026 it is still the record. [Added 2026-11-10]

7.3 The prime hunt

Here is an experiment from the companion site. Read the first n beats of a number's rhythm as a binary number, a hit as 1 and a rest as 0, and test it for primality as n grows. Does the rhythm of π attract primes?

Table 7.2. The prime hunt: probable primes among the binary prefixes of the first 1,500 beats of each rhythm. "Expected" is $\sum 2/\ln v_n$ over odd prefixes (the site's simpler $\sum 2/(n \ln 2)$ in parentheses). The last two columns are Poisson probabilities of a count at least as low, or at least as high, as the one found.

number	found	expected	$P(X \le \text{found})$	$P(X \ge \text{found})$
$\pi - 3$	1	2.4 (2.4)	0.30	0.91
$e - 2$	14	14.8 (14.3)	0.48	0.62
$\sqrt{2} - 1$	11	8.3 (8.8)	0.86	0.22
$\varphi - 1$	13	12.3 (11.8)	0.64	0.46

Method. For each of $\alpha = \pi - 3$, $e - 2$, $\sqrt{2} - 1$ and $\varphi - 1$ we built the first 1,500 beats with Theorem 3.5, formed the prefixes v_n (the binary number of the first n beats) for $n = 2, \ldots, 1500$, and tested every odd prefix with the Baillie–PSW probable-prime test (Appendix B). A prefix is odd exactly when beat n is a hit. For a random odd number of size v, the prime number theorem gives a probability of about $2/\ln v$ of being prime, so the expected count is $\sum 2/\ln v_n$ over the odd prefixes. The site uses the slightly cruder $\sum 2/(n \ln 2)$, which treats every prefix as an n-digit number; both are reported.

Results. Table 7.2 and Fig. 7.1 give the counts. The rhythm of $\pi - 3$ produced a single probable prime, at beat 49, where about 2.4 were expected; its rhythm is mostly rests, so its prefixes are odd only once every seven beats or so, and the expected count is small. The other three constants found close to their expected counts. In total the four rhythms gave 39 probable primes against 37.9 expected. No count is unusual: every Poisson tail probability in the table is above 0.2.

Interpretation. The rhythm of π does not attract primes, and neither does any other rhythm we tried. This is a null result, and we report it as one. The comparison is with Mersenne numbers $2^p - 1$, whose binary form is the all-hit rhythm: they *are* prime far more often than random numbers of their size, because every prime factor of $2^p - 1$ must be $\equiv 1 \pmod{2p}$, which removes all the small potential divisors. A construction that clearly beat its chance count in this test would deserve a closer look; none did. The same verdict, by a different route, comes from the statistics of π's terms in Chapter 10 and the prime convergents in Chapter 8.

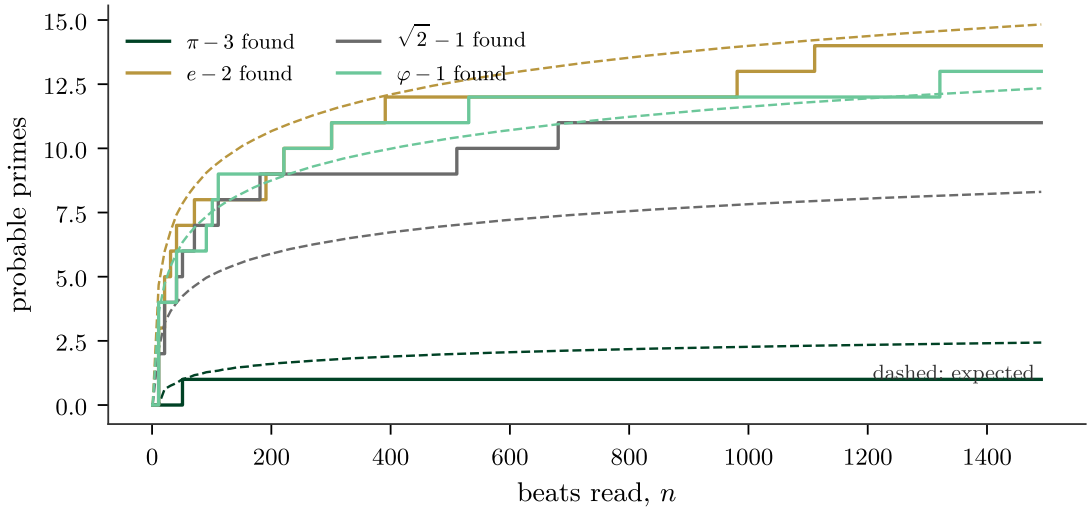


Figure 7.1. The prime hunt: cumulative probable primes among binary prefixes of each rhythm (steps) against the expected count $\sum 2/\ln v_n$ (dashed), for the first 1,500 beats.

CHAPTER 8

Thread (a): prime convergents

8.1 The question

The three classical fractions for π, 22/7, 223/71 and 355/113, all have prime denominators, and two of them are convergents. Among the convergents p_n/q_n of π, the prime denominators begin

$$q_1 = 7, \quad q_3 = 113, \quad q_8 = 265{,}381, \quad q_{32} = 842{,}468{,}587{,}426{,}513{,}207,$$

a list recorded in the OEIS [106]; the prime numerators begin $3, 103993, 833719, 4272943, \ldots$ [105]. [Note: the 3 here is p_0; every count of prime numerators later in this paper runs over $n \geq 1$ and leaves it out. [Added 2026-11-10]] Does the list go on forever? Nobody knows. The same question for the golden ratio asks whether there are infinitely many Fibonacci primes, and for $\sqrt{2}$ whether there are infinitely many Pell primes; both are famous open problems.

We cannot answer the question. What we can do is the next best thing: derive a heuristic prediction, with an explicit constant, for how many prime denominators a *typical* real number should have among its first N convergents; test the prediction on random numbers; and see where π, e, $\sqrt{2}$, φ and $\log_2(3/2)$ fall. This chapter does all three.

8.2 A heuristic with an explicit constant

Step 1: size. By Lévy's theorem (Theorem 2.7), for almost every x the denominators grow like

$$\ln q_n \sim Ln, \qquad L = \frac{\pi^2}{12 \ln 2} = 1.18657\ldots \tag{8.1}$$

Step 2: the naive guess. By the prime number theorem, a "random" integer near q is prime with probability about $1/\ln q$. If the q_n behaved like random integers of their size, the expected number of prime q_n with $n \leq N$ would be about $\sum_{n \leq N} 1/(Ln) \approx (1/L) \ln N$.

Step 3: local corrections. The q_n are not random integers. Consecutive denominators are coprime by (2.5), so, for example, two consecutive q_n are never both even. The right description of their local behavior is the following equidistribution theorem.

Table 8.1. The share of convergent denominators q_n divisible by a prime ℓ: for π ($n \le 10{,}000$), for 200 random reals ($n \le 2{,}000$ each), the prediction $1/(\ell+1)$ of Corollary 8.2, and $1/\ell$ for random integers.

ℓ	π	random reals	$1/(\ell+1)$	$1/\ell$
2	0.3344	0.3331	0.3333	0.5000
3	0.2510	0.2499	0.2500	0.3333
5	0.1620	0.1666	0.1667	0.2000
7	0.1284	0.1243	0.1250	0.1429
11	0.0820	0.0833	0.0833	0.0909
13	0.0680	0.0708	0.0714	0.0769

Theorem 8.1 (Szüsz; Moeckel; Jager and Liardet)**.** *Let $m \ge 2$. For almost every x, the pairs (q_{n-1}, q_n) reduced modulo m are equidistributed among the pairs (u, v) of residues with $\gcd(u, v, m) = 1$.*

Szüsz proved the statement for pairs of denominators in 1962 [127, Satz 3.2]; Moeckel obtained it up to sign through the geodesic flow on modular surfaces [94]; Jager and Liardet proved equidistribution of the whole matrix $\left(\begin{smallmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{smallmatrix}\right)$ modulo m [64]. Borda's recent paper surveys this history and extends it [19]. The case $m = 2$ says that, for almost every x, exactly one third of the denominators are even.

Corollary 8.2. *For almost every x and every prime ℓ, the proportion of n with $\ell \mid q_n$ is $1/(\ell+1)$, not the $1/\ell$ of random integers.*

Proof. Modulo a prime ℓ, the admissible pairs are the $\ell^2 - 1$ nonzero pairs (u, v). Those with $v = 0$ are the $\ell - 1$ pairs $(u, 0)$, $u \neq 0$. By Theorem 8.1 the proportion is $(\ell-1)/(\ell^2-1) = 1/(\ell+1)$. □

Table 8.1 and Fig. 8.1 check the corollary on the first 10,000 convergents of π and on 400,000 convergents of 200 random reals. Both match $1/(\ell+1)$ to within sampling error; π is typical here too.

Step 4: the correction factor is $\zeta(2)$. In the usual way of refining the prime number theorem for structured sequences, the probability that q_n is prime is $1/\ln q_n$ multiplied, for each prime ℓ, by the ratio of the chance that $\ell \nmid q_n$ to the chance that ℓ does not divide a random integer:

$$\mathfrak{S} = \prod_{\ell} \frac{1 - \frac{1}{\ell+1}}{1 - \frac{1}{\ell}} = \prod_{\ell} \frac{\ell}{\ell+1} \cdot \frac{\ell}{\ell-1} = \prod_{\ell} \frac{1}{1-\ell^{-2}} = \zeta(2) = \frac{\pi^2}{6}. \tag{8.2}$$

The product converges absolutely, so the correction is a single constant. It is a pleasant surprise that the constant is $\pi^2/6$, the same number as in Chapter 5, and for a related reason: the correction measures how much more often than random a denominator avoids each prime, and the reason it does is the coprimality of neighbors.

Step 5: the count. Putting the steps together, for a typical x,

$$\mathbb{E}\,\#\{n \le N : q_n \text{ prime}\} \approx \sum_{n \le N} \frac{\zeta(2)}{\ln q_n} \approx \frac{\zeta(2)}{L} \ln N = \frac{\pi^2}{6} \cdot \frac{12 \ln 2}{\pi^2} \ln N = 2 \ln 2 \cdot \ln N. \tag{8.3}$$

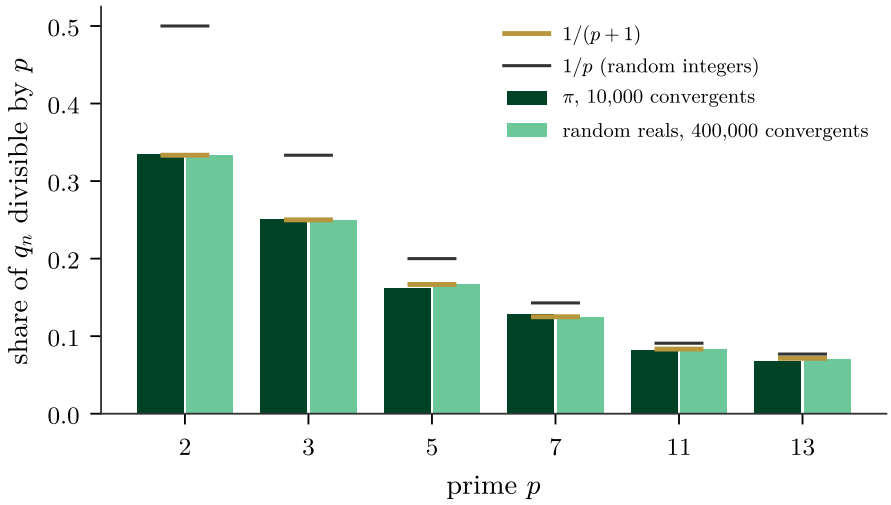


Figure 8.1. Convergent denominators avoid small prime factors: the share of q_n divisible by ℓ follows $1/(\ell+1)$ (gold), not $1/\ell$ (black), both for π and for random reals.

Heuristic 8.3. For almost every real x, the number of prime denominators among the first N convergents grows like $c \ln N$ with

$$c = 2 \ln 2 = 1.3863\ldots$$

The same holds for prime numerators, because Theorem 8.1 in the matrix form of Jager and Liardet gives the numerators the same local behavior.

Three remarks. First, the constant is independent of everything except the two laws used, which is why it is so clean: $\zeta(2)$ from the arithmetic and $\pi^2/(12 \ln 2)$ from the dynamics, and the two π^2 cancel. Second, a count growing like $\ln N$ is very slow: it predicts infinitely many prime denominators for a typical number, but only about one new one each time N is multiplied by $e^{1/c} \approx 2.06$, roughly one per doubling. Third, this is a heuristic. It treats primality of different q_n as independent events with the stated probabilities. It says nothing about any particular number such as π unless π is typical, which is unproved.

8.3 The test on random reals

To test Heuristic 8.3 we need many truly random real numbers with known continued fractions. We draw a random integer r with $D = 1{,}620$ digits and expand $r/10^D$ by exact integer arithmetic. Every real number in the interval $[r/10^D, (r+1)/10^D)$ shares the first terms, and since the error of the nth convergent shrinks like $q_n^{-2} \approx e^{-2Ln}$, about $D \ln 10/(2L) \approx 0.97D$ terms are shared, far more than the 1,000 we use. So the first 1,000 terms are those of a uniformly random real. We drew 400 such reals and counted probable primes among $q_1, \ldots, q_{1000}$ and $p_1, \ldots, p_{1000}$ (Appendix B).

Results. The mean number of prime denominators was 9.90 (standard deviation 2.87), and of prime numerators 9.68 (2.99). The growth is the test of the constant: between $n = 100$ and $n = 1000$ the mean count rose by 3.29, against the predicted $2 \ln 2 \cdot \ln 10 = 3.19$; between $n = 200$ and $n = 1000$ it rose by 2.22 against 2.23. A least-squares fit of the mean count

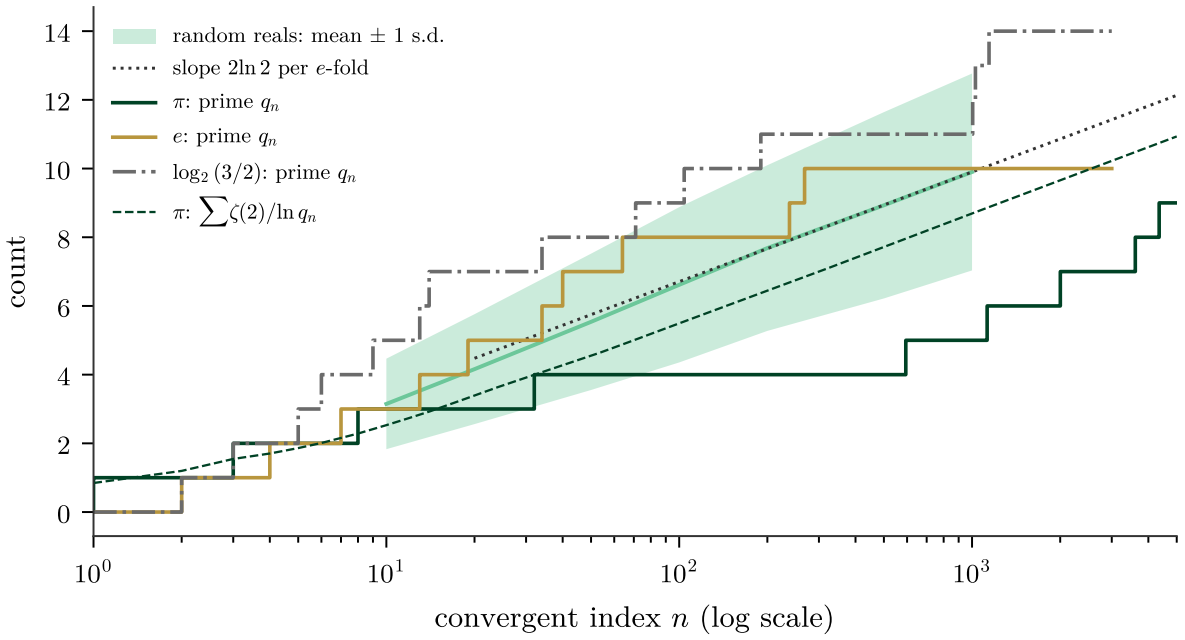


Figure 8.2. Prime convergent denominators, counted cumulatively against the index n (log scale). The shaded band is the mean ± one standard deviation over 400 random reals; the dotted line has the predicted slope $2\ln 2$ per e-fold. Steps: π (to $n = 5{,}000$), e and $\log_2(3/2)$ (to 3,000); dashed: π's own expected count $\sum\zeta(2)/\ln q_n$.

against $\ln n$ over $50 \le n \le 1000$ gives slope 1.45 ± 0.04, slightly above $2\ln 2$ because the earliest convergents, where q_n is small, still carry some weight. The absolute level has an additive constant that the heuristic does not fix: the pathwise sum $\sum\zeta(2)/\ln q_n$ averages 10.78 at $N = 1000$, a little above the observed mean, because $1/\ln q$ is a poor guide for the first few, very small denominators. The heuristic's *slope* is what matters, and it fits.

8.4 Is pi typical?

Table 8.2 gives the counts for the five constants of the Suite, and Figs. 8.2 and 8.3 compare them with the random sample.

Pi. Among its first 5,000 convergents, π has 9 prime denominators, at $n = 1$, 3, 8, 32, 594, 1126, 2002, 3610, 4355, against 10.9 expected; the last has 2,239 digits. It has 13 prime numerators, at $n = 4$, 8, 10, 15, 86, 229, 333, 593, 839, 852, 1148, 2725, 3787 (the largest has 1,949 digits). The first four denominators and the first numerators agree with the OEIS lists [105, 106]. At $n \le 1000$, π's count of 5 prime denominators sits at the 6th percentile of the random sample: on the low side, but well inside the range of ordinary numbers, and its count of prime numerators is about average. Its even denominators make up 0.3330 of the first 10,000 (prediction 1/3) and those divisible by 3 make up 0.2484 (prediction 1/4). By every measure here, π looks like a typical real number. Heuristic 8.3 predicts that it has infinitely many prime denominators, with the next few spread out roughly one per doubling of n. [Update: the five found since, at n = 6,925, 25,332, 27,651, 32,394 and 37,721 (the last four probable primes;

Table 8.2. Probable primes among convergents. Columns: the number of convergents N examined; prime denominators q_n, $n \le N$, and the heuristic expectation $\sum_{n \le N} \zeta(2)/\ln q_n$; prime numerators; the counts of prime q_n and p_n for $n \le 1000$ with the expectation; and the percentile of the q_n count at $n \le 1000$ among 400 random reals (share of random reals with at most as many). For $\sqrt{2}$ and φ the typical-number heuristic does not apply.

		$n \le N$			$n \le 1000$			
number	N	q_n	exp.	p_n	q_n	p_n	exp.	pct.
π	5,000	9	10.9	13	5	10	8.7	6
e	3,000	10	12.4	10	10	10	11.6	61
$\log_2(3/2)$	3,000	14	13.9	11	11	10	12.4	74
$\sqrt{2}$	3,000	23	16.8	18	17	16	14.7	–
φ	3,000	22	27.7	22	21	21	23.9	–

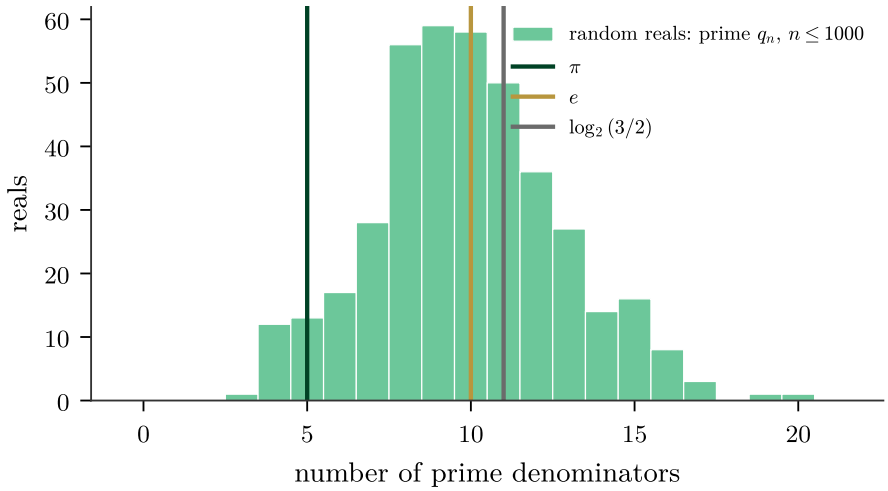


Figure 8.3. The number of prime denominators among the first 1,000 convergents, for 400 random reals (histogram), with π, e and $\log_2(3/2)$ marked.

Section 17.5), fit that rate; so does a sixteenth prime numerator, the probable prime p_{42037}. [Added 2026-11-10]]

The fifth and e. The number $\log_2(3/2)$ has 14 prime denominators up to $n = 3{,}000$ against 13.9 expected; e has 10 against 12.4. Both are unremarkable, although e is not a typical number (its terms grow, and its denominators are even exactly one third of the time because of its pattern, not because of Theorem 8.1). [Update: the typical-number heuristic does not apply to e at all: its denominators can be prime only at $n = 2$ or $n \equiv 1 \pmod 3$ (Theorem 17.9). [Added 2026-11-10]]

Quadratic irrationals. For φ, q_n is the Fibonacci number F_{n+1}; for $\sqrt{2}$ it is the Pell number P_{n+1}. Our counts recover the known lists of indices of Fibonacci and Pell primes [103, 108] up to index 3,001: the Fibonacci indices are 3, 4, 5, 7, 11, 13, 17, 23, 29, 43, 47, 83, 131, 137, 359, 431, 433, 449, 509, 569, 571, 2971, and the Pell indices are 2, 3, 5, 11, 13, 29, 41, 53, 59, 89, 97, 101, 167, 181, 191, 523, 929, 1217, 1301, 1361, 2087, 2273, 2393. These sequences break the

typical-number heuristic in a structured way. They are divisibility sequences ($F_m \mid F_n$ whenever $m \mid n$), so F_n can be prime only when n is prime (or $n = 4$), and their denominators are not equidistributed modulo m: half of the Pell denominators are even, not a third. Grantham and Granville [52] develop heuristics for primes in such linear recurrences; for Fibonacci numbers they predict about $e^{\gamma} \log_{\varphi} N$ primes F_n with $n \le N$, which is 29.6 for $N = 3{,}001$, against the 22 found (they note that the Fibonacci data run below the prediction). Whether either list is infinite is open.

8.5 Do the site's tools help?

The owner of the companion site asked whether the tools assembled there help with this question. Here is a frank assessment.

- *Rhythms and Sturmian words.* The denominators are block lengths (Theorem 3.5), which makes the question easy to state musically: does π's form contain infinitely many sections whose length in bars is prime? But the rhythm adds no information about primality beyond the recurrence. [Update: still true for proofs, but the recurrence run modulo small primes, the rhythm read one prime at a time, is now a working search tool (the rhythm sieve, Section 17.6). [Added 2026-11-10]]
- *Coprimality and $6/\pi^2$.* This is the tool that does real work: the coprimality of neighboring blocks is exactly what produces the correction factor $\zeta(2)$ in (8.2).
- *Necklaces, diffraction, the prime hunt.* These do not bear on the question. The prime hunt is a different experiment with the same verdict: rhythms of π neither attract nor repel primes.
- *Self-similarity.* For quadratic irrationals, exact self-similarity means a linear recurrence, and the recurrence's divisibility structure is exactly what governs Fibonacci and Pell primes. For typical numbers, self-similarity in rule but not in shape is what makes the random model plausible.

In short, the tools organize the heuristic and explain its constant. They do not bring a proof closer. A proof that π has infinitely many prime denominators would require, at the least, control of π's terms that nobody has: it is not even known that they are unbounded.

8.6 What is known, and what is not

Known: the equidistribution of convergents modulo m for almost every x (Theorem 8.1); Lévy's theorem; the published lists of prime numerators and denominators for π and of Fibonacci and Pell primes; the Grantham–Granville heuristics for linear recurrences. Open: whether π, e, φ or $\sqrt{2}$ has infinitely many prime convergent denominators. We did not find in the literature we searched the explicit heuristic constant $2 \ln 2$ for typical reals, or a test of it on a large random sample; both are offered here as a heuristic and an experiment, not a theorem.

CHAPTER 9

Thread (b): one spectrum, from crystal to randomness

9.1 The question

Chapter 6 ended with the Montgomery–Dyson observation: the zeros of the zeta function seem to be spaced like the eigenvalues of large random Hermitian matrices, the model physicists use for energy levels of heavy nuclei. The companion site lists this as an open thread and asks whether the tools assembled here help. They cannot prove anything about the zeros. What they can do is supply a scale on which to place the zeros: at one end a perfect crystal, next the rhythm of π (a quasicrystal), then the zeros and random matrices, and at the other end pure randomness. This chapter computes the statistics that define that scale, for every point set on it, with one set of tools.

9.2 The point sets

All point sets are *unfolded*, that is, rescaled so that the mean spacing is 1, before any statistic is computed.

- *Crystal:* the integers.
- *Quasicrystal:* the hits of π's rhythm in the first 33,102 beats (Theorem 3.5), multiplied by $\pi - 3$. The spacings take two values, $7(\pi - 3) = 0.9911$ and $8(\pi - 3) = 1.1327$. The three-gap points $\{n(\pi-3)\}$ of Theorem 3.7 belong here too; for $N = 1{,}000$ their spacings, in units of $1/N$, take the values $0.030, 8.610, 8.640$.
- *Zeta zeros:* the imaginary parts $\gamma_1 < \gamma_2 < \cdots$ of the first 5,000 nontrivial zeros, up to height 5447.86, computed with mpmath and checked against the zero-counting function. They are unfolded with the smooth part of the Riemann–von Mangoldt counting formula, $\bar{N}(T) = \frac{T}{2\pi}\ln\frac{T}{2\pi e} + \frac{7}{8}$; the unfolded spacings have mean 1.0000. Odlyzko's tables [100] reach far higher; we used only zeros we computed ourselves.
- *GUE:* the eigenvalues of 16 random 1000×1000 complex Hermitian matrices with independent Gaussian entries, from the central half of each spectrum, unfolded with the semicircle law [92]; mean spacing 0.9999.

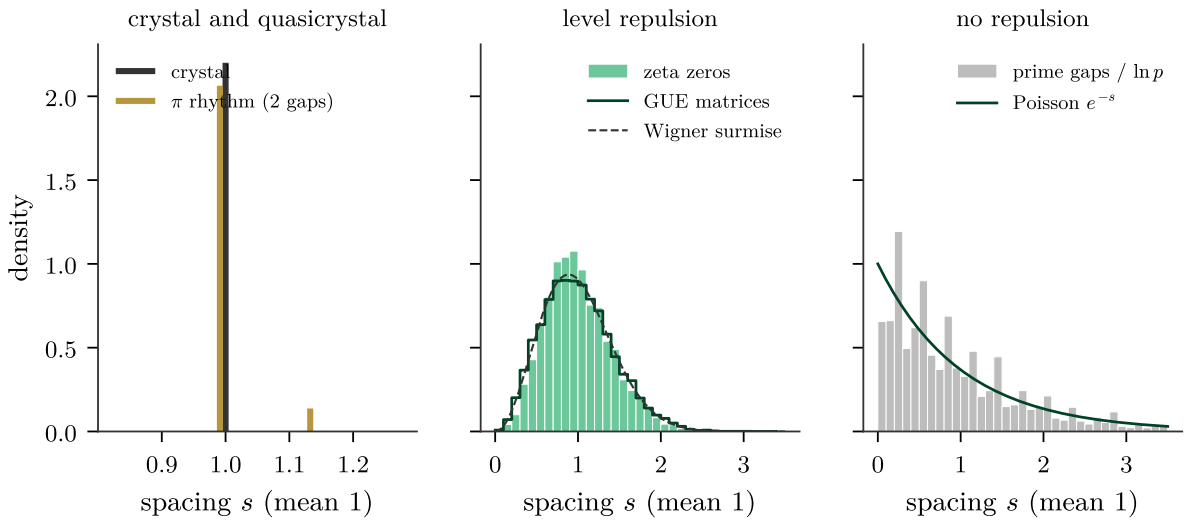


Figure 9.1. Nearest-neighbor spacing distributions, all with mean spacing 1. Left: a crystal (one spacing) and π's rhythm (two spacings, heights proportional to their frequencies; note the zoomed axis). Center: 5,000 zeta zeros (bars) and GUE eigenvalues (steps), with the Wigner surmise. Right: prime gaps divided by $\ln p$ near 10^9, with the Poisson law e^{-s}.

- *Primes:* the gaps between the 14,491 primes in $[10^9, 10^9 + 3 \cdot 10^5)$, divided by $\ln p$ and normalized to mean 1.
- *Poisson:* independent exponential spacings, the model of points with no correlation at all.

9.3 Nearest-neighbor spacings

Figure 9.1 shows the distribution of the spacings s between neighbors. The crystal has a single spacing and the quasicrystal two. The zeros and the GUE eigenvalues share a distribution that vanishes like s^2 at $s = 0$: close pairs are rare, a phenomenon called *level repulsion*. The curve drawn with them is the Wigner surmise $P(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$, a simple and very accurate approximation to the exact GUE spacing law [92]. The prime gaps, by contrast, look like the Poisson law e^{-s}, with the most likely spacing near zero.

A simple number captures the repulsion: the share of spacings below $1/4$. It is 0.009 for the zeros, 0.016 for the GUE matrices and 0.016 for the Wigner surmise, but 0.212 for the Poisson process and 0.129 for the normalized prime gaps (whose discreteness, all gaps being even, distorts the smallest bins). The Kolmogorov–Smirnov distance between the zero spacings and the GUE spacings is 0.036, against 0.305 between zeros and Poisson. The small remaining difference between zeros and GUE is expected: at the low heights we can reach, the zeros behave like random matrices of a modest effective size, a finite-height effect analyzed by Bogomolny, Bohigas, Leboeuf and Monastra [16]. Correspondingly, the variance of the zero spacings, 0.152, is a little below the GUE value 0.183.

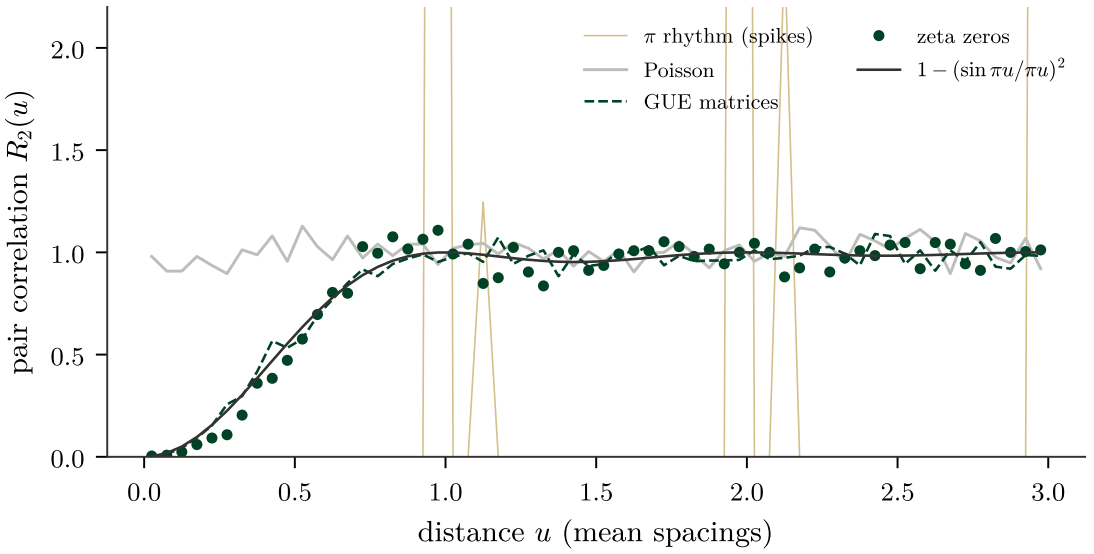


Figure 9.2. Pair correlation $R_2(u)$: 5,000 zeta zeros (dots), GUE eigenvalues (dashed), Poisson points (grey), π's rhythm (gold spikes), and Montgomery's function (9.1) (black).

9.4 Pair correlation

The pair correlation $R_2(u)$ measures how likely it is to find another point at distance u from a given point, relative to a uniform density. Montgomery's conjecture [96] is that for the unfolded zeros

$$R_2(u) = 1 - \left(\frac{\sin \pi u}{\pi u}\right)^2, \tag{9.1}$$

which is exactly the GUE pair correlation in the limit of large matrices [36, 92]. Figure 9.2 shows our estimate from the first 5,000 zeros, the GUE sample, a Poisson sample (flat at 1) and π's rhythm (sharp spikes at the sums of its two spacings). The root-mean-square difference between the zeros' histogram and (9.1) over $0 \le u \le 3$ is 0.063, comparable to the sampling noise of the histogram.

9.5 Number variance: a scale of rigidity

The most telling statistic is the *number variance* $\Sigma^2(L)$: drop a window of length L (in mean spacings) at a random place and count the points inside; $\Sigma^2(L)$ is the variance of that count. It measures how rigid the arrangement is over a range of L spacings.

- For a Poisson process, $\Sigma^2(L) = L$: the counts fluctuate as much as they can.
- For the GUE, $\Sigma^2(L) \approx \frac{1}{\pi^2}\left(\ln(2\pi L) + \gamma + 1\right)$ for large L, with γ Euler's constant [92]: the fluctuations grow only logarithmically.
- For a crystal, $\Sigma^2(L) = \{L\}(1-\{L\}) \le \frac{1}{4}$, bounded. Point sets with bounded number variance are called *hyperuniform* [130]; crystals and many quasicrystals, including Sturmian ones like π's rhythm, are of this kind.

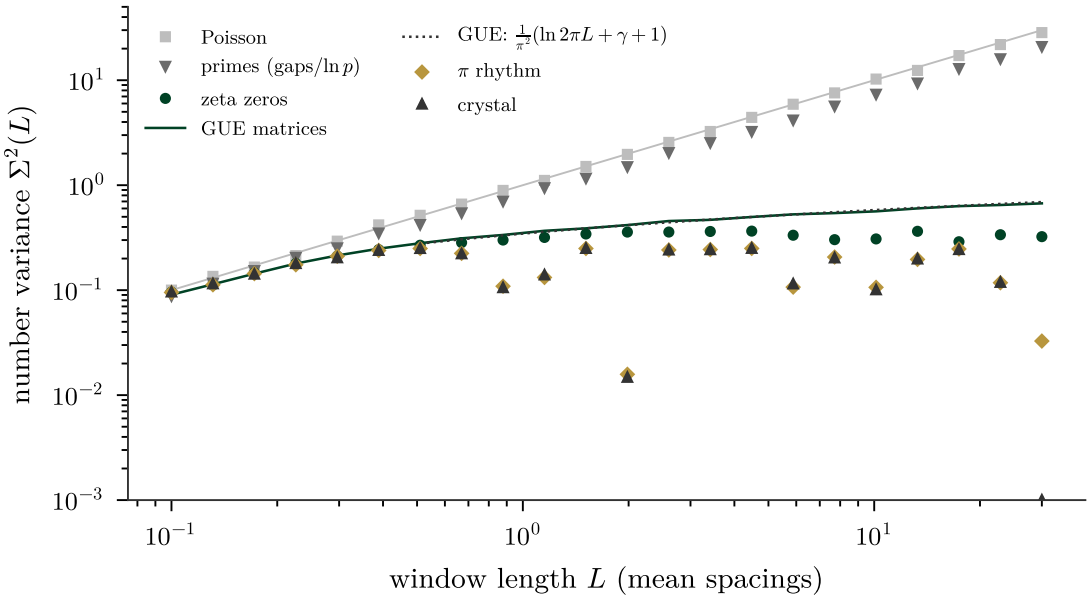


Figure 9.3. Number variance $\Sigma^2(L)$ on log–log axes for six point sets. From bottom to top: crystal and π's rhythm (bounded, with dips where L is close to a period), zeta zeros and GUE (logarithmic growth; dotted: the asymptotic GUE formula), normalized prime gaps and Poisson (linear growth).

For the zeros, Berry predicted that $\Sigma^2(L)$ follows the GUE law for small L and then saturates at a level set by the height of the zeros [14]. Figure 9.3 shows all six point sets on one logarithmic plot. At $L = 10.1$ we find $\Sigma^2 = 0.10$ for the crystal, 0.11 for π's rhythm, 0.31 for the zeros, 0.56 for the GUE sample (asymptotic formula 0.58), 7.36 for the normalized prime gaps and 10.26 for Poisson. The zeros lie below the GUE curve at this L, consistent with Berry's saturation at our low heights, while the GUE sample matches its formula.

9.6 The spectrum, and Dyson's bridge

The three statistics put the point sets in one order of decreasing rigidity:

$$\text{crystal} \;\to\; \text{quasicrystal } (\pi\text{'s rhythm}) \;\to\; \text{GUE and zeta zeros} \;\to\; \text{Poisson (and the primes at short range).}$$

The crystal has one spacing and bounded fluctuations. The rhythm of π has two spacings, at most three on the circle, and still bounded fluctuations: it is ordered without being periodic. The zeros have a continuous spacing law but repel each other and fluctuate only logarithmically. Poisson points do not interact at all, and the primes, seen through their gaps at short range, look Poisson-like, although Chapter 6 showed their long-range diffraction is far from random.

Where do the zeros belong, then? Freeman Dyson, in his lecture "Birds and Frogs," observed that if the Riemann hypothesis is true then the zeros form a one-dimensional quasicrystal in a precise sense, and suggested that classifying one-dimensional quasicrystals might lead to a proof [35]. His sense is about the Fourier transform of the zeros, not their spacings, and the explicit formula (6.2) explains it: the sum of the waves $\cos(\gamma t)$ over the zeros is concentrated at $t = \ln p^m$, the logarithms of prime powers, just as the diffraction of a quasicrystal is concentrated on a discrete set. Figure 9.4 shows this with our 5,000 zeros: the sum $\sum_\gamma \cos(\gamma t)$,

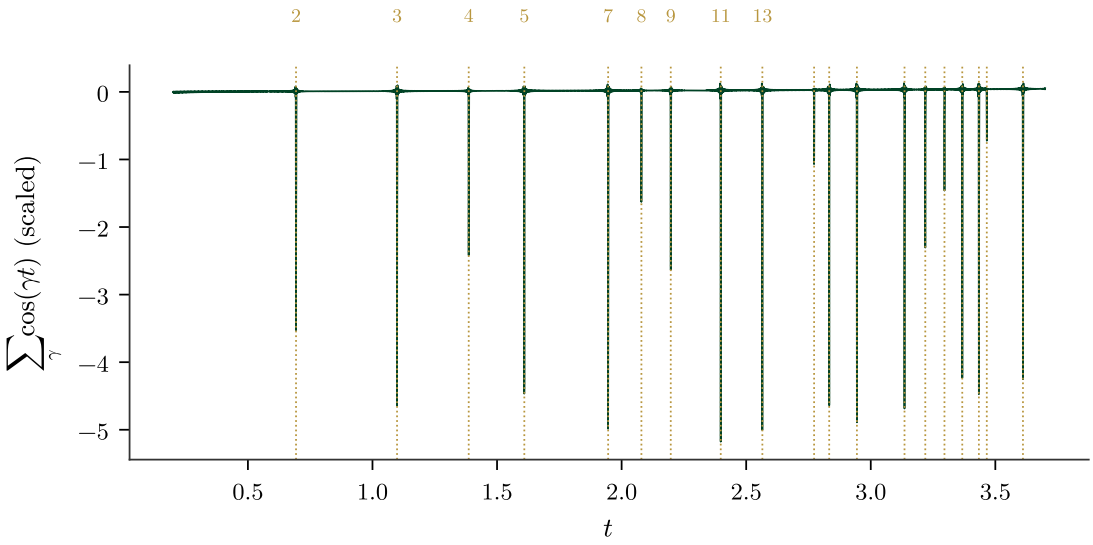


Figure 9.4. The zeros "diffract" at logarithms of prime powers: $\sum_\gamma \cos(\gamma t)$ over the first 5,000 zeros with a smooth cutoff, scaled by $1/\sqrt{\text{number of zeros}}$. Dotted gold lines mark $t = \ln n$ for the prime powers $n \le 40$; the dips sit exactly there, as the explicit formula predicts.

smoothly cut off, has sharp dips at $\ln 2$, $\ln 3$, $\ln 4$, $\ln 5$, $\ln 7$, … (depths -3.5 and -4.6 at $\ln 2$ and $\ln 3$ on the plotted scale, against a background fluctuation of about 0.01). So the zeros are GUE-like at short range and quasicrystal-like in their Fourier transform. The rhythm of π is quasicrystal-like in both. That is the bridge Dyson pointed to, and this picture organizes it; it does not cross it.

9.7 What this does and does not show

Everything in this chapter is a measurement of known phenomena with our own data, arranged so that a reader can see a crystal, π's rhythm, the zeros, random matrices and the primes on one scale. It illuminates the Montgomery–Dyson connection and makes Dyson's remark concrete. It proves nothing about the Riemann hypothesis, about the pair-correlation conjecture, or about any operator whose eigenvalues might be the zeros. Of the site's tools, the ones that help are the rhythm (as the quasicrystal end of the scale) and the diffraction picture (as the language for Dyson's remark); the others play no role.

CHAPTER 10

Self-similarity and fractals

10.1 Teddy Rook's picture

Teddy Rook pictured π as "a number, then a slightly smaller number after it, at least a decimal place over, then a smaller one after that, and so on." Does that idea fit with fractals? It does, and the precise version is one line long. By (2.1),

$$x = a_0 + \frac{1}{x_1}, \qquad x_1 = [a_1; a_2, a_3, \dots].$$

The "smaller number after it," the tail $1/x_1$, is built from a continued fraction made by exactly the same rule; so is its tail, and so on forever. An object that contains a copy of itself, made by the same rule, at every scale is *self-similar*, and self-similarity is the defining feature of a fractal. And each step really is smaller: by (2.7) the nth convergent misses by less than $1/q_n^2$. For π the complete quotients are $x_1, \dots, x_6 = 7.0625, 15.9966, 1.0034, 292.6346, 1.5758, 1.7367$ (Fig. 10.1).

Exact self-similarity, and self-similarity of rule. When the terms repeat, the tail is literally the number again: $\varphi = 1 + 1/\varphi$, so every complete quotient of φ is φ, and every complete quotient of $\sqrt{2} = [1; 2, 2, \dots]$ after the first is $1 + \sqrt{2}$. Every picture we draw from such a number repeats exactly under zoom. For π the terms $7, 15, 1, 292, 1, 1, 1, 2, \dots$ follow no known pattern, so the *rule* is the same at every level but the *shape* is new each time. That is not a flaw in the picture; it is one of the most interesting facts about π.

A twist: patterned fractions for π. If the numerators of a continued fraction are allowed to be numbers other than 1, π does have beautiful patterns, and then Teddy Rook's "never-ending string of continued fractions" holds with a clean rule all the way down. Brouncker found

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cdots}}} \tag{10.1}$$

in 1655, at the request of John Wallis, who published it in his *Arithmetica Infinitorum*; Khrushchev has reconstructed Brouncker's proof [73]. Lange found $\pi = 3 + 1^2/(6 + 3^2/(6 +$

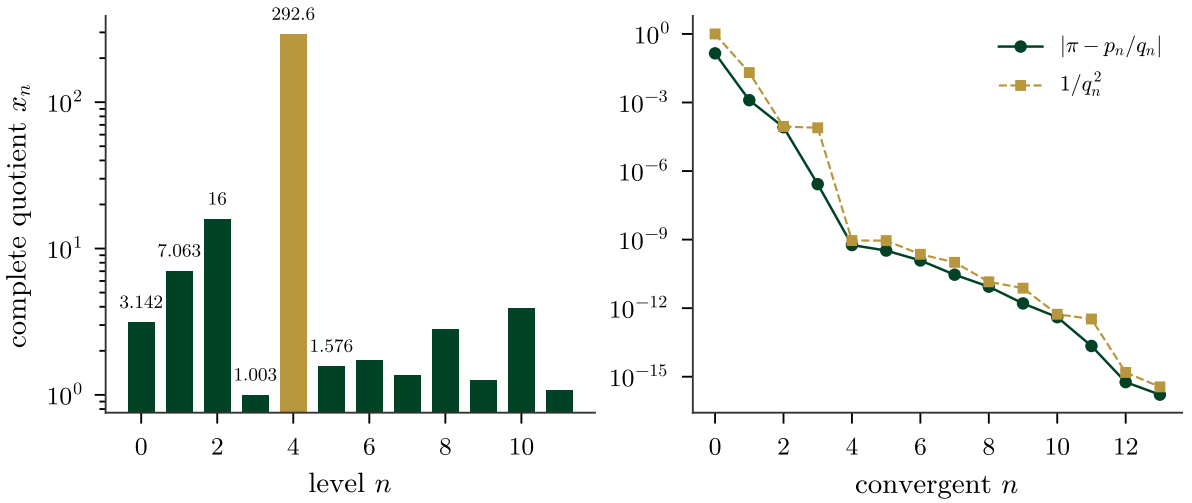


Figure 10.1. The nested tail of π. Left: the complete quotients x_n (log scale); each is the number "seen from" level n, and its whole part is the next term. The gold bar is $x_4 = 292.63$. Right: the error of each convergent (green) stays below $1/q_n^2$ (gold).

$5^2/(6+\cdots)))$ [79], and the continued fraction of the arctangent gives $\pi = 4/(1+1^2/(3+2^2/(5+3^2/(7+\cdots))))$. They converge at three very different speeds. After 10, 100 and 1000 levels Brouncker's fraction misses π by 0.0907, 0.0099 and 0.000999; its stopping points are exactly the partial sums of the Leibniz series $4(1-\frac{1}{3}+\frac{1}{5}-\cdots)$, as our script confirms with exact fractions. Lange's misses by 0.000186, $2.43e-7$ and $2.49e-10$ (its stopping points are the partial sums of Nilakantha's series). The arctangent fraction misses by 0.00013, $1.9e-8$ and $4.3e-16$ after 5, 10 and 20 levels; each level shrinks the error by a factor $0.1716 = (\sqrt{2}-1)^2$.

The rest of this chapter turns the rule "a number contains a copy of itself after every term" into pictures. Each is a classical fractal; for each we say who found it and draw it for π.

10.2 Ford circles

In 1938 Lester Ford drew, for every fraction p/q in lowest terms, a circle $C(p/q)$ tangent to the real axis at p/q with radius $1/(2q^2)$ [44].

Proposition 10.1 (Ford)**.** *Two Ford circles $C(p/q)$ and $C(r/s)$ never overlap. They are tangent if and only if $|ps-qr|=1$, that is, if p/q and r/s are neighbors in some Farey sequence; and between two tangent circles, the largest circle tangent to both is $C\big((p+r)/(q+s)\big)$.*

Proof of the first two statements. The squared distance between the centers minus the square of the sum of the radii is

$$\Big(\frac{p}{q}-\frac{r}{s}\Big)^2+\Big(\frac{1}{2q^2}-\frac{1}{2s^2}\Big)^2-\Big(\frac{1}{2q^2}+\frac{1}{2s^2}\Big)^2=\frac{(ps-qr)^2-1}{q^2s^2},$$

which is ≥ 0 for distinct fractions, with equality exactly when $|ps-qr|=1$. □

Zoom in anywhere and the same pattern of circles appears again; the picture is self-similar under the maps $x \mapsto (ax+b)/(cx+d)$ with integer entries and determinant ± 1. Ford circles turn Legendre's theorem into geometry. A vertical line through x passes through $C(p/q)$ exactly

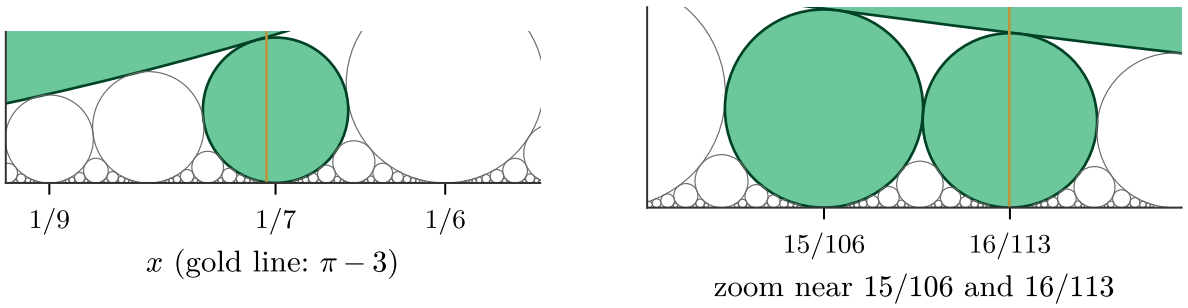


Figure 10.2. Ford circles near $\pi - 3$ (gold line), drawn to scale. Filled circles belong to convergents of $\pi - 3$. Left: the circle at $1/7$ with its neighbors. Right: a zoom by a factor of about 300, where the circles at $15/106$ and $16/113$ touch; the line passes through the second and misses the first.

when $|x - p/q| < 1/(2q^2)$, so by Theorem 2.4 every circle it pierces belongs to a convergent; and of any two consecutive convergents, at least one is pierced [71, Ch. II]. For $\pi - 3$ our script finds the pierced circles among the first convergents to be $0/1, 1/7, 16/113, 4703/33215, 14093/99532$, while $15/106$ and $4687/33102$ are convergents whose circles the line just misses (Fig. 10.2). Falling down the line toward $\pi - 3$, one drops through circle after circle, and after $16/113$ the fall is very long, because $355/113$ is so close: that is the 292 again.

10.3 The Stern–Brocot tree

Start with $0/1$ and $1/0$ and repeatedly insert the *mediant* $(p + r)/(q + s)$ between neighbors p/q and r/s. The result, described by Moritz Stern in 1858 [125] and independently by the clockmaker Achille Brocot in 1861 [21], who used it to design gear trains, is a binary tree that contains every positive fraction exactly once, in lowest terms [51, §4.5]. To locate a number, start at $1/1$ and go left (L) if the number is smaller and right (R) if larger. The path is a sequence of runs, and the run lengths are the terms of the continued fraction. For $355/113 = [3; 7, 16]$ the path is $R^3L^7R^{15}$ (the last run is one short because the path stops on the fraction itself), and for π our script finds $R^3L^7R^{15}LR^{292}LR\ldots$, with the 292 as a run of 292 Rs. The golden ratio zigzags $RLRL\ldots$ forever and $\sqrt{2}$ goes $R, LL, RR, LL, \ldots$. Every subtree is a copy of the whole tree with the fractions changed, which is self-similarity again (Fig. 10.3).

10.4 Minkowski's question-mark function

In 1904 Hermann Minkowski introduced a function that turns continued fractions into binary expansions [93]. For $x = [0; a_1, a_2, a_3, \ldots]$,

$$?(x) = 2\left(2^{-a_1} - 2^{-(a_1+a_2)} + 2^{-(a_1+a_2+a_3)} - \cdots\right), \tag{10.2}$$

which in binary is $a_1 - 1$ zeros, then a_2 ones, then a_3 zeros, and so on: each term becomes one run of binary digits. The function is continuous and strictly increasing from $?(0) = 0$ to $?(1) = 1$, maps rationals to dyadic rationals and quadratic irrationals to rationals, and yet its derivative is zero almost everywhere: it is a "singular" function, all of whose increase happens on a set of measure zero [32, 117]. It satisfies $?(x/(1 + x)) = ?(x)/2$, so the left half of its

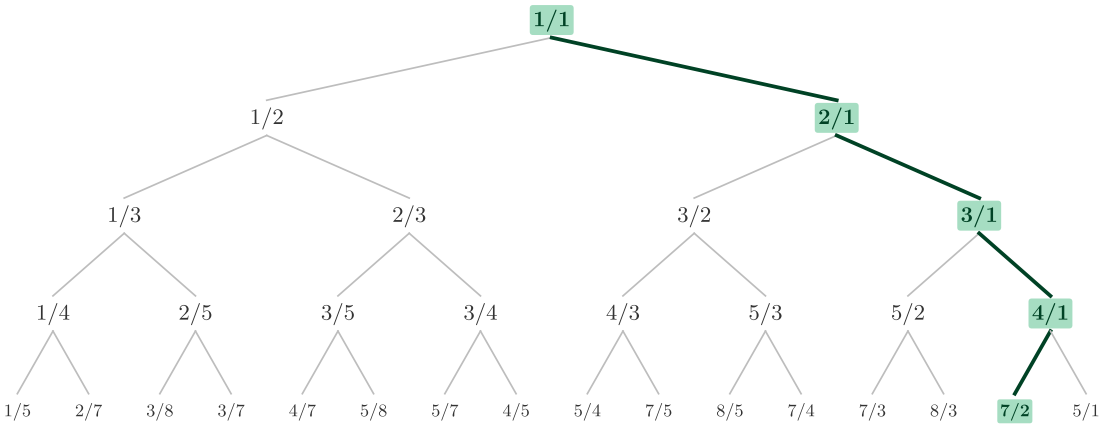


Figure 10.3. The first five levels of the Stern–Brocot tree, with the beginning $RRRL$ of the path to π highlighted: $1/1 \to 2/1 \to 3/1 \to 4/1 \to 7/2$. The path continues with six more Ls to $22/7$.

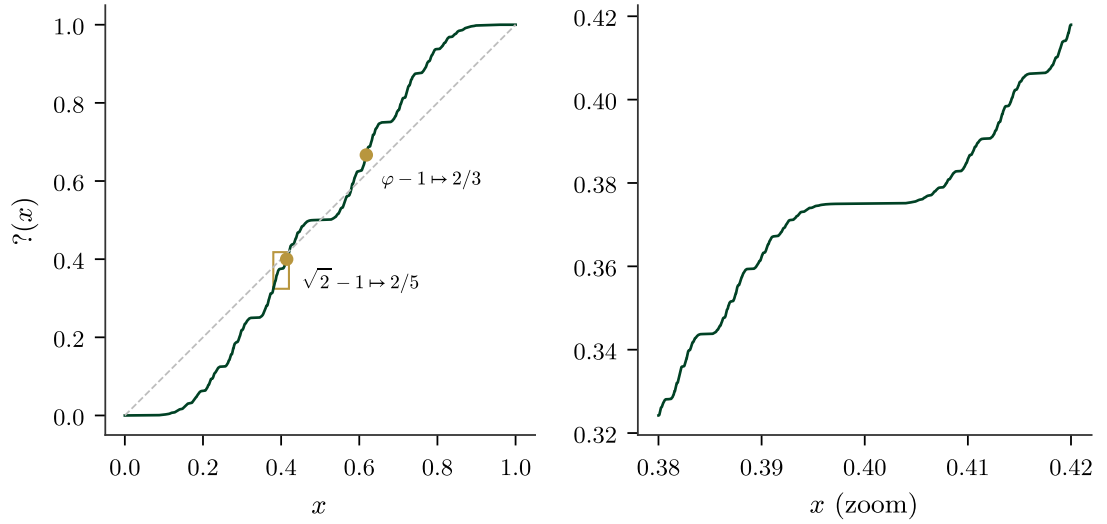


Figure 10.4. Minkowski's $?(x)$. Left: the whole graph with the diagonal dashed; quadratic irrationals land on fractions. Right: the boxed region magnified, showing flat stretches and bursts of growth at every scale.

graph is the whole graph squashed by a factor 2. Our script confirms $?(\varphi - 1) = 2/3$ and $?(\sqrt{2} - 1) = 2/5$, and finds $?(\pi - 3) = 0.01562476\ldots$, just under $1/64$: six zeros, then fifteen ones (Fig. 10.4).

10.5 The devil's staircase

Return to the circle map (4.1) of Chapter 4, at full coupling $K = 1$. Plot the rotation number W against Ω: the result is a staircase with a flat step at every fraction (Fig. 10.5). At $K = 1$ the steps fill the whole interval except a Cantor set of measure zero, whose fractal dimension

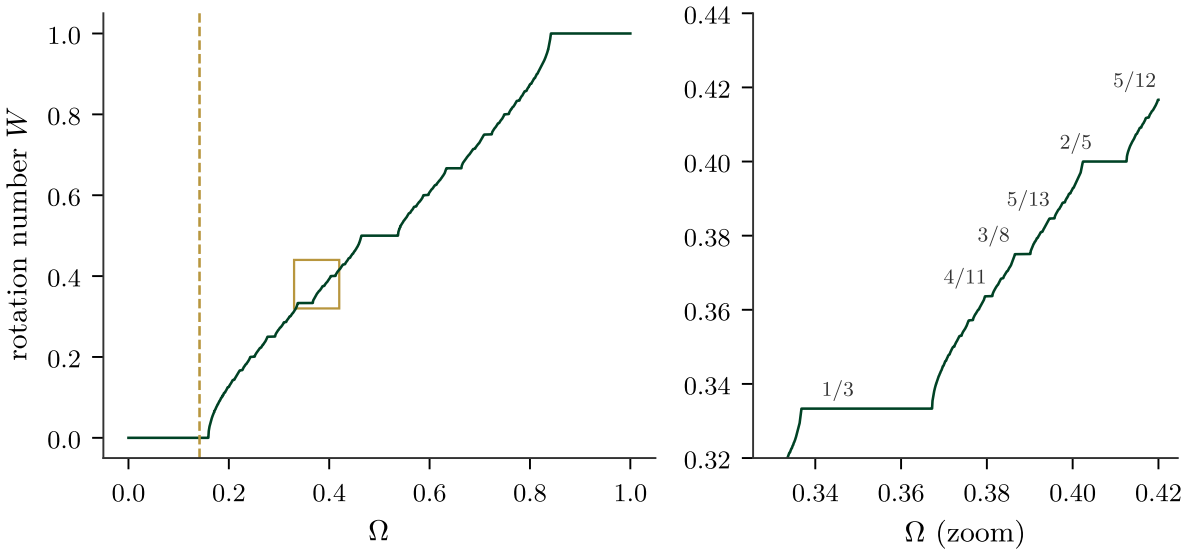


Figure 10.5. The devil's staircase: rotation number W of the sine circle map at $K = 1$ against Ω (2,520 iterations per point). The dashed line is $\Omega = \pi - 3$, inside the wide 0/1 step. Right: the boxed region, with steps labeled by their fractions; between 1/3 and 2/5 the widest step is 3/8, between 3/8 and 2/5 it is 5/13.

Jensen, Bak and Bohr measured as about 0.87 [67]. Between two adjacent steps p/q and r/s the widest step is the one at the mediant $(p+r)/(q+s)$, the Stern–Brocot rule once more, so zooming between two steps reproduces the whole staircase in miniature.

10.6 Cantor dust: numbers with small terms

Which numbers in $(0, 1)$ have continued fractions that use only the terms 1 and 2? Fixing the first term to be 1 or 2 keeps two intervals; fixing the second keeps two sub-intervals of each, and so on. What survives forever is a Cantor-like "dust" E_2 (Fig. 10.6). It has length zero, but more than a point's worth of size: its Hausdorff dimension is $\dim_H E_2 = 0.5312805062772051\ldots$. I. J. Good bounded it between 0.5306 and 0.5320 in 1941 [50]; Jenkinson and Pollicott computed it to 25 digits with a transfer-operator method [65] and later to 100 rigorously certified digits [66].

The method is elegant. The dimension of the set E_A of numbers whose terms all lie in a finite set A is the unique s for which the transfer operator

$$(\mathcal{L}_s f)(x) = \sum_{a \in A} (a + x)^{-2s} f\left(\frac{1}{a + x}\right)$$

has spectral radius 1. We discretize $\mathcal{L}_s$ on Chebyshev nodes and bisect on s (Appendix B). With 60 nodes our script gives $\dim_H E_{\{1,2\}} = 0.531280506277$, agreeing with the published value in all twelve printed digits. It also gives $\dim_H E_{\{1,2,3\}} = 0.70566$, $\dim_H E_{\{1,2,3,4\}} = 0.78895$, $\dim_H E_{\{1,\ldots,6\}} = 0.86762$ and $\dim_H E_{\{2,3\}} = 0.33744$. Since $\pi - 3$ starts with a 7 and contains a 292, it lies in none of these small-term dusts; the golden ratio's $\varphi - 1$ (all 1s) and $\sqrt{2} - 1$ (all 2s) are single points of $E_{\{1,2\}}$.

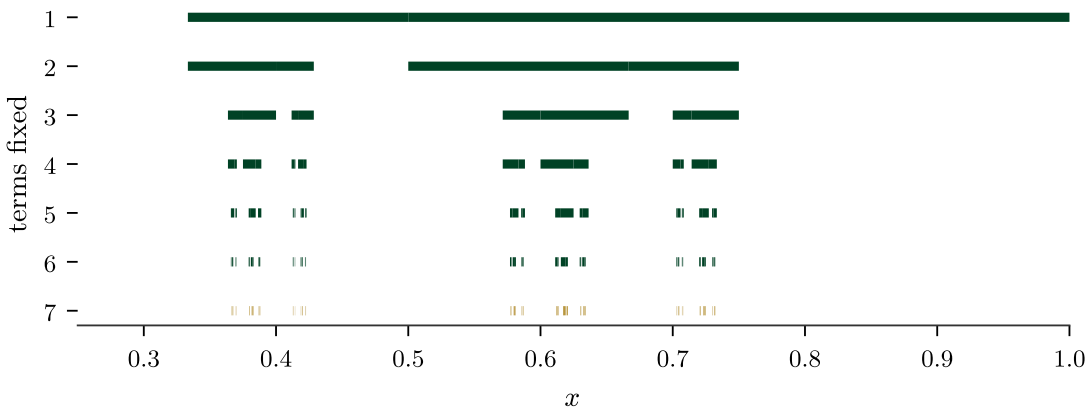


Figure 10.6. The numbers whose first k terms are all 1 or 2, for $k = 1, \ldots, 7$. Each row keeps two pieces of each piece of the row above; the limit is a Cantor set of dimension 0.5313.

10.7 The Mandelbrot set's Farey bulbs

The Mandelbrot set consists of the complex c for which $z \mapsto z^2 + c$, started at $z = 0$, stays bounded. Its heart-shaped main body is fringed with bulbs, and each bulb carries a fraction: the bulb attached at the point $c = e^{2\pi ip/q}/2 - e^{4\pi ip/q}/4$ of the main cardioid is the p/q bulb, and inside it the orbit settles on an attracting cycle of period q. Devaney explained how these fractions are arranged [33]: between the p/q and r/s bulbs, the largest bulb (measured by the share of the outside that leads into it) is the $(p + r)/(q + s)$ bulb, the mediant once more, which puts the Fibonacci fractions $1/2, 1/3, 2/5, 3/8, 5/13, \ldots$ on a path toward the golden-mean point. Our script places a test point just inside each bulb with $q \leq 12$ and finds an attracting cycle of period exactly q in every case (0 exceptions). Figure 10.7 labels the larger bulbs. Walking $\pi - 3 = 0.14159$ of the way around the cardioid lands just short of the $1/7$ bulb, since $\pi - 3 < 1/7$, and π's convergents $15/106$ and $16/113$ are bulbs much smaller and much closer to that spot.

10.8 The Fibonacci word fractal

In 2009 Alexis Monnerot-Dumaine drew the Fibonacci word as a path [95]: take one unit step for each letter, and after the kth letter, if it is a 0, turn a quarter turn, left if k is even and right if k is odd. The Fibonacci word 0100101001001… is exactly the rhythm of $\varphi - 1$ from Chapter 3 with hits and rests exchanged (our script checks this over 30,000 letters), so this is the golden rhythm, walked. The path is a fractal curve made of smaller copies of itself, of dimension $3 \ln \varphi / \ln(1 + \sqrt{2}) = 1.6379$ [95]. Figure 10.8 draws the same rule for three numbers: the golden rhythm makes the fractal, $\sqrt{2} - 1$ weaves a grid, and $e - 2$ draws a zigzag of growing arms. For $\pi - 3$ the turns nearly cancel within each 7- or 8-beat gap, and over the first 33,102 beats the path is a long, thin zigzag.

10.9 Six numbers, six pictures

The Continued-Fraction Suite of Chapter 11 uses five numbers. One construction draws them all on the same footing: cut a $1 \times x$ rectangle into as many squares as fit (that count is the first

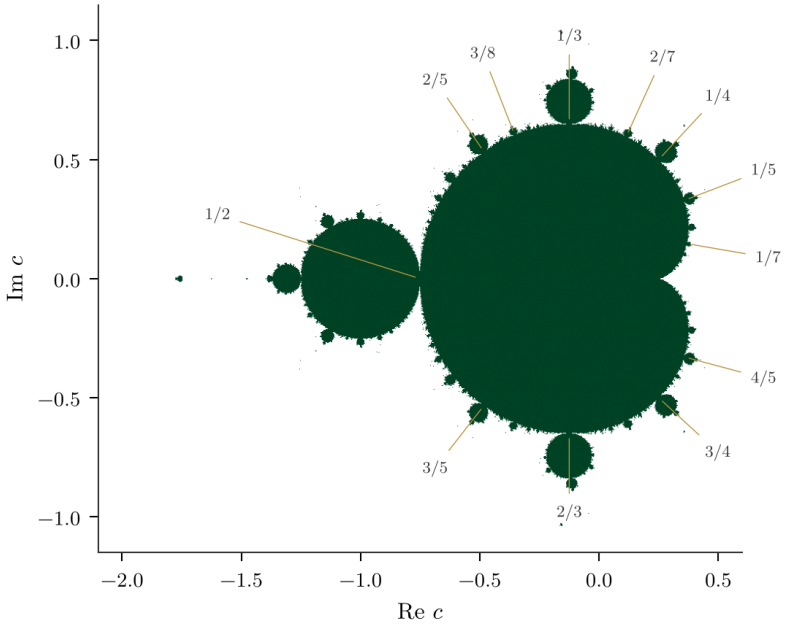


Figure 10.7. The Mandelbrot set with its main bulbs labeled p/q by the point where they attach to the cardioid. Inside the p/q bulb the orbit of 0 settles on a cycle of q points.

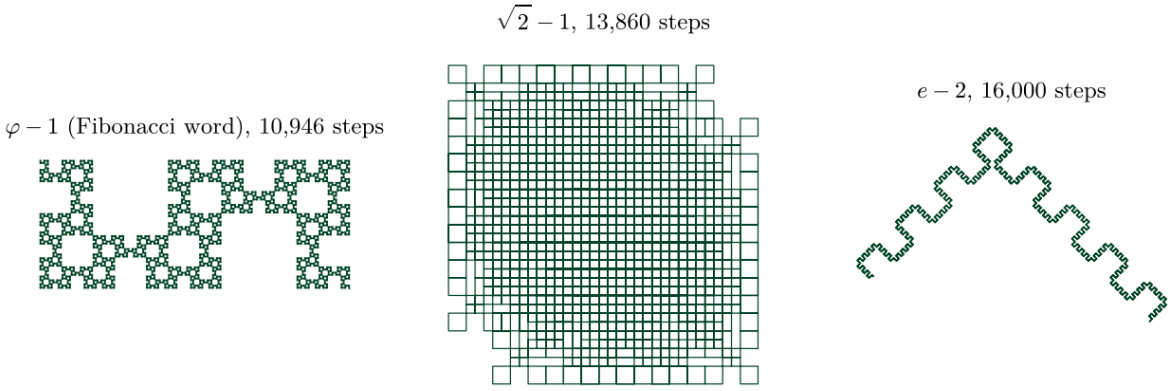


Figure 10.8. Rhythms walked as paths with Monnerot-Dumaine's rule. Left: the golden rhythm, the Fibonacci word fractal. Center: $\sqrt{2} - 1$. Right: $e - 2$.

term), turn the leftover a quarter turn, and repeat. Figure 10.9 shows the first six levels of each number, each level's rectangle stretched to the same height so that thin levels stay visible. The golden ratio cuts one square at every level and the silver ratio two: exactly self-similar. The fifth and e change their counts, e in its stretching pattern $1, 2, 1, 1, 4, 1, 1, 6$. Pi cuts 7, then 15, then 1, then 292 squares, the last a dense gold band. The sixth panel zooms to level 4 of π, the 292, which is where the sixth movement lives.

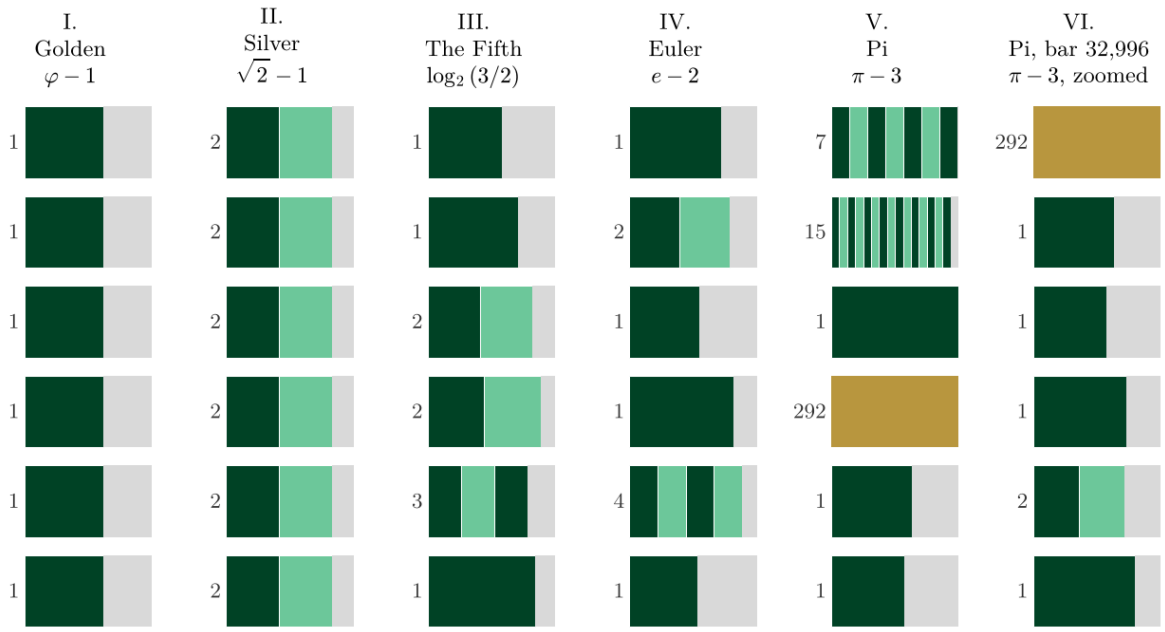


Figure 10.9. The Suite's numbers as square-cutting ladders. Each row is one level: the number of squares cut (printed at the left) is the term, and the grey sliver is what passes to the next level. Rows are stretched to equal height. A count above 50 is drawn as a gold band.

10.10 Is pi typical?

The Gauss map (2.2) reads off the terms, and the Gauss–Kuzmin law (Theorem 2.6) says what a typical number's terms look like. Here is a formal test of π's first 10,000 terms against it. Group the terms as $1, 2, \ldots, 12$ and 13+; the expected count in each group is 10,000 times its Gauss–Kuzmin probability. The chi-square statistic is $\chi^2 = 13.5$ on 12 degrees of freedom, with $p = 0.33$. With the finer grouping $1, \ldots, 20$ and 21+, $\chi^2 = 25.3$ on 20 degrees of freedom, $p = 0.19$. Either way the counts are fully consistent with the law (Fig. 10.10): 4,206 ones (42.1%, law 41.5%), 1,672 twos and 882 threes. The geometric mean of the terms is 2.6831 after 100 terms, 2.6627 after 1,000 and 2.6632 after 10,000, wandering around Khinchin's $K_0 = 2.685452001$ as a random number's would; and $\ln q_{10000}/10000 = 1.17941$ against Lévy's 1.18657 (Fig. 10.11). The largest term so far is 20,776, at position 431.

Nobody has proved that π obeys these laws, or even that its terms are unbounded, but by every count we can make it behaves like a typical number. This fits the null result of the prime hunt (Section 7.3) and the typical behavior of its prime convergents (Chapter 8). The number e, by contrast, fails the test on purpose: its terms $1, 2, 1, 1, 4, 1, 1, 6, \ldots$ have far too many ones and a geometric mean that grows without bound.

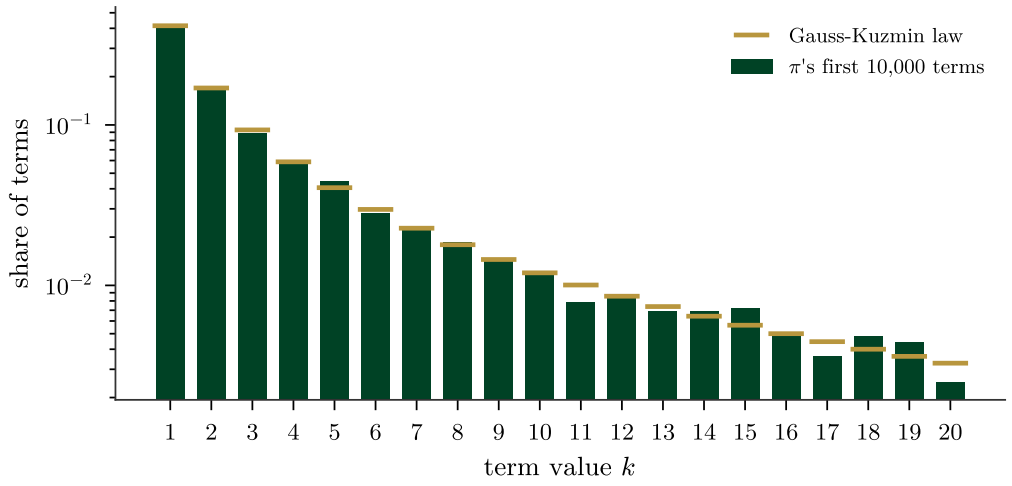


Figure 10.10. The shares of the values $1, \ldots, 20$ among π's first 10,000 terms (bars) against the Gauss–Kuzmin law (gold).

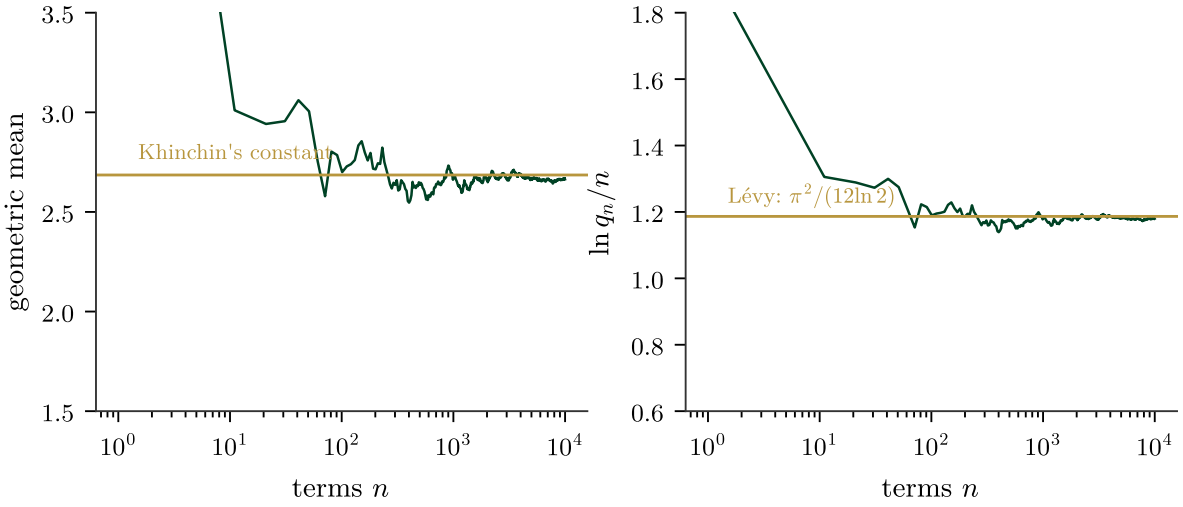


Figure 10.11. Running geometric mean of π's terms against Khinchin's constant (left) and $\ln q_n/n$ against Lévy's constant $\pi^2/(12 \ln 2)$ (right).

CHAPTER 11

Composition: the Continued-Fraction Suite as a method

11.1 The work

The *Continued-Fraction Suite* is an original composition by Teddy Rook: six short pieces for piano, one number each, engraved with LilyPond and published as a free score with a recorded performance [116]. It grew out of the Composer and "Compose a piece" instruments of the companion site, which play a number's rhythm with one voice per level. Whatever the prior art on its method (Chapter 13), the Suite is a new artwork. This chapter describes it as a method, in words and form diagrams; it does not reproduce the score. The musical facts below (keys, meters, tempos, bar counts, rehearsal marks) were read from the score by the site's builders and recorded in the site's data, and our script checks every bar count against the continued fractions.

11.2 The method

Each movement applies the same five decisions to its number.

1. *Choose the number and its levels.* Compute the terms $a_1, a_2, \ldots$ and the block lengths q_n (Theorem 3.5).

2. *Choose which level is a bar.* The block lengths then become lengths in bars of phrases, sections and larger units. Levels that are too short to be phrases are passed over: Golden starts its phrases at 3 and 5 bars, Silver at 2 and 5, Pi at 7.

3. *Lay out the form as the standard word.* At each level the long block appears a_{n+1} times and is closed by one short block. The score marks the long unit A, the short one B, and a unit that the number brings in only rarely as a "rare block."

4. *Put the rhythm inside each bar.* The left hand carries accents on the eighth notes where the number's own beat word has a hit (with α replaced by $1 - \alpha$ when $\alpha > \frac{1}{2}$, so the accents stay sparse). So the same number shapes the music at the smallest scale and the largest.

Table 11.1. The six movements of the Continued-Fraction Suite. "Bars" is the length of the movement as printed; Movement VI begins at bar 32,968.

movement	number	key	meter, tempo	bars
I. Golden	$\varphi - 1 = [0; 1, 1, 1, \dots]$	G major	3/4, Andante cantabile	89
II. Silver	$\sqrt{2} - 1 = [0; 2, 2, 2, \dots]$	D major	4/4, Adagio	70
III. The Fifth	$\log_2 \frac{3}{2} = [0; 1, 1, 2, 2, 3, \dots]$	F major	4/4, Allegretto	53
IV. Euler	$e - 2 = [0; 1, 2, 1, 1, 4, \dots]$	A minor	3/4, Tempo di valzer	71
V. Pi	$\pi - 3 = [0; 7, 15, 1, 292, \dots]$	C major	4/4, Moderato	113
VI. Pi, bar 32,996	$\pi - 3$, at level 4	C major	4/4, Moderato	50

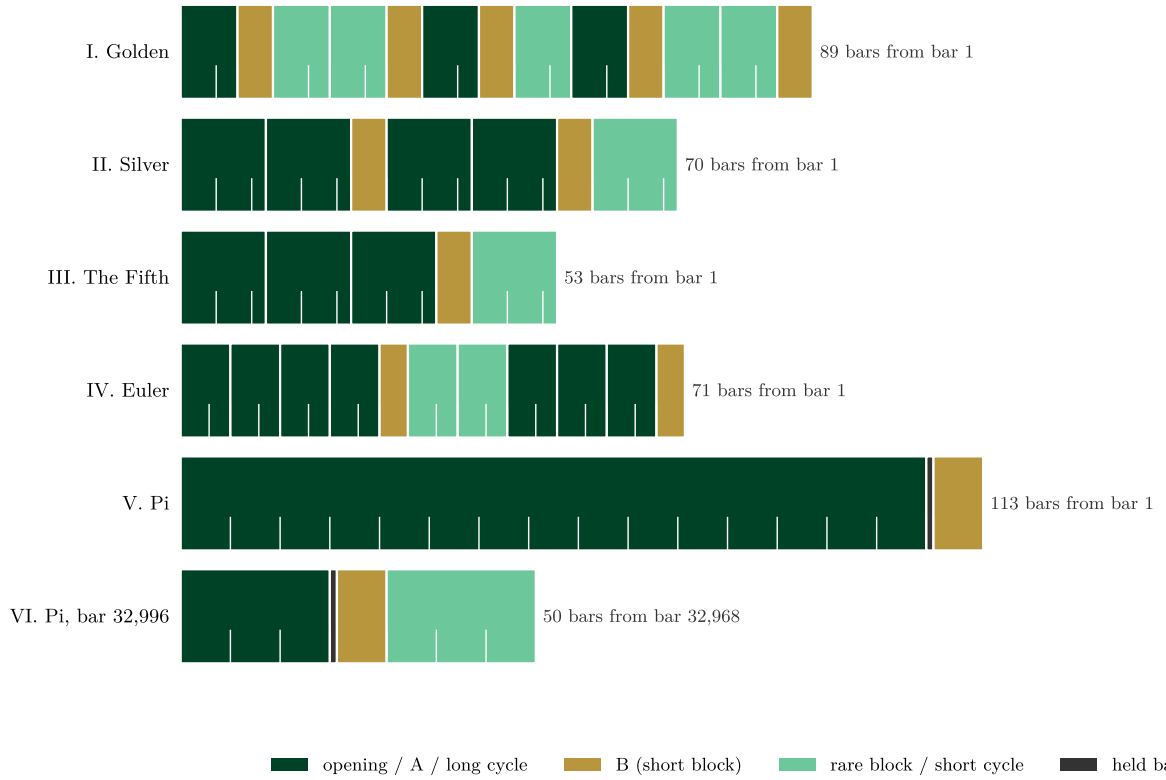


Figure 11.1. The forms of the six movements, drawn to scale (one unit per bar) from the section list read from the score. Colors follow the rehearsal marks: green for the opening and A sections (and Movement VI's long cycle), gold for B, mint for rare blocks and the short cycle, dark grey for a held bar. White ticks mark phrase boundaries.

5. *Let the melody walk the digits.* The score's preface says that the melody follows the number's decimal digits; this is a free compositional choice and carries no structural weight in what follows.

Figure 11.1 draws the six forms to scale, and the rest of this section reads each one.

I. Golden. Every term of $\varphi - 1$ is 1, so each block is the two blocks before it joined: the Fibonacci rule. Phrases are 5 and 3 bars, sections are $8 = 5 + 3$ and 5 bars, and the sections follow the Fibonacci word: $8, 5, 8, 8, 5, 8, 5, 8, 8, 5, 8, 8, 5$, which is eight 8s and five 5s, $64 + 25 =$

89 bars, itself a Fibonacci number. Our script checks that this sequence is the golden rhythm, letter for letter. Because every term is 1, no section ever comes three times in a row.

Golden also shows a subtlety of labeling that we call *two-level labels*. In the other movements the short block is truly rare. Here every level is "one long block, then one short block," and the sizes slide up the Fibonacci numbers: an 8-bar unit is the *long* block at the level of sections (8 against 5) and the *short* block at the level above ($13 = 8 + 5$ against 8). The score names each 8-bar unit by whichever level's boundary falls there, so the same 8 bars of music are marked A in some places and "rare block" in others. That is the self-similarity of the golden ratio, written into rehearsal marks.

II. Silver. Every term of $\sqrt{2} - 1$ is 2: "two of these, then one of those." Phrases are 5 and 2 bars, sections $12 = 5 + 5 + 2$ and 5 bars, and the whole is $29 + 29 + 12 = 70$ bars, where $29 = 12 + 12 + 5$. The numbers $2, 5, 12, 29, 70$ are the Pell numbers, the denominators of $\sqrt{2}$.

III. The Fifth. The terms of $\log_2(3/2)$ change: $1, 1, 2, 2, 3, 1, 5, \ldots$ Phrases are 5 and 2 bars and sections $12 = 5{+}5{+}2$; then the 3 takes over, three 12-bar sections and a 5 make 41, and one more 12 makes 53. The section sequence $12, 12, 12, 5, 12$ builds the denominators $5, 12, 41, 53$ of the best fractions for the fifth (Chapter 4) directly into the form.

IV. Euler. The terms of $e - 2$ grow in a pattern: $1, 2, 1, 1, 4, 1, 1, 6, \ldots$ Phrases are 4 and 3 bars, sections $7 = 4 + 3$ and 4; the 4 in the continued fraction makes a 32-bar block of four 7s and a 4; then $39 = 32 + 7$ and $71 = 39 + 32$. The denominators $7, 32, 39, 71$ are those of the convergents 5/7, 23/32, 28/39, 51/71 of $e - 2$.

V. Pi. The first three terms of $\pi - 3$ build a 113-bar cycle: fifteen 7-bar phrases, one held bar, and a last 7-bar phrase marked B, so $15 \times 7 + 1 + 7 = 113$. The held bar is the "one extra rest" of the standard word $s_2 = s_1^{15} s_0$, and the B phrase is the closing s_1 of $s_3 = s_2 s_1$.

VI. Pi, bar 32,996. The fourth term, 292, says that the 113-bar cycle repeats 292 times before the first short cycle of 106 bars. If Movement V never stopped, that moment would come after $292 \times 113 = 32{,}996$ bars, about 22.9 hours at the marked tempo (2.5 seconds per bar). The last movement jumps there. It plays bars 32,968 to 33,017: the end of the 292nd cycle (three phrases, the held bar and the B phrase, here in F major) and the opening three phrases of the short cycle, which begins at the printed bar 32,997. A short cycle is a long one without its closing phrase, so on paper it starts exactly like a long one; the composer marks the moment instead with a change of key and a hush. Section 3.4 explains why "beat 32,996" on the site and "bar 32,997" in the score name the same moment.

11.3 What the method gives a composer

The method gives proportions without dictating content. A number fixes how long each unit is and when contrast arrives; the composer still chooses every note. Quadratic irrationals give forms that repeat exactly at every level (Golden, Silver); numbers with changing terms give forms in which each level follows a different rule (The Fifth, Euler); and a number with one huge term gives a form with a surprise that is postponed almost indefinitely (Pi). In the language of Chapter 10, the first kind is self-similar in shape and the others only in rule.

CHAPTER 12

Connections: music, patterns, math and chemistry

The Rhythm of Numbers is the deep dive of a larger page on the companion site, a four-way Venn diagram of music, patterns, mathematics and chemistry whose fifteen regions each show a place where the subjects meet [114]. This chapter places the work of the earlier chapters in that diagram, briefly and only with claims we have checked.

12.1 Music and math: ratios, clocks and rhythms

This region is the home of most of this work. Simple frequency ratios (2:1, 3:2, 4:3) sound smooth together; the twelve notes form a clock in which the step of a fifth, 7, is coprime to 12 and so visits every note; and the comma that keeps the clock from closing exactly is the irrationality of $\log_2(3/2)$ (Chapter 4). Euclidean rhythms, the drum machine and the Suite belong here too.

12.2 Music and chemistry: octaves of elements

In 1865 John Newlands arranged the known elements by atomic weight and noticed that chemically similar elements recurred every eighth place, which he compared with the octave of a musical scale and called the law of octaves [86]. He presented the arrangement to the Chemical Society of London on March 1, 1866, which declined to publish it [86]; the periodic table of Mendeleev, with gaps for undiscovered elements, followed in 1869. His "octave" was a way of counting, not a sound. Molecules, on the other hand, do have characteristic vibration frequencies, far above hearing; infrared spectra are their fingerprints.

12.3 Music and patterns: sand that draws sound

In 1787 Ernst Chladni bowed sand-covered metal plates and recorded the patterns the sand made when the plate rang [27]. The sand collects on the nodal lines of a standing wave. For a square plate a classical approximation to the modes is

$$z(x,y) = \cos(n\pi x)\cos(m\pi y) \pm \cos(m\pi x)\cos(n\pi y), \qquad 0 \le x, y \le 1,$$

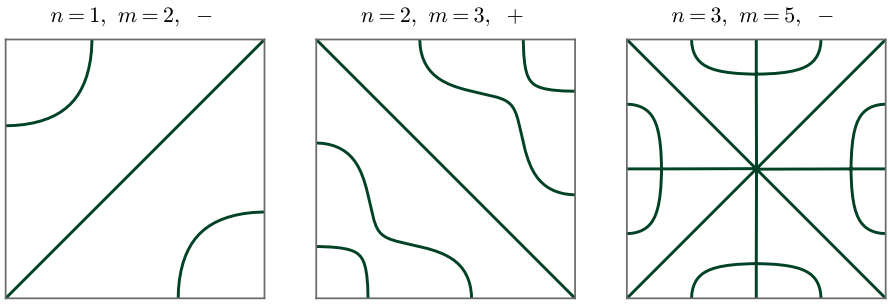


Figure 12.1. Chladni figures: the nodal curves $z = 0$ of the approximate square-plate modes for three choices of n, m and the sign.

with the sand on the curve $z = 0$ (Fig. 12.1). Real plates differ in detail near the edges. Melodies have symmetries of the same kind: transposition, inversion and retrograde are the translations and reflections of a pitch–time grid.

12.4 Math and patterns, patterns and chemistry

Periodic patterns in the plane and in space are classified by their symmetry groups, and crystals by the symmetries of their unit cells. Quasicrystals (Chapter 6) do not fit that classification: they are ordered, with sharp diffraction, but have no repeating cell [118]. On the site, the stained glass of cathedral windows shows the chemistry of color: dissolved metal ions absorb some wavelengths, and tiny metal particles scatter others.

12.5 Math and chemistry: why 2, 8, 18, 32

The electron shell n of an atom has subshells $l = 0, \dots, n-1$, each with $2l+1$ orbitals holding two electrons apiece. Since $1 + 3 + \cdots + (2n-1) = n^2$, shell n holds $2n^2$ electrons: $2, 8, 18, 32$. The squares here are the same squares whose reciprocals sum to $\pi^2/6$ in Chapter 5; as there, this is a shared formula, not shared physics. Section 18.1 gives a link that is shared physics: the same inverse-square symmetry that keeps planets on closed ellipses is why shell n holds $2n^2$ states. [Added 2026-11-10]

12.6 All four: waves and symmetry

At the center of the diagram is the standing wave. A string's modes are whole-number multiples of its lowest frequency, which is why it has a clear pitch. A circular drum's modes are not: their frequencies are proportional to zeros of Bessel functions, and the lowest four are in the ratios $1 : 1.59 : 2.14 : 2.30$, which is why a drum thumps rather than sings. An electron in an atom is a standing wave in three dimensions; its orbitals sort by their number of nodal surfaces just as drum modes sort by nodal lines, and their counts, filled in order, give the shape of the periodic table.

12.7 Where the Rhythm of Numbers sits

The Rhythm of Numbers lives mainly in the music–math region and reaches into the others: tuning and rhythm (music and math), coprimality and diffraction (math and patterns, patterns and chemistry), the $1/n^2$ of hydrogen and the n^2 of shells (math and chemistry), and self-similarity, which is the subject of patterns itself. What the diagram adds is a reminder that the same few structures (ratios, standing waves, symmetry, and the continued fraction that measures how far a ratio is from simple) appear in each subject for different reasons.

Chapter 18 adds a fifth subject to the diagram, physics and astronomy: the atom and the solar system as one problem, orbital resonance as rhythm, and the energy bands of a crystal in a magnetic field as the form of a number. [Added 2026-11-10]

CHAPTER 13

What is new here, and what is re-found

A work that walks through this much classical mathematics owes its readers a careful account of what is old. This chapter audits every idea in the work and in the companion site against the literature, and sorts it into four classes:

Classical: a standard result with a long history; we cite a primary source or a standard text.

Specialist: known in a research literature, though not in standard texts; we cite it.

New presentation: a new arrangement, picture, synthesis or computation of known results.

New: something we did not find anywhere: an original artifact, a computation, or a heuristic that we could not locate in the literature we searched.

The searches covered the standard texts cited throughout, the music-theory literature on maximally even sets, well-formed scales, Christoffel words and Sturmian rhythms (searching for, among others, "Sturmian words music," "continued fractions musical form," "Christoffel words rhythm"), the OEIS, and the number-theory literature on convergent denominators. No search of this kind is complete. When we say "we did not find prior work," we mean exactly that, and nothing stronger.

13.1 The audit

Table 13.1. The prior-art audit, idea by idea.

idea	class	notes and sources
The guess $\pi \approx 3+k/(7k+1)$ and $k = 16$	classical	The guess is the continued fraction of π stopped after two terms (Chapter 1); 355/113 is Zu Chongzhi's Milü [91]; 223/71 and 22/7 are Archimedes' bounds. Re-found independently by Teddy Rook.
Continued fractions, convergents, best approximation, Legendre, Lagrange	classical	Hardy and Wright [56], Khinchin [71].

idea	class	notes and sources
Gauss–Kuzmin, Khinchin, Lévy	classical	Khintchine [72], Kuzmin [77], Lévy [84, 85].
The "rhythm of a number"	classical	It is the characteristic Sturmian word of α, the cutting sequence of a line of slope α [4, 15, 88]. The rhythm of π in particular is the characteristic word of $\pi - 3$; we found no claim that it is new and make none.
Nested blocks: block lengths are the q_n	classical	Standard words; Brown [22], Fraenkel et al. [45]; our proof in Chapter 3 is a direct version for completeness.
The beats 32,996 and 33,102	new presentation	Elementary consequences of the standard word $s_4 = s_3^{292} s_2$; computing and reconciling the two counts is ours.
The continued fraction as musical *form* (bar, phrase, section, movement)	specialist / new presentation	Closely related work exists. Carey and Clampitt studied self-similar pitch structures and their rhythmic analogues [25, 26]; Clampitt and Noll use Sturmian morphisms and Christoffel words for modes [28, 34]; Callender built rhythmic canons from Sturmian words whose voices and relations are set by the continued fraction of the slope [23, 24]; Fibonacci proportions in form have a long analytic literature [83]. We did not find the specific reading of a number's *whole* continued fraction as a nested hierarchy of formal units, each term a repeat count with a closing contrasting unit, used to lay out complete pieces.
The Composer and "Compose a piece" instruments	new (artifact)	Software on the companion site.
Euclid's algorithm and Euclidean rhythms	specialist	Clough and Douthett [29], Gómez-Martín et al. [49], Toussaint [132]; Bjorklund's 2003 algorithm as reported by Toussaint. That $E(16, 113)$ is π's level-3 block is an observation that follows at once.
Tuning from $\log_2(3/2)$, the comma, 41 and 53	classical	Benson [12].
Circle map, Arnold tongues, devil's staircase	classical	Jensen et al. [67]; Huygens's clocks [11].
$6/\pi^2$, the Euler product, $\zeta(2)$	classical	Euler [40, 41], Hardy and Wright [56].
Correlation of pulse trains, $(\gcd - 1)/\sqrt{(a-1)(b-1)}$	new presentation	An elementary computation. We did not find it stated in this form, but expect it is known; the "orchard as correlation matrix" reading is a presentation, not a theorem.

idea	class	notes and sources
Hydrogen's $1/n^2$ and $\pi^2/6$	classical, with a caveat	Balmer [9]; a shared formula, not shared physics (Chapter 5).
Residue bias of consecutive primes	specialist	Lemke Oliver and Soundararajan [82]; our counts re-done.
The prime sunflower of π	new presentation	Dirichlet's theorem and wheel sieves are classical [56]; seeding the $\pi-3$ spiral with primes is a picture of them.
Diffraction of π's rhythm and of the primes	specialist	Sturmian diffraction [8]; primes [131]. The Ramanujan-sum explanation of the square-free rule is standard reasoning; our computation re-done.
Explicit formula and zeros	classical	Riemann [113], von Mangoldt [134].
Continued-fraction factoring	classical	Lehmer and Powers [81], Morrison and Brillhart [98].
Mersenne necklaces and Fermat's little theorem	classical	Golomb [48].
The prime hunt in rhythm prefixes	new (experiment)	We did not find prior work testing binary prefixes of Sturmian words for primality. The outcome is a null result (Chapter 7).
The prime-convergent heuristic with $c = 2\ln 2$	new (heuristic)	Built entirely from known theorems [64, 85, 94, 127]. We did not find the constant or the identification of the correction factor as $\zeta(2)$ in the literature we searched. Related heuristics for linear recurrences: Grantham and Granville [52].
The random-real test of the heuristic, and π's placement	new (computation)	Our computation (Chapter 8); the lists for π match the OEIS [105, 106].
The spacing spectrum (crystal $\to$ quasicrystal $\to$ GUE $\to$ Poisson)	new presentation	Every ingredient is known: Berry [14], Bogomolny et al. [16], Dyson [36], Mehta [92], Montgomery [96], Odlyzko [100]; ordering point patterns by number variance is the hyperuniformity program of Torquato and Stillinger [130]; Dyson's quasicrystal remark is in Dyson [35]. Putting π's rhythm, the zeros, GUE and the primes on one scale with one set of tools is our synthesis.
Teddy Rook's picture of the nested tail	classical	The self-similarity of continued fractions; Chapter 10.
Patterned continued fractions for π	classical	Brouncker [73]; Lange [79].

idea	class	notes and sources
Ford circles, Stern–Brocot, $?(x)$, Cantor dust, Farey bulbs, Fibonacci word fractal	classical / specialist	Brocot [21], Denjoy [32], Devaney [33], Ford [44], Good [50], Jenkinson and Pollicott [65, 66], Minkowski [93], Monnerot-Dumaine [95], Salem [117], Stern [125]. Our dimension computation re-does the published method.
The six-numbers pictures	new presentation	Square cutting is a classical picture of Euclid's algorithm; applying one construction to the Suite's numbers is ours.
Gauss–Kuzmin test of π's first 10,000 terms	new presentation	Statistical checks of π's terms are a known kind of computation; the specific statistics are ours.
The Continued-Fraction Suite	new (artwork)	An original composition by Teddy Rook [116].
Golden's two-level labels	new presentation	An observation about labeling form in a self-similar piece.
Twelve million certified terms of π and calibrated tests of the Gauss–Kuzmin, Khinchin, Lévy and Galambos laws [Added 2026-11-05]	new presentation (computation)	Far longer expansions exist [101, 136]; our certified divide-and-conquer expansion, the simulation-calibrated p-values and the pair and block tests are ours (Chapter 15).
Prime convergents of π to $n = 20{,}000$ [Added 2026-11-05]	new (computation)	Denominators agree with OEIS A259490 [110]; the Poisson-binomial comparison is ours.
The exact expected number of n with $q_n(x) = q$, and the expected prime count $2\ln 2 \cdot \ln N + O(1)$ [Added 2026-11-05]	new (theorem)	Built on the classical cylinder lengths [71] and Mertens' theorem; we did not find either statement in the literature we searched.
Almost-everywhere bounds $\ln N$ and $2\ln N$; finitely many n with p_n, q_n both prime [Added 2026-11-05]	new (corollaries)	Immediate from deep theorems of Erdős [38] and Koukoulopoulos et al. [75], with Legendre's theorem and Lévy's theorem; the last part is an elementary Borel–Cantelli argument.
Divisibility of e's denominators by small primes [Added 2026-11-05]	new (observation)	Periodicity modulo ℓ is elementary; the densities and their correlation are our computation.
Structure of the correlation matrix (Gram form, determinant, mean and variance, top-eigenvalue bound) [Added 2026-11-05]	new presentation / new	The Gram form and determinant are the pattern of Smith's determinant [119]; the moment and eigenvalue statements we did not find for this matrix. GCD matrices: Hong and Loewy [62], Lindqvist and Seip [87].
Ranking of equal temperaments, and prime sizes [Added 2026-11-05]	new presentation	Measures from Erlich [39]; equidistribution over primes from Vinogradov [133].

idea	class	notes and sources
The count of prime q_n is $2\ln 2\cdot\ln N\,(1+o(1))$ for almost every x [Added 2026-11-06]	new (theorem), as far as we found	The overlap estimates follow the Duffin–Schaeffer tradition; the variance lemma is from Harman [57] and the equidistribution of inverses from Weil's bound [135]. We did not find the statement, and specialists in metric Diophantine approximation may know it (Chapter 16).
The period of e's denominators modulo ℓ divides 6ℓ [Added 2026-11-06]	new (proof), as far as we found	An elementary trace argument; we did not find it stated. [Superseded by Proposition 17.8: the period is exactly 6ℓ. [Added 2026-11-10]]
The top eigenvalue of the correlation matrix outgrows every power of $\ln m$; near-copies of small gcd matrices in its spectrum [Added 2026-11-06]	new presentation / new	The square-free construction behind the lower bound is the one used for GCD sums [2, 17]; the near-copies we did not find.
Why the 171-note temperament wins [Added 2026-11-06]	specialist	Known to tuning theorists: 171 tempers out the schisma, breedsma and ragisma [137, 138]; the adjugate identity for the errors is Cramer's rule.
Doeblin–Lenstra and serial-correlation tests of π's terms [Added 2026-11-06]	new presentation (computation)	Classical laws [20]; the calibrated test on 12 million terms is ours.
$\ln\lambda_{\max}(C_m)$ and $\ln(1/\lambda_{\min}(C_m))$ are $(2+o(1))\sqrt{\ln m/\ln\ln m}$ [Added 2026-11-10]	new (theorems), as far as we found	The resonance method of Soundararajan [122], with a Schur test for the upper bounds; the counterpart of his Theorem 2.1. Specialists in GCD sums may know it; Hilberdink [59] is the place to check (Chapter 17).
A limiting law for the spectrum of C_m; the mean of $1/\lambda$ [Added 2026-11-10]	new (theorem), as far as we found	A truncation argument with the Hoffman–Wielandt inequality [60]; the inverse is the Möbius form of Smith's determinant [119].
e: sign flip modulo ℓ, exact periods, forced divisors, primes only at $n\equiv 1 \pmod 3$ [Added 2026-11-10]	new (theorem), as far as we found	From the classical closed forms [30]. Not in OEIS A094007 [107] or the papers it cites; may be known.
π's convergents to $n = 50{,}000$; the eleventh to fourteenth prime denominators and the sixteenth prime numerator; the notes and the cluster [Added 2026-11-10]	new (computation)	OEIS A259490 lists ten terms and a limit of 25,000 [110]; OEIS A086785 lists fifteen numerators [105]. Probable primes, found by Teddy Rook; a proof of the eleventh is in progress.

idea	class	notes and sources
π's own constant against the uncorrected heuristic; sieve survivors against $\zeta(2)$; the voices test on two million convergents [Added 2026-11-10]	new (computation), as far as we found	Teddy Rook's tests (Section 17.7); Mertens' theorem; Corollary 8.2 and Jager and Liardet [64].
The family $(22k+3)/(7k+1)$ of the opening guess [Added 2026-11-10]	new presentation	Dirichlet's theorem; the Bateman–Horn conjecture [10].
The rhythm sieve [Added 2026-11-10]	new presentation	Sieving a recurrence modulo small primes is standard practice in prime searching; the reading as a polyrhythm is ours.
Prime intermediate fractions and the $(\ln N)^2$ heuristic [Added 2026-11-10]	new (heuristic and computation), as far as we found	Primes in arithmetic progressions; Khinchin's law for the sum of the terms [71].
Kepler's law in the atom; the $2n^2$ shells from the Runge–Lenz symmetry [Added 2026-11-10]	classical	Pauli, Fock. The reading as a repair of the "shared formula" caveat of Chapters 5 and 12 is ours.
Atoms with $n \approx 1009$ and the fourth harmonic [Added 2026-11-10]	specialist	Stepkin et al.; the arithmetic of the harmonic is our check.
Orbital resonances, Kirkwood gaps, KAM [Added 2026-11-10]	classical	Standard celestial mechanics.
The big-term test and Table 18.1 [Added 2026-11-10]	new presentation	Continued fractions of period ratios are an old device; the test against random ratios is our computation.
The sign mirror, Proposition 18.2 [Added 2026-11-10]	new, as far as we found	A two-line consequence of Proposition 17.4. May well be known for GCD matrices.
The one-prime chain and its mirror map [Added 2026-11-10]	classical	The Kac–Murdock–Szegő matrix and the Poisson kernel.
Hofstadter's butterfly, Hall integers, Cantor spectrum [Added 2026-11-10]	specialist	Hofstadter; Thouless et al.; Avila and Jitomirskaya.
The butterfly read as a number's musical form; the instrument [Added 2026-11-10]	new presentation	Our computation and instrument.
The harmonic form, Proposition 18.4 [Added 2026-11-10]	new, as far as we found	One line from Proposition 15.6. The harmonic numbers make it worth stating.

idea	class	notes and sources
The ratio of the two ends, Tables 18.2 and 18.3 [Added 2026-11-10]	new computation	To $m = 3 \times 10^7$. The exact cases of Proposition 18.5 are elementary.
Conjunction figures, Proposition 18.6 [Added 2026-11-10]	new presentation	The cycles are ancient and the three-gap theorem is classical; reading the trigon as a convergent and its drift as the next term is ours.
Kepler's harmonies tested, Table 18.6 [Added 2026-11-10]	new computation	Others have re-examined them; the tests against random ratios, the comparison of 1600 with 2000, and the test on the planets found later are ours.
Constructible polygons and $\varphi(n)$; quasicrystals; sphere packing [Added 2026-11-10]	classical	Gauss, Wantzel; Penrose, Shechtman; Hales. Only the pointers to this paper are ours.
The second law in Bohr's atom; Sommerfeld's ellipses [Added 2026-11-10]	classical	Bohr 1913, Sommerfeld 1916.

13.2 A reader's map of what is proved

Added 2026-11-10.

The paper now states about two dozen results that are proved here rather than quoted. Table 13.2 lists them with what each proof rests on, so that a reader can check them one at a time. Results quoted from the literature (with a citation in their heading) and heuristics are not listed.

Table 13.2. Results proved in this work, where, and what the proof rests on.

statement	where	rests on
Floors near a convergent; near-periodicity of the rhythm	Lemmas 3.3 and 3.4	Proposition 2.2
Nested blocks: the blocks of the rhythm have lengths q_n (our direct proof of a classical theorem)	Theorem 3.5	the two lemmas above
Bjorklund's Euclidean rhythm is a Christoffel word	Proposition 4.1	the Euclidean algorithm
Correlation of two pulse trains, $(\gcd(a,b)-1)/\sqrt{(a-1)(b-1)}$	Proposition 5.2	counting
e's denominators are periodic modulo every prime	Proposition 15.1	finite-state argument
Exact expected number of n with $q_n(x) = q$	Theorem 15.3	cylinder lengths [71]
Expected prime count $2\ln 2 \cdot \ln N + O(1)$	Corollary 15.4	Theorem 15.3, Mertens
Almost-everywhere bounds; finitely many n with p_n, q_n both prime	Theorem 15.5	Erdős [38], Koukoulopoulos et al. [75], Lévy; Borel–Cantelli

statement	where	rests on
Gram form, determinant and geometric mean of C_m	Proposition 15.6	Gauss's identity $\sum_{d\mid g}\varphi(d)=g$
Mean 1 and variance 2 of the spectrum of C_m	Theorem 15.7	Euler's $\sum H_a/a^2=2\zeta(3)$
$\lambda_{\max}(C_m)\geq(2/\pi^2)\ln^2 m\,(1+o(1))$	Theorem 15.8	a test vector, Möbius inversion
The pieces of E_q and their overlaps	Lemmas 16.1 and 16.2	Weil's bound [135], Erdős–Turán, Koksma
Heuristic 8.3 for almost every number	Theorem 16.4	the overlaps, Harman's variance lemma, Lévy
Period of e's denominators divides 6ℓ	Proposition 16.5	a trace argument
Top eigenvalue between $e^{c\sqrt{\ln m}/\ln\ln m}$ and $m^{o(1)}$	Theorem 16.6	square-free test vector; Schur's test
Near-copies of small gcd matrices in the spectrum	Proposition 16.7	min-max principle
$\ln\lambda_{\max}(C_m)=(2+o(1))\sqrt{\ln m/\ln\ln m}$	Theorem 17.1	Schur's test; Soundararajan's resonator; prime number theorem
Mirror law for $\lambda_{\min}(C_m)$	Theorem 17.5	the Möbius form of C_m^{-1} (Proposition 17.4); as above
A limiting law for the spectrum of C_m	Theorem 17.6	Hoffman–Wielandt; moments of truncations
Mean of $1/\lambda$ tends to $\prod_p(1+2/(p(p-1)))$	Proposition 17.7	Proposition 17.4; Dirichlet convolution
Sign flip and exact periods 6ℓ, 3ℓ for e	Proposition 17.8	closed forms (17.2) [30]
Forced divisors of e's convergents; primes only at $n=2$ or $n\equiv1 \pmod 3$	Theorem 17.9	closed forms (17.2)
Sign pattern of C_m^{-1}: the smallest eigenvalue as a Perron eigenvalue [Added 2026-11-10]	Proposition 18.2	Proposition 17.4; Perron–Frobenius
The harmonic form: C_m has the eigenvalues of a non-negative matrix of common multiples [Added 2026-11-10]	Proposition 18.4	Proposition 15.6
The ratio of the two ends on the divisors of a square-free number and on short prime chains [Added 2026-11-10]	Proposition 18.5	Kronecker products
Conjunction figures: the figure is a convergent, and the next term counts its turns [Added 2026-11-10]	Proposition 18.6	the three-gap theorem; Proposition 2.2

13.3 The two questions the owner asked

Is the rhythm of π new? No. The rhythm of any number, in the sense of Chapter 3, is its characteristic Sturmian word, a classical object, and the nested-block structure is a classical theorem. The rhythm of π is one instance. What is less common is to *listen* to it and to name its levels as musical form; that reading has close relatives in the music-theory literature, notably the Sturmian canons of Callender, in which the continued fraction governs the relations between voices, and the work of Carey, Clampitt and Noll on Sturmian structures in scales and rhythms. We did not find the full form reading (every term a repeat count, from bar to movement) laid out as a method for composing complete pieces.

Is the new music new? Yes, as music. The Continued-Fraction Suite is an original work by Teddy Rook, and its originality as an artwork does not depend on the novelty of its method, any more than a fugue's does on the novelty of counterpoint. Of its method, the Fibonacci proportions of Golden have a long history in analysis and composition; Sturmian rhythms and continued-fraction-governed canons exist in the specialist literature. We did not find a prior composition whose whole form is laid out level by level from a number's continued fraction, nor one that sets out to reach the 292nd return of π's 113-bar cycle.

CHAPTER 14

Discussion

14.1 What the work shows

One guess about π led, through one algorithm, to a surprising amount of mathematics and music. The guess $3+k/(7k+1)$ is a continued fraction; a continued fraction is a rhythm whose levels are its terms (Theorem 3.5); the same algorithm builds tunings, Euclidean rhythms, mode-locking staircases and a factoring method; coprimality, which every level of the rhythm obeys, is where $6/\pi^2$ enters; and the self-similarity of the continued fraction is the common root of a family of classical fractals. Along the way we found the precise meaning of two numbers that the site's sources disagreed about (32,996 and 33,102), and we checked every numerical claim by script.

The two research threads gave concrete answers. For prime convergents, a heuristic built from Lévy's theorem and the equidistribution of convergents modulo m predicts $2\ln 2 \cdot \ln N$ prime denominators among the first N convergents of a typical number, with the correction factor turning out to be $\zeta(2)$; a sample of random reals fits the predicted growth, and π is typical. For the Montgomery–Dyson connection, one set of statistics places a crystal, π's rhythm, the zeta zeros, random matrices, the primes and Poisson points on a single scale, and the explicit formula makes Dyson's quasicrystal remark visible. Neither thread proves a theorem about π or the zeta function, and we have been careful to say so.

14.2 Limitations

- *Heuristics are not theorems.* Heuristic 8.3 treats primality of different denominators as independent events; the agreement with data supports it but does not prove it. Chapter 15 proves it in expectation, not for almost every number. [Added 2026-11-05] Chapter 16 proves it for almost every number; for π itself it is still a heuristic. [Added 2026-11-06]

- *Finite samples.* The prime-convergent test uses 400 random reals and 1,000 convergents each; the slope fit has a standard error of 0.04 and is slightly above the prediction because of early terms. A larger sample, and longer expansions, would sharpen the test.

- *Low zeros.* We used the first 5,000 zeros, which we computed ourselves. At these heights finite-size corrections are visible [14, 16]; Odlyzko's tables of zeros at much greater heights [100] would show closer agreement with the GUE.

- *Probable primes.* Large "primes" are Baillie–PSW probable primes. No composite is known to pass that test, but it is not a proof of primality.
- *Score facts.* The facts about the Suite come from the site's data, which its builders read from the engraved score; we checked the bar arithmetic but not each note.
- *Literature search.* The audit of Chapter 13 is as careful as we could make it, but the music-theory literature in particular is large, and related work may exist that we did not find.

14.3 Open questions

Six questions were left open when this work was first written. Chapter 15 reports what further work on them produced; the status after that work is given in brackets, and later updates follow it. [Added 2026-11-05]

1. Does π have infinitely many prime convergent denominators? Infinitely many prime numerators? [Open. We tested q_n and p_n to $n = 20{,}000$ using 12 million certified terms; the counts stay inside the heuristic's 95% range (Section 15.2).] [Added 2026-11-05] [Update: the period of e's denominators modulo a prime ℓ divides 6ℓ, now proved (Proposition 16.5).] [Added 2026-11-06] [Update: numerators and denominators searched to $n = 50{,}000$. Four new probable-prime denominators, at $n = 25{,}332$, 27,651, 32,394 and 37,721, and a new probable-prime numerator, at $n = 42{,}037$, all found by Teddy Rook. They form a cluster, and the prime convergents lean toward one note; both look like chance, and neither says where the next one is (Sections 17.5 and 17.6). For e, the period modulo ℓ is exactly 6ℓ and prime denominators can occur only at $n = 2$ or $n \equiv 1 \pmod 3$ (Proposition 17.8 and Theorem 17.9).] [Added 2026-11-10]
2. Are there infinitely many Fibonacci primes? Pell primes? [Open. The known lists agree reasonably with the Grantham–Granville heuristic (Section 15.3).] [Added 2026-11-05]
3. Can Heuristic 8.3 be proved in a model? For example, for almost every x, is the number of $n \le N$ with q_n prime asymptotic to $2\ln 2 \cdot \ln N$? Even an upper bound of that order for almost every x would be interesting. [Partly answered. For a random x the expected count is $2\ln 2 \cdot \ln N + O(1)$, a theorem (Corollary 15.4); for almost every x the count lies between $\ln N$ and $2\ln N$ asymptotically, using recent work on the Duffin–Schaeffer conjecture (Theorem 15.5). The almost-everywhere constant $2\ln 2$ is still a conjecture.] [Added 2026-11-05] [Update: answered for almost every x. The count is $2\ln 2 \cdot \ln N\,(1+o(1))$ for almost every x, with an explicit error term (Theorem 16.4). Whether π is one of these numbers is open.] [Added 2026-11-06] [Update: for π itself, its prime convergents to $n = 50{,}000$ now favor the constant $2\ln 2$ over the value without the factor $\zeta(2)$ by about 30 to 1 (evidence, Section 17.7).] [Added 2026-11-10]
4. Are π's terms unbounded? Do they obey the Gauss–Kuzmin law? [Open. Calibrated tests on 12 million terms are all consistent with typical behavior (Section 15.5).] [Added 2026-11-05] [Update: two more tests, the Doeblin–Lenstra law and the serial correlation of the terms, are also consistent (Section 16.5).] [Added 2026-11-06] [Update: every prime below 1,000 divides π's first two million denominators at its typical rate $1/(\ell+1)$, and the share of denominators with no prime factor below 10^7 matches the typical-number prediction (Section 17.7).] [Added 2026-11-10]

5. How does the spectrum of the arithmetic correlation matrix of Section 5.2 compare with random-matrix benchmarks, and does $6/\pi^2$ appear in its bulk? [Partly answered. Its eigenvalues have mean 1 and variance tending to 2, and its determinant and geometric mean are explicit (proved); $6/\pi^2$ appears in a lower bound $(2/\pi^2)\ln^2 m$ for the top eigenvalue, not in the first two moments; its spacings look like the GOE's (Section 15.6).] [Added 2026-11-05] [Update: the top eigenvalue grows faster than any power of $\ln m$ and at most like $m^{o(1)}$ (Theorem 16.6); the spectrum contains near-copies of the small gcd matrices, one family for each value of $\lfloor m/p \rfloor$ (Proposition 16.7), and with these set aside the spacing ratio falls from the GOE value toward the middle, so the GOE-like look was partly an artifact.] [Added 2026-11-06] [Update: the extremes are settled to leading order: $\ln \lambda_{\max}$ and $\ln(1/\lambda_{\min})$ are both $(2+o(1))\sqrt{\ln m/\ln\ln m}$ (Theorems 17.1 and 17.5). The eigenvalue distribution converges to a limiting law (Theorem 17.6), whose known exact constants do not involve $6/\pi^2$. The shape of that law is open.] [Added 2026-11-10] [Update: the inverse has the sign pattern of Liouville's function, so both extreme eigenvalues are Perron eigenvalues of non-negative matrices built from common multiples (Propositions 18.2 and 18.4). The ratio of $1/\lambda_{\min}$ to $\lambda_{\max}$ is about 2.6 at $m = 3 \times 10^7$ and near 2.57 for the gcd matrix; whether it has a limit is open (Section 18.3).] [Added 2026-11-10]

6. Rank all equal temperaments up to 200 notes by how well they approximate several just intervals at once. Do prime sizes stand out? [Answered: no, by eight measures and by an equidistribution theorem (Section 15.7).] [Added 2026-11-05] [Update: why 171 wins: it is the equal temperament fixed by three tiny commas (Section 16.4), a fact known to tuning theorists.] [Added 2026-11-06]

14.4 Future work

On the mathematical side: find the second-moment estimate that would prove Heuristic 8.3 for almost every x (Section 15.4); find the exact growth of the top eigenvalue of the correlation matrix; and repeat the spacing statistics with zeros at large height. [Added 2026-11-05] [Update: the second-moment estimate is now proved (Theorem 16.4); the top eigenvalue's exact growth is still open, between the two bounds of Theorem 16.6.] [Added 2026-11-06] [Update: the growth of the top eigenvalue is now known to leading order (Theorem 17.1); a second-order term, and a description of the limiting bulk law, are the open problems.] [Added 2026-11-10] [Update: on the computational side, certificates of primality for the five new probable primes of Section 17.5, and a search of π's convergents beyond $n = 50{,}000$, are the next steps.] [Added 2026-11-10] On the musical side: the method of Chapter 11 invites pieces in tunings whose number of notes is itself a convergent denominator (a pi movement in a 113-note tuning, for instance), and pieces built from numbers with chosen terms, whose form a composer could design by picking the continued fraction. The companion site's instruments already do much of this; the score of each new piece is a continued fraction made audible.

CHAPTER 15

Working on the open questions

Added 2026-11-05.

Section 14.3 ended with six open questions. This chapter reports what happened when we went back to them. Most of them are famous problems, and we did not solve them. Whether π has infinitely many prime convergent denominators, whether there are infinitely many Fibonacci or Pell primes, and whether π's terms obey the Gauss–Kuzmin law are all still open. What we can offer is new computation, sharper heuristics, and some results that we can actually prove.

How to read the labels. In this chapter, a *Theorem* or *Proposition* with no citation in its heading is proved here, in full, from results that are cited. One with a citation is quoted from the literature. When a proof leans on a deep cited theorem, we say so in the statement. A *Heuristic* predicts without proving. Numbers from computation are *numerical evidence*: they can support a conjecture, never prove it. "Prime" for a large number means a Baillie–PSW probable prime (Appendix B); no composite number is known to pass that test, but passing it is not a proof.

15.1 Twelve million terms of pi

Every question about π starts with its terms. Chapter 2 used the first 10,000. Here we use the first 12,000,000, enough to reach the term 878,783,625 at position 11,504,930.

Method. We take π from mpmath, accurate to the last binary place, as an interval $[(m-2)/2^b, (m+2)/2^b]$ with b about 43 million bits. A divide-and-conquer version of Euclid's algorithm then keeps exactly the terms that every real number in the interval shares. It expands the interval cut to its top half, which gives a prefix of the terms and the matrix $\left(\begin{smallmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{smallmatrix}\right)$ of (2.4); it applies the inverse of that matrix to the full interval; and it repeats on what is left. Since the set of reals that share a given prefix is an interval, a term shared by both endpoints is a term of π. The computation is done twice, at two precisions 4,096 bits apart, and the two lists must agree. It takes under two minutes on one core.

Checks. The first 10,000 terms agree with those of Chapter 2, which were found by a different method. The record terms of our list, and their positions (292 at position 4, then 436, 20,776,

Table 15.1. Probable primes among π's convergents, $n \le N$: the observed count, the expected count $\sum \zeta(2)/\ln$ of Heuristic 8.3, the central 95% range of the Poisson-binomial law with those chances, and the share of that law at or below the observed count.

	denominators q_n				numerators p_n			
N	obs.	exp.	95% range	share	obs.	exp.	95% range	share
1,000	5	8.7	4–14	0.11	10	8.0	3–14	0.82
5,000	9	10.9	5–17	0.34	13	10.3	5–17	0.85
20,000	10	12.9	7–20	0.25	15	12.2	6–19	0.83

78,629, 179,136, 528,210, 12,996,958 and 878,783,625 at positions 307, 431, 28,421, 156,381, 267,313, 453,293 and 11,504,930), agree with the OEIS tables of record terms [104], which come from much longer computations. The script stops if any of these checks fails.

Prior computations. We are far from the first. Gosper computed about 17 million terms in 1977; Havermann reached 20 million in 1999 and 180 million in 2002; Bickford 458 million in 2010; Weisstein 15 billion by 2013; Fahad 30 billion in 2021; and Frank 653.52 billion in 2025 [101, 136]. Our twelve million are a small, independent, self-checking sample, enough for the statistics below and for every prime-convergent search we can afford.

15.2 Question 1: the prime convergents of pi

The search. We tested q_n and p_n for every $n \le 20{,}000$; the largest of these numbers have about 10,268 digits. Each number first goes through trial division by the primes below 10^5 and then the Baillie–PSW test. The prime denominators occur at

$$n = 1, 3, 8, 32, 594, 1{,}126, 2{,}002, 3{,}610, 4{,}355, 6{,}925,$$

so 10 in all, and none between $n = 6{,}926$ and $n = 20{,}000$. [Update: the next are at $n = 25{,}332$, 27,651, 32,394 and 37,721, probable primes (Section 17.5). [Added 2026-11-10]] This agrees with OEIS A259490, whose index k is our $n + 1$ and which records that there are no more below $k = 25{,}000$ [110]. The prime numerators occur at

$$n = 4, 8, 10, 15, 86, 229, 333, 593, 839, 852, 1{,}148, 2{,}725, 3{,}787, 7{,}441, 8{,}750,$$

15 in all, and none between $n = 8{,}751$ and $n = 20{,}000$. The first thirteen are the ones of Chapter 8; the two beyond $n = 5{,}000$ are at $n = 7{,}441, 8{,}750$. The OEIS list of prime numerators, A086785, holds 15 terms [105], the same number as ours; we compared the first thirteen values.

The comparison. Heuristic 8.3 gives each q_n a chance $\zeta(2)/\ln q_n$ of being prime, and each p_n a chance $\zeta(2)/\ln p_n$. If these were independent events, the count would follow a Poisson-binomial law, which we compute exactly. Table 15.1 and Fig. 15.1 compare. Both counts sit inside the central 95% range of the heuristic at every checkpoint. The denominators run on the low side (the share of the law at or below the observed count is between 0.1 and 0.35), and the numerators on the high side (about 0.83). A low count for one and a high count for the other is what chance alone often produces, and nothing in the table is unusual.

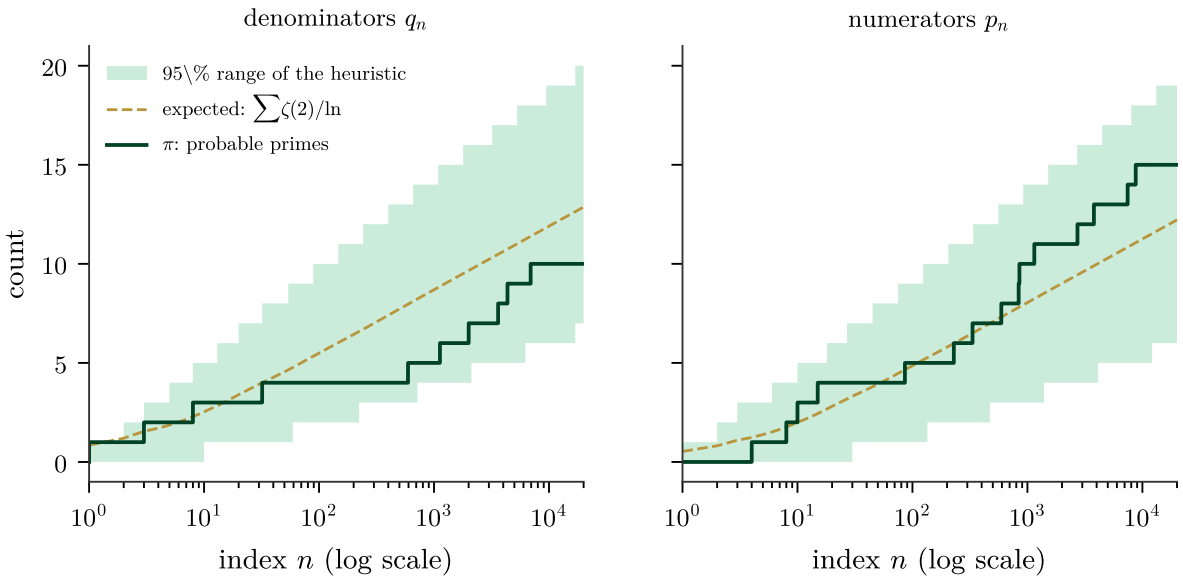


Figure 15.1. Probable-prime denominators (left) and numerators (right) of π's convergents up to $n = 20{,}000$ (green steps), the expected count of Heuristic 8.3 (gold, dashed) and the central 95% range of the heuristic (shaded).

Both at once. Only $n = 8$ gives a convergent, 833,719/265,381, whose numerator and denominator are both prime. For a typical number this is expected to happen only finitely often; Theorem 15.5(iv) below proves it for almost every real number. The same local reasoning as in Heuristic 8.3 gives a correction factor $\zeta(2)$ for the pair (the pair (p_n, q_n) modulo a prime ℓ is equidistributed over the $\ell^2 - 1$ nonzero pairs, by the theorem of Jager and Liardet [64]), and the expected number of further cases beyond $n = 8$ is then only about 0.11. [Update: the searches to $n = 50{,}000$ add none; p_{25332}, p_{27651}, p_{32394} and p_{37721} are composite, and so is q_{42037}, whose numerator is a new probable prime (Section 17.5). [Added 2026-11-10]]

The number e. The earlier count for e (Chapter 8) was 10 prime denominators up to $n = 3{,}000$, the last at $n = 268$. The terms of e are not typical, so Heuristic 8.3 does not apply. What replaces it? One piece can be proved.

Proposition 15.1. *Let q_n be the convergent denominators of e and ℓ a prime. The sequence $q_n \bmod \ell$ is purely periodic. Hence the share f_ℓ of n with $\ell \mid q_n$ exists and equals the share of zeros in one period.*

Proof. The terms of e are $a_{3k-2} = a_{3k} = 1$ and $a_{3k-1} = 2k$, so $a_n \bmod \ell$ depends only on $n \bmod 3\ell$. The state $(n \bmod 3\ell,\ q_{n-1} \bmod \ell,\ q_n \bmod \ell)$ therefore determines the next state, by $q_{n+1} = a_{n+1}q_n + q_{n-1}$. This map is a bijection of a finite set (it can be run backward, since $q_{n-1} = q_{n+1} - a_{n+1}q_n$), so every orbit is a cycle. □

For every prime $\ell < 3{,}000$ our script finds that the period divides 6ℓ (we had not proved this in general; Proposition 16.5 now does [Added 2026-11-06]), and computes f_ℓ exactly: $f_2 = 1/3$, $f_3 = 1/3$, $f_5 = 1/5$, $f_7 = 5/21$, $f_{11} = 5/33$, $f_{13} = 7/39$, $f_{17} = 1/17$. A typical number would have $f_\ell = 1/(\ell + 1)$ (Corollary 8.2). For e, the average of ℓf_ℓ over the odd primes below 3,000 is 2.00: its denominators are divisible by a given small prime about twice as often as

random integers. If the events "$\ell \mid q_n$" were independent across ℓ, the correction factor would be $\prod_\ell (1 - f_\ell)/(1 - 1/\ell)$, which falls from 0.66 (primes below 100) to 0.45 (below 1,000) and 0.38 (below 3,000) and does not settle. But they are far from independent. Among e's first 3,000 denominators, the share with no prime factor below 1,000 is 0.093, against 0.036 if the events were independent and 0.081 for random integers (below 100: 0.160, 0.080 and 0.120). The same denominators are divisible by many small primes at once, and the rest are almost as free of small factors as random integers. Using the observed ratio 1.14 as the correction gives 8.6 expected prime denominators up to $n = 3{,}000$, against 10 found; the typical-number factor $\zeta(2)$ gives 12.4, and the independent product gives 5.0. We offer the first as a rough guide, not a heuristic with a constant. [Update: this discussion of correction factors is superseded. Proposition 17.8 and Theorem 17.9 account for half of the factor of about two (three zeros in every 3ℓ indices are forced) and show that e's denominators can be prime only at $n = 2$ or $n \equiv 1 \pmod 3$; OEIS A094007 reports no prime denominator of e beyond $n = 268$ up to about $n = 50{,}000$ [107] (Section 17.4). [Added 2026-11-10]]

The growth of e's denominators is easy to pin down: since $\prod_{k\le n} a_k \le q_n \le \prod_{k\le n}(a_k + 1)$, we get $\ln q_n = \frac{n}{3} \ln n \, (1 + o(1))$, much faster than Lévy's linear rate. Any heuristic of the form "prime with chance $c/\ln q_n$" then predicts about $3c \ln\ln N$ prime denominators up to N. So e should have infinitely many, but very few: with $c = 1.14$, about two more between $n = 3{,}000$ and $n = 10^6$.

Status. Open. The new computation quadruples the range searched; everything still looks typical.

15.3 Question 2: Fibonacci and Pell primes

For $\varphi = [1; 1, 1, \dots]$ the convergent denominators are the Fibonacci numbers, $q_n = F_{n+1}$; for $\sqrt{2} = [1; 2, 2, \dots]$ they are the Pell numbers, $q_n = P_{n+1}$ ($P_1 = 1$, $P_2 = 2$, $P_{n+1} = 2P_n + P_{n-1}$). Whether either sequence has infinitely many primes is a famous open problem.

Proposition 15.2 (classical)**.** *If F_n is prime then $n = 4$ or n is prime. If P_n is prime then n is prime.*

Proof. Both sequences ($u_n = F_n$ or $u_n = P_n$, with $u_0 = 0$, $u_1 = 1$) satisfy $u_{m+k} = u_m u_{k+1} + u_{m-1} u_k$ for $m \ge 1$, $k \ge 0$: it holds for $k = 0$ and $k = 1$, and both sides obey the sequence's recurrence in k. Taking $k = m$ and m replaced by jm gives $u_{(j+1)m} = u_{jm} u_{m+1} + u_{jm-1} u_m$, so by induction on j, $u_m \mid u_{jm}$. If $n = jm$ with $1 < m < n$, then u_m divides u_n. For Fibonacci numbers $1 < F_m < F_n$ unless $m \le 2$; a composite $n \ne 4$ has a divisor m with $3 \le m < n$, so F_n is composite. For Pell numbers $P_m > 1$ for every $m \ge 2$, and every composite n has such a divisor. □

The proposition is why the typical-number heuristic fails for these two numbers: their denominators are divisibility sequences, not random-looking integers. Grantham and Granville give the heuristic that replaces it. For a Lucas sequence $u_n = (\alpha^n - \beta^n)/(\alpha - \beta)$ with $\alpha > |\beta|$, they predict

$$\#\{n \le N : u_n \text{ prime}\} \sim e^\gamma \log_\alpha N \tag{15.1}$$

[52]. The Fibonacci numbers have $\alpha = \varphi$. The Pell numbers are the Lucas sequence with $\alpha = 1 + \sqrt{2}$, so the same formula applies to them, though Grantham and Granville do not discuss them by name.

Table 15.2. The number of Fibonacci primes F_n and Pell primes P_n with $n \le N$, from the OEIS lists, against the Grantham–Granville prediction $e^{\gamma}\log_{\alpha} N$ with $\alpha = \varphi$ and $\alpha = 1+\sqrt{2}$. The Pell list is complete only below 10^5.

	Fibonacci		Pell	
N	found	predicted	found	predicted
100	12	17.0	11	9.3
1,000	21	25.6	17	14.0
10,000	26	34.1	24	18.6
100,000	33	42.6	31	23.3
1,000,000	43	51.1	–	–
3,340,367	51	55.6	–	–

What is known. OEIS A001605 lists 51 indices of Fibonacci primes up to $n = 3{,}340{,}367$; the first 37 (up to $n = 201{,}107$) are proven primes and the rest probable primes, and later searches added five more and then $n = 11{,}964{,}299$ in 2025 [103]. OEIS A096650 lists 31 indices of Pell primes below 10^5, and records that there are no others below 10^5; those above 523 are probable primes [108]. Our own lists, up to index 3,001, agree with both (Chapter 8). Table 15.2 compares the counts with (15.1).

The Fibonacci counts run steadily below the prediction, by about a sixth at the larger N, as Grantham and Granville note themselves; the Pell counts run above it, by about a third at 10^5. For counts this small and this slowly growing, such gaps are not alarming, and the constant in such heuristics is the least certain part. Both lists are consistent with infinitely many primes and a count that grows like $\log N$; that is as far as the data go.

Status. Open, and not touched by anything in this work; the heuristic and the data agree to within the accuracy one can expect.

15.4 Question 3: a theorem in the model

Here we can prove something. Pick x at random, uniformly in $(0,1)$, and write $q_n = q_n(x)$. Recall that the set of x whose first n terms are $a_1,\dots,a_n$ is an interval (a *cylinder*) of length

$$\frac{1}{q_n(q_n+q_{n-1})} \tag{15.2}$$

[71, Ch. III], and that, by (2.4), $q_n/q_{n-1} = [a_n; a_{n-1},\dots,a_1]$ (the "mirror formula": $q_n/q_{n-1} = a_n + q_{n-2}/q_{n-1}$, and induct).

Theorem 15.3 (the exact expected count)**.** *For every integer $q \ge 2$,*

$$\mathbb{E}\,\#\{n \ge 1 : q_n(x) = q\} = \frac{2}{q}\sum_{\substack{1\le r<q\\ \gcd(r,q)=1}} \frac{1}{q+r}. \tag{15.3}$$

For a prime p this is $\frac{2}{p}(H_{2p-1}-H_p)$, where $H_m = 1+\frac12+\cdots+\frac1m$. Consequently

$$\mathbb{E}\,\#\{n \ge 1 : q_n(x) \text{ is prime and } q_n(x) \le X\} = \sum_{p\le X}\frac{2}{p}\,(H_{2p-1}-H_p) = 2\ln 2\cdot\ln\ln X + B_0 + o(1) \tag{15.4}$$

as $X \to \infty$, with $B_0 = 2\ln 2 \cdot M - C_1 = -0.2945\ldots$, where $M = 0.2615\ldots$ is the Meissel–Mertens constant and $C_1 = \sum_p \frac{2}{p}(\ln 2 - H_{2p-1} + H_p) = 0.6570\ldots$

Proof. Sequences $(a_1, \ldots, a_n)$ of positive integers correspond to pairs (q_{n-1}, q_n). By the mirror formula, the sequence read backward is a continued fraction expansion of the rational number q_n/q_{n-1}, all of whose terms are at least 1. Conversely, every rational q/r with $q \geq 2$, $1 \leq r < q$ and $\gcd(r, q) = 1$ has exactly two such expansions, one ending in a term at least 2 and one ending in 1 (for instance $7/2 = [3; 2] = [3; 1, 1]$), and each, read backward, is a sequence $(a_1, \ldots, a_n)$ with $q_{n-1} = r$ and $q_n = q$ (the two lengths differ by one). So the cylinders with $q_n = q$, for all n together, are exactly two for each r, each of length $1/(q(q + r))$ by (15.2). Summing the lengths over n gives the expected number of n with $q_n = q$, which is (15.3). For a prime p every r from 1 to $p - 1$ is coprime to p, and $\sum_{r=1}^{p-1} 1/(p + r) = H_{2p-1} - H_p$.

For the asymptotic, $H_{2p-1} - H_p = \ln 2 + O(1/p)$, so the series defining C_1 converges absolutely, and $\sum_{p \leq X} \frac{2}{p}(H_{2p-1} - H_p) = 2\ln 2 \sum_{p \leq X} \frac{1}{p} - C_1 + O(1/X)$. Mertens' theorem $\sum_{p \leq X} 1/p = \ln\ln X + M + o(1)$ [56, Thm. 427] finishes the proof. □

Our script checks (15.3) against a direct search over all sequences with $q_n = q$ for every $q < 40$, in exact rational arithmetic, and computes C_1 from the primes below 10^7 (the omitted tail is below 10^{-7}).

Where $\zeta(2)$ comes from. Theorem 15.3 shows the correction factor of Chapter 8 at work, now as an identity. A prime q gets weight about $2\ln 2/q$, because all $q - 1$ numbers $r < q$ are coprime to it. A general q gets about $\frac{\varphi(q)}{q} \cdot \frac{2\ln 2}{q}$, and $\varphi(q)/q$ averages $6/\pi^2$. Relative to all integers, primes are favored by exactly $\pi^2/6 = \zeta(2)$. (Summed over $q \leq X$, the general weight gives about $\frac{12\ln 2}{\pi^2}\ln X = (\ln X)/L$ convergents up to X, which matches Lévy's theorem.)

Corollary 15.4. *For x uniform in $(0, 1)$,*

$$\mathbb{E}\,\#\{n \leq N : q_n(x) \text{ prime}\} = 2\ln 2 \cdot \ln N + O(1).$$

Proof. Lower bound. Since $q_n \geq F_{n+1} \geq \varphi^{n-1}$, any n with $q_n \leq \varphi^{N/2}$ has $n \leq N/2 + 1 \leq N$ (for $N \geq 2$). So the count is at least the number of n with q_n prime and at most $X = \varphi^{N/2}$, and by (15.4) its expectation is at least $2\ln 2 \cdot \ln(N \ln\varphi/2) + B_0 + o(1)$.

Upper bound. Let $C = 2\ln\mu + 2 = 4.92\ldots$, where $\mu = \sum_{k \geq 1} 2/(k\sqrt{k+1}) = 4.312\ldots$. If $n \leq N$ and q_n is prime, then either $q_n \leq e^{CN}$, or $q_N \geq q_n > e^{CN}$. Hence the count is at most $\#\{n : q_n \text{ prime}, q_n \leq e^{CN}\} + N \cdot \mathbf{1}[q_N > e^{CN}]$. The first term has expectation $2\ln 2 \cdot \ln(CN) + B_0 + o(1)$ by (15.4). For the second, the cylinder lengths give, for the conditional chance of the next term,

$$\Pr(a_{n+1} = k \mid a_1, \ldots, a_n) = \frac{1 + s}{(k + s)(k + 1 + s)} \leq \frac{2}{k(k+1)}, \qquad s = q_{n-1}/q_n \in [0, 1],$$

so $\mathbb{E}[(1 + a_{n+1})^{1/2} \mid a_1, \ldots, a_n] \leq \mu$. Since $q_N \leq \prod_{k \leq N}(1 + a_k)$, conditioning one term at a time gives $\mathbb{E}\, q_N^{1/2} \leq \mu^N$, and Markov's inequality gives $\Pr(q_N > e^{CN}) \leq \mu^N e^{-CN/2} = e^{-N}$. So the second term has expectation at most Ne^{-N}. □

So, in expectation, Heuristic 8.3 is a theorem, with exactly the constant $2\ln 2$. To the best of our search we did not find Theorem 15.3 or Corollary 15.4 stated in the literature, although the counting behind them is classical. What about almost every x, which is what the heuristic actually claims? Here deep recent theorems carry us part of the way. Write $\theta_n = q_n^2\,|x - p_n/q_n|$;

every convergent has $\theta_n < 1$, and by Legendre's theorem, any fraction a/q in lowest terms with $|x - a/q| < 1/(2q^2)$ is a convergent of x [56, Thm. 184].

Theorem 15.5 (almost every x; parts (i)–(iii) rest on cited theorems)**.** *For almost every real x:*

(i) infinitely many q_n are prime, and infinitely many p_n are prime;

(ii) the number of $n \le N$ with q_n prime and $\theta_n < 1/2$ is $(1 + o(1)) \ln N$;

(iii) hence $1 \le \liminf_{N\to\infty} \dfrac{\#\{n \le N : q_n \text{ prime}\}}{\ln N} \le \limsup_{N\to\infty} \dfrac{\#\{n \le N : q_n \text{ prime}\}}{\ln N} \le 2$;

(iv) only finitely many n have both p_n and q_n prime.

Proof. (i) Erdős proved that if $n_1 < n_2 < \cdots$ are integers with $\sum \varphi(n_i)/n_i^2 = \infty$, then for almost every x infinitely many n_i are convergent denominators of x [38]. For the primes, $\sum_p (p-1)/p^2 = \infty$. For numerators, note that the convergents of $1/x$ are the reciprocals of those of x (if $x > 1$, then $1/x = [0; a_0, a_1, \dots]$; if $0 < x < 1$, then $1/x = [a_1; a_2, \dots]$), so the numerators of x are, up to a shift of index, the denominators of $1/x$; and the map $x \mapsto 1/x$ sends sets of measure zero to sets of measure zero.

(ii) Koukoulopoulos, Maynard and Yang proved the following quantitative form of the Duffin–Schaeffer conjecture [75], extending [74]. Let ψ take the positive integers to $[0, 1/2]$, and suppose that

$$\Psi(Q) = \sum_{q \le Q} \frac{2\varphi(q)\psi(q)}{q} \to \infty.$$

Then for almost every x the number of coprime pairs (a, q) with $q \le Q$ and $|x - a/q| < \psi(q)/q$ is $\Psi(Q)(1 + o(1))$. Take $\psi(q) = 1/(2q)$ when q is prime and 0 otherwise; then $\Psi(Q) = \sum_{p \le Q} (p-1)/p^2 = \ln\ln Q + O(1)$. By Legendre's theorem the pairs counted are exactly the convergents with q_n prime, $q_n \le Q$ and $\theta_n < 1/2$, one for each such n. To pass from $q_n \le Q$ to $n \le N$, use Lévy's theorem: for almost every x and any $\varepsilon > 0$, $e^{(L-\varepsilon)n} \le q_n \le e^{(L+\varepsilon)n}$ for all large n, with $L = \pi^2/(12 \ln 2)$. Then for large N, every n with $q_n \le e^{(L-\varepsilon)N}$ has $n \le N$, and every $n \le N$ has $q_n \le e^{(L+\varepsilon)N}$; and $\ln\ln e^{(L\pm\varepsilon)N} = \ln N + O(1)$.

(iii) The lower bound is (ii). For the upper bound, use the same theorem with $\psi(q) = 1/q$ for prime q (so $\Psi(Q) = 2\ln\ln Q + O(1)$): every convergent with q_n prime has $|x - p_n/q_n| < 1/q_n^2$, so it is one of the pairs counted. Convert from Q to N as before.

(iv) This part is elementary, and we give it for $0 < x < 1$ (for other x, replace the bound on $\pi(q)$ below by the Brun–Titchmarsh inequality [97]). If $p_n = a$ and $q_n = q$ are both prime, then x lies within $1/q^2$ of a/q with $a < q$ prime. For each prime q, the set of such x has measure at most $2\pi(q)/q^2 \le C'/(q \ln q)$ by Chebyshev's bound $\pi(q) = O(q/\ln q)$ [56, Thm. 7], and $\sum_p 1/(p \ln p) < \infty$. By the Borel–Cantelli lemma almost every x lies in only finitely many of these sets, so only finitely many primes q can occur, and each value q_n occurs for at most two n. □

Where the argument stops. The conjecture behind Question 3 sits strictly inside Theorem 15.5(iii): $2\ln 2 \approx 1.3863$ is between 1 and 2. And it matches the theorem in a precise way. For almost every x, the share of n with $\theta_n < 1/2$ is $1/(2\ln 2)$ (the Doeblin–Lenstra law, proved by Bosma, Jager and Wiedijk [20]), and $2\ln 2 \times 1/(2\ln 2) = 1$, which is exactly the constant in (ii). The missing piece is the convergents with $\theta_n \ge 1/2$. The set of x for

Table 15.3. Simulation of 20,000 uniform random reals. Top: the mean number of prime $q_n \le X$, with its standard error, against (15.4); and the same for prime $q_n \le X$ with $\theta_n < 1/2$, against $\sum_{p \le X} (p-1)/p^2$. Bottom: the mean number of prime q_n with $n \le N$, its standard error and variance, and $2 \ln 2 \cdot \ln N$.

X	mean	s.e.	exact	mean, $\theta_n < \frac{1}{2}$	s.e.	exact
10^5	3.084	0.011	3.093	2.235	0.009	2.253
10^{10}	4.049	0.012	4.054	2.929	0.011	2.946
10^{20}	5.008	0.014	5.015	3.626	0.012	3.639
10^{50}	6.279	0.016	6.285	4.543	0.014	4.555
10^{100}	7.230	0.018	7.246	5.230	0.015	5.248
10^{200}	8.196	0.019	8.207	5.925	0.016	5.942

N	mean	s.e.	variance	$2 \ln 2 \cdot \ln N$
10	3.079	0.010	1.82	3.192
20	4.063	0.012	2.78	4.153
50	5.346	0.014	4.13	5.423
100	6.311	0.016	5.15	6.384
200	7.270	0.017	6.12	7.345
400	8.237	0.019	7.17	8.306

which a given fraction a/q is a convergent is an interval around a/q whose two halves have lengths $1/(q(q+r))$ and $1/(q(2q-r))$, where r is determined by a ($ar \equiv \pm 1 \pmod q$). The Duffin–Schaeffer machinery counts approximations whose target interval depends only on q; here it depends on a too, through a modular inverse. A second-moment bound for these events would give the full almost-everywhere statement, and that is what we cannot supply. [Update: Theorem 16.4 supplies it, by a different route that avoids the a-dependence. [Added 2026-11-06]] Note also that Corollary 15.4 alone, with Markov's inequality and the Borel–Cantelli lemma along $N_k = e^{2^k}$, gives the weaker but elementary bound $O(\ln N \, (\ln \ln N)^{1+\varepsilon})$ for almost every x, without the deep theorem.

The numbers. Table 15.3 and Fig. 15.2 check Theorem 15.3 on 20,000 random reals, each given by a random fraction with 2,600 binary digits. The simulated means agree with the exact values within two standard errors at every X (the six values are cumulative, so their deviations are strongly correlated), for all prime denominators and for those with $\theta_n < 1/2$; the ratio of the two is close to $1/(2 \ln 2) \approx 0.72$, as the Doeblin–Lenstra law predicts. The counts by index (second part of the table) grow by $2 \ln 2$ per e-fold, as Corollary 15.4 says; their level lies a constant below $2 \ln 2 \cdot \ln N$, as the $O(1)$ allows. The variance is a little below the mean, as for a sum of independent events with these chances (a Poisson-binomial count has variance below its mean).

Status. Partly answered. In expectation the heuristic's constant is now a theorem (Corollary 15.4). For almost every x the count is between $\ln N$ and $2 \ln N$ asymptotically, using the recent quantitative Duffin–Schaeffer theorem; that it is $2 \ln 2 \cdot \ln N$ remains a conjecture. [Update: it is now a theorem for almost every x (Theorem 16.4). [Added 2026-11-06]] Nothing here applies to π, which is one number and may or may not be typical.

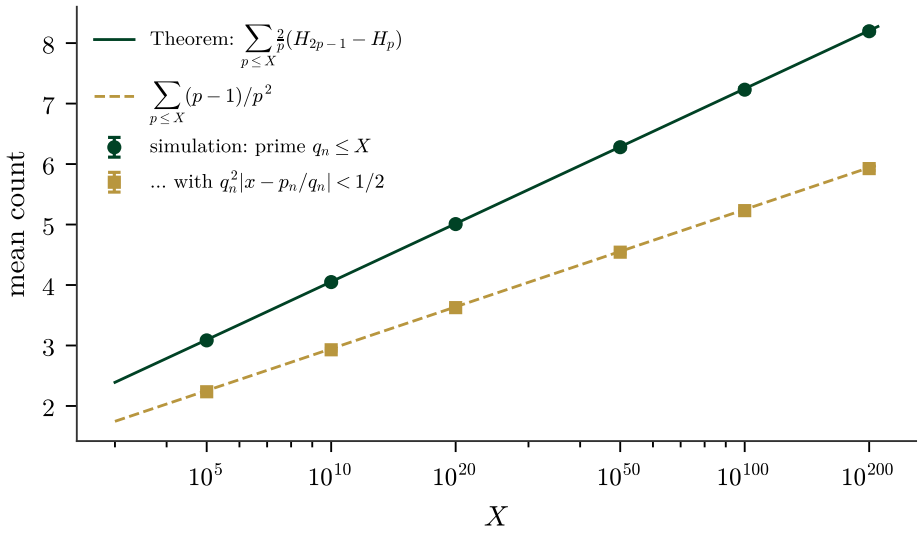


Figure 15.2. Theorem 15.3 against simulation: the mean number of prime convergent denominators up to X (green, with two standard errors) and of those with $\theta_n < 1/2$ (gold), for 20,000 random reals, with the exact expectations.

15.5 Question 4: is pi typical?

The Gauss–Kuzmin law, Khinchin's constant and Lévy's constant are laws for almost every number. Whether π obeys them is open; it is not even known whether π's terms are bounded. All we can do is test, and every test has to be careful about two things. First, the terms of a typical number are weakly dependent, so textbook chi-square p-values, which assume independence, are only approximate. We therefore calibrated every statistic on 1,000 random reals with 100,000 terms each, expanded exactly. Second, a p-value is not a probability that π is typical; a pass is evidence of the absence of a large deviation, not proof of typicality.

Results. On the first 12,000,000 terms (Fig. 15.3):

- *Values 1 to 100 and above 100* (101 cells): $\chi^2 = 100.9$ on 100 degrees of freedom, textbook $p = 0.45$, calibrated $p = 0.45$. The ones make up 41.489%, against 41.504% for the law.
- *The tail*, in the dyadic cells $[2^j, 2^{j+1})$, $j = 0, \dots, 19$, and $[2^{20}, \infty)$: $\chi^2 = 20.6$ on 20 degrees of freedom, $p = 0.42$.
- *Pairs* (a_n, a_{n+1}) with values $1, \dots, 5$ and "6 or more" (36 cells, probabilities from the Gauss measure of the two-term cylinders): $\chi^2 = 26.0$ on 35 degrees of freedom, textbook $p = 0.86$, calibrated $p = 0.82$. The calibration matters here: for random reals the textbook test rejects at the 5% level 0.077 of the time, more than it should (for single values, 0.044).
- *Blocks.* Cut into 120 blocks of 100,000 terms, the 120 block p-values look uniform (Kolmogorov–Smirnov $p = 0.30$), as they should if no stretch of π's expansion is unusual.

Table 15.4. The largest of π's first n terms, its position, and its percentile under Galambos's law $e^{-n/(M_n \ln 2)}$.

n	largest term M_n	position	percentile
1,000	20,776	431	0.93
10,000	20,776	431	0.50
100,000	78,629	28,421	0.16
1,000,000	12,996,958	453,293	0.89
10,000,000	12,996,958	453,293	0.33
12,000,000	878,783,625	11,504,930	0.98

- *Khinchin.* The geometric mean of the terms is 2.68655, against $K_0 = 2.685452\ldots$; measured in the standard deviation that random reals show at this length, the deviation is 1.29.
- *Lévy.* $\ln q_n/n = 1.186889$ at $n = 12$ million, against $\pi^2/(12\ln 2) = 1.186569\ldots$, a deviation of 1.21 standard deviations. The two standardized curves in Fig. 15.4 move almost together, because $\ln q_n$ is $\sum_{k\le n} \ln a_k$ plus a sum of corrections $\ln(1 + q_{k-2}/(a_k q_{k-1}))$, each between 0 and $\ln 2$, so the two sums rise and fall with the same large terms.
- *Largest terms.* Galambos proved that for almost every x, the largest of the first n terms, M_n, satisfies $\Pr(M_n \le nt/\ln 2) \to e^{-1/t}$ [46]. Table 15.4 gives the percentile $e^{-n/(M_n \ln 2)}$ at several n. They wander over the whole range, as they should; the term 878,783,625 at position 11,504,930 is large (98th percentile at $n = 12$ million), but not extreme.
- *Missing values.* The smallest positive integer that does not occur among the first 12 million terms is 2,150. A value near 2,000 is expected about $12 \cdot 10^6/(k^2 \ln 2) \approx 4$ times, so a first gap in this range is ordinary.

The much longer computations behind the record tables and MathWorld's plots of the running Khinchin and Lévy means [104, 136] show the same picture, and we know of no published statistic of π's terms that looks atypical.

Status. Open. All statistics we computed on 12 million terms are consistent with π being a typical number for the Gauss measure. That is evidence, and the evidence is good; it is not a proof, and no method we know of could turn it into one.

15.6 Question 5: the spectrum of the correlation matrix

The matrix of Section 5.2, for periods $2, \ldots, m$, is

$$C_m = \left[\frac{\gcd(a,b) - 1}{\sqrt{(a-1)(b-1)}} \right]_{a,b=2}^{m},$$

an $(m-1)\times(m-1)$ matrix with ones on the diagonal (Section 5.2 used $m = 40$). Its structure is cleaner than its look.

Proposition 15.6 (structure)**.** *Let V be the $(m-1)\times(m-1)$ matrix with $V_{ad} = 1$ if $d \mid a$ and 0 otherwise $(a, d = 2, \ldots, m)$, $\Phi = \mathrm{diag}(\varphi(d))$ and $D = \mathrm{diag}(1/\sqrt{a-1})$. Then*

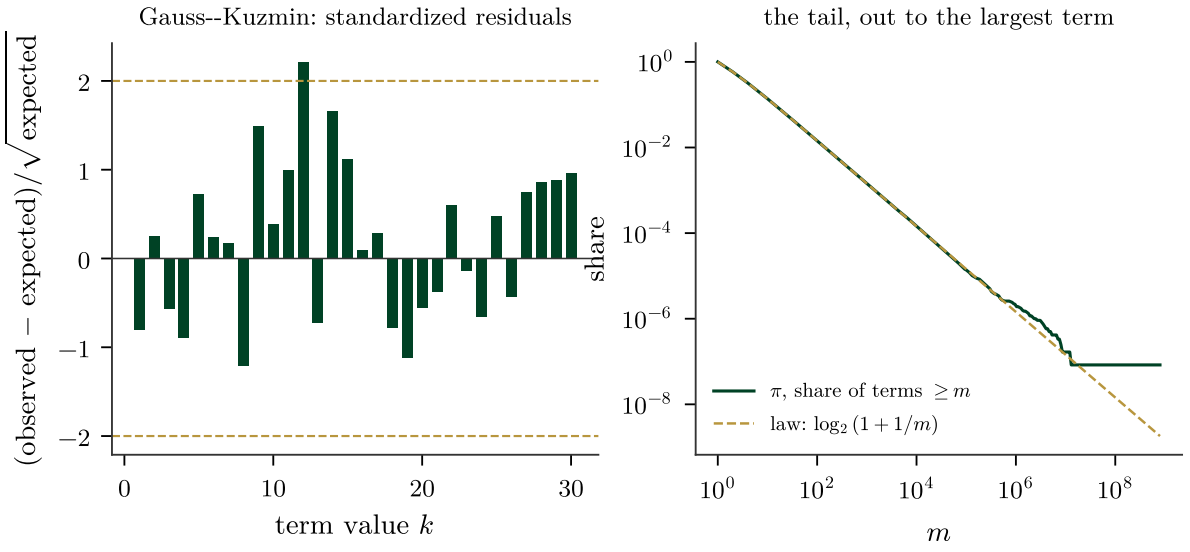


Figure 15.3. π's first 12,000,000 terms against the Gauss–Kuzmin law. Left: standardized differences between observed and expected counts for the values 1 to 30; under the law they behave roughly like standard normal variables. Right: the share of terms at least m, out to the largest term, against the law's $\log_2(1 + 1/m)$.

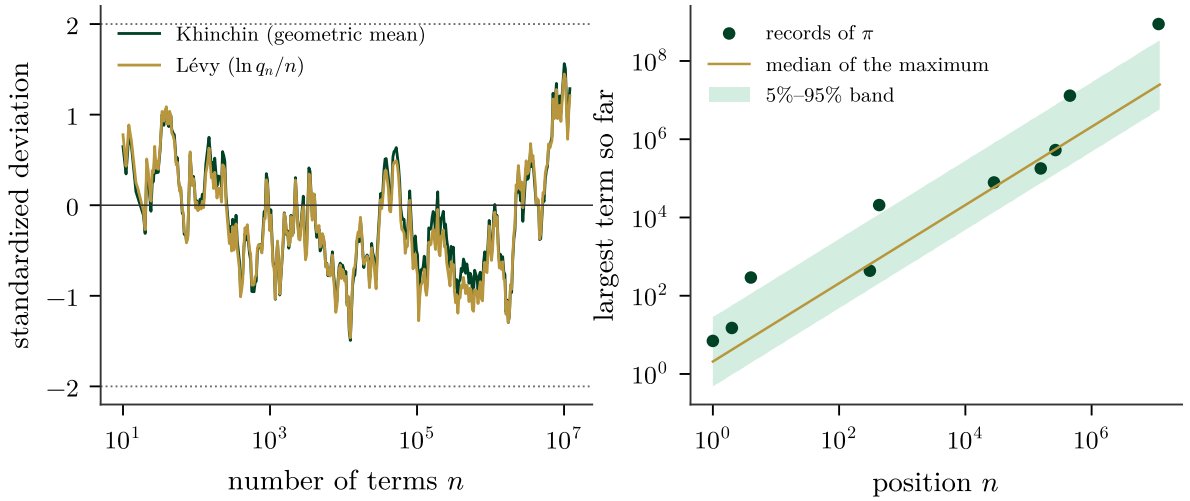


Figure 15.4. Left: the running deviation of the geometric mean of π's terms from Khinchin's constant, and of $\ln q_n/n$ from Lévy's constant, in units of the standard deviation that random reals show after n terms; the dotted lines are ± 2. Right: the record terms of π against the median and the 5%–95% band of Galambos's law for the largest term.

(i) $C_m = DV\Phi V^{\mathsf{T}}D$, *so* C_m *is positive definite;*

(ii) $\det C_m = \prod_{a=2}^{m} \varphi(a)/(a-1)$;

(iii) $(\det C_m)^{1/(m-1)}$, *the geometric mean of the eigenvalues, tends to* $\exp\bigl(\sum_p \frac{1}{p}\ln(1-\frac{1}{p})\bigr) = 0.5599\ldots$;

(iv) for every prime p with $m/2 < p \le m$, the unit vector e_p is an eigenvector with eigenvalue 1.

Proof. (i) Gauss's identity $\sum_{d|g}\varphi(d) = g$ gives $\gcd(a,b) - 1 = \sum_{d|a,\ d|b,\ d\ge 2}\varphi(d) = (V\Phi V^{\mathsf T})_{ab}$. The matrix V is lower triangular with ones on the diagonal (a divisor d of a is at most a), so it is invertible, and Φ has positive entries; hence C_m is a Gram matrix of independent vectors. (ii) $\det V = 1$, so $\det C_m = \det(D)^2 \prod \varphi(d)$. This is the analogue of Smith's determinant $\det[\gcd(i,j)] = \prod \varphi(k)$ [119]. (iii) Since $\ln(\varphi(a)/a) = \sum_{p|a}\ln(1-1/p)$ and a prime p divides a share $1/p$ of integers, the average of $\ln(\varphi(a)/a)$ over $a \le m$ tends to $\sum_p p^{-1}\ln(1-1/p)$; the extra factor $\prod a/(a-1) = m$ contributes $\ln m/(m-1) \to 0$. (iv) For such p, the only multiple of p in range is p itself, so $\gcd(p,b) = 1$ for $b \ne p$, and row p of C_m is e_p. □

For $m = 40$, $\det C_{40} = 6.22 \times 10^{-9}$. Part (iv) gives at least $\pi(m) - \pi(m/2)$ eigenvalues equal to 1, and in every case we computed the multiplicity of the eigenvalue 1 is exactly that.

Theorem 15.7 (the bulk: mean 1 and variance 2)**.** $\operatorname{tr} C_m = m-1$, *and* $\operatorname{tr} C_m^2 = 3m + O(\ln^3 m)$. *So the eigenvalues of C_m have mean 1 and, in the limit, mean square 3 and variance 2.*

Proof. The trace is clear. For the square, let $T(m) = \sum_{a,b\le m}\gcd(a,b)^2/(ab)$. Writing $a = ga'$, $b = gb'$ with $\gcd(a',b') = 1$,

$$T(m) = \sum_{\gcd(a',b')=1} \frac{1}{a'b'}\left\lfloor \frac{m}{\max(a',b')}\right\rfloor = m \sum_{\substack{\gcd(a',b')=1\\ \max(a',b')\le m}} \frac{1}{a'b'\max(a',b')} + O(\ln^2 m).$$

Over all pairs, $\sum_{a,b\ge 1} 1/(ab\max(a,b)) = \zeta(3) + 2\sum_a H_{a-1}/a^2 = \zeta(3) + 2\zeta(3) = 3\zeta(3)$, by Euler's identity $\sum_a H_a/a^2 = 2\zeta(3)$ [42]. The summand is homogeneous of degree -3, so restricting to coprime pairs divides by $\zeta(3)$ (Möbius inversion), giving 3; the pairs with $\max(a',b') > m$ contribute $O(\ln m/m)$. So $T(m) = 3m + O(\ln^2 m)$. Finally, $\operatorname{tr} C_m^2 = \sum_{a,b=2}^{m}(\gcd(a,b)-1)^2/((a-1)(b-1))$ differs from $T(m)$ by $O(\ln^3 m)$: expanding, the difference is bounded by a constant times $\sum \gcd(a,b)/(ab) + \sum 1/(ab) + \sum \gcd(a,b)^2(a+b)/(a^2b^2)$ over $a,b \le m$, and these sums are $O(\ln^3 m)$, $O(\ln^2 m)$ and $O(\ln^2 m)$ by the same substitution. □

Numerically $\operatorname{tr} C_m^2/(m-1)$ is 2.95 at $m = 6{,}000$ and 2.9783 at $m = 20{,}000$, closing in on 3 at the rate the error term allows. Note that $\zeta(3)$ appears and cancels; $6/\pi^2$ does not appear in these moments.

Theorem 15.8 (the top eigenvalue: a lower bound)**.** $\lambda_{\max}(C_m) \ge \dfrac{2}{\pi^2}\ln^2 m\,(1 + o(1)) = \dfrac{1}{3}\cdot\dfrac{6}{\pi^2}\ln^2 m\,(1+o(1))$.

Proof. Take the test vector $x_a = 1/\sqrt{a-1}$. Then $x^{\mathsf T}x = H_{m-1} = \ln m + O(1)$, and

$$x^{\mathsf T}C_m x = \sum_{a,b=2}^{m} \frac{\gcd(a,b)-1}{(a-1)(b-1)} \ge \sum_{a,b\le m}\frac{\gcd(a,b)}{ab} - H_m^2 - 2H_m.$$

With $a = ga'$, $b = gb'$ as before, $\sum_{a,b\le m}\gcd(a,b)/(ab) = \sum_{g\le m} g^{-1}S(m/g)$, where, by Möbius inversion,

$$S(y) = \sum_{\substack{a',b'\le y\\ \gcd(a',b')=1}} \frac{1}{a'b'} = \sum_{d\le y}\frac{\mu(d)}{d^2}H^2_{\lfloor y/d\rfloor} = \frac{6}{\pi^2}\ln^2 y + O(\ln y + 1).$$

Table 15.5. The spectrum of C_m (dense diagonalization): $\operatorname{tr} C_m^2/(m-1)$ (limit 3), the geometric mean of the eigenvalues (limit 0.5599), the smallest eigenvalue, the multiplicity of the eigenvalue 1 and the lower bound $\pi(m)-\pi(m/2)$, the largest eigenvalue, and the mean spacing ratio $\langle r\rangle$ in the central 80% of the spectrum (Poisson 0.386, GOE 0.531).

m	$\operatorname{tr} C^2/(m-1)$	geo. mean	$\lambda_{\min}$	mult. of 1	bound	$\lambda_{\max}$	$\langle r\rangle$
500	2.755	0.5672	0.0246	42	42	14.887	0.493
1,000	2.838	0.5640	0.0196	73	73	18.773	0.515
2,000	2.895	0.5622	0.0159	135	135	23.279	0.539
4,000	2.934	0.5611	0.0130	247	247	28.458	0.529
6,000	2.950	0.5607	0.0117	353	353	31.822	0.525

Table 15.6. The largest eigenvalue of C_m, computed by the Lanczos method with a matrix-vector product of cost $O(m\log m)$ from Proposition 15.6(i); the Rayleigh quotient of the test vector of Theorem 15.8; and $(2/\pi^2)\ln^2 m$.

m	$\lambda_{\max}$	test vector	$(2/\pi^2)\ln^2 m$
1,000	18.773	10.680	9.670
3,000	26.221	13.952	12.990
10,000	36.429	18.091	17.190
30,000	47.879	22.377	21.536
100,000	63.078	27.632	26.860
300,000	79.678	32.938	32.230
1,000,000	101.252	39.312	38.678

Then $\sum_{g\le m} g^{-1}\frac{6}{\pi^2}\ln^2(m/g) = \frac{6}{\pi^2}\cdot\frac{\ln^3 m}{3} + O(\ln^2 m)$. Dividing by $x^{\mathsf{T}}x$ gives the bound, since $\lambda_{\max} \ge x^{\mathsf{T}}C_m x/x^{\mathsf{T}}x$. □

Comparison with random-matrix benchmarks. Tables 15.5 and 15.6 and Fig. 15.5 give the numbers.

- *Not Marchenko–Pastur.* A sample correlation matrix of independent noise follows the Marchenko–Pastur law [90], with bounded support. With the same mean 1 and variance 2, that law would put half of all eigenvalues at 0 and none above $(1+\sqrt{2})^2 \approx 5.8$. Our matrix has no eigenvalue at 0 (the smallest is about 0.01 at $m = 6{,}000$ and shrinks slowly), about two thirds of its eigenvalues below 1, the eigenvalue 1 exactly at the large primes, and a top eigenvalue that grows without bound. It is not a noisy correlation matrix, and the comparison mainly shows how much arithmetic structure it carries.

- *Not a semicircle.* The GOE's semicircle is symmetric about its center; this spectrum is positive and strongly skewed.

- *But GOE-like locally.* The mean ratio of consecutive spacings, $\langle r\rangle$, needs no unfolding and is about 0.386 for uncorrelated (Poisson) levels and 0.531 for the GOE [5]. In the bulk of our spectrum it is 0.51 to 0.54 for $m \ge 1{,}000$ (0.525 at $m = 6{,}000$), close to the GOE value: neighboring eigenvalues repel each other as in a random symmetric matrix. We report this as an observation; we have no explanation for it. [Update: much of it is an artifact of near-degenerate families of eigenvalues (Proposition 16.7). [Added 2026-11-06]]

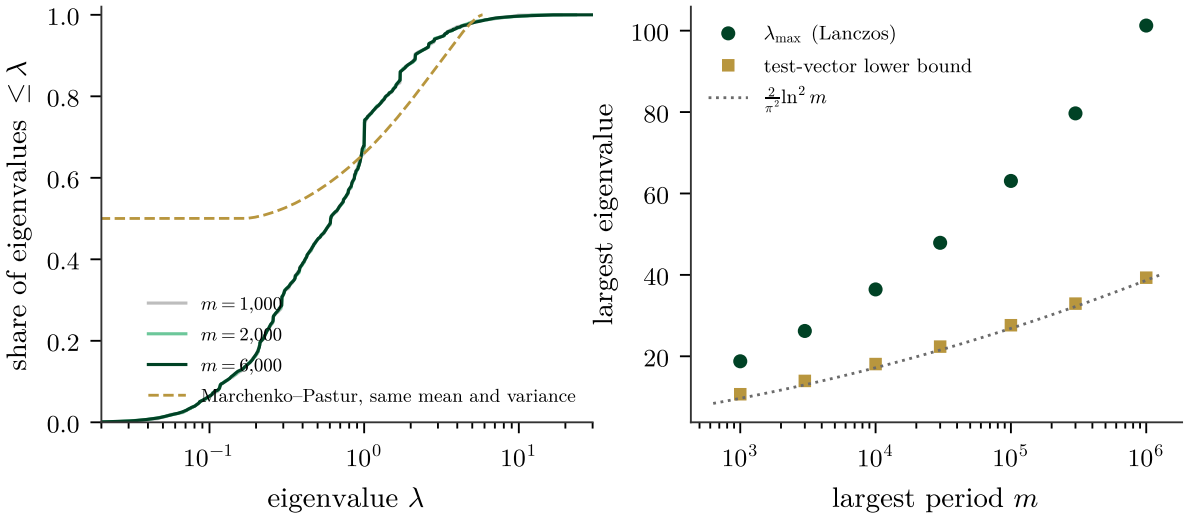


Figure 15.5. Left: the distribution of the eigenvalues of C_m for three sizes (the curves almost coincide), against the Marchenko–Pastur law with the same mean and variance (which puts half its mass at 0). Right: the largest eigenvalue up to $m = 10^6$, the proven lower bound from the test vector, and $(2/\pi^2)\ln^2 m$.

- *The top eigenvalue.* It grows faster than our lower bound: the ratio $\lambda_{\max}/\ln^2 m$ rises from about 0.39 at $m = 1{,}000$ to 0.53 at $m = 10^6$. The test vector's quotient tracks $(2/\pi^2)\ln^2 m$ closely, as the proof says, but the true top eigenvector puts extra weight on numbers with many small factors. GCD matrices of this kind are studied in the literature on GCD sums and on eigenvalues of GCD matrices [62, 87]; we did not find the asymptotic for this particular matrix, and we do not know it. [Update: it grows faster than any power of $\ln m$ (Theorem 16.6). [Added 2026-11-06]] [Update: its growth is $\exp((2+o(1))\sqrt{\ln m/\ln\ln m})$ (Theorem 17.1). [Added 2026-11-10]]

Status. Partly answered. The matrix is a Gram matrix with an exact determinant; its spectrum has mean 1 and variance tending to 2 (proved); its geometric mean converges to an explicit constant (proved); its top eigenvalue is at least $(2/\pi^2)\ln^2 m$, which is where $6/\pi^2$ appears (proved), but grows faster in our data. It is far from the Marchenko–Pastur and semicircle laws, yet its level spacings look like those of the GOE (numerical evidence). Whether $6/\pi^2$ appears in the limiting bulk distribution is open; it does not appear in its first two moments.

15.7 Question 6: ranking the equal temperaments

The measures. For each n from 5 to 200 we measured how well the n-note equal temperament approximates two sets of just intervals: the 5-limit consonances 3/2, 5/4, 6/5, and the 7-limit set that adds 7/4, 7/5, 7/6 (their inversions have the same errors). We used four measures:

RMS: the root-mean-square error in cents, each interval approximated by its nearest step;

Table 15.7. The top 20 equal temperaments between 5 and 200 notes for each measure, best first. Prime sizes are in bold.

measure	best 20
RMS, 5-limit	171, 118, 152, 183, 190, **137**, 186, **193**, 140, 164, **53**, 106, 159, 99, 198, **149**, 87, 174, 65, 130
RMS, 7-limit	171, 140, 99, 198, 130, **193**, 183, **167**, 152, 186, 190, 161, **181**, 175, **197**, 156, **103**, **149**, 187, 72
TOP, 5-limit	171, 118, 152, 183, 190, **53**, 106, 159, 164, 99, 198, **193**, **137**, 186, 140, 65, 130, 195, 178, **149**
TOP, 7-limit	171, 99, 198, **193**, 183, 152, 140, 130, 190, **149**, 159, 118, 200, **197**, 186, 176, **179**, 125, **181**, 72
relative, 5-limit	118, 171, **53**, 152, 65, 34, 87, 99, **19**, 183, 84, 106, **137**, **31**, 12, 140, 190, 72, 46, 186
relative, 7-limit	171, 99, **31**, 140, 72, 130, 68, **41**, **103**, 198, **53**, 152, 27, 118, 183, **167**, 84, **193**, 62, 46
TOP relative, 5-limit	171, 118, **53**, 65, 106, 152, 99, 183, 34, 12, 87, **19**, 72, 159, 46, 84, 164, **137**, 190, 130
TOP relative, 7-limit	171, 99, 72, 130, **31**, **41**, 140, 152, 118, **53**, 198, **149**, 22, 183, **193**, **19**, 12, 46, 159, 27

relative: the RMS error in units of one step, that is RMS $\times\, n/1200$. Large n wins any absolute measure simply by having small steps; the relative error asks how good the temperament is for its size;

TOP: Erlich's Tenney-optimal error [39]: map each prime $2, 3, 5$ (and 7) to its nearest number of steps, weight the error at the prime p by $1/\log_2 p$, allow the octave to be stretched, and take the smallest possible maximum weighted error, in cents per octave;

TOP relative: TOP in units of one step.

The ranking. Table 15.7 lists the top 20 for each measure. The familiar names are all there: 171, 118, 99, 72, 53, 41, 31, 19 and 12, and 140 and 130. Of the 16 sizes between 5 and 200 that Gene Ward Smith singled out by the peaks of the Riemann zeta function on the critical line (5, 7, 10, 12, 19, 22, 27, 31, 41, 53, 72, 99, 118, 130, 152, 171; 109), 5 are in the top 20 of the 7-limit RMS list and most are high in the relative lists. The winner on almost every measure is 171.

Do prime sizes stand out? No. Of the 196 sizes, 44 are prime, so a random top 20 would hold about 4.5 primes. Table 15.8 compares the mean score of prime and composite sizes with a permutation test and a Mann–Whitney test, and counts primes in each top 20 against the hypergeometric law. No test comes near significance for the scale-free (relative) measures. On the absolute measures the primes look slightly worse on average, but that is because primes are more common among small n, whose steps are large; the difference is not significant either. Figure 15.6 shows the relative error for every size.

There is a theorem behind this answer. The relative errors of n-EDO depend only on the fractional parts of $n\log_2 3$, $n\log_2 5$ and $n\log_2 7$. The numbers $1, \log_2 3, \log_2 5, \log_2 7$ are linearly independent over the rationals (a relation would give $2^a 3^b 5^c 7^d = 1$), and by Vinogradov's theorem on exponential sums over primes, the vectors $(p\log_2 3, p\log_2 5, p\log_2 7) \bmod 1$, p prime,

Table 15.8. Prime against composite sizes $n = 5, \ldots, 200$: mean score of each group (smaller is better), two-sided p-values of a permutation test (100,000 permutations) and of the Mann–Whitney test, the number of primes in the top 20, and the chance of at least that many in a random 20.

measure	primes	composites	p (perm.)	p (M–W)	top-20 primes	p
relative, 5-limit	0.283	0.273	0.51	0.46	4	0.70
relative, 7-limit	0.283	0.282	0.91	0.37	6	0.27
TOP relative, 5-limit	0.112	0.110	0.80	0.82	3	0.87
TOP relative, 7-limit	0.128	0.128	1.00	0.91	6	0.27
RMS, 5-limit	8.433	5.889	0.13	0.27	4	0.70
RMS, 7-limit	8.770	5.945	0.08	0.41	6	0.27

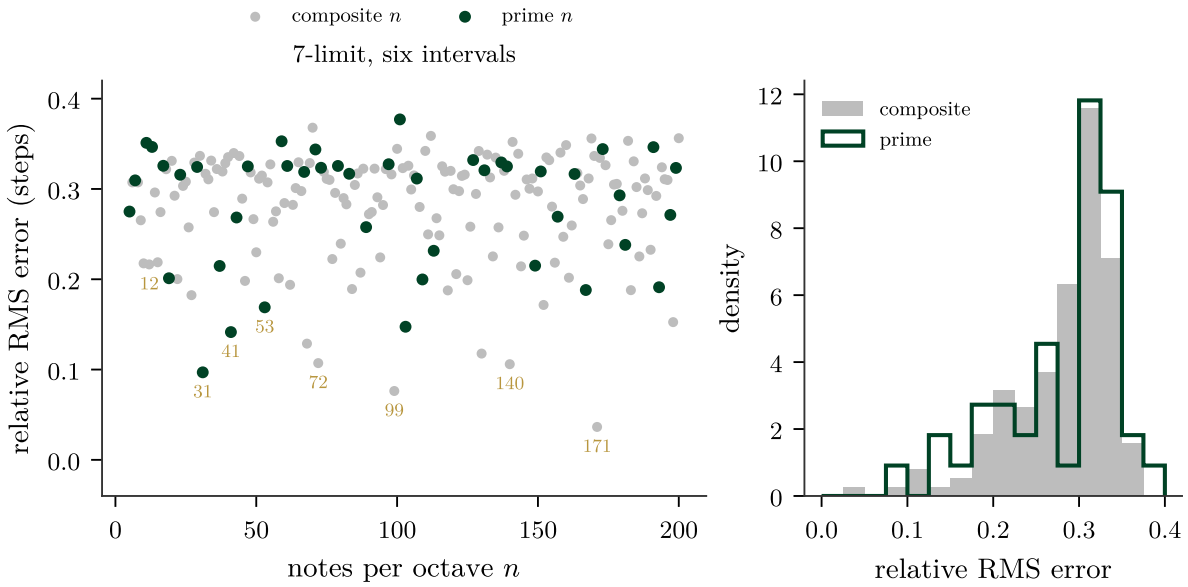


Figure 15.6. The relative 7-limit error (in steps) of every equal temperament from 5 to 200 notes, prime sizes in green (left), and the distributions for prime and composite sizes (right).

are equidistributed in the cube [133], exactly as they are for all integers n by Weyl's theorem. So in the long run, the relative errors of prime-sized temperaments have the same distribution as those of all temperaments. Prime sizes do have one real musical property, noted in Chapter 4: every step size generates the whole scale. That is a fact about the scale's structure, not about its tuning accuracy.

Status. Answered: no. Prime sizes do not stand out in tuning accuracy, by any of the eight measures, and an equidistribution theorem says they cannot stand out in the long run. [Update: Section 16.4 explains why 171 wins. [Added 2026-11-06]]

15.8 Status after this work

Table 15.9 sums up where the six questions stood after this chapter; Table 16.4 gives the state after Chapter 16 [Added 2026-11-06].

Table 15.9. The six questions of Section 14.3 after this chapter.

question	what we did	status
1. Prime convergents of π	12 million certified terms; q_n and p_n tested to $n = 20{,}000$; 15 prime numerators, 10 denominators; counts inside the heuristic's 95% range; e's divisibility analyzed	open; data typical
2. Fibonacci and Pell primes	known lists compared with the Grantham–Granville heuristic; divisibility facts proved	open
3. The heuristic in a model	exact expectation proved, so the constant $2\ln 2$ holds in expectation; for almost every x the count lies between $\ln N$ and $2\ln N$ (using deep cited theorems); finitely many n with p_n, q_n both prime (proved)	partly answered
4. Gauss–Kuzmin for π	calibrated tests on 12 million terms: values, tail, pairs, blocks, Khinchin, Lévy, records	open; data typical
5. Spectrum of the correlation matrix	Gram structure, determinant, mean and variance, geometric mean, lower bound for the top eigenvalue proved; GOE-like spacings observed	partly answered
6. Prime-sized temperaments	eight measures, top-20 lists, permutation tests	answered: no

CHAPTER 16

A second look at the open questions

Added 2026-11-06.

Chapter 15 left several loose ends: a second-moment estimate that would settle Question 3 for almost every number, an observation about e that was checked but not proved, a top eigenvalue whose growth we could bound only from below, a level-spacing statistic we could not explain, and a winning temperament we did not account for. This chapter goes back to each of them. The labels mean what they meant in Chapter 15: a Theorem or Proposition without a citation in its heading is proved here from cited results; numbers from computation are numerical evidence.

The main result is Theorem 16.4: Heuristic 8.3 is true for almost every real number. It says nothing about π, which is a single number and may or may not be typical.

16.1 Question 3: the heuristic for almost every number

For an integer $q \geq 2$ let E_q be the set of $x \in (0,1)$ for which q is a convergent denominator, and write λ for length (Lebesgue measure). Theorem 15.3 says that $\lambda(E_q) = \frac{2}{q}\sum_r \frac{1}{q+r}$, the sum over $1 \leq r < q$ with $\gcd(r,q) = 1$. Chapter 15 stopped because the pieces of E_q are not all the same size. The way forward is to describe the sizes exactly and show that they are spread out evenly.

Lemma 16.1 (the pieces of E_q)**.** *Let $0 < a < q$ with $\gcd(a,q) = 1$, and let s be the denominator of the convergent just before a/q in the expansion of a/q whose last term is at least 2. The set of $x \in (0,1)$ that have a/q as a convergent is an interval with a/q inside it, of length*

$$\frac{1}{q(q+s)} + \frac{1}{q(2q-s)} = \frac{g(s/q)}{q^2}, \qquad g(u) = \frac{1}{1+u} + \frac{1}{2-u}.$$

Here $as \equiv \pm 1 \pmod q$, and since $g(u) = g(1-u)$, the length depends only on the inverse of a modulo q. The pieces for different a do not overlap, and E_q is their union.

Proof. The rational a/q has two expansions, $[0; c_1, \ldots, c_k]$ with $c_k \geq 2$ and $[0; c_1, \ldots, c_k - 1, 1]$ (the proof of Theorem 15.3). In the first, the last two denominators are s and q; in the second they are $q - s$ and q. By (15.2) the two cylinders have lengths $1/(q(q+s))$ and $1/(q(2q-s))$. A cylinder of length n consists of the points $(p_n + tp_{n-1})/(q_n + tq_{n-1})$ with $0 \leq t < 1$, so it has

p_n/q_n at one end and lies on the side given by the sign of $p_{n-1}q_n - p_nq_{n-1} = (-1)^n$. The two expansions have lengths of opposite parity, so the two cylinders lie on opposite sides of a/q and together form an interval. The relation $p_kq_{k-1} - p_{k-1}q_k = \pm 1$ gives $as \equiv \pm 1 \pmod q$. Finally, q_n increases strictly with $n \geq 1$, so a given x has at most one convergent with denominator q. □

Summing the lengths over a gives Theorem 15.3 again. The function g lies between $4/3$ and $3/2$, so the pieces differ in size by at most 12%, and its average over $(0,1)$ is $\int_0^1 g = 2\ln 2$.

Lemma 16.2 (overlaps)**.** *Let $q < r$ be primes. Then*

(i) $\lambda(E_q \cap E_r) \leq 8/(qr)$*;*

(ii) $\bigl|\lambda(E_q \cap E_r) - \lambda(E_q)\lambda(E_r)\bigr| \leq C\,q\,r^{-3/2}\ln^2 r$*, with an absolute constant C.*

Proof. (i) Every convergent satisfies $|x - a/q| < 1/q^2$, so the piece for a/q lies within $1/q^2$ of a/q, and the piece for b/r within $1/r^2$ of b/r. Two such pieces can meet only if $|a/q - b/r| < 2/q^2$, that is $|ar - bq| < 2r/q$. The integer $k = ar - bq$ is not 0 (since $a/q \neq b/r$), and it determines the pair: $a \equiv kr^{-1} \pmod q$ fixes a, and then $b = (ar-k)/q$. So at most $4r/q$ pairs of pieces meet, each in a set of length at most $2/r^2$.

(ii) Fix a piece I of E_q. The integers b with $b/r \in I$ form a run of $N = r\lambda(I) + O(1)$ consecutive integers, all prime to r. By Lemma 16.1 the piece of E_r at b/r has length $g(\bar b/r)/r^2$, where $\bar b$ is the inverse of b modulo r. Weil's bound for Kloosterman sums [135], by the standard completion argument [63, Ch. 11], gives $\bigl|\sum_b e^{2\pi i h\bar b/r}\bigr| \leq 2\sqrt r\,(1+\ln r) + 1$ for every run of b and every h not divisible by r. The Erdős–Turán inequality with $H = r-1$ then shows that the N points $\bar b/r$ have discrepancy at most $C\sqrt r\ln^2 r/N$, and Koksma's inequality (the total variation of g is $1/3$) gives $\sum_b g(\bar b/r) = 2\ln 2\cdot N + O(\sqrt r\ln^2 r)$ [76, Ch. 1–2]. At most two pieces of E_r cross an end of I, each of length at most $3/(2r^2)$. Hence

$$\lambda(I \cap E_r) = \frac{2\ln 2\cdot r\lambda(I)}{r^2} + O\!\left(\frac{\sqrt r\ln^2 r}{r^2}\right).$$

Summing over the $q-1$ pieces of E_q gives $\lambda(E_q\cap E_r) = (2\ln 2/r)\,\lambda(E_q) + O(q\sqrt r\ln^2 r/r^2)$, and $\lambda(E_r) = (2/r)(H_{2r-1} - H_r) = 2\ln 2/r + O(1/r^2)$. □

So the events "q is a convergent denominator" and "r is a convergent denominator" are nearly independent when r is much larger than q, and never strongly dependent. That is exactly what a second-moment argument needs. We use it in the following form.

Lemma 16.3 (a variance lemma; 57, Lemma 1.5)**.** *Let $f_k \geq 0$ be measurable functions on a probability space, and $0 \leq \varphi_k \leq 1$ numbers with $\int f_k = \varphi_k$. Write $\Phi(N) = \sum_{k\leq N}\varphi_k$. If there is a constant C with $\int\bigl(\sum_{M<k\leq N}(f_k - \varphi_k)\bigr)^2 \leq C\sum_{M<k\leq N}\varphi_k$ for all $M < N$, then for every $\varepsilon > 0$ and almost every point, $\sum_{k\leq N} f_k = \Phi(N) + O\bigl(\Phi(N)^{1/2}(\ln\Phi(N))^{3/2+\varepsilon} + 1\bigr)$.*

(Harman states the lemma with $0 \leq f_k \leq \varphi_k \leq 1$ for the means and a general measure; our case, indicator functions and their measures, is the simplest one.)

Theorem 16.4 (Heuristic 8.3 for almost every number)**.** *Let $\Psi(Q) = \sum_{p\leq Q}\lambda(E_p) = 2\ln 2\cdot\ln\ln Q + B_0 + o(1)$ (Theorem 15.3). For every $\varepsilon > 0$ and almost every real x:*

(i) the number of primes $p \leq Q$ that are convergent denominators of x is

$$\Psi(Q) + O\bigl(\Psi(Q)^{1/2}(\ln\Psi(Q))^{3/2+\varepsilon}\bigr);$$

(ii) the number of $n \le N$ with q_n prime is

$$2\ln 2 \cdot \ln N + O\big((\ln N)^{1/2}(\ln\ln N)^{3/2+\varepsilon}\big);$$

(iii) the same holds for the numerators: the number of $n \le N$ with p_n prime has the same asymptotic formula.

Proof. It is enough to take $x \in (0,1)$, since adding an integer to x does not change the denominators.

(i) Number the primes $p_1 < p_2 < \cdots$ and apply Lemma 16.3 with f_k the indicator of E_{p_k} and $\varphi_k = \lambda(E_{p_k})$. Expanding the square, the integral in the lemma is $\sum_k \varphi_k(1-\varphi_k) + 2\sum_{j<k}\big(\lambda(E_{p_j} \cap E_{p_k}) - \varphi_j\varphi_k\big)$, both sums over $M < j < k \le N$. Fix $q = p_j$. For the primes r with $q < r \le q^5$, Lemma 16.2(i) bounds the terms by $\sum_{q<r\le q^5} 8/(qr) \le 8c_0/q$, where $c_0 = \sup_q \sum_{q<r\le q^5} 1/r$ is finite by Mertens' theorem ($\sum_{q<r\le q^5} 1/r \to \ln 5$). For $r > q^5$, Lemma 16.2(ii) bounds them by $Cq\sum_{r>q^5} r^{-3/2}\ln^2 r \le C'q^{-3/2}\ln^2 q$. Both are at most a constant times $1/q$, and $\lambda(E_q) \ge \frac{2}{q}\cdot\frac{q-1}{2q-1} \ge \frac{2}{3q}$, so both are at most a constant times φ_j. Summing over j gives the hypothesis of the lemma, and its conclusion is (i).

(ii) The denominators q_n, $n \ge 1$, increase strictly and $q_0 = 1$ is not prime, so the primes among $q_1, \dots, q_N$ are exactly the primes $p \le q_N$ that are convergent denominators. By Lévy's theorem [85], $\ln q_N = LN(1+o(1))$ for almost every x, with $L = \pi^2/(12\ln 2)$, so $\Psi(q_N) = 2\ln 2\cdot(\ln N + \ln L + o(1)) + B_0 + o(1) = 2\ln 2\cdot\ln N + O(1)$. Put $Q = q_N$ in (i).

(iii) As in the proof of Theorem 15.5(i), the numerators of x are, up to a shift of index, the denominators of $1/x$, and $x \mapsto 1/x$ preserves sets of measure zero. □

What this settles. Theorem 16.4 answers Question 3 in the form it was asked: for almost every x, the number of $n \le N$ with q_n prime is asymptotic to $2\ln 2\cdot\ln N$, with the heuristic's constant, and the same for p_n. It sharpens Theorem 15.5(iii) from "between $\ln N$ and $2\ln N$" to the exact constant, and it does not need the quantitative Duffin–Schaeffer theorem. The deep inputs are Weil's bound for Kloosterman sums, Lévy's theorem and the variance lemma. Combined with Theorem 15.5(ii), it also shows that for almost every x a share $1/(2\ln 2) \approx 72\%$ of the prime denominators have $\theta_n < 1/2$, the same share as for all convergents (the Doeblin–Lenstra law).

What it does not settle. It does not tell us anything about π, e, $\sqrt{2}$ or any other single number: a set of measure zero can contain all of them. For φ and $\sqrt{2}$ the count is a different problem (Section 15.3), and for e it grows only like $\ln\ln N$ (Section 15.2). Question 1, about π, is as open as before.

Prior work. Overlap estimates of this kind are the bread and butter of metric Diophantine approximation [57], and the equidistribution of modular inverses is classical. We did not find Theorem 16.4, or the expected-value version Corollary 15.4, stated anywhere, but the argument is short enough that specialists may well know it, and we would be glad to learn of a reference.

The numbers. Table 16.1 computes every overlap $\lambda(E_q \cap E_r)$ exactly (as lengths of intervals) for the 92,235 pairs of primes $q < r < 3{,}000$, grouped by the ratio r/q. The script also checks Lemma 16.1 piece by piece and the bound of Lemma 16.2(i) for every pair. When $r < 2q$ the two events repel each other (the ratio is about 0.69); from $r > 4q$ on, the ratio is within about

Table 16.1. Exact overlaps for the 430 primes below 3,000: for each range of r/q, the number of pairs of primes $q < r$ and the ratio $\sum \lambda(E_q \cap E_r) / \sum \lambda(E_q)\lambda(E_r)$. A ratio of 1 means independence.

r/q	pairs	ratio
$[1, 2)$	40,825	0.6865
$[2, 4)$	22,525	0.9359
$[4, 8)$	12,473	0.9983
$[8, 16)$	6,991	1.0116
$[16, 32)$	3,931	1.0120
$[32, 64)$	2,293	1.0052
$[64, 128)$	1,319	1.0032
≥ 128	1,878	1.0013

1% of 1, as Lemma 16.2(ii) predicts. Over all these primes, $\Psi = 2.593$ and the sum of the covariances is -0.184: the count has variance below its mean, as the simulation of Chapter 15 also showed.

Status. Answered for almost every number (theorem). For π itself, open.

16.2 Question 1: e modulo a prime

Section 15.2 proved that e's convergent denominators are periodic modulo every prime ℓ, and observed that the period divides 6ℓ. Here is a proof.

Proposition 16.5. *Let q_n be the convergent denominators of e and ℓ a prime. Then $q_{n+6\ell} \equiv q_n \pmod{\ell}$ for every $n \geq 0$. The same holds for the numerators p_n.*

Proof. Write $v_n = (q_n, q_{n-1})^{\mathsf{T}}$ (with $q_{-1} = 0$) and $A(a) = \left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so that $v_n = A(a_n)v_{n-1}$. The terms come in blocks $(a_{3k-2}, a_{3k-1}, a_{3k}) = (1, 2k, 1)$, so $v_{3k} = B(k)v_{3k-3}$ with

$$B(k) = A(1)A(2k)A(1) = \begin{pmatrix} 2k+2 & 2k+1 \\ 2k+1 & 2k \end{pmatrix}, \qquad \det B(k) = -1.$$

For $\ell = 2$, $B(k) \equiv \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, whose square is the identity, so $v_{3k+6} \equiv v_{3k}$. Let ℓ be odd, and work modulo ℓ, where $B(k)$ depends only on $k \bmod \ell$. For c modulo ℓ let $P(c) = B(c)B(c+1)\cdots B(c+\ell-1)$, a product over one full cycle.

(a) *The trace of $P(c)$ does not depend on c.* Since $B(c+\ell) = B(c)$, we have $P(c+1) = B(c)^{-1}P(c)B(c)$, and conjugate matrices have equal traces. Call the common trace κ.

(b) $\kappa = 0$. With $W = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ one checks $B(-1-k) = -WB(k)W$. Multiplying over $k = c, c+1, \ldots, c+\ell-1$ gives $B(-1-c)B(-2-c)\cdots B(-\ell-c) = (-1)^\ell WP(c)W$. The matrices B are symmetric, so the left side is the transpose of $B(-\ell-c)\cdots B(-1-c) = P(-\ell-c)$, whose trace is κ. The right side has trace $-\kappa$, since ℓ is odd. So $\kappa = -\kappa$, and $\kappa = 0$.

(c) $P(c)^2 = I$. By the Cayley–Hamilton theorem, $P^2 = (\operatorname{tr} P)P - (\det P)I = 0 - (-1)^\ell I = I$.

The matrix carrying v_{3k} to $v_{3k+3\ell}$ is $B(k+\ell)\cdots B(k+1)$, the transpose of $P(k+1)$, so its square is also I, and $v_{3k+6\ell} \equiv v_{3k}$ for every $k \geq 0$. The terms a_n modulo ℓ repeat with period

3ℓ (for $n \geq 1$), so the steps from v_{3k} to v_{3k+1} and v_{3k+2} repeat too, and $v_{n+6\ell} \equiv v_n$ for every $n \geq 0$. The numerators satisfy the same recurrence, with other starting values, so the argument applies to them unchanged. □

The script confirms the proposition for every prime $\ell < 3{,}000$ and checks step (b) directly for every starting block and every prime below 400. In every case computed, the period is exactly 6ℓ for odd ℓ (and 6 for $\ell = 2$); we have not proved that it is never shorter. [Update: Proposition 17.8 proves that it is exactly 6ℓ. [Added 2026-11-10]]

Status. Unchanged: open for π. The structure of e is now fully proved; the densities f_ℓ of Section 15.2 are exact counts over one period of length 6ℓ.

16.3 Question 5: the correlation matrix again

The top eigenvalue. Chapter 15 proved $\lambda_{\max}(C_m) \geq (2/\pi^2)\ln^2 m\,(1+o(1))$ and found the true value larger. In fact no power of $\ln m$ is enough.

Theorem 16.6 (the top eigenvalue)**.** *As $m \to \infty$,*

$$\exp\left((2+o(1))\frac{\sqrt{\ln m}}{\ln\ln m}\right) \leq \lambda_{\max}(C_m) \leq \exp\left((\ln 2+o(1))\frac{\ln m}{\ln\ln m}\right).$$

So the top eigenvalue grows faster than every power of $\ln m$ and slower than every power of m.

Proof. Lower bound. Let y be the largest prime with $P = \prod_{p\leq y} p \leq m$, and let $k = \pi(y)$. For square-free d, e dividing P,

$$\frac{\gcd(d,e)}{\sqrt{de}} = \prod_{p\leq y} t_p, \qquad t_p = \begin{cases} 1 & \text{if } p \text{ divides both or neither,} \\ p^{-1/2} & \text{if } p \text{ divides exactly one,} \end{cases}$$

so the matrix $[\gcd(d,e)/\sqrt{de}]$ over the 2^k divisors of P is a tensor product of the 2×2 matrices $\left(\begin{smallmatrix} 1 & p^{-1/2} \\ p^{-1/2} & 1 \end{smallmatrix}\right)$, and the sum of its entries is $\prod_{p\leq y}(2+2p^{-1/2}) = 2^k\Pi$, with $\Pi = \prod_{p\leq y}(1+p^{-1/2})$. Take the test vector that is 1 on the divisors $d \geq 2$ of P and 0 elsewhere. Since $(d-1)(e-1) \leq de$,

$$\sum_{\substack{d,e\mid P \\ d,e\geq 2}} \frac{\gcd(d,e)-1}{\sqrt{(d-1)(e-1)}} \geq \sum_{\substack{d,e\mid P \\ d,e\geq 2}} \frac{\gcd(d,e)-1}{\sqrt{de}} = 2^k\Pi - \Pi^2,$$

because the terms with $d = 1$ or $e = 1$ are zero and $\sum_{d,e\mid P} 1/\sqrt{de} = \Pi^2$. Dividing by the squared length $2^k - 1$ gives $\lambda_{\max}(C_m) \geq \Pi\,(1 - \Pi/2^k)$, which is $\Pi\,(1-o(1))$ because $\Pi/2^k = \prod_{p\leq y}\frac{1+p^{-1/2}}{2} \to 0$. Finally $\ln\Pi = \sum_{p\leq y} p^{-1/2} + O(1) = (2+o(1))\sqrt{y}/\ln y$ by the prime number theorem, and $y \sim \ln m$ because $\sum_{p\leq y}\ln p \sim y$.

Upper bound. For $a \geq 2$, $a - 1 \geq a/2$, so every entry of C_m is at most $2\gcd(a,b)/\sqrt{ab}$. Both matrices have nonnegative entries, so the largest eigenvalue of C_m is at most twice that of $G_m = [\gcd(a,b)/\sqrt{ab}]_{a,b\leq m}$ (Perron–Frobenius). For G_m use Schur's test with the weights $b^{-1/2}$: by Gauss's identity of Proposition 15.6,

$$\sum_{b\leq m} \frac{\gcd(a,b)}{\sqrt{ab}}\,b^{-1/2} = a^{-1/2}\sum_{d\mid a}\varphi(d)\sum_{j\leq m/d}\frac{1}{dj} \leq a^{-1/2}(1+\ln m)\sum_{d\mid a}\frac{\varphi(d)}{d} \leq a^{-1/2}(1+\ln m)\,\tau(a),$$

Table 16.2. The largest eigenvalue of C_m (from Table 15.6) against $(2/\pi^2)\ln^2 m \cdot \ln\ln m$. The fit is empirical and cannot hold for all m (Theorem 16.6); the true growth is $\exp((2+o(1))\sqrt{\ln m/\ln\ln m})$ (Theorem 17.1) [Added 2026-11-10].

m	$\lambda_{\max}$	$(2/\pi^2)\ln^2 m\,\ln\ln m$	ratio
1,000	18.77	18.69	1.005
3,000	26.22	27.02	0.970
10,000	36.43	38.17	0.954
30,000	47.88	50.24	0.953
100,000	63.08	65.63	0.961
300,000	79.68	81.69	0.975
1,000,000	101.25	101.56	0.997

where $\tau(a)$ is the number of divisors of a. So $\lambda_{\max}(C_m) \le 2(1+\ln m)\max_{a\le m}\tau(a)$, and $\tau(a) \le 2^{(1+o(1))\ln a/\ln\ln a}$ [56, Thm. 317]. □

The lower bound uses the same square-free construction that drives the theory of GCD sums, where sums of $\gcd(m,n)/\sqrt{mn}$ over arbitrary sets of N integers are known to grow roughly like $e^{c\sqrt{\ln N}}$ up to iterated logarithms [2, 17]. Here the set is fixed (all integers up to m), and the true order lies somewhere between our two bounds; we do not know where. [Update: Theorem 17.1 finds it: $\exp((2+o(1))\sqrt{\ln m/\ln\ln m})$. [Added 2026-11-10]]

The bounds say little at sizes we can compute. At $m = 10^6$ the lower bound is 11.09 (with $y = 17$), the upper bound is 2,570, and the true value is 101.25. Over the range we computed, the data follow $(2/\pi^2)\ln^2 m \cdot \ln\ln m$ closely (Table 16.2: the ratio stays between 0.953 and 1.005). That is a description of the data, not a law: by Theorem 16.6 it must fail for large m, though only when $\ln m$ is in the thousands.

Near-copies in the spectrum. Proposition 15.6(iv) found the eigenvalue 1 once for each prime between $m/2$ and m. That is the first case of a general pattern. Let

$$M_K = \left[\frac{\gcd(k,k')}{\sqrt{kk'}}\right]_{k,k'\le K},$$

the $K \times K$ corner of the matrix G_m above.

Proposition 16.7 (near-copies)**.** *Fix $K \ge 1$. Let F be the set of primes $p > \sqrt{m}$ with $\lfloor m/p\rfloor = K$, let p_0 be the smallest of them, and let μ be an eigenvalue of M_K of multiplicity ν. Then C_m has at least $\nu(|F|-K)$ eigenvalues within c_K/p_0 of μ, where c_K depends only on K.*

Proof. For $p \in F$ the multiples of p up to m are kp, $1 \le k \le K$, and $K < \sqrt{m} < p$, so $p \nmid k$. Take an eigenvector v of M_K for μ, and numbers α_p ($p \in F$) with $\sum_{p\in F}\alpha_p(kp-1)^{-1/2} = 0$ for $k = 1,\dots,K$: these K conditions leave a space of dimension at least $|F|-K$. Put $u = \sum_{p\in F}\sum_{k\le K}\alpha_p v_k\, e_{kp}$. If b is not a multiple of any $p \in F$, then $\gcd(kp,b) = \gcd(k,b)$, so

$$(C_m u)_b = \sum_k v_k \frac{\gcd(k,b)-1}{\sqrt{b-1}} \sum_p \frac{\alpha_p}{\sqrt{kp-1}} = 0.$$

If $b = k'p'$ with $p' \in F$, the terms with $p = p'$ give

$$\alpha_{p'}\sum_k v_k \frac{p'\gcd(k,k')-1}{\sqrt{(kp'-1)(k'p'-1)}} = \alpha_{p'}\big(\mu v_{k'} + O_K(1/p')\big),$$

Table 16.3. The spectrum of C_m with the eigenvalue 1 removed: the number of eigenvalues, how many eigenvectors are localized (over 90% of the weight on indices with a prime factor above $\sqrt{m}$), the mean spacing ratio $\langle r\rangle$ in the central 80% for all eigenvalues and for the delocalized ones only (Poisson 0.386, GOE 0.531), and the number of spacings below 10^{-6}.

m	eigenvalues	localized	$\langle r\rangle$, all	$\langle r\rangle$, rest	spacings $< 10^{-6}$
2,000	1,864	679	0.539	0.523	46
4,000	3,752	1,502	0.529	0.503	224
6,000	5,646	2,367	0.525	0.500	529

and the terms with $p \neq p'$, where $\gcd(kp, k'p') = \gcd(k, k')$, give

$$\frac{1}{\sqrt{k'p'-1}} \sum_k v_k\big(\gcd(k,k')-1\big) \sum_{p\neq p'} \frac{\alpha_p}{\sqrt{kp-1}} = -\frac{1}{\sqrt{k'p'-1}} \sum_k v_k\big(\gcd(k,k')-1\big) \frac{\alpha_{p'}}{\sqrt{kp'-1}}$$
$$= O_K\left(\frac{\alpha_{p'}}{p'}\right),$$

by the choice of α. So $\|(C_m - \mu)u\| \leq (c_K/p_0)\|u\|$ on a space of dimension at least $\nu(|F| - K)$ (one copy for each of ν orthogonal eigenvectors v), and the min-max principle gives the eigenvalues. For $K = 1$ the error terms vanish and $\mu = 1$, which is Proposition 15.6(iv). □

So the spectrum of C_m contains, for every K, about $|F| \approx m/(K(K+1)\ln m)$ near-copies of each eigenvalue of the small matrix M_K: the matrix contains many nearly decoupled small copies of its own corner. For $m = 4{,}000$ and $K = 2$, there are 86 such primes, and we find 87 and 85 eigenvalues within 0.00073 of $1 \mp 1/\sqrt{2}$, the eigenvalues of M_2; for $K = 3$ there are 49 primes and 59, 51, 49 eigenvalues near $0.2578, 0.6051, 2.1372$.

The spacing ratio revisited. These families are clusters of nearly equal eigenvalues, the opposite of level repulsion, so they distort any spacing statistic. Table 16.3 separates them out. We call an eigenvector *localized* if more than 90% of its weight sits on indices a with a prime factor above $\sqrt{m}$ (the indices of the families); about 40% of all eigenvectors are. Among the rest, the mean spacing ratio $\langle r\rangle$ is 0.503 at $m = 4{,}000$ and slowly falling, between the Poisson value 0.386 and the GOE value 0.531. The near-GOE value for the whole spectrum (0.529) comes partly from mixing the families with the rest: inside a tight cluster, consecutive ratios of tiny spacings take all values. So the GOE-like look reported in Section 15.6 was, at least in part, an artifact. What remains is a real but weaker repulsion among the delocalized eigenvalues, which is what one expects of a real symmetric matrix with no remaining symmetry; we have no finer explanation, and we cannot tell from these sizes whether it tends to the GOE value, to the Poisson value or to something in between. This is numerical evidence only.

Status. Still partly answered, with more proved. The top eigenvalue grows faster than every power of $\ln m$ and slower than every power of m (proved); its exact order is open. The spectrum contains near-copies of every small gcd matrix (proved), and the GOE-like spacing was partly caused by them (numerical evidence). Whether $6/\pi^2$ appears in the limiting bulk distribution is still open.

16.4 Question 6: why 171 wins

Section 15.7 found that the 171-note equal temperament wins seven of the eight measures. Tuning theorists know this temperament well: the Xenharmonic Wiki notes that 171 has a lower relative error in the 7-limit than any smaller equal temperament, and that it tempers out three tiny intervals (commas) [137, 138]. Those three commas explain the win completely.

A comma is a ratio of small primes very close to 1. Write it by its exponents of $2, 3, 5, 7$: the schisma $32805/32768 = 3^8 \cdot 5/2^{15}$ is $(-15, 8, 1, 0)$ and measures 1.954 cents; the breedsma $2401/2400 = 7^4/(2^5 \cdot 3 \cdot 5^2)$ is $(-5, -1, -2, 4)$ and measures 0.721 cents; the ragisma $4375/4374 = 5^4 \cdot 7/(2 \cdot 3^7)$ is $(-1, -7, 4, 1)$ and measures 0.396 cents. An n-note temperament maps each prime p to a whole number of steps v_p, and it *tempers out* a comma when the steps add up to zero. Put the three commas and the octave $(1, 0, 0, 0)$ in the rows of a 4×4 matrix M. Then $\det M = 171$, and the adjugate of M is

$$\operatorname{adj} M = \begin{pmatrix} 0 & 0 & 0 & 171 \\ 18 & 1 & -4 & 271 \\ 27 & -8 & 32 & 397 \\ 18 & 39 & 15 & 480 \end{pmatrix}.$$

Its last column, $(171, 271, 397, 480)$, is the 171-note temperament: $171 \log_2 p$ rounded for $p = 2, 3, 5, 7$. It is, up to multiples, the only way to map the four primes to steps that tempers out all three commas.

Why is it so accurate? Let $J = (\log_2 2, \log_2 3, \log_2 5, \log_2 7)$ and let $\varepsilon_1, \varepsilon_2, \varepsilon_3$ be the sizes of the three commas in octaves. Then $MJ = (\varepsilon_1, \varepsilon_2, \varepsilon_3, 1)$, so $171\, J = \operatorname{adj} M\, (\varepsilon_1, \varepsilon_2, \varepsilon_3, 1)$, which gives the identity

$$171 \log_2 p - v_p = (\text{row } p \text{ of } \operatorname{adj} M) \cdot (\varepsilon_1, \varepsilon_2, \varepsilon_3, 0).$$

The errors, in steps, are small whole-number combinations of three tiny numbers: for 3, $18\varepsilon_1 + \varepsilon_2 - 4\varepsilon_3 = 0.0286$ steps; for 5, 0.0497; for 7, 0.0577. All three have the same sign and are nearly proportional to $\log_2 p$ (divided by $\log_2 p$ they are 0.0180, 0.0214 and 0.0206), so a slightly stretched octave, which the TOP measure allows, absorbs most of what is left.

This is the same mechanism as a convergent of a continued fraction, one dimension up. A convergent p_n/q_n is good because it is pinned by a small quantity, $|q_n\alpha - p_n|$; here the temperament is pinned by three small commas. The family even grows by adding: the mappings of 27, 72 and 99 notes satisfy $27 + 72 = 99$ and $72 + 99 = 171$ in every coordinate, like mediants in the Stern–Brocot tree. All of this is known in regular temperament theory; what we add is the explicit adjugate identity for the errors, which is Cramer's rule.

Status. Answered (no, prime sizes do not stand out), and the winner is explained. The explanation is known to tuning theorists.

16.5 Question 4: two more tests of pi

Chapter 15 tested the values of π's terms, their pairs, their tail, Khinchin's and Lévy's constants (the growth of $q_n^{1/n}$) and the records. Two typical-number laws were not yet tested.

- *The Doeblin–Lenstra law.* For almost every x, the share of n with $\theta_n = q_n^2 |x - p_n/q_n| \le t$ tends to $t/\ln 2$ for $t \le 1/2$ and to $(1 - t + \ln 2t)/\ln 2$ for $1/2 \le t \le 1$ [20]. This is the law behind the ratio $1/(2\ln 2)$ in Section 15.4. For the first 11,999,959 convergents of π,

in 20 equal bins of θ: $\chi^2 = 17.2$ on 19 degrees of freedom, textbook $p = 0.57$, calibrated $p = 0.58$ (from 300 random reals with 100,000 terms each; for them the textbook test rejects at the 5% level a share 0.060 of the time). The share with $\theta_n < 1/2$ is 0.72144, against $1/(2\ln 2) = 0.72135$, a deviation of 1.02 standard deviations.

- *Serial correlation.* Under the Gauss measure, consecutive terms are weakly dependent: the correlation of $\ln a_n$ and $\ln a_{n+1}$ is about -0.08765 (estimated from the random reals, standard error 0.00018), and at lag 2 about 0.03056. For π the two correlations are -0.08756 and 0.03014, deviations of 0.30 and -1.47 standard deviations.

Status. Open; both new tests are consistent with π being typical. As before, this is evidence, not proof.

16.6 Status after the second look

Table 16.4. The six questions of Section 14.3 after Chapter 16. Proved results are labeled as such; everything else is numerical evidence or known.

question	new in this chapter	status
1. Prime convergents of π	period of e's denominators modulo ℓ divides 6ℓ (proved)	open
2. Fibonacci and Pell primes	nothing new	open
3. The heuristic in a model	$2\ln 2 \cdot \ln N$ prime denominators and numerators for almost every x, with an error term (proved)	answered for almost every x
4. Gauss–Kuzmin for π	Doeblin–Lenstra law and serial correlation tested	open; data typical
5. Spectrum of the correlation matrix	top eigenvalue between $e^{c\sqrt{\ln m}/\ln\ln m}$ and $m^{o(1)}$ (proved); near-copies of small gcd matrices (proved); GOE-like spacing partly an artifact (evidence)	partly answered
6. Prime-sized temperaments	171 explained by three commas (known to tuning theorists)	answered: no

CHAPTER 17

A third look at the open questions

Added 2026-11-10.

Chapter 16 ended with three loose ends that looked reachable: the true growth of the top eigenvalue of the correlation matrix, whether the period 6ℓ for e can be shorter, and a search for prime convergents of π that stopped at $n = 20{,}000$. This chapter closes the first two with proofs and extends the third. It also takes up a practical question: is there a better way to find prime convergents than testing them one at a time, and a better way to confirm them? It follows Teddy Rook's notes and scripts (November 2026): the statements and proofs are his, edited into the style of this paper, and every one was checked again for this chapter, line by line and with independent scripts (`c12_third_*.py`, Appendix C). Questions 2 and 6 are untouched; the new primes bear on Questions 3 and 4 only as evidence (Section 17.7).

- *Question 5, proved.* The largest eigenvalue of the correlation matrix satisfies $\ln \lambda_{\max}(C_m) = (2 + o(1))\sqrt{\ln m / \ln\ln m}$. Theorem 16.6 had left the order open between $\sqrt{\ln m}/\ln\ln m$ and $\ln m/\ln\ln m$.

- *Question 5, proved.* The smallest eigenvalue obeys the mirror law $\ln(1/\lambda_{\min}(C_m)) = (2 + o(1))\sqrt{\ln m / \ln\ln m}$.

- *Question 5, proved.* The eigenvalue distribution converges to a limiting law as $m \to \infty$, and the mean of $1/\lambda$ tends to $\prod_p (1 + 2/(p(p-1))) = 3.27958\ldots$ None of the four exact bulk constants now known is $6/\pi^2$.

- *Question 1, proved (for e).* Modulo an odd prime ℓ, e's denominators change sign every 3ℓ steps and its numerators repeat, so the periods are exactly 6ℓ and 3ℓ. Every integer m divides q_{3m-1}, q_{3m-3} and q_{3m-7}, so a denominator of e can be prime only at $n = 2$ or $n \equiv 1 \pmod 3$.

- *Question 1, evidence (for π).* The denominators of π's convergents number 25,332, 27,651, 32,394 and 37,721 are probable primes of 13,017, 14,190, 16,695 and 19,465 digits, found by Teddy Rook: the eleventh to fourteenth prime denominators, the first beyond the limit recorded in the OEIS. So is the numerator number 42,037, with 21,667 digits, the first new prime numerator since $n = 8{,}750$. His search covers numerators and denominators to $n = 50{,}000$ (ours: denominators to 38,571 and numerators to 42,037). It uses a "rhythm sieve" that runs the recurrence modulo small primes. The four new prime denominators

are bunched, and the prime convergents lean toward one note; both look like chance. Primes turn out to be far more plentiful among the intermediate fractions that belong to π's largest terms, and the opening guess of Chapter 1 belongs to such a family.

- *Questions 3 and 4, evidence (for π).* With the new primes, π's own count of prime convergents favors the constant $2\ln 2$ of Heuristic 8.3 over the value without its correction factor $\zeta(2)$ by about 30 to 1; the sieve survivors match the corrected prediction, and every small prime divides π's denominators at its proper rate over two million convergents.

Nothing here proves anything about π. The labels mean what they meant in Chapter 15: a Theorem, Proposition or Lemma without a citation in its heading is proved here from cited results; *Known* marks what others found first; numbers from computation are numerical evidence.

17.1 Question 5: the largest eigenvalue, settled to leading order

Added 2026-11-10.

Theorem 16.6 trapped the top eigenvalue of $C_m = [(\gcd(a,b)-1)/\sqrt{(a-1)(b-1)}]_{a,b=2}^{m}$ between $\exp(2\sqrt{\ln m}/\ln\ln m)$ and $\exp(\ln 2\cdot \ln m/\ln\ln m)$. The true rate is strictly between the two, and both the order and the constant can be pinned down. Write $G_m = [\gcd(a,b)/\sqrt{ab}]_{a,b=1}^{m}$, as in that theorem.

Theorem 17.1. *As $m \to \infty$,*

$$\ln \lambda_{\max}(C_m) = (2+o(1))\sqrt{\frac{\ln m}{\ln\ln m}},$$

and the same holds for G_m.

So the exponent is larger than the old lower bound by a factor $\sqrt{\ln\ln m}$, and far below the old upper bound. The proof has two halves that meet: a Schur test with multiplicative weights from above, and a multiplicative test vector (a "resonator") from below. We use standard consequences of the prime number theorem [56, Ch. XXII]: as $y\to\infty$ and $\ln z/\ln y \to \infty$,

$$\sum_{y<p\le z}\frac{1}{p\ln p}\sim\frac{1}{\ln y},\qquad \sum_{y<p\le z}\frac{1}{p\ln^2 p}\sim\frac{1}{2\ln^2 y},\qquad \sum_{y<p\le z}\frac{1}{p}=\ln\frac{\ln z}{\ln y}+o(1), \tag{17.1}$$

and the first two hold with $z=\infty$ as well.

The upper bound.

Lemma 17.2 (valid for every m)**.** *Let $\sigma>0$ and $y\ge 1$ be such that $p(p^\sigma-1)>1$ for every prime $p>y$. Then*

$$\lambda_{\max}(G_m) \le m^\sigma \prod_{p\le y}\bigl(1-p^{-1/2}\bigr)^{-2}\prod_{p>y}\Bigl(1-\frac{1}{p\,(p^\sigma-1)}\Bigr)^{-1}.$$

Proof. G_m is symmetric with non-negative entries, so for any positive weights w, $\lambda_{\max}(G_m)\le \max_{a\le m}(G_m w)_a/w_a$ (Schur's test:

$$|x^{\mathsf T}Ax| \le \sum_{a,b} A_{ab}\tfrac12\bigl(x_a^2 w_b/w_a + x_b^2 w_a/w_b\bigr) = \sum_a x_a^2 (Aw)_a/w_a).$$

Take w completely multiplicative with $w(p)=t_p$, where $t_p=1$ for $p\le y$ and $t_p=p^{-1/2}/(1-p^{-\sigma})$ for $p>y$. Since $(m/b)^\sigma\ge 1$ for $b\le m$, extending the sum to all b gives, for $a=\prod p^{e_p}$,

$$\frac{(G_m w)_a}{w(a)}\le m^\sigma\prod_p E_p(e_p),\qquad E_p(e)=\frac{1}{(\sqrt{p}\,t_p)^e}\sum_{k\ge 0}p^{\min(k,e)}\bigl(t_p p^{-1/2-\sigma}\bigr)^k.$$

Write $s=p^{-\sigma}$ and $x_p=t_p p^{-1/2-\sigma}$. Splitting the sum at $k=e$, $E_p(e)=\sum_{k\le e}(\sqrt{p}\,t_p)^{k-e}s^k+s^e/(1-x_p)$. For $p>y$ the choice of t_p makes $1/(\sqrt{p}\,t_p)=1-s$, so $E_p(e)\le\frac{1}{1-x_p}\sum_{k=0}^{e}\binom{e}{k}(1-s)^{e-k}s^k=\frac{1}{1-x_p}$ with $x_p=1/(p(p^\sigma-1))$: every such prime dividing a contributes no more than if it did not. For $p\le y$, $x_p\le p^{-1/2}$ and the first sum is at most $\sum_{j\ge1}p^{-j/2}$, so $E_p(e)\le(1+p^{-1/2})/(1-p^{-1/2})\le(1-p^{-1/2})^{-2}$. Multiplying over p gives the lemma. □

Upper bound in Theorem 17.1. Put $L=\sqrt{\ln m\cdot\ln\ln m}$, $\sigma=1/L$ and $y=L^2/(\ln L)^4$. For $p>y$, $p^\sigma-1\ge\sigma\ln p$, so $1/(p(p^\sigma-1))\le L/(p\ln p)\le L/(y\ln y)\to 0$, and by (17.1) the second product is $\exp((1+o(1))L/\ln y)=\exp((1+o(1))L/(2\ln L))$. The first product is $\exp(O(\sqrt{y}/\ln y))=\exp(O(L/\ln^3 L))$, and $m^\sigma=\exp(\ln m/L)$. Since $\ln L=(\tfrac12+o(1))\ln\ln m$, both $\ln m/L$ and $L/(2\ln L)$ equal $(1+o(1))\sqrt{\ln m/\ln\ln m}$. Finally every entry of C_m is at most twice the matching entry of G_m (proof of Theorem 16.6), so $\lambda_{\max}(C_m)\le 2\lambda_{\max}(G_m)$. □

The lower bound.

Lemma 17.3 (valid for every m)**.** *Let f be multiplicative, supported on square-free integers, with $f(p)=f_p\ge 0$ and only finitely many $f_p\ne 0$. For every $\sigma>0$, $\lambda_{\max}(C_m)$ is at least*

$$\prod_p\Bigl(1+\frac{2f_p}{\sqrt{p}\,(1+f_p^2)}\Bigr)\Bigl[1-\frac{2}{m^\sigma}\prod_p\Bigl(1+(p^\sigma-1)\Bigl(f_p^2+\frac{f_p}{\sqrt{p}}\Bigr)\Bigr)-\prod_p\frac{(1+f_p/\sqrt{p})^2}{1+f_p^2+2f_p/\sqrt{p}}\Bigr].$$

Proof. Use the test vector $x_a=f(a)$, $2\le a\le m$. Because $\gcd(a,b)-1\ge 0$ and $(a-1)(b-1)\le ab$,

$$x^{\mathsf T}C_m x\ge\sum_{a,b\le m}f(a)f(b)\frac{\gcd(a,b)-1}{\sqrt{ab}}=\sum_{a,b\le m}f(a)f(b)\frac{\gcd(a,b)}{\sqrt{ab}}-\Bigl(\sum_{a\le m}\frac{f(a)}{\sqrt{a}}\Bigr)^2$$

(the terms with $a=1$ or $b=1$ vanish). Over all $a,b\ge1$ the first sum is the Euler product $S_\infty=\prod_p(1+f_p^2+2f_p/\sqrt{p})$, because $\gcd(a,b)/\sqrt{ab}$ factors over primes for square-free a,b. The pairs with $a>m$ contribute at most $m^{-\sigma}\sum a^\sigma f(a)f(b)\gcd(a,b)/\sqrt{ab}=m^{-\sigma}\prod_p(1+p^\sigma f_p^2+(1+p^\sigma)f_p/\sqrt{p})$, which is at most $m^{-\sigma}S_\infty\prod_p(1+(p^\sigma-1)(f_p^2+f_p/\sqrt{p}))$; the same holds for $b>m$. The subtracted square is at most $\prod_p(1+f_p/\sqrt{p})^2$, and $x^{\mathsf T}x<\prod_p(1+f_p^2)$. Divide. □

Lower bound in Theorem 17.1. Fix $0<c<1$, put $L=\sqrt{c\ln m\cdot\ln\ln m}$, and take $f_p=L/(\sqrt{p}\ln p)$ for $L^2<p\le\exp(\ln^2 L)$ and $f_p=0$ otherwise; let $\sigma=(\ln L)^{-3}$. This is Soundararajan's resonator [122]. On this range $f_p\le 1/(2\ln L)$, and:

- the main product has logarithm $(1+o(1))\sum 2L/(p\ln p)=(1+o(1))L/\ln L=(2\sqrt{c}+o(1))\sqrt{\ln m/\ln\ln m}$, by (17.1);
- $\sigma\ln p\le 1/\ln L\to 0$ on the range, so $p^\sigma-1\le(1+o(1))\sigma\ln p$, and the second product is at most $\exp((1+o(1))\sigma L^2/(2\ln L))=\exp((c+o(1))\sigma\ln m)$ (the part with $f_p/\sqrt{p}$ adds only $O(\sigma L\ln\ln L)$); so $m^{-\sigma}$ times it tends to 0, because $\sigma\ln m\to\infty$;

Table 17.1. The largest eigenvalue of C_m (Lanczos method): the best Rayleigh quotient of a resonator test vector (a lower bound); $\lambda_{\max}$; $\exp(2\sqrt{L/\ln L})$ with $L = \ln m$; twice Schur's ratio for G_m evaluated by computer, and twice the bound of Lemma 17.2 (both upper bounds); the ratios $\ln\lambda_{\max}/\sqrt{L/\ln L}$ and $\ln(1/\lambda_{\min})/\sqrt{L/\ln L}$, which both tend to 2; and $\operatorname{tr} C_m^{-1}/(m-1)$, which tends to $\kappa = 3.27958\ldots$

m	resonator	$\lambda_{\max}$	$e^{2\sqrt{L/\ln L}}$	Schur	Lemma 17.2	$\frac{\ln\lambda_{\max}}{\sqrt{L/\ln L}}$	$\frac{\ln(1/\lambda_{\min})}{\sqrt{L/\ln L}}$	$\frac{\operatorname{tr} C^{-1}}{m-1}$
1,000	7.97	18.77	43.87	85	1,870	1.55	2.08	3.21078
10,000	13.61	36.43	58.76	196	4,653	1.77	2.25	3.26848
100,000	21.29	63.08	76.81	239	10,575	1.91	2.36	3.27793
1,000,000	31.20	101.25	98.26	415	17,795	2.01	2.43	3.27935

- the third product is at most $\exp(-\sum f_p^2(1-1/p)/(1+f_p^2+2f_p/\sqrt{p})) \le \exp(-\sum f_p^2/8) \to 0$, because $\sum f_p^2$ is of order $L^2/\ln^2 L$.

So the bracket tends to 1, and letting $c \to 1$ gives the lower bound. The same vector gives it for G_m. □

Known, and how this relates to it. The method is the resonance method. Soundararajan's Theorem 2.1 evaluates the analogous maximum for the one-sided form $\sum r(m)r(mk)/\sqrt{k}$ over $mk \le N$ as $\exp((1+o(1))\sqrt{\ln N/\ln\ln N})$ [122]. G_m is, up to a logarithmic weight, that divisor matrix times its transpose, which is why the exponent here is exactly twice his. For arbitrary sets of N integers instead of an interval, the largest possible GCD sum is bigger, of order $\exp(A\sqrt{\ln N\cdot\ln\ln\ln N/\ln\ln N})$ [2, 17, 18]. Hilberdink stated that $\ln\lambda_{\max}(G_m)$ lies between two constant multiples of $\sqrt{\ln m/\ln\ln m}$, without giving the constants [59]. We did not find Theorem 17.1 stated for this matrix, but a specialist would regard it as a corollary of those techniques.

Evidence, with a warning. At $m = 10^6$ the formula $\exp(2\sqrt{\ln m/\ln\ln m})$ gives 98.3 against the true 101.25, but that agreement is an accident. At sizes one can compute, the primes below about 30 contribute most of the eigenvalue, and they sit inside the $o(1)$; the ratio $\ln\lambda_{\max}/\sqrt{\ln m/\ln\ln m}$ climbs from 1.55 at $m = 1{,}000$ to 2.01 at $m = 10^6$ (Table 17.1). The explicit bound of Lemma 17.2 at $m = 10^4$, optimized over σ and y, is 2,326 for G_m, against a true value of 40.5; evaluating Schur's ratio for the same kind of weights by computer does much better ($\lambda_{\max}(C_m) \le 415$ at $m = 10^6$, a bound checked by computation, not by hand). Lemma 17.3, evaluated for a small family of resonators at the sizes of the table, never exceeds 4.17, while $\lambda_{\max} \ge 18$ there: the asymptotic regime starts far beyond what we can compute. The empirical fit $(2/\pi^2)\ln^2 m\cdot\ln\ln m$ of Table 16.2 remains the better description of the computable range; Theorem 17.1 says what replaces it eventually.

17.2 Question 5: the smallest eigenvalue mirrors the largest

Added 2026-11-10.

Chapter 15 noted that the smallest eigenvalue "shrinks slowly". It shrinks at exactly the rate at which the largest grows.

Proposition 17.4 (the inverse)**.** *For $2 \le a, b \le m$,*

$$(C_m^{-1})_{ab} = \sqrt{(a-1)(b-1)} \sum_{\substack{d \le m \\ a \mid d,\ b \mid d}} \frac{\mu(d/a)\,\mu(d/b)}{\varphi(d)}.$$

Proof. By Proposition 15.6, $C_m = DV\Phi V^{\mathsf T} D$, and the inverse of the divisor matrix V is the Möbius matrix with entries $\mu(a/d)$ for $d \mid a$ (restricting to $2, \dots, m$ does not change this, because the index 1 sits in a separate block of the triangular matrix). □

Theorem 17.5 (the mirror law)**.** *As $m \to \infty$,*

$$\ln \frac{1}{\lambda_{\min}(C_m)} = (2 + o(1)) \sqrt{\frac{\ln m}{\ln \ln m}},$$

so the condition number of C_m is $\exp((4 + o(1))\sqrt{\ln m / \ln \ln m})$.

Proof of the upper bound for $\lambda_{\max}(C_m^{-1}) = 1/\lambda_{\min}(C_m)$. The absolute values of the entries of C_m^{-1} are at most $N_{ab} = \sqrt{ab} \sum_d \mu^2(d/a)\mu^2(d/b)/\varphi(d)$, the sum over $d \le m$ divisible by a and b, so $\lambda_{\max}(C_m^{-1}) \le \lambda_{\max}(N)$ (for any x, $|x^{\mathsf T} C_m^{-1} x| \le |x|^{\mathsf T} N |x|$), and Schur's test applies to N. Take w completely multiplicative, $w(p) = t_p$, and use $(m/d)^\sigma \ge 1$. If p^e exactly divides a, only the exponents e and $e+1$ of p in d occur, and for each, only the exponents $k-1$ and k in b, because d/a and d/b must be square-free. With $s = p^{-\sigma}$ and $c = p^{-1/2}/t_p$ the local factor is

$$E_p(0) = 1 + \frac{s}{p-1} + \frac{s}{(p-1)c}, \qquad E_p(e) = s^e \frac{p(1+c)}{p-1} + s^{e+1}\Big(\frac{1}{p-1} + \frac{1}{(p-1)c}\Big) \quad (e \ge 1).$$

For $p > y$ choose $c = p^\sigma(1 - 1/p) - 1$, which is positive once $(p-1)\sigma \ln p > 1$. Then $s\,p(1+c)/(p-1) = 1$, so $E_p(e) \le s^{e-1} + s^{e+1}(E_p(0) - 1)/s \le E_p(0)$ for all $e \ge 1$, and $c \ge (1 - o(1))\sigma \ln p$. For $p \le y$ take $t_p = 1$; then $E_p(e)$ and $E_p(0)$ are at most $(1 + p^{-1/2})/(1 - p^{-1/2}) = 1 + O(p^{-1/2})$. With $\sigma = 1/L$, $y = L^2/(\ln L)^4$ and $L = \sqrt{\ln m \cdot \ln \ln m}$ as before, the sum of $s/((p-1)c)$ over $p > y$ is at most $(1 + o(1))L/\ln y$, exactly as for the largest eigenvalue, and the only new term is $\sum_{p>y} p^{-\sigma}/(p-1) = O(\ln L)$, which is negligible. The total is again $\exp((2 + o(1))\sqrt{\ln m / \ln \ln m})$. □

Proof of the lower bound. By Proposition 17.4,

$$x^{\mathsf T} C_m^{-1} x = \sum_{d \le m} \frac{1}{\varphi(d)} \Big(\sum_{a \mid d} \mu(d/a) \sqrt{a-1}\, x_a \Big)^2.$$

Take $x_a = \mu(a) f(a) \sqrt{a/(a-1)}$ with the same f as in the proof of Theorem 17.1. For square-free d built from the primes carrying f, every divisor $a \ge 2$ of d is in range and $\mu(d/a)\mu(a) = \mu(d)$, so the inner sum is $\mu(d)(F(d) - 1)$ with $F(d) = \prod_{p \mid d}(1 + f_p\sqrt{p})$. Keeping only those d, and using $(F-1)^2 \ge F^2 - 2F$,

$$x^{\mathsf T} C_m^{-1} x \ge \sum_d \frac{F(d)^2 - 2F(d)}{\varphi(d)}, \qquad \text{where} \quad \frac{(1 + f_p\sqrt{p})^2}{p-1} \ge f_p^2 + \frac{2 f_p}{\sqrt{p}}.$$

The cut $d \le m$ and the subtracted sum are handled exactly as in Lemma 17.3 (the subtracted sum is small compared with the main one because $f_p\sqrt{p} = L/\ln p \to \infty$ on the range), and $x^{\mathsf T} x \le 2 \prod (1 + f_p^2)$. The quotient is at least $(\tfrac12 - o(1)) \prod_p (1 + 2f_p/(\sqrt{p}(1 + f_p^2)))$, the same product as before. □

Table 17.2. The largest and smallest eigenvalues of C_m (Lanczos method, for C_m and for C_m^{-1} in the form of Proposition 17.4), the ratio $(1/\lambda_{\min})/\lambda_{\max}$, and the mean of $1/\lambda$, $\operatorname{tr} C_m^{-1}/(m-1)$. The $\lambda_{\max}$ column reproduces Table 15.6.

m	$\lambda_{\max}$	$\lambda_{\min}$	$1/\lambda_{\min}$	ratio	mean of $1/\lambda$
500	14.887	0.024604	40.64	2.730	3.16404
1,000	18.773	0.019617	50.98	2.715	3.21078
10,000	36.429	0.010219	97.86	2.686	3.26848
100,000	63.078	0.005984	167.10	2.649	3.27793
1,000,000	101.252	0.003770	265.28	2.620	3.27935
3,000,000	124.402	0.003080	324.62	2.609	3.27949

Our checks. The script evaluates Schur's ratio for N with these weights: at $m = 10^4$ it is 180, between $1/\lambda_{\min} = 97.86$ and the product bound 537, and at $m = 10^5$ it is 324, between 167.10 and 973. It also checks the formula of Proposition 17.4 against a dense inverse at $m = 1{,}000$.

Evidence. The two extremes track each other closely at computable sizes (Table 17.2): $1/\lambda_{\min}$ is about 2.61 to 2.73 times $\lambda_{\max}$ from $m = 500$ to 3×10^6, the ratio drifting down slowly. We have no proof that the ratio stays bounded; Theorems 17.1 and 17.5 only say that the two logarithms are asymptotically equal.

17.3 Question 5: the bulk has a limiting law

Added 2026-11-10.

The curves in Fig. 15.5 "almost coincide" because they converge.

Theorem 17.6 (a limiting law)**.** *Let ν_m be the distribution that puts mass $1/(m-1)$ on each eigenvalue of C_m. There is a probability distribution ν on $[0,\infty)$ such that $\nu_m \to \nu$ as $m \to \infty$, both weakly and with convergence of second moments. It has mean 1 and variance 2.*

Proof. The key is that $\gcd(a,b)/\sqrt{ab}$ depends only on the ratio b/a: if $b/a = u/v$ in lowest terms, the entry is $\psi(u/v) = 1/\sqrt{uv}$. Three steps.

1. *Close matrices have close spectra.* For symmetric $n \times n$ matrices A and B with eigenvalues in increasing order, $\sum_i (\lambda_i(A) - \lambda_i(B))^2 \le \|A - B\|_F^2$ [60]. So the Wasserstein-2 distance between the two eigenvalue distributions is at most $\|A - B\|_F/\sqrt{n}$.

2. *Truncate the ratios.* C_m differs from G_m (rows and columns 2 to m) by $O(\ln^2 m)$ in squared Frobenius norm: the difference of the entries is $\gcd(a,b)\big(((a-1)(b-1))^{-1/2} - (ab)^{-1/2}\big) - ((a-1)(b-1))^{-1/2}$, and the squares sum to $O(\ln^2 m)$ by the substitution $a = ga'$, $b = gb'$ of the proof of Theorem 15.7. For a finite set S of ratios, closed under inversion, let G_m^S keep the entries of G_m with $b/a \in S$ and set the rest to 0. The ratio u/v occurs for the pairs $a = vk$, $b = uk$ with $k \le m/\max(u,v)$, so

$$\frac{1}{m}\|G_m - G_m^S\|_F^2 \le \sum_{u/v \notin S} \frac{1}{uv\max(u,v)} =: \varepsilon(S),$$

the sum over coprime pairs, and $\varepsilon(S) \to 0$ as S grows, because the full sum over coprime pairs is 3 (proof of Theorem 15.7). The bound does not depend on m.

3. *Each truncation converges.* G_m^S has at most $|S|$ non-zero entries per row, each at most 1, so its eigenvalues lie in $[-|S|, |S|]$. Its kth moment is $1/m$ times a sum over closed walks $a \to aq_1 \to aq_1q_2 \to \cdots$ with $q_i \in S$ and $q_1 \cdots q_k = 1$. A given walk is allowed exactly when a is divisible by the least common denominator V of the partial products and a times the largest partial product Q is at most m, which holds for $m/(VQ) + O(1)$ values of a. So every moment converges, and since the spectra are uniformly bounded, the eigenvalue distribution of G_m^S converges to a limit ν^S.

By steps 1 and 2 the ν^S form a Cauchy family in Wasserstein-2 distance, with a limit ν, and ν_m is within $\sqrt{\varepsilon(S)} + o(1)$ of ν^S for every S. The mean and variance come from Theorem 15.7, since Wasserstein-2 convergence carries the first two moments. □

The proof also says what ν is: the average spectral measure, at the starting point, of the operator "convolve with ψ on the positive rationals", cut down to the rationals q for which aq is an integer at most m, for a random integer $a \le m$. The cut at m is what stops the limit from being an Euler product: without it, each prime would contribute an independent factor.

Proposition 17.7 (the mean of $1/\lambda$)**.**

$$\frac{1}{m-1}\operatorname{tr} C_m^{-1} = \kappa + O\Big(\frac{\ln^2 m}{m}\Big), \qquad \kappa = \prod_p \Big(1 + \frac{2}{p(p-1)}\Big) = 3.27958\ldots$$

Proof. By Proposition 17.4 the diagonal entry of the inverse at a is $(a-1)\sum \mu^2(k)/\varphi(ak)$ over $k \le m/a$. Summing over a and writing $d = ak$,

$$\operatorname{tr} C_m^{-1} = \sum_{d=2}^{m} \frac{1}{\varphi(d)} \sum_{\substack{a \mid d \\ d/a \text{ square-free}}} (a-1) = \sum_{d=2}^{m} \Big(h(d) - \frac{2^{\omega(d)}}{\varphi(d)}\Big), \qquad h(d) = \prod_{p \mid d} \frac{p+1}{p-1},$$

because $\sum a = d\prod_{p\mid d}(1 + 1/p)$ over the a with d/a square-free, there are $2^{\omega(d)}$ such a, and the term $a = 1$ contributes 0 either way. The second sum is $O(\ln^2 m)$. Write $h = 1 * g$ (Dirichlet convolution), where g is supported on square-free numbers with $g(p) = 2/(p-1)$, so $g(k) = 2^{\omega(k)}/\varphi(k)$ for square-free k. Then

$$\sum_{d \le m} h(d) = \sum_{k \le m} g(k) \Big\lfloor \frac{m}{k} \Big\rfloor = m \sum_{k \ge 1} \frac{g(k)}{k} - m \sum_{k > m} \frac{g(k)}{k} + O\Big(\sum_{k \le m} g(k)\Big),$$

with $\sum_k g(k)/k = \prod_p (1 + g(p)/p) = \kappa$; since $\sum_{k \le x} g(k) = O(\ln^2 x)$, partial summation bounds the other two terms by $O(\ln^2 m)$. □

Teddy Rook's notes reach the same constant by a different route: for square-free k, the mean of $a/\varphi(ak)$ over a is $\varphi(k)^{-1}\prod_{p\nmid k}(1 + 1/(p(p-1)))$ (a prime dividing k contributes 1 whether or not it divides a), and summing over $a \le m/k$ and then k gives $m\kappa$. The version above, written out in checking, gives the error term. Numerically $\operatorname{tr} C_m^{-1}/(m-1)$ is 3.2685 at $m = 10^4$, 3.2779 at 10^5 and 3.27949 at 3×10^6 (Table 17.2).

Where this leaves $6/\pi^2$. Four exact statistics of the eigenvalues of C_m now have known limits, and each is a different product over primes:

statistic of the eigenvalues	limit	value
mean	1 (Theorem 15.7)	1
mean square	$3\zeta(3)/\zeta(3)$ (Theorem 15.7)	3
geometric mean	$\prod_p (1-1/p)^{1/p}$ (Proposition 15.6)	0.5599
mean of $1/\lambda$	$\prod_p (1+2/(p(p-1)))$ (Proposition 17.7)	3.27958

None is $6/\pi^2 = \prod_p (1-1/p^2)$ or a simple function of it. So the answer to "does $6/\pi^2$ shape the bulk?" is now: a limiting bulk law exists, and $6/\pi^2$ enters the matrix as the share of zero entries and as the constant in the first test vector (Theorem 15.8), but it does not appear in any exact bulk statistic found so far.

One caution, added in checking. The first two statistics are statistics of ν itself (Theorem 17.6). Weak convergence does not carry the last two over, because $\ln\lambda$ and $1/\lambda$ are unbounded near 0, where the smallest eigenvalues go. For ν itself Fatou's lemma gives only $\int \lambda^{-1}\, d\nu \le \kappa$; equality would need the eigenvalues near 0 to carry a vanishing share of the sum of the $1/\lambda$. So the last two rows are exact limits of statistics of C_m, and statistics of ν only if that holds. Still open: a formula for ν, whether ν has a density, and whether the weak level repulsion of Section 16.3 survives in the limit.

Status of Question 5. The extremes are answered to leading order (Theorems 17.1 and 17.5, proved), a limiting law exists (Theorem 17.6, proved), and the mean of $1/\lambda$ tends to κ (Proposition 17.7, proved). The shape of the bulk law is open.

17.4 Question 1: the number e, completed

Added 2026-11-10.

Section 16.2 showed that e's denominators repeat modulo a prime ℓ with a period dividing 6ℓ, and left open whether the period can be shorter. It cannot, and the reason is a sign flip that also explains why e's denominators are so often divisible by small primes. This section follows Teddy Rook's notes. Notation as in the paper: $e = [2; 1, 2, 1, 1, 4, 1, \dots]$, so $a_{3k-2} = a_{3k} = 1$ and $a_{3k-1} = 2k$, with convergents p_n/q_n, $q_0 = q_1 = 1$, $p_{-1} = 1$, $q_{-1} = 0$.

The closed forms (known). The tool is the classical closed form for e's convergents, which comes from Hermite's integrals [58] as used in Cohn's short proof of Euler's expansion of e [30]. For $n \ge 1$,

$$
\begin{aligned}
p_{3n-2} &= \sum_{j=0}^{n} \binom{n}{j} \frac{(n+j)!}{n!}, & q_{3n-2} &= \sum_{j=0}^{n} (-1)^{n+j} \binom{n}{j} \frac{(n+j)!}{n!}, \\
p_{3n-1} &= \sum_{j=0}^{n} \binom{n}{j} \frac{(n+1+j)!}{n!}, & q_{3n-1} &= \sum_{j=0}^{n+1} (-1)^{n+1+j} \binom{n+1}{j} \frac{(n+j)!}{n!}, \\
p_{3n} &= \sum_{j=0}^{n+1} \binom{n+1}{j} \frac{(n+j)!}{n!}, & q_{3n} &= \sum_{j=0}^{n} (-1)^{n+j} \binom{n}{j} \frac{(n+1+j)!}{n!}.
\end{aligned}
\tag{17.2}
$$

For completeness, here is the derivation. Let A_n, B_n, C_n be $\int_0^1 x^n(x-1)^n e^x\,dx/n!$, $\int_0^1 x^{n+1}(x-1)^n e^x\,dx/n!$ and $\int_0^1 x^n(x-1)^{n+1}e^x\,dx/n!$. Cohn's Proposition 1, translated to our indexing (his expansion starts $[1;0,1,1,2,\dots]$, so his index $3n$ is our $3n-2$), says $A_n = q_{3n-2}e - p_{3n-2}$, $B_n = p_{3n-1} - q_{3n-1}e$ and $C_n = p_{3n} - q_{3n}e$. Integration by parts gives $\int_0^1 x^k e^x\,dx = e\,u_k + (-1)^{k+1}k!$ and $\int_0^1 x^k e^{-x}\,dx = k! - e^{-1}w_k$ with integers u_k, w_k. Since e is irrational, a number $r + se$ with integers r, s determines r and s. Expanding $(x-1)^n$ by the binomial theorem gives the integer parts, hence the p's; substituting $x \mapsto 1-x$ first (for instance $A_n = (-1)^n e \int_0^1 x^n(1-x)^n e^{-x}\,dx/n!$) gives the coefficients of e, hence the q's. The numbers $p_1, p_4, p_7, \dots = 3, 19, 193, \dots$ are the values $y_n(2)$ of the Bessel polynomials, and $q_{3n-2} = (-1)^n y_n(-2)$; the first sequence is OEIS A001517 [102]. Our script checks all six formulas for $n < 200$.

Proposition 17.8 (the sign flip)**.** *Let ℓ be an odd prime. For every $n \geq 0$,*

$$q_{n+3\ell} \equiv -q_n \quad \textit{and} \quad p_{n+3\ell} \equiv p_n \pmod{\ell}.$$

The period of $q_n \bmod \ell$ is exactly 6ℓ, and the period of $p_n \bmod \ell$ is exactly 3ℓ. For $\ell = 2$ both periods are 6.

Proof. Put $n = \ell$ in (17.2). Modulo ℓ the binomial coefficient $\binom{\ell}{j}$ vanishes unless $j = 0$ or ℓ, and $(\ell+j)!/\ell!$ vanishes when $j = \ell$. So $p_{3\ell-2} \equiv 1 = p_{-2}$ and $q_{3\ell-2} \equiv (-1)^\ell = -1 = -q_{-2}$, where $p_{-2} = q_{-2} = 1$ come from running the recurrence backward (Cohn's form of the expansion supplies these terms). In the same way $p_{3\ell-1} \equiv 1 = p_{-1}$ and $q_{3\ell-1} \equiv 0 = q_{-1}$. The terms of the expansion repeat modulo ℓ with period 3ℓ (only the terms $2k$ change, by 2ℓ), and the recurrence $v_{n+1} = \left(\begin{smallmatrix} a_{n+1} & 1 \\ 1 & 0 \end{smallmatrix}\right) v_n$ for $v_n = (q_n, q_{n-1})$ is linear, so two consecutive matches propagate to all n.

For exactness: 3ℓ is not a period of q, since $q_{3\ell} \equiv -1 \not\equiv q_0$. If T is a period, then comparing $q_{j+T} = a_{j+T}q_{j-1+T} + q_{j-2+T}$ with $q_j = a_j q_{j-1} + q_{j-2}$ gives $a_{j+T}q_{j-1} \equiv a_j q_{j-1}$ for all $j \geq 2$; at $j = 2$ ($a_2 = 2$, $q_1 = 1$) this fails for $T = 2\ell$ and for $T = 6$, because $a_{2+2\ell}$ is 1 for $\ell \neq 3$ (its index is not $\equiv 2 \pmod 3$) and $a_8 = 6 \not\equiv 2$. Every proper divisor of 6ℓ divides 3ℓ, 2ℓ or 6, so the period is 6ℓ. The numerators are handled the same way with $p_1 = 3 \not\equiv 0$: $a_{2+T} \equiv 2$ fails for $T = 1$ ($a_3 = 1$), $T = 3$ ($a_5 = 4$) and $T = \ell$ ($a_{2+\ell} = 1$). For $\ell = 3$ use $j = 5$, where $p_4 = 19 \not\equiv 0$ and $a_8 = 6 \not\equiv a_5 = 4 \pmod 3$; the cases $\ell = 3$ and $\ell = 2$ are also confirmed by direct computation. □

This sharpens Proposition 16.5: the transfer matrix over 3ℓ steps has eigenvalues $+1$ and -1, the numerators live in the first eigenspace and the denominators in the second.

Theorem 17.9 (forced divisors)**.** *For every integer $m \geq 1$, m divides q_{3m-1} and q_{3m-3}; for $m \geq 3$ it divides q_{3m-7}; for $m \geq 2$ it divides p_{3m-4}. Consequently q_n is composite for every $n \geq 3$ with $n \equiv 0$ or $2 \pmod 3$: a denominator of e can be prime only if $n = 2$ or $n \equiv 1 \pmod 3$. Likewise p_n is composite for every $n \geq 5$ with $n \equiv 2 \pmod 3$.*

Proof. Each claim is a closed form read modulo the right number; note that $(m+i)!/m!$ is a product of i consecutive integers.

- q_{3m-1}, modulo m: by the formula for q_{3n-1} with $n = m$, and $(m+i)!/m! \equiv i!$, $q_{3m-1} \equiv \pm\sum_i (-1)^i \binom{m+1}{i} i! = \pm D_{m+1}$, the number of derangements of $m+1$ objects, and $D_{m+1} = m(D_m + D_{m-1})$.
- q_{3m-3}: the formula for q_{3n} with $n = m-1$; every term contains $(n+1+j)!/n!$, a multiple of $n+1 = m$.

- q_{3m-7}: the formula for q_{3n-1} with $n = m-2$, modulo $n+2 = m$; the terms with $j \geq 2$ contain the factor $n+2$, and the terms $j = 0$ and $j = 1$ add up to $(-1)^{n+1}\big(1-(n+1)^2\big) = (-1)^{m-1}m(2-m) \equiv 0$.

- p_{3m-4}: the formula for p_{3n-1} with $n = m-1$; every term is a multiple of $n+1 = m$.

For the consequences: $n = 3m-1$ with $m = (n+1)/3 \geq 2$ covers $n \equiv 2 \pmod 3$, $n \geq 5$; $n = 3m-3$ with $m = (n+3)/3 \geq 2$ covers $n \equiv 0$, $n \geq 3$; $n = 3m-4$ with $m \geq 3$ covers p_n for $n \equiv 2$, $n \geq 5$. In each case q_n (or p_n) is larger than the divisor $m > 1$. □

A second proof of the first claim, found in checking: $q_{3m-1} = q_{3m} - q_{3m-2}$ because $a_{3m} = 1$, and by (17.2) this is $\equiv \sum_j (-1)^{m+j}\binom{m}{j}\big((j+1)! - j!\big) = \sum_j (-1)^{m+j}\, j\binom{m}{j}\, j! \pmod m$, where $j\binom{m}{j} = m\binom{m-1}{j-1} \equiv 0$. The three forced zeros are the zeros of the backward continuation, $q_{-1} = q_{-3} = q_{-7} = 0$, carried forward by Proposition 17.8. The script confirms all four divisibilities for every $m < 1{,}000$, and Proposition 17.8, including the exact periods, for every odd prime below 3,000.

What this explains. Section 15.2 found that e's denominators are divisible by a small prime ℓ about twice as often as random integers (the average of ℓf_ℓ was 2.00) and that the divisibility comes in clumps. Both follow. In each window of 3ℓ indices there are three forced multiples of ℓ, and on average over the 429 odd primes below 3,000 there are 3.014 more, about the number chance would give ($3\ell - 3$ further residues, each 0 with chance $1/\ell$); so $f_\ell = (3 + \text{extra})/(3\ell) \geq 1/\ell$, with equality for $\ell = 5, 17, 19, 29$ and for 89 of the primes below 3,000 in all. And q_{3m-1} is divisible by every prime factor of m and of $m+2$ at once. That the further zeros average 3 is numerical evidence, not a theorem.

Evidence. All ten known prime denominators of e sit where Theorem 17.9 allows: $n = 2, 4, 7, 13, 19, 34, 40, 64, 238, 268$, and every one after $n = 2$ is $\equiv 1 \pmod 3$ (our script finds exactly these for $n \leq 3{,}000$). OEIS A094007, whose index is our $n+1$, records that there are no more up to about $n = 50{,}000$ [107], well past the $n = 3{,}000$ of Chapter 8. Teddy Rook's notes add a heuristic count: restricted to $n \equiv 1 \pmod 3$, the denominators are never divisible by 2, 3 or 5 and are 3.4 times as likely as random integers to have no prime factor below 1,000; with that factor, the count predicted for $4 \leq n \leq 50{,}000$ is $3.4 \times \sum 1/\ln q_n = 3.4 \times 2.57 \approx 8.8$, against 9 found. The sum grows like $\ln\ln N$, so the forecast of Section 15.2 stands: infinitely many, but very few. This is a heuristic, not a theorem, and its numbers are from his notes (we did not recompute the factor 3.4).

Known? We did not find Proposition 17.8 or Theorem 17.9 stated in the OEIS entries or in the papers they cite, but the closed forms are classical and the facts may well be known.

Status. The arithmetic of e's convergents modulo primes is now completely proved. What remains open for e is Question 1 itself: whether infinitely many q_n with $n \equiv 1 \pmod 3$ are prime, which is as hard as the Fibonacci prime problem.

17.5 Question 1: five new prime convergents for pi

Added 2026-11-10.

Table 17.3. The eleventh to fourteenth prime denominators of π's convergents, and the sixteenth prime numerator. All five are probable primes. The OEIS index is $n+1$; the note is the remainder modulo 12, as in Section 6.1.

	n	digits	begins	ends	note
11th	25,332	13,017	27592568545368344821647067544 4	60691105041036014942809094 9703	B
12th	27,651	14,190	30535723507832454974931670505 1	36550844484511009313143309672 9729	F
13th	32,394	16,695	19657074450635224428636658351 1	88187689677809171096749992078 7	G
14th	37,721	19,465	18836415951238280222855878106 4	91325300643075380619883906111 9	B
num. 16th	42,037	21,667	41920286229331659941956881005 5	38782569212580126280918087952 1	F

The finds (evidence). Section 15.2 tested π's convergents to $n = 20{,}000$ and found ten prime denominators, the last at $n = 6{,}925$, and fifteen prime numerators, the last at $n = 8{,}750$. Teddy Rook has now tested every denominator and every numerator to $n = 50{,}000$, with scripts of his own (`c12_third_pisearch.py` and `c12_third_sieve.py` are cleaned versions). There are four new probable-prime denominators and one new probable-prime numerator (Table 17.3).

Each of the five survives the sieve of Section 17.6 (no prime factor below 10^7) and passes the strong Baillie–PSW test, strong probable-prime tests to the bases 2, 3, 5 and 7, and further Miller–Rabin rounds. For each, the terms of π up to its index were computed twice, from mpmath and from MPFR's π as an interval, and agree. In each case the other part of the fraction is composite, so $n = 8$ remains the only convergent with both parts prime.

Our checks. These statements were checked independently for this chapter. Each of the five numbers passes the strong Baillie–PSW test and 25 Miller–Rabin rounds with random bases in our runs, the other part of its fraction fails the Baillie–PSW test, and the digits in the files Teddy Rook sent agree with ours digit for digit. Our own searches cover the denominators for $20{,}000 < n \le 38{,}571$ (`c12_third_pisearch.py` to 25,332, then the rhythm sieve, `c12_third_sieve.py`) and the numerators for $20{,}000 < n \le 42{,}037$, and find exactly the primes listed here and nothing else; our numerator search reached p_{42037} on its own. For the denominators with $38{,}571 < n \le 50{,}000$ and the numerators with $42{,}037 < n \le 50{,}000$ the statement that there are no others rests on his search alone. His runs used trial division (later the sieve to 10^7) and the tests above; his second search rediscovered the eleventh independently, and the seam between two of his runs ($n = 26{,}255$ to 26,270) was retested.

So, if all five are prime, the prime denominators of π's convergents up to $n = 50{,}000$ are at

$$n = 1, 3, 8, 32, 594, 1{,}126, 2{,}002, 3{,}610, 4{,}355, 6{,}925, 25{,}332, 27{,}651, 32{,}394, 37{,}721,$$

14 in all, and the prime numerators are at

$$n = 4, 8, 10, 15, 86, 229, 333, 593, 839, 852, 1{,}148, 2{,}725, 3{,}787, 7{,}441, 8{,}750, 42{,}037,$$

16 in all. OEIS A259490, whose index is $n+1$, lists ten prime denominators and states that the eleventh has index above 25,000 in its numbering [110]. So, as far as that record goes, these are the eleventh to fourteenth, and the eleventh sat just past the previous search limit. Its companion list of prime numerators, OEIS A086785, holds fifteen terms [105], as many as we had found to $n = 20{,}000$, so p_{42037} would be the sixteenth. We cannot rule out that someone has found any of these and not reported it.

How they were checked. Our script `c12_third_pi.py` computes π's terms from mpmath's π at two precisions and from MPFR's π (through gmpy2) as an interval at 100,000 bits, and the lists agree through $n = 26{,}080$; `c12_third_newprimes.py`, `c12_third_fourteenth.py` and `c12_third_fifty.py` repeat this at higher precision, and the lists agree through $n = 50{,}080$. Each number is rebuilt by the recurrence $q_n = a_n q_{n-1} + q_{n-2}$ (or the same for p_n), compared digit by digit with the copy published on the companion site, and tested again (the Baillie–PSW test alone takes about 38, 51, 69, 133 and 191 seconds for the five, on one core). The SHA-256 hashes of their decimal digits (no spaces or line breaks) are

q_{25332}	`73f98bd668e4c94394d07e7846ca7643d6ecf617317850bcbff60ac260ad7974`
q_{27651}	`4373b2822612a676bfbbeec29a0ea3e56aac31467a99903804990469e10b659d`
q_{32394}	`08bfe3ab0ebbd69b1945ad957a2fbae729aaca5a3e26ef00a80c9650711eb4a6`
q_{37721}	`e400a170de02c0578e23b9d3b2e2ca73f2345ba9a4232cfee009c148d15af77d`
p_{42037}	`ca837a3355299394dacfb035d2b07ad092625da952405d18ef3a1b2389510098`

The terms around them are $a_{25330}, \ldots, a_{25334} = 3, 12, 2, 4, 1$; $a_{27649}, \ldots, a_{27653} = 2, 7, 2, 1, 2$; $a_{32392}, \ldots, a_{32396} = 2, 1, 2, 3, 1$; $a_{37719}, \ldots, a_{37723} = 1, 8, 35, 1, 7$; and $a_{42035}, \ldots, a_{42039} = 7, 6, 7, 6, 2$. All their digits are printed in Appendix D, and the companion site has them as plain text files [115].

Status: probable primes. No composite number is known to pass the Baillie–PSW test, but the test is not a proof. A proof for numbers of this size needs an elliptic-curve certificate (6; Section 17.6), which anyone can check much faster than it was found.

Proofs of primality. A probable prime of this size can be proved prime by the elliptic curve method (ECPP). Teddy Rook is running Enge's implementation of fastECPP [37] on the eleventh, q_{25332}, on a six-core laptop. The first two of roughly a thousand steps of its first phase took 26 and 34 minutes; the cost of a step falls roughly with the cube of the size of the number, which puts the whole run at about a week. The cost of ECPP grows with about the fourth power of the number of digits, so the twelfth, thirteenth and fourteenth would take about 1.4, 2.7 and 5.0 times as long, and the new numerator about 7.7 times. *Until a certificate exists and has been verified, each of the five is a probable prime and nothing more*, and so is every statement below that counts them.

Against the heuristic. Heuristic 8.3 gives q_n a chance $\zeta(2)/\ln q_n$ of being prime; for q_{25332} that is about one in 18,220. Table 17.4 continues Table 15.1, and Table 17.5 extends Table 15.1 with the same exact Poisson-binomial law. At $n = 50{,}000$ the heuristic expects 14.1 prime denominators (central 95% range 8–22) against 14 found, and 13.5 numerators (range 7–21) against 16. Both counts stay well inside the 95% ranges. (Teddy Rook's table gives 13.4, 13.9 and 14.2 for the denominators at $n = 28{,}000$, 40,000 and 50,000; the sums are 13.34, 13.83 and 14.14, which round to 13.3, 13.8 and 14.1.)

A cluster. The totals are where the heuristic puts them: 14 denominators against 14.1, and 16 numerators against 13.5. But the four new prime denominators are bunched. In the heuristic time $\Lambda(n) = \sum_{k \le n} \zeta(2)/\ln q_k$, in which a typical number's prime denominators arrive at rate 1, the stretch $25{,}000 < n \le 40{,}000$ has length 0.65, and it holds four primes; a Poisson count with mean 0.65 reaches four with probability 0.0045. The window was chosen after the primes were seen, which makes that figure too small by some unknown factor; the stretch before it ran low (10 found to $n = 20{,}000$ against 12.9), and the stretch after it, $40{,}000 < n \le 50{,}000$, holds none.

Table 17.4. Table 15.1 continued: probable primes among π's convergents with $n \le N$, and the expected counts $\sum \zeta(2)/\ln$ of Heuristic 8.3. Teddy Rook has tested numerators and denominators to $n = 50{,}000$; our own searches reach $n = 42{,}037$ for numerators and $n = 38{,}571$ for denominators.

	denominators q_n		numerators p_n	
N	found	expected	found	expected
20,000	10	12.9	15	12.2
28,000	12	13.3	15	12.7
32,394	13	13.5	15	12.9
40,000	14	13.8	15	13.2
50,000	14	14.1	16	13.5

Table 17.5. Probable-prime denominators among π's convergents, $n \le N$: the observed count, the expected count $\sum \zeta(2)/\ln q_n$ of Heuristic 8.3, the central 95% range of the Poisson-binomial law, and the share of that law at or below the observed count. The first three rows repeat Table 15.1.

N	observed	expected	95% range	share
1,000	5	8.7	4–14	0.11
5,000	9	10.9	5–17	0.34
20,000	10	12.9	7–20	0.25
25,332	11	13.2	7–20	0.32
28,000	12	13.3	7–21	0.42
32,394	13	13.5	7–21	0.51
37,721	14	13.7	7–21	0.60
40,000	14	13.8	7–21	0.59
50,000	14	14.1	8–22	0.56

Over the whole range $30 \le n \le 50{,}000$ there are 11 primes against 10.3 expected, and their positions are consistent with random arrivals (Kolmogorov–Smirnov $p = 0.20$; the version of the test that keeps every prime and measures time from $n = 1$ gives $p = 0.19$; Section 17.6). The new numerator, by contrast, ends a long drought: nothing between $n = 8{,}750$ and $n = 42{,}037$, a gap of 2.18 in heuristic time, which an exponential law exceeds with probability 0.11. We report the cluster as one of the two most lopsided things we have seen in π's convergents; the other is next.

The primes as notes. Section 6.1 plays a prime as the note given by its remainder modulo 12, so every prime above 3 lands on C♯, F, G or B (the remainders 1, 5, 7, 11; together the four notes form the chord G–B–D♭–F, a dominant seventh with a flattened fifth in C major, the key of the Pi movements). Played this way the 14 prime denominators are

G F C♯ G G G G G C♯ F B F G B.

The first ten use only three of the four notes; the eleventh is the first to land on B. That much is ordinary: ten primes miss one of four notes with probability 0.22. The lean toward G is less ordinary (Table 17.6).

If the four notes were equally likely, some note would reach 15 of 30 with probability 0.011; leaving out the small early primes ($n < 30$), G has 12 of 23, and some note reaches 12 of 23

Table 17.6. The notes of π's prime convergents, and three comparisons. The last row is the exact expected count on each note for a typical number, among its prime denominators above 3 up to the size of q_{37721} (Theorem 15.3); up to the size of q_{32394} it is 13.7 primes, split 3.29, 3.55, 3.48 and 3.38.

	C♯	F	G	B
14 prime denominators	2	3	7	2
16 prime numerators	3	4	8	1
together (30)	5	7	15	3
all denominators coprime to 12, $n \le 33{,}000$	4,147	4,131	4,111	4,136
prime intermediate denominators, levels 4 to 3,000	15	15	22	23
typical number, expected among 13.9 primes	3.34	3.60	3.54	3.44

with probability 0.019 (G in particular, 0.005; these are exact multinomial sums). Three things say this is chance all the same. First, the denominators that *could* be prime show no preference at all (fourth row; chi-square test $p = 0.98$): only the ones that turned out prime lean. Second, for a typical number Theorem 15.3 gives the expected count on each note exactly, and the small primes 5 and 7 tilt it by about a tenth of a prime, slightly toward F (last row). Third, an independent family of primes tied to π, the intermediate denominators of Section 17.6, shows no lean toward G (fifth row, $p = 0.39$). We noticed the lean when there were 27 primes; the three found since are one G, one B and one F, which is what chance predicts. The statistic was picked out after looking, in a work that has by now run several dozen tests of π. We record it, with the cluster, as something to watch: if it is chance, about a quarter of future prime convergents will be G.

Can the lean speed up the search? No. The note of every candidate is known for free from the sieve, so testing only the G candidates would cut the work to a quarter; but if the lean is chance it also finds only a quarter of the primes, and it can never certify the *next* prime, because it skips three candidates in four. Of the five new primes one is G. The only safe use is ordering: test the G candidates of a range first and the rest afterwards, which costs nothing and loses nothing. The contrast with e is instructive. There the restriction to $n \equiv 1 \pmod 3$ is a theorem (Theorem 17.9) and a true threefold saving. For π the note depends on the number and not on its index, and the indices of the prime denominators show no preference modulo 2, 3 or 4.

The primes in the form. In the reading of Chapter 3 a prime denominator is a block whose length in beats is prime: it cannot be cut into equal sub-phrases, any regular pulse laid against it comes back into step only at its end, and all of its rotations are different rhythms (the necklace argument of Chapter 7). At thirteen thousand digits and more that is a statement about form, not about anything that could be played. The fourteen-note motif of the denominators can be.

What it means for Question 1. Nothing is settled: five more primes are what a typical number would be expected to show sooner or later, and say nothing about infinitely many.

17.6 Finding prime convergents faster

Added 2026-11-10.

This section follows Teddy Rook's text. The new prime convergents were found by testing one convergent after another. Is there a pattern that would say where to look, or a way to find many at once? There is no shortcut along the convergents themselves, there are two real speed-ups for the search, and there is a neighboring family, tied to the largest terms of π, in which primes do come many at a time.

17.6.1 The rhythm sieve

Proposition 17.10 (the rhythm sieve)**.** *For every prime ℓ the residues $q_n \bmod \ell$ obey the recurrence of the denominators, $q_{n+1} \equiv a_{n+1} q_n + q_{n-1} \pmod{\ell}$. Hence the set of pairs (n, ℓ) with $n \le N$, $\ell < B$ and $\ell \mid q_n$ can be computed in $O(N\pi(B))$ operations on machine words, without forming any q_n. Moreover ℓ never divides two consecutive denominators.*

Proof. The first statement is the recurrence of Proposition 2.1 reduced modulo ℓ; the last is $\gcd(q_n, q_{n+1}) = 1$, from (2.5). □

In the language of Chapter 3, each prime ℓ is a voice that sounds on the beats n where ℓ divides q_n. No voice sounds on two beats running; for a typical number each voice sounds on a share $1/(\ell+1)$ of the beats (Corollary 8.2); for e each voice is strictly periodic, with period 3ℓ and three forced hits (Proposition 17.8 and Theorem 17.9). A prime denominator is a beat on which every voice is silent. The sieve of Eratosthenes is this polyrhythm for the sequence $1, 2, 3, \ldots$; Proposition 17.10 is the same sieve for the denominators of a continued fraction. The companion site plays it [115].

In practice the sieve replaces the trial division of Section 15.2. All primes below $B = 10^7$ (664,579 of them) are carried in one array, and one pass over a few tens of thousands of n takes about three minutes (177 seconds in our runs), against hours for the probable-prime tests that follow. By Mertens' theorem and the correction factor of Chapter 8, the share of denominators that survive is about $\zeta(2)e^{-\gamma}/\ln B = 5.7\%$ for $B = 10^7$. In our passes over $25{,}332 < n \le 37{,}721$, 718 of the 12,389 denominators survived (5.8%), and over $20{,}000 < n \le 32{,}394$, 701 of the 12,394 numerators (5.7%); Teddy Rook's pass over $25{,}000 < n \le 32{,}000$ left 391 of 7,000 denominators (5.6%) and 412 of 7,000 numerators (5.9%). In his search all survivors, numerators and denominators, have been tested up to $n = 50{,}000$; in ours, up to $n = 38{,}571$ and $n = 42{,}037$.

17.6.2 A cheaper test, and what the search costs

Every prime passes a strong probable-prime test to base 2, and it costs one modular exponentiation. The Baillie–PSW test adds a Lucas test that costs several more. At 13,017 digits Teddy Rook measured 6.2 seconds for the base-2 test and 37.7 seconds for Baillie–PSW with gmpy2. So the survivors of the sieve are tested to base 2 first, and only those that pass (in this search: the primes, and nothing else) go on to Baillie–PSW. Together with the deeper sieve this is roughly eight times less work per index than the procedure of Section 15.2 (trial division to 10^5, then Baillie–PSW on every survivor).

The cost of going further is easy to state. Among the indices in $(N, 2N]$ about $2 \times 5.7\%$ of the numerators and denominators need a test, the numbers have about $0.515\,n$ digits, and with gmpy2 the time of a test grows like the 2.6th power of the number of digits. So searching $(N, 2N]$ costs about twelve times as much as searching $(N/2, N]$, and by Heuristic 8.3 it is expected to yield $2\ln 2 \cdot \ln 2 \approx 0.96$ new prime denominators. Each new prime costs about ten times as much as the one before. For almost every real number this scarcity is a theorem

(Theorem 16.4); for π it is what the data show. No arrangement of the search can produce many prime convergents of one number at once, because there are not many at any one size to find.

17.6.3 Is there a pattern in where they fall?

If Heuristic 8.3 is right, the prime denominators arrive like a Poisson process in the heuristic time $\Lambda(n)$ of Section 17.5. Teddy Rook tests this for the primes with $n \geq 30$, leaving out the first few denominators, for which $1/\ln q$ is too large to be a probability. Up to $n = 50{,}000$ the denominators give 11 primes against 10.3 expected, with positions consistent with uniform in heuristic time (Kolmogorov–Smirnov $p = 0.20$); the numerators give 12 against 10.2 ($p = 0.76$). Our first version of the test kept every prime convergent and measured heuristic time from $n = 1$; it gives $p = 0.19$ and $p = 0.91$. Both versions are valid tests of the same hypothesis, and they differ only in which small primes they keep; his is the better-founded one, since the heuristic is not meant for the first few n. The gaps between successive prime denominators, in heuristic time (the first measured from $n = 30$), are

$$0.13, 3.96, 0.89, 0.81, 0.82, 0.26, 0.65, 1.81, 0.12, 0.22, 0.21.$$

For an exponential law with mean 1 the long gap before the eleventh prime (1.81) has probability 0.16, and short gaps are the most likely gaps; it is the run of three short ones that makes the cluster of Section 17.5. The indices show no preference modulo 2, 3 or 4 (chi-square p-values by simulation, 0.80, 1.00 and 0.91 for the denominators), and the terms next to a prime denominator ($a_n = 7, 1, 2, 6, 4, 60, 1, 1, 1, 31, 2, 2, 2, 35$; $a_{n+1} = 15, 292, 1, 99, 2, 3, 6, 1, 5, 3, 4, 1, 3, 1$) show nothing we can use. With fourteen and sixteen data points these tests are weak. Two statistics stand out, the cluster and the lean toward G, and neither predicts an index. *So there is no way to say where the next prime denominator is.* The heuristic gives odds only. Seen from $n = 32{,}394$, the chance of a fourteenth by $n = 40{,}000$ was about 25%; one came at $n = 37{,}721$. Starting from $n = 50{,}000$, the chance of a fifteenth prime denominator by $n = 62{,}500$ is about 27%, by $n = 100{,}000$ about 62%, and by $n = 200{,}000$ about 85%; the odds for a seventeenth prime numerator are the same (27, 62 and 85%). (Poisson model, with $\ln q_n$ continued beyond $n = 50{,}000$ at the slope measured from $n = 20{,}000$.)

17.6.4 Where primes do come many at a time: intermediate fractions

Between the convergents numbered $n-1$ and $n+1$ lie the intermediate fractions

$$\frac{p_{n-1} + kp_n}{q_{n-1} + kq_n}, \qquad k = 1, \ldots, a_{n+1} - 1,$$

the best approximations from one side (Chapter 4 met one of them, the fifth 17/29). In the rhythm, $q_{n-1} + kq_n$ is the length of k copies of the level-n block closed by one level-$(n-1)$ block: the section stopped after k phrases. A large term a_{n+1}, a phrase repeated many times, therefore gives a long arithmetic progression of denominators of nearly the same size.

The work began with one such progression. The guess (1.1), $3 + k/(7k+1)$, runs through the intermediate fractions of level 1: its denominator is $q_0 + kq_1 = 1 + 7k$ for $k \leq 14$. The gold rings of Fig. 1.1 with $k \leq 14$ are the prime intermediate denominators of that level, 29, 43 and 71, and the last of them is Archimedes' 223/71.

The family of the guess. Written out, $3 + k/(7k+1) = (22k+3)/(7k+1)$. Because $22 \cdot 1 - 3 \cdot 7 = 1$ the fraction is always in lowest terms, and by Dirichlet's theorem each of the two progressions $7k+1$ and $22k+3$ contains infinitely many primes. For $k \le 16$, the range in which the guess follows π's own ladder, the denominator is prime for $k = 4, 6, 10, 16$ $(29, 43, 71, 113)$ and the numerator for $k = 2, 5, 7, 8, 10, 14$ $(47, 113, 157, 179, 223, 311)$. Both are prime only for $k = 10$: Archimedes' lower bound $223/71$ is the one fraction on this stretch of the ladder with a prime on top and a prime below. (Zu Chongzhi's $355/113$ has a prime only below.) Beyond $k = 16$ the family leaves π and drifts back toward $22/7$, but as a family of prime pairs it goes on: both parts are prime for $k = 10, 28, 34, 40, 70, 100, 184, \ldots$, giving $223/71$, $619/197$, $751/239$, $883/281$, … Whether there are infinitely many such pairs is unknown, as for twin primes. The Bateman–Horn conjecture [10] predicts

$$\#\{k \le X : \ 7k+1 \text{ and } 22k+3 \text{ both prime}\} \sim C \int_1^X \frac{dt}{\ln(7t+1)\,\ln(22t+3)},$$

with

$$C = \prod_p \frac{1 - \nu(p)/p}{(1-1/p)^2} = 1.7604\ldots,$$

where $\nu(p)$, the number of solutions of $(7k+1)(22k+3) \equiv 0 \pmod p$, is 1 for $p = 2, 7, 11$ and 2 otherwise. The counts are 6, 28, 157, 1,040, 7,613 for $X = 10^2, \ldots, 10^6$, against 5.1, 26.5, 159.4, 1,064.1, 7,621.0 predicted. (Our script recounts the pairs with a sieve, finds $\nu(p)$ by solving the two congruences, and takes the product over the primes below 10^7; it agrees with Teddy Rook's figures.) The same is true at every level of every continued fraction: the intermediate fractions of level n are a pair of linear forms $p_{n-1} + kp_n$, $q_{n-1} + kq_n$ with determinant ± 1.

Proposition 17.11 (one sieve for a whole level)**.** *Fix $n \ge 1$ and a prime ℓ. If $\ell \mid q_n$, then ℓ divides none of the numbers $q_{n-1} + kq_n$. Otherwise it divides exactly those with $k \equiv -q_{n-1}q_n^{-1} \pmod \ell$. Hence the whole progression is sieved by the primes below B in $O(\pi(B) + a_{n+1} \ln\ln B)$ operations, given the residues of Proposition 17.10.*

Proof. If ℓ divided both q_n and $q_{n-1}+kq_n$ it would divide q_{n-1}, contradicting $\gcd(q_{n-1}, q_n) = 1$. The rest is the solution of a linear congruence. □

Heuristic 17.12. The number of primes among $q_{n-1} + kq_n$, $1 \le k < a_{n+1}$, is about

$$\frac{q_n}{\varphi(q_n)} \cdot \frac{a_{n+1} - 1}{\ln q_{n+1}}.$$

For a typical number the average of $q_n/\varphi(q_n)$ over the levels is $\prod_\ell (1 + 1/(\ell^2 - 1)) = \zeta(2)$, the constant of Heuristic 8.3, and the number of prime intermediate denominators up to level N is of order $(\ln N)^2$, against $2\ln 2 \cdot \ln N$ for the convergents themselves.

The first statement is the usual count of primes in an arithmetic progression with difference q_n, applied to a range of k far too short for any theorem. For the second, a prime ℓ divides q_n for a share $1/(\ell+1)$ of the levels (Corollary 8.2), which gives the product; and the sum of the first N terms of a typical number is about $N \log_2 N$ (in probability, by a theorem of Khinchin [72]), which turns $\sum_n (a_{n+1} - 1)\zeta(2)/(Ln)$ into $(\ln N)^2$. The count is dominated by the few largest terms, so it fluctuates far more than the count for convergents.

Table 17.7 gives the counts for π. The famous 292 is a vein of 33 primes of the form $106 + 113k$; the term 20,776 at position 431 gives 33 primes of about 220 digits from a single

Table 17.7. Probable-prime denominators among the intermediate fractions of π, $q_{n-1} + kq_n$, $1 \le k < a_{n+1}$: the level n, the term a_{n+1}, the number of candidates, their size in digits, the number of probable primes, and the expected number $\sum(q_n/\varphi(q_n))/\ln x$ over the candidates x, with $q_n/\varphi(q_n)$ computed from the primes below the sieve limit (Heuristic 17.12).

level n	term	candidates	digits	primes	expected
1	15	14	1–2	3	4.5
3	292	291	3–5	33	31.5
430	20,776	20,775	216–221	33	41.1
15,542	19,055	19,054	7,993–7,997	2	1.4
all levels $1, \ldots, 3{,}000$		41,390		111	122.3

sieve and about a thousand quick tests, which took under a second. (The candidates have 216 to 221 digits; Teddy Rook's table gives 217 for the smallest, but $q_{429} + q_{430}$, the candidate with $k = 1$, has 216.) Up to level 3,000 there are 111 prime intermediate denominators (122.3 expected), against the 7 prime convergent denominators in the same range (10.2 expected). Deeper in the expansion the veins are thinner but still pay: the term 19,055 at position 15,543 gives 19,054 candidates of nearly 8,000 digits, of which 905 survive a sieve to 10^7 and 2 are probable primes ($k = 4{,}703$ and $k = 17{,}981$). The next veins of this kind are the terms 19,308 at position 23,398 (about 12,014 digits, 0.7 primes expected before the local factor, 2.2 with it) and 78,629 at position 28,421 (about 14,598 digits, 2.3 expected, 4.7 with the local factor).

Two cautions. These are primes in arithmetic progressions; nothing about π guarantees them, and each candidate still has the chance $1/\ln$ of its size, so the gain is only that one level supplies thousands of candidates of one size and one sieve serves them all. And they are intermediate fractions, not convergents: they answer a neighboring question, not Question 1.

17.6.5 Confirming primes

For numbers of a few hundred digits a proof of primality takes seconds with standard software (for example `isprime` in PARI/GP), so every prime of Table 17.7 below a thousand digits can be proved at once; we report them as probable primes because we did not run those proofs here. For the new prime denominators the only practical method is ECPP [6, 37], whose cost grows with about the fourth power of the number of digits (so the twelfth, thirteenth and fourteenth would take about 1.4, 2.7 and 5.0 times as long as the eleventh), and no method is known that does better for a number with no special form: the fast proofs for Mersenne or Fermat-type numbers need most of $N - 1$ or $N + 1$ to be factored, and a convergent denominator offers no such factorization. The certificate, once produced, can be checked in a small fraction of the time it took to find. So of the two bottlenecks, the search can be made about eight times faster by the two devices above, and the proof cannot be shortened except by more processors.

17.7 What the new primes say about the six questions

Added 2026-11-10.

This section follows Teddy Rook's text. Five more probable primes cannot settle a question about infinitely many. They do sharpen three things, all on the side of π behaving like a typical number, and all evidence, not proof. Questions 2, 5 and 6 are untouched by them.

Table 17.8. Pi's prime convergents with $n \geq 30$ (numerators and denominators together) against Heuristic 8.3 with and without the factor $\zeta(2)$. The estimate of the constant c in "$c \ln N$" and its 95% range treat the count as Poisson; the heuristic value is $2\ln 2 = 1.386$ and the uncorrected value $12\ln 2/\pi^2 = 0.843$. The counts beyond our own search (Section 17.5) are Teddy Rook's.

search limit	found	expected with $\zeta(2)$	expected without	chance of so many without $\zeta(2)$	estimate of c (95% range)	likelihood ratio
20,000	18	17.9	10.9	0.030	1.39 (0.82 to 2.20)	7 to 1
50,000	23	20.5	12.5	0.005	1.56 (0.99 to 2.34)	30 to 1

Pi's own constant (Questions 1 and 3). Theorem 16.4 says that almost every number has $2\ln 2 \cdot \ln N$ prime denominators among its first N convergents. Does π? Treat the constant as unknown and estimate it from π's data. Leaving out $n < 30$, where $1/\ln q$ is not yet a probability, π has 11 prime denominators and 12 prime numerators with $30 \leq n \leq 50{,}000$. Heuristic 8.3 expects 20.5; the same heuristic *without* its correction factor $\zeta(2)$ (Step 2 of its derivation in Chapter 8, which treats the q_n as random integers) expects 12.5. Table 17.8 compares the two before and after the new primes.

At $n = 20{,}000$ the data could not tell the two constants apart: the uncorrected value 0.843 lay inside the 95% range. With the five new primes it lies outside. So for π itself, and not only for almost every number, the count of prime convergents now favors the corrected constant by about 30 to 1, and is consistent with exactly $2\ln 2$. This is the first place in this work where π's prime convergents discriminate between two candidate laws. The caveats: the two counts are pooled as if independent, twenty-three events are few, and the comparison is between two specific constants, not a proof of either.

The correction factor, tested directly (Questions 1 and 4). The rhythm sieve gives a far stronger test of the same factor, one that uses thousands of convergents and not a handful of primes. By Mertens' theorem a random integer has no prime factor below $B = 10^7$ with probability $e^{-\gamma}/\ln B = 3.48\%$; Heuristic 8.3 says that for a convergent denominator the figure is $\zeta(2)$ times larger, 5.73%. Among the 23,732 indices with $26{,}268 < n \leq 50{,}000$ the sieve leaves 1,360 denominators and 1,391 numerators. The prediction with the factor is 1,360 of each; without it, 827. This extends Table 8.1, which checked the small primes on the first 10,000 convergents, to every prime below ten million. Given the survivors, each has chance $e^{\gamma}\ln B/\ln q_n$ of being prime, which predicts 1.8 primes in that range; there are 4 (probability 0.11 of four or more, the cluster of Section 17.5 seen another way).

A new test of typicality (Question 4). The sieve suggests a test that the earlier chapters did not run at scale: is the voice of each prime sounding at its proper rate? For almost every number the share of n with $\ell \mid q_n$ is $1/(\ell+1)$ (Corollary 8.2), and q_n modulo 12 has the distribution given by the theorem of Jager and Liardet (Theorem 8.1). Teddy Rook computed 2,000,000 certified terms of π and ran the recurrence modulo each of the 168 primes below 1,000. The statistic for a prime is the standardized count of hits; its natural spread was measured on 800 random reals of 100,000 terms each, expanded exactly, because the hits are far from independent coin flips (no voice sounds twice running, and for $\ell = 2$ the count fluctuates only 0.42 times as much as a binomial count would).

Every statistic in Table 17.9 is ordinary. For $\ell = 2$ the share of even denominators is 0.333295 over two million convergents, against 1/3. So the local half of Heuristic 8.3, the part

Table 17.9. The voices test on 2,000,000 convergents of π (Teddy Rook's run): the share of n with $\ell \mid q_n$ against $1/(\ell+1)$, and summary statistics over the 168 primes below 1,000, with p-values calibrated on 800 random reals.

ℓ	share for π	$1/(\ell+1)$	standardized deviation
2	0.333295	0.333333	−0.28
3	0.250157	0.250000	+0.92
5	0.166717	0.166667	+0.29
7	0.124899	0.125000	−0.60
11	0.083145	0.083333	−1.20
13	0.071356	0.071429	−0.47

sum of squared deviations, denominators: 181.8 (random reals average 168.2), $p = 0.22$
sum of squared deviations, numerators: 176.1 (average 168.3), $p = 0.33$
largest deviation: 3.24 at $\ell = 641$ (denominators, $p = 0.18$); 2.77 at $\ell = 271$ (numerators, $p = 0.59$)
q_n modulo 12 against Jager–Liardet: $\chi^2 = 11.8$ on 11 degrees of freedom, calibrated $p = 0.24$

that produces $\zeta(2)$, holds for π as far as we can measure, in two independent ways: prime by prime for the small primes, and in bulk for all primes below 10^7.

Our checks. `c12_third_rate.py` recomputes Table 17.8 and the sieve counts: it runs the rhythm sieve afresh over every $n \le 50{,}000$, for numerators and denominators, and its survivors agree with those of our search runs wherever they overlap. It repeats the voices test on the first 2,000,000 of the twelve million certified terms of Chapter 15, with a calibration of its own: 800 different random reals of 100,000 terms, expanded exactly. The six shares in Table 17.9 come out the same (0.333295, 0.250157, 0.166717, 0.124899, 0.083145 and 0.071356). For $\ell = 2$ our sample puts the spread at 0.40 times the binomial one, and the standardized deviations for $\ell = 2$ to 13 are −0.29, +0.94, +0.29, −0.64, −1.25 and −0.48. The sums of squares are 183.6 for the denominators (random reals average 168.2, $p = 0.20$) and 177.2 for the numerators (168.2, $p = 0.29$); the largest deviations are 3.10 at $\ell = 641$ and 2.77 at $\ell = 271$; and the χ^2 for q_n modulo 12 is 11.8 ($p = 0.23$). The small differences from Table 17.9 come from the different random reals; the conclusion is the same.

What they do not say. The cluster of four denominators and the lean toward one note (Section 17.5) remain the two unusual statistics, each at about the one-in-a-hundred level and each chosen after looking. None of the five new primes has a prime partner, so $n = 8$ is still the only convergent with both parts prime, in line with the 0.11 further cases expected in Section 15.2. And nothing here bears on whether the list is infinite: a number could agree with every test above for its first two million terms and still stop producing primes.

17.8 Status after the third look

Added 2026-11-10.

Table 17.10. The six questions of Section 14.3 after this chapter. "Probable prime" means that a primality proof is pending.

question	new in this round	status
1. Prime convergents of π	search to $n = 50{,}000$ with the rhythm sieve; four new probable-prime denominators, at n = 25,332, 27,651, 32,394 and 37,721, and a new probable-prime numerator at n = 42,037, found by Teddy Rook; a cluster and a lean toward one note, both probably chance; many primes among intermediate fractions (evidence). For e: sign flip, exact periods 6ℓ and 3ℓ, forced divisors, primes only at $n = 2$ or $n \equiv 1 \pmod 3$ (proved)	open
2. Fibonacci and Pell primes	nothing	open
3. The heuristic for almost every number	Theorem 16.4 stands; for π itself the data now favor the constant $2\ln 2$ over the uncorrected 0.843 by about 30 to 1 (evidence)	answered a.e.
4. Is π typical?	the rate of every small prime among two million convergents, and the share with no factor below 10^7, match a typical number (evidence)	open
5. The correlation matrix	$\ln \lambda_{\max}$ and $\ln(1/\lambda_{\min})$ both $(2 + o(1))\sqrt{\ln m / \ln\ln m}$; a limiting bulk law exists; the mean of $1/\lambda$ tends to $\prod_p (1 + 2/(p(p-1)))$ (all proved)	partly answered
6. Prime-sized scales	nothing	answered: no

Still open, in order of how reachable they look.

- A second-order term for $\lambda_{\max}(C_m)$. The proofs as written give a relative error no better than about $\ln\ln\ln m / \ln\ln m$; Soundararajan's analysis suggests $1/\sqrt{\ln\ln m}$ is attainable, and the data ask for a description of the small-prime factor that dominates at computable sizes.
- Whether $(1/\lambda_{\min})/\lambda_{\max}$ tends to a constant. It is 2.61 to 2.73 for every m in Table 17.2.
- A description of the limiting law ν: a density, its behavior near 0, its tail, and whether its mean of $1/\lambda$ equals κ.
- For e, an asymptotic for the count of prime denominators among $n \equiv 1 \pmod 3$. This is as hard as the Fibonacci prime problem.
- A rigorous version of Heuristic 17.12 for almost every number.
- Anything at all about π itself. Nothing here, or anywhere, touches it.

A caution on novelty. Theorems 17.1 and 17.5 use the resonance method exactly as number theorists use it for the zeta function, and Theorem 17.6 is a standard truncation argument. We searched and did not find these statements for this matrix, nor Theorem 17.9 for e, but results of this kind are often folklore. Before any of them is called new, a specialist in GCD sums should be asked; Hilberdink's paper [59] is the first place to look.

Acknowledgment. The searches that found q_{25332}, q_{27651}, q_{32394}, q_{37721} and p_{42037}, Teddy Rook's finished text of this chapter, on which this version is based, and the statements and proofs of this chapter, are from Teddy Rook's notes and scripts of November 2026. The scripts `c12_third_*.py` are cleaned versions of his, extended and rerun for this chapter; the independent checks, the second proofs noted in the text, the caution on the limiting law's mean of $1/\lambda$, the repeated sieve and voices test of Section 17.7, and the corrections noted in Sections 17.5 and 17.6 were added in checking.

CHAPTER 18

Large and small: the same patterns in atoms and in orbits

Added 2026-11-10.

The Connections diagram of Chapter 12 has four subjects: music, patterns, mathematics and chemistry. This chapter adds physics and astrophysics, and asks one question of them: where do the structures of the earlier chapters appear both in very small things and in very large ones? We found four places where the answer is exact or measurable, and a fifth section returns to Kepler, whose work runs through all of them. None of the physics is ours. What we add is the reading through the continued fraction, several computations, and two new exact statements about the correlation matrix (Propositions 18.2 and 18.4).

A warning first. This subject is full of numerology: the Titius–Bode "law" of planetary distances and Dirac's large-number coincidences are the classic cases of patterns that explain nothing. The discipline of the earlier chapters applies. A pattern is reported with its test, and a coincidence is called one.

18.1 The atom and the solar system are one problem

Section 5.3 called the link between hydrogen's $1/n^2$ and $\zeta(2)$ "a shared formula, not shared physics," and Section 12.5 said the same of the shell sizes $2, 8, 18, 32$. There is a neighbouring link that *is* shared physics, and it strengthens the chemistry corner of the diagram.

Gravity and the electric force both fall off as the inverse square of the distance. So a planet round the Sun and, in Bohr's model, an electron round a proton obey the same law of periods.

Proposition 18.1 (Kepler's third law, twice; classical)**.** *For a circular orbit of radius r and period T,*

$$T^2 = \frac{4\pi^2}{GM}\, r^3 \quad \textit{(planet round a mass } M\textit{)}, \qquad T^2 = \frac{4\pi^2 m_e}{ke^2}\, r^3 \quad \textit{(electron round a proton)}.$$

In Bohr's model the allowed orbits have $r_n = a_0 n^2$ and therefore $T_n = T_1 n^3$, with $a_0 = 5.292 \times 10^{-11}\ m$ and $T_1 = 1.520 \times 10^{-16}\ s$.

Proof. Set the attracting force equal to the centripetal force $m(2\pi/T)^2 r$, with force GMm/r^2 in one case and ke^2/r^2 in the other. Bohr's condition, angular momentum $n\hbar$, gives $r_n = n^2\hbar^2/(m_e k e^2)$. □

Our script checks both. For the Earth, T^2/r^3 and $4\pi^2/(GM_\odot)$ are both 2.9747×10^{-19} in SI units; for the electron in its lowest orbit both sides are 0.15588. The two constants differ by a factor of 5×10^{17}, and the law is the same.

The rhythm of an atom, heard by a radio telescope. An electron in a very high orbit behaves almost classically, and the light it emits when it drops k levels has k times its orbital frequency: the k-th harmonic of its rhythm. Astronomers have detected carbon atoms in interstellar space with $n \approx 1009$, through transitions with $k = 4$ near 26 MHz [124]. The numbers fit. At $n = 1009$ the orbit has radius $a_0 n^2 = 54\,\mu$m, so the atom is about a tenth of a millimetre across, a million times the size of an ordinary atom; its orbital period is $T_1 n^3 = 1.56 \times 10^{-7}$ s, an orbital frequency of 6.41 MHz; and the transition $n = 1013 \to 1009$ has frequency 25.5 MHz, which is 4.000 times the orbital frequency at $n = 1011$. A dish on Earth is listening to the fourth harmonic of an electron's orbit.

The hidden symmetry, and the periodic table. An inverse-square force has a conserved quantity beyond energy and angular momentum, the Laplace–Runge–Lenz vector, which points along the long axis of the ellipse. It is the reason planetary orbits are closed ellipses that do not precess. Pauli (1926) and Fock (1935) showed that the same conserved vector gives the hydrogen atom a four-dimensional rotational symmetry, and that this symmetry is why all n^2 states of level n have the same energy [43, 111]. With two spin states each, that is $2n^2$: the shell sizes $2, 8, 18, 32$. So the squares of Section 12.5 come from the symmetry that keeps the planets on closed ellipses. In atoms with many electrons the other electrons screen the nucleus, the force is no longer exactly inverse-square, and the symmetry is broken; the rows of the periodic table then have lengths $2, 8, 8, 18, 18, 32, 32$, each shell size after the first appearing twice. The periodic table is hydrogen's symmetry, broken.

A historical echo, and a caution. Kepler published his third law in *Harmonices Mundi* (1619), a book whose subject is the music of the planets: he assigned each planet a melody from the ratio of its fastest and slowest speeds. And the exponent in $T \propto r^{3/2}$ is the ratio of a musical fifth. That last fact is a coincidence of small numbers and nothing more.

18.2 Resonance as rhythm

Section 4.4 treated mode locking: coupled oscillators fall into step at simple frequency ratios. Orbits do the same. Two bodies whose periods are near a ratio p/q meet in the same configuration every q turns of one and p of the other, and the repeated tug can lock the ratio. The table of convergents of Chapter 2 turns out to be a detector for this.

The big-term test. Expand the ratio of two periods as a continued fraction. If it is locked to p/q, then p/q is a convergent and *the next term is large*, for the reason that 292 follows 355/113 in π. Table 18.1 applies this to four systems. The planets' periods are their long-term mean periods [123], the moons' are from NASA's fact sheets [99], and the others are from the discovery papers [1, 89].

What the table shows. The true resonances announce themselves with large terms: 53 for Pluto and Neptune (locked at 3/2), 137, 68 and 160 for Jupiter's moons, and from 180 to over 3000 for HD 110067. (Of the three for Jupiter, 137 and 68 belong to the locked trio

Table 18.1. Neighbouring period ratios (outer over inner): the expansion, the nearest fraction with denominator at most 8, the normalized error $q^2|r - p/q|$ of Table 2.1, and the term that follows p/q. The last line of each block compares the mean normalized error with that of random ratios drawn log-uniformly from $[1.25, 3]$.

pair	ratio	expansion	nearest	$q^2\times$error	next term
Venus/Mercury	2.5543	$[2; 1, 1, 4, \ldots]$	5/2	0.217	4
Earth/Venus	1.6255	$[1; 1, 1, 1, 2, \ldots]$	13/8	0.033	29
Mars/Earth	1.8808	$[1; 1, 7, 2, \ldots]$	2/1	0.119	7
Jupiter/Mars	6.3067	$[6; 3, 3, 1, \ldots]$	19/3	0.239	3
Saturn/Jupiter	2.4833	$[2; 2, 14, 2, \ldots]$	5/2	0.067	14
Uranus/Saturn	2.8521	$[2; 1, 5, 1, \ldots]$	3/1	0.148	5
Neptune/Uranus	1.9614	$[1; 1, 24, 1, \ldots]$	2/1	0.039	24
Pluto/Neptune	1.5046	$[1; 1, 1, 53, \ldots]$	3/2	0.018	53
Solar system: mean 0.110*; random* 0.146*; chance of being this close* 0.14					
Europa/Io	2.0073	$[2; 137, \ldots]$	2/1	0.007	137
Ganymede/Europa	2.0147	$[2; 68, \ldots]$	2/1	0.015	68
Callisto/Ganymede	2.3326	$[2; 3, 160, \ldots]$	7/3	0.006	160
Jupiter's moons: mean 0.009*; chance* 0.0001					
TRAPPIST-1 c/b	1.6031	$[1; 1, 1, 1, \ldots]$	8/5	0.076	12
d/c	1.6719	$[1; 1, 2, 20, \ldots]$	5/3	0.047	20
e/d	1.5067	$[1; 1, 1, 36, \ldots]$	3/2	0.027	36
f/e	1.5092	$[1; 1, 1, 26, \ldots]$	3/2	0.037	26
g/f	1.3416	$[1; 2, 1, 12, \ldots]$	4/3	0.074	12
h/g	1.5198	$[1; 1, 1, 12, \ldots]$	3/2	0.079	12
TRAPPIST-1: mean 0.057*; chance* 0.005					
HD 110067 c/b	1.5003	$[1; 1, 1, \ldots]$	3/2	0.001	> 300
d/c	1.5007	$[1; 1, 1, \ldots]$	3/2	0.003	> 300
e/d	1.5007	$[1; 1, 1, \ldots]$	3/2	0.003	> 300
f/e	1.3334	$[1; 2, 1, \ldots]$	4/3	0.000	> 300
g/f	1.3339	$[1; 2, 1, \ldots]$	4/3	0.006	180
HD 110067: mean 0.0025*; chance below* 0.0001					

Io, Europa and Ganymede. Callisto's 160 marks a very near miss of 7/3 that is not known to be a dynamical lock, a reminder that a large term is evidence and not proof.) Three pairs of Saturn's moons do the same: Tethys/Mimas $[2; 318, \ldots]$, Dione/Enceladus $[1; 1, 388, \ldots]$ and Hyperion/Titan $[1; 2, 1, 109, \ldots]$. The planets of the solar system taken together are *not* significantly closer to simple fractions than random ratios would be ($p = 0.14$); the chains of TRAPPIST-1 and HD 110067 are, beyond doubt. Two near-misses in the solar system are famous for their consequences and not locked: Saturn/Jupiter, $[2; 2, 14, \ldots]$, near 5/2, drives a swing in the motions of the two planets with a period of 883 years, known as the great inequality, and Neptune/Uranus sits near 2/1. Earth/Venus is $[1; 1, 1, 1, 2, \ldots]$, the start of the golden ratio's expansion, with convergents 8/5 and 13/8: the reason Venus traces a five-petalled figure in our sky every eight years. It is not a resonance in the dynamical sense. (The 137 of Io and Europa has nothing to do with the 137 of atomic physics.)

Three bodies in step. For Io, Europa and Ganymede the mean motions (degrees per day) satisfy $n_1 - 3n_2 + 2n_3 = -0.00003$, zero to the accuracy of the data, while the two-body near-misses $n_1 - 2n_2$ and $n_2 - 2n_3$ are both 0.7395: the two pairs drift off the exact ratio 2/1 at the same rate, so the three-body pattern repeats exactly. This is Laplace's resonance. As a polyrhythm it is 4 against 2 against 1 with a slow common drift.

As music. The resonant chains are stacked intervals in the sense of Section 4.1. HD 110067 is three fifths and two fourths $(3/2, 3/2, 3/2, 4/3, 4/3)$: while the outermost planet goes round 9 times the others go round 12, 16, 24, 36 and 54 times. TRAPPIST-1 is looser: while its outermost planet goes round twice the others make 3.04, 4.08, 6.15, 9.27, 15.5 and 24.9 turns, close to $3 : 4 : 6 : 9 : 15 : 24$. Either can be loaded into the polyrhythm instrument of the companion site as it stands.

Where resonance removes instead of locks. Kepler's third law puts the orbits whose periods are 1/3, 2/5, 3/7 and 1/2 of Jupiter's at 2.500, 2.824, 2.957 and 3.277 astronomical units. Those are the Kirkwood gaps of the asteroid belt, where almost no asteroids are found. (A commonly quoted set of positions is 2.502, 2.825, 2.958 and 3.279. It comes from scaling Jupiter's distance instead of using its period, which adds about 0.0015 units: Jupiter's own mass enters its orbit but not the asteroid's, and the value taken for Jupiter's distance varies. A resonance is a statement about periods, so we give the period-based figures.) In the same way an orbit with half the period of Saturn's moon Mimas lies, by Kepler's law, 116,780 km from Saturn's centre. The bright B ring ends and the Cassini Division begins at 117,580 km, 0.7 per cent further out. The difference is Saturn's flattened shape, which Kepler's law ignores: with the measured flattening put into the calculation the resonance moves to 117,530 km, within 50 km of the edge. So simple fractions sometimes lock an orbit (Pluto, the moons) and sometimes empty it (the gaps). The theorem behind the second case is that of Kolmogorov, Arnold and Moser: an orbit survives small disturbances when its frequency ratio is hard to approximate by fractions, that is, when its continued fraction has small terms. The golden ratio, all of whose terms are 1, is the hardest number of all to approximate (Chapter 2); a number with a 292 in it is easy to approximate by the fraction that precedes the 292.

18.3 The mirror in the correlation matrix

Theorems 17.1 and 17.5 say that the largest eigenvalue of the correlation matrix C_m and the reciprocal of the smallest grow at the same rate. Is there an exact symmetry behind that? There is, in two parts.

Proposition 18.2 (the sign mirror)**.** *Let $\lambda(a) = (-1)^{\Omega(a)}$ be Liouville's function, where $\Omega(a)$ counts the prime factors of a with multiplicity. For $2 \le a, b \le m$,*

$$\left(C_m^{-1}\right)_{ab} = \lambda(a)\lambda(b)\,\sqrt{(a-1)(b-1)} \sum_{\substack{d \le m,\ \operatorname{lcm}(a,b) \mid d \\ d/a,\ d/b \text{ square-free}}} \frac{1}{\varphi(d)}.$$

Hence, with $S = \operatorname{diag}(\lambda(a))$, every entry of $SC_m^{-1}S$ is non-negative; $1/\lambda_{\min}(C_m)$ is the Perron eigenvalue of that non-negative matrix; and an eigenvector of C_m for its smallest eigenvalue can be chosen of the form $\lambda(a)w_a$ with every $w_a \ge 0$.

Proof. In the formula of Proposition 17.4 a term is non-zero only when d/a and d/b are square-free, and then $\mu(d/a) = \lambda(d/a) = \lambda(d)\lambda(a)$, because λ is completely multiplicative and $\lambda(a)^2 = 1$. So $\mu(d/a)\mu(d/b) = \lambda(a)\lambda(b)$ for every term. The rest is the Perron–Frobenius theorem. □

So the two ends of the spectrum are tied by a change of sign. The top eigenvector has no negative entries: all the pulse trains in phase. The bottom one is a positive vector with the

sign $\lambda(a)$: every pulse train flipped once for each prime factor. And the two matrices are built in mirrored ways. An entry of C_m is a sum of $\varphi(d)$ over the *common divisors* d of a and b (Proposition 15.6); an entry of the inverse is, up to sign, a sum of $1/\varphi(d)$ over their *common multiples* up to m. What two rhythms share below, against where they meet again above: gcd against lcm.

Proposition 18.3 (one prime at a time; classical)**.** *Let $G = [\gcd(a,b)/\sqrt{ab}]$, the matrix of Chapter 17 that differs from C_m only in the shift from* gcd *to* gcd -1*. Restricted to the powers $1, p, p^2, \dots$ of one prime p, it is $K = [r^{|i-j|}]$ with $r = p^{-1/2}$, the correlation matrix of a chain with correlation r between neighbours. Its inverse is tridiagonal. On the infinite chain its spectrum is the interval $[1/M, M]$ with $M = (\sqrt{p}+1)/(\sqrt{p}-1)$, and the map*

$$\lambda \longmapsto \frac{\lambda}{2c\lambda - 1}, \qquad c = \frac{p+1}{p-1},$$

carries the spectrum to itself and exchanges its two ends.

Proof. On the infinite chain K is convolution with $r^{|k|}$, whose Fourier series is the Poisson kernel $P(\theta) = (1-r^2)/(1-2r\cos\theta + r^2)$, with range $[(1-r)/(1+r), (1+r)/(1-r)]$. Then $1/P(\theta) + 1/P(\theta+\pi) = 2(1+r^2)/(1-r^2) = 2c$, which is the stated map, and $1/P$ has only the frequencies $0, \pm 1$, which is the tridiagonal inverse. □

The shift $\theta \mapsto \theta + \pi$ is multiplication by $(-1)^k$, Liouville's sign on the powers of p. So prime by prime the mirror is exact, and it is a fractional-linear map, the same kind of map that drives continued fractions. Because $\gcd(a,b)/\sqrt{ab}$ is multiplicative, the full matrix G is the product of these chains over all primes, cut off at m. Without the cut-off the top would be $\prod_p M_p = \infty$ and the bottom 0; the cut-off makes both finite, and Theorems 17.1 and 17.5 say that for C_m it treats them alike to leading order. It does not treat them exactly alike: $1/\lambda_{\min}$ is about 2.6 times $\lambda_{\max}$ (Table 18.2 below), and the mean of the eigenvalues is 1 while the mean of their reciprocals tends to 3.28 (Proposition 17.7).

The numbers. At $m = 400$ the smallest entry of $SC_m^{-1}S$ is -9×10^{-15}, zero to rounding, and the formula of Proposition 18.2 matches a dense inverse to 10^{-14}. At $m = 20{,}000$ the bottom eigenvector times $\lambda(a)$ is non-negative on every index. The two vectors live in different places: the top one is largest at $a = 60, 120, 180, 24, 12, \dots$, numbers with many repeated small factors, and the bottom one at $a = 30, 6, 60, 42, 12, 210, \dots$, led by the products of the first few distinct primes.

In physics. The chain of Proposition 18.3 is the correlation matrix of the one-dimensional Ising model of magnetism, and of a damped random walk; a tridiagonal inverse is the statement that each site depends only on its neighbours. So the arithmetic correlation matrix is a lattice of such chains, one direction for each prime, and its inverse is local: the entry for a and b vanishes unless they have a common multiple $d \le m$ with d/a and d/b square-free. Physics has exact small-to-large symmetries of this kind, the best known being the duality that exchanges high and low temperature in the two-dimensional Ising model. We claim only the analogy.

Both ends, written with common beats. Proposition 18.2 wrote the bottom of the spectrum as a sum over common multiples. The top can be written the same way.

Proposition 18.4 (the harmonic form)**.** *C_m has the same eigenvalues as the non-negative matrix*

$$T_m = \Big[\sqrt{\varphi(a)\varphi(b)} \sum_{\substack{n\le m \\ \operatorname{lcm}(a,b)\mid n}} \frac{1}{n-1}\Big]_{2\le a,b\le m},$$

and $G_m = [\gcd(a,b)/\sqrt{ab}]_{1\le a,b\le m}$ has the same eigenvalues as

$$\Big[\frac{\sqrt{\varphi(a)\varphi(b)}}{\operatorname{lcm}(a,b)}\, H_{\lfloor m/\operatorname{lcm}(a,b)\rfloor}\Big]_{1\le a,b\le m}, \qquad H_k = 1+\tfrac{1}{2}+\cdots+\tfrac{1}{k}.$$

Proof. By Proposition 15.6, $C_m = DV\Phi V^T D = B^T B$ with $B = \Phi^{1/2}V^T D$, a square matrix. So C_m and BB^T have the same eigenvalues, and the (a,b) entry of BB^T is $\sqrt{\varphi(a)\varphi(b)}\sum_n [a \mid n][b \mid n]/(n-1)$. For G_m the weight $1/(n-1)$ becomes $1/n$, and the sum of $1/n$ over the multiples of L up to m is $H_{\lfloor m/L\rfloor}/L$. □

The number $\operatorname{lcm}(a,b)$ is the beat on which the pulse trains a and b first coincide, the length of their polyrhythm (Section 4.3), and its multiples are their later coincidences. So both ends of the spectrum are Perron eigenvalues of matrices that count coincidences up to the horizon m. For the top every coincidence n counts, with weight $1/(n-1)$. For the bottom (Proposition 18.2) the weight is $1/\varphi(n)$, and a coincidence counts only if n/a and n/b are square-free. In the gcd matrix the first of these is the harmonic series itself: the entry for two rhythms is a harmonic number, H_k, where k is how many times they coincide before the horizon.

How far apart are the two ends? With these forms the two ends can be computed for much larger matrices (the divisor sums are done one prime at a time). Table 18.2 gives the result.

Table 18.2. The two ends of the spectrum and their ratio $(1/\lambda_{\min})/\lambda_{\max}$, for the correlation matrix C_m and, in the last column, for the gcd matrix G_m.

m	$\lambda_{\max}(C_m)$	$1/\lambda_{\min}(C_m)$	ratio for C_m	ratio for G_m
10^2	8.096	21.967	2.713	2.515
10^3	18.773	50.976	2.715	2.550
10^4	36.429	97.857	2.686	2.570
10^5	63.078	167.104	2.649	2.572
10^6	101.252	265.276	2.620	2.570
10^7	154.040	400.477	2.600	2.568
3×10^7	185.434	480.554	2.592	2.565

Three things stand out. First, for the gcd matrix every value we computed from $m = 2\times10^4$ to 3×10^7, some fifty sizes in all, lies between 2.565 and 2.573, while both eigenvalues grew fourfold. Second, the ratio for C_m is falling towards the ratio for G_m: the difference between them is 0.165, 0.117, 0.077, 0.050, 0.032 and 0.026 down the table from 10^3. So the question is really one about G_m. Third, neither ratio has settled: since 10^5 the ratio for G_m has been drifting down by about 0.002 for each tenfold increase in m.

Proposition 18.5 (two cases where the ratio is exact)**.** *(i) Let n be square-free. On the divisors of n the gcd matrix has $\lambda_{\max} = \prod_{p\mid n}(1+p^{-1/2})$ and $\lambda_{\min} = \prod_{p\mid n}(1-p^{-1/2})$, so*

$$\frac{1/\lambda_{\min}}{\lambda_{\max}} = \prod_{p\mid n}\frac{p}{p-1} = \frac{n}{\varphi(n)}.$$

(ii) On the powers $1, p, \dots, p^{k-1}$ *of one prime the ratio is* $p/(p-1)$ *for* $k = 2$ *and for* $k = 3$*, and it tends to* 1 *as* $k \to \infty$*.*

Proof. (i) The matrix is the Kronecker product of the 2×2 matrices with rows $(1, r)$ and $(r, 1)$, $r = p^{-1/2}$, whose eigenvalues are $1 \pm r$. (ii) For $k = 3$ the eigenvalues of Proposition 18.3's matrix are $1 - r^2$ and $\frac{1}{2}\big(2 + r^2 \pm r\sqrt{r^2+8}\big)$; the last two are the extreme ones and their product is $1 - r^2$. As $k \to \infty$ the extremes tend to M and $1/M$. □

So the ratio measures how short the chains are. Where each prime is used once, it is the totient ratio $n/\varphi(n)$ of Chapter 5; where a prime is used to a high power, it falls back to 1. Computing with only the numbers built from the primes up to a bound y shows how the two effects combine (Table 18.3).

Table 18.3. The gcd matrix on the numbers up to 10^7 whose prime factors are all at most y. With a fixed set of primes both ends have the same ceiling, $\prod_{p \le y} M_p$.

y	numbers	$\lambda_{\max}$	$1/\lambda_{\min}$	ratio	ceiling
2	24	5.335	5.805	1.088	5.83
3	190	15.735	21.056	1.338	21.8
5	768	29.841	50.513	1.693	56.9
7	2 155	45.430	94.498	2.080	126.2
13	8 289	69.097	175.848	2.545	415.5
29	37 627	96.862	261.104	2.696	2 418
97	269 882	127.264	338.751	2.662	1.3×10^5
997	2 028 358	150.352	391.162	2.602	
all	10^7	157.790	405.157	2.568	∞

The table shows the mechanism. *The bottom reaches its ceiling sooner than the top.* With the powers of 2 alone, 24 levels bring the bottom to within half a per cent of the ceiling $3 + 2\sqrt{2} = 5.83$, while the top is still more than eight per cent short. Each further small prime adds to the bottom's lead, up to about 2.7 near $y = 29$. The large primes then take a little of it back, because they still add to the top while the bottom has less left to gain. The same picture shows in the eigenvectors. At $m = 10^6$ half the weight of the top vector lies on indices above 47,000, and half the weight of the bottom vector on indices below 600: the bottom is made by a few small rhythms, the top by many large ones.

For any fixed set of primes the ratio falls to 1 as m grows: with powers of 2 only it is 1.200, 1.117 and 1.088 at $m = 10^4$, 10^6 and 10^7. With all primes it stays near 2.57, because new primes keep arriving as the old ones saturate. Whether that balance has a limit is the open question, and the computation cannot decide it. A limit near 2.5 and a very slow decline are both consistent with Table 18.2. We did not recognize 2.57 as any known constant, and at these sizes it would be numerology to try.

18.4 The butterfly of a number

An electron moving on a square crystal lattice in a magnetic field obeys Harper's equation

$$\psi_{k+1} + \psi_{k-1} + 2\cos(2\pi k\alpha + \nu)\,\psi_k = E\,\psi_k,$$

where α is the magnetic flux through one cell of the lattice in natural units. Hofstadter (1976) found that when $\alpha = p/q$ the allowed energies form q bands, and that the picture over all α

is a fractal, the "butterfly" [61]. It was observed in graphene superlattices in 2013 [31]. For irrational α the spectrum is a Cantor set [7], like the dust of Section 10.6.

The spectrum of α is approached through the convergents of α, so the continued fraction organizes it. We computed the bands at the convergents of the five numbers of the Suite (Table 18.4).

Table 18.4. Energy bands of Harper's equation at convergents p/q: the sizes of the groups into which the widest gaps divide the q bands, and the total width of the bands against Thouless's estimate $32G/(\pi q) = 9.33/q$ (G is Catalan's constant).

number	p/q	groups	total width	$9.33/q$
$\pi - 3$	1/7	1, 1, 1, 1, 1, 1, 1	1.2845	1.3329
	15/106	15, 15, 15, 16, 15, 15, 15	0.0888	0.0880
	16/113	16, 16, 16, 17, 16, 16, 16	0.0825	0.0826
$\varphi - 1$	8/13	5, 3, 5	0.7116	0.7177
	13/21	8, 5, 8	0.4438	0.4443
	21/34	13, 8, 13	0.2753	0.2744
$\sqrt{2} - 1$	12/29	5, 7, 5, 7, 5	0.3216	0.3217
	29/70	12, 17, 12, 17, 12	0.1333	0.1333
$e - 2$	51/71	20, 31, 20	0.1314	0.1314
$\log_2(3/2)$	7/12	2, 3, 2, 3, 2	0.7832	0.7775
	31/53	9, 13, 9, 13, 9	0.1760	0.1760

For π the 113 bands fall into seven groups of 16, with one of 17 in the middle: the bar of 7 and the phrase of 113 of Chapter 3, as an energy spectrum. For the golden ratio the groups are Fibonacci numbers, for $\sqrt{2}$ they are built from the Pell numbers $2, 5, 12, 29$ (with $7 = 5 + 2$ and $17 = 12 + 5$), and for the fifth the twelve bands of 7/12 divide as $2, 3, 2, 3, 2$. The total width shrinks like $9.33/q$ [128], which is the Cantor dust appearing.

Each gap carries a whole number. If r bands lie below a gap, there is exactly one pair of integers (s, t) with $|t| \le q/2$ and

$$r = q\,s + p\,t,$$

found by Euclid's algorithm, and t is the Hall conductance of the electron gas when the gap is at the Fermi energy, in units of e^2/h [129]. For the six widest gaps of 16/113 these integers are $1, 2, 3, -3, -2, -1$. So the oldest algorithm in the book produces a quantity that laboratories measure to parts in a billion.

An interactive version, in which any continued fraction can be entered, its levels marked on the butterfly and its bands played from low energy to high, accompanies this chapter.

18.5 Kepler, continued

Kepler has now appeared twice: his third law in Section 18.1 and his harmony of the world in Section 18.2. Three more pieces of his work meet the methods of this paper, and in each the continued fraction adds something he did not have.

The trigon is a continued fraction. On 19 July 1595 Kepler drew for his class in Graz the places in the zodiac where Jupiter and Saturn meet, about every twenty years [47]. Each meeting falls roughly two thirds of the way round from the last, so three of them make a triangle, the *trigon*, and the triangle slowly turns. The drawing gave him the idea for his first book [68]. It is a picture of a continued fraction.

Proposition 18.6 (conjunction figures)**.** *Let two bodies move on circular orbits in one plane with periods $T_1 < T_2$, and let $r = T_2/T_1$. Successive conjunctions are separated by the time $T_1T_2/(T_2 - T_1)$, and each lies*

$$\alpha = \frac{1}{r-1}$$

of a turn beyond the last. Hence:
(i) the first N conjunctions cut the circle into arcs of at most three different lengths, for every N;
(ii) if p/q is a convergent of the fractional part of α and a is the term that follows it, then every q conjunctions nearly close a figure with q points, which has turned by $q\alpha - p$ of a circle; and the figure has turned through the spacing of its own points, $1/q$ of a circle, after $1/(q^2|\alpha - p/q|)$ rounds, a number between a and $a + 2$.

Proof. Between conjunctions the inner body gains one turn on the outer, so the time t satisfies $t/T_1 - t/T_2 = 1$, and in that time the outer body moves $t/T_2 = 1/(r-1)$ turns. The conjunction points are therefore the orbit of a rotation by α. (i) is the three-gap theorem of Section 3.6, and (ii) is the bound $1/(q(q' + q)) < |\alpha - p/q| < 1/(qq')$ of Chapter 2, where $q' = aq + q''$ is the next denominator and $q'' \le q$ the previous one. □

So the figure is the convergent, and *the next term counts how many times the figure is drawn before it has turned by one point.* This is the normalized error of Table 2.1 again. Table 18.5 applies it to the planets, with long-term mean periods; real orbits are not circles, so real conjunctions wander a few degrees about these places.

Table 18.5. Conjunction figures from mean periods. Each line gives the step α between meetings, its expansion, and the cycles that the convergents produce. An opposition of an outer planet is a conjunction of the Earth with it, seen from the Sun.

pair	step α	cycles
Jupiter, Saturn	$0.6742 = [0; 1, 2, 14, 2, \ldots]$	2/3: a triangle every 59.6 years, turned by 8.1°; by 120° after 14.8 rounds, 883 years. 29/43: 853.9 years.
Venus, Earth	$0.5987 = [0; 1, 1, 2, 29, 2, \ldots]$	3/5: a five-pointed star every 7.99 years, turned by 2.4°; by 72° after 29.9 rounds. 88/147: 235.0 years. 91/152: 243.00 years.
Earth, Mars	$0.1353 = [0; 7, 2, 1, 1, 3, \ldots]$	7 oppositions in 14.95 years; 15 in 32.0; 22 in 47.0; 37 in 79.01.
Earth, Jupiter	$0.0921 = [0; 10, 1, 6, 4, \ldots]$	11 oppositions in 12.01 years; 76 in 83.00.
Earth, Saturn	$0.0351 = [0; 28, 2, 5, \ldots]$	28 oppositions in 29.0 years; 57 in 59.00.
Mercury, Earth	$0.3172 = [0; 3, 6, 1, 1, 2, 1, \ldots]$	22 meetings in 6.98 years; 41 in 13.01; 104 in 32.99; 145 in 46.00.

The first line is Kepler's trigon. The convergent 2/3 is the triangle; the term 14 that follows says the triangle is drawn about fifteen times before it has turned onto itself, which takes 883 years. That is exactly the period of the great inequality of Section 18.2: the turning of the trigon and the slow swing of Jupiter and Saturn are one beat, with frequency $5n_S - 2n_J$ in the two mean motions, seen in two ways. This is the "great cycle" of the conjunctions, which medieval astrologers put at 960 years and Kepler at 794. Kepler's figure is forty conjunctions (794.4 years), and 27/40 is an intermediate fraction, the last step before the next convergent 29/43. The term 14 is delicate. Jupiter disturbs Saturn's motion, and with the period of Saturn's present orbit in place of its long-term mean the term is 13 and the turn takes 842

years. The second line is the five-pointed star of Venus, with the large term 29 behind it; the fraction 91/152, the first intermediate step beyond the convergent 88/147, gives 243.00 years, the period after which transits of Venus across the Sun repeat. Kepler predicted the transit of Venus of 1631. The remaining lines reproduce periods that Babylonian astronomers already used to predict the planets: 8 years for Venus, 46 for Mercury, 47 and 79 for Mars, 83 for Jupiter and 59 for Saturn [53]. Their second period for Jupiter, 71 years, is 65 oppositions: the fraction 6/65 is not a convergent but an intermediate fraction, the fifth of the six steps that lead from 1/10 to 7/76, on one of the veins of Section 17.6.

The figures themselves are Kepler's too. In the first two books of *Harmonices Mundi* he studied the regular polygons, star polygons among them, and how they fit together; the figure for a convergent p/q is the star polygon with q points joined p apart. He had both halves, the stars and the conjunctions; the continued fraction is what joins them. For Jupiter's moons the same proposition gives the resonance of Section 18.2: the meeting points of Io with Europa, and of Europa with Ganymede, step backwards round Jupiter at the same rate, 0.7395° a day, once round in 487 days.

An interactive version, "Kepler's Figures", accompanies this section: any two bodies can be chosen and their meetings watched as they draw the figure and turn it. With Kepler's ellipses in place of circles it puts the meeting of Venus and the Earth at 2012.43, the transit of June 2012, and the meeting of Jupiter and Saturn as seen from the Sun at 2020.84, seven weeks before the two were seen together from the Earth.

Kepler's harmonies, tested. In Book V of *Harmonices Mundi* [70] Kepler compared each planet's fastest angular speed as seen from the Sun, at perihelion, with its slowest, at aphelion, and heard a musical interval in each ratio. For an orbit of eccentricity e the ratio is $((1+e)/(1-e))^2$. He printed the speeds he used (for Saturn $1'46''$ and $2'15''$ of arc a day, for the Earth $57'3''$ and $61'18''$), and he printed his own misses beside them: Saturn's ratio exceeds a major third by 53 : 54, Mars falls short of a fifth by 29 : 30. Table 18.6 sets three versions side by side: his own figures, the orbits as they were in 1600, and the orbits of today [123].

Table 18.6. Kepler's intervals for the single planets, and by how much the ratio of fastest to slowest angular speed misses each one, in cents (a semitone of the piano is 100 cents). The orbits of 1600 and 2000 are mean orbits from one formula valid from 3000 BC to 3000 AD.

planet	Kepler's interval	his figures	1600	2000	e in 2000
Saturn	5 : 4, major third	+32	+7	−2	0.0555
Jupiter	6 : 5, minor third	+32	+16	+21	0.0485
Mars	3 : 2, fifth	−60	−56	−53	0.0934
Earth	16 : 15, semitone	+13	+5	+4	0.0167
Venus	25 : 24, diesis	−21	−22	−24	0.0068
Mercury	12 : 5, octave + minor third	−43	−72	−71	0.2056
mean miss		33	30	29	
chance of doing as well		0.22	0.15	0.14	

The last line is the test of Section 18.2: six ratios drawn at random between a unison and a twelfth, each matched to the nearest of sixteen intervals of the just scale (the kinds Kepler drew on), miss by 47 cents on average and do as well as the planets about one time in six. By that test the harmonies of the single planets are not established, in his figures or in ours. The Earth's semitone, the "mi, fa, mi" that Kepler heard as misery and famine, and Saturn's major third are the close fits; Mars and Mercury miss by half a semitone or more.

Did they fit better in his day? No. The sky of 1600 held the same six planets on almost the same orbits: the eccentricities have changed in the fourth decimal place, and the fastest mover, Saturn's interval, drifts by about 2 cents a century. The columns for 1600 and 2000 differ by 9 cents for Saturn and by 5 or less for the rest. Where Kepler's own column differs from the orbit of 1600 the cause is his data, and it cuts both ways. His figure for Saturn's slowest motion, $1'46''$ where the orbit of 1600 gives $1'48''$, cost him 25 cents. His eccentricity for Mercury, 0.2095 against a true 0.2056, made Mercury fit 29 cents better than it should. For Mars, the Earth and Venus his speeds are within a quarter of a per cent of the modern reconstruction, a tribute to Tycho Brahe's observations. (One caution about Saturn: Jupiter disturbs its orbit so strongly that its eccentricity depends on the span averaged over. A fit to the years 1800 to 2050 alone gives a miss of −13 cents where the long formula gives −2.)

Why they cannot hold. Nothing fixes an eccentricity at a musical value. The planets trade eccentricity among themselves over tens and hundreds of thousands of years. The Earth's has ranged from nearly 0 to 0.068 [80], which carries its "semitone" from a unison to within a quarter-tone of a perfect fourth. On the present trend the Earth will sing a true 16 : 15 around the year 3600, and Saturn's major third is passing through exactness about now. Period ratios are different: a resonance holds them (Section 18.2). The harmony Kepler was looking for exists where there is something to hold it. One such place is the family of Jupiter's moons, which he was among the first to study and for which he coined the word *satellite*.

Two planets at a time. Kepler set more store by the intervals between neighbours: the slowest motion of the outer planet against the fastest of the inner ("diverging") and the fastest of the outer against the slowest of the inner ("converging"), ten intervals in all, such as 1 : 2 for Saturn and Jupiter converging and 2 : 3 for Mars and the Earth. Here the test comes out differently. With the orbits of 2000 the ten intervals miss his consonances by 33 cents on average, where random intervals miss the nearest consonance by 57; the chance of doing as well is about 0.03. It is 0.015 if the same six eccentricities are dealt to the planets in a different order (all 720 orders), and 0.03 if the eccentricities are drawn at random. As Kepler himself concluded, the converging intervals fit best (23 cents) and the diverging ones worst (42). Mars and the Earth converging make a fifth to within 2 cents.

That would be a modest result in Kepler's favour, if it were the end of the evidence. It is not, because planets were found after he died, and they are a test his scheme never had: Uranus (1781), Neptune (1846), and the smaller bodies Ceres (1801) and Pluto (1930). As single voices Uranus and Neptune do about as well as Jupiter and Venus (9 cents from a minor third and from a diesis), while Ceres and Pluto miss by 47 and 62. But the four new intervals between neighbours, for Saturn with Uranus and Uranus with Neptune, miss the nearest consonance by 86, 55, 154 and 90 cents, worse than chance. With all fourteen intervals the mean miss is 51 cents and the chance of doing as well is 0.29. Kepler also arrived at the angular speeds seen from the Sun only after trying and rejecting distances, periods and linear speeds, which weakens any single test. So the fair verdict is that his two-planet harmonies were a real feature of the six planets he knew, at about the level of one chance in thirty, and that the planets he did not know do not continue it. (A further planet far beyond Neptune has been proposed since 2016. None had been found when this was written, so it cannot be tested.)

The second law in the atom. Kepler's second law, equal areas in equal times, is the conservation of angular momentum: a body of mass m and angular momentum L sweeps area at the rate $L/(2m)$. Bohr's rule for the atom is $L = n\hbar$. So the electron in orbit n sweeps area

at

$$n \cdot \frac{\hbar}{2m_e} = n \times 5.79 \times 10^{-5}\ \mathrm{m}^2/\mathrm{s},$$

a whole number of times one fixed rate. (The Earth sweeps $2.2 \times 10^{15}\,\mathrm{m}^2/\mathrm{s}$.) Kepler's second law is what Bohr quantized. In Sommerfeld's refinement of 1916 [120] the electron moves on Kepler ellipses, and the ellipses allowed at level n are those whose short and long axes are in the ratio k/n, $k = 1, \ldots, n$: the shapes of the orbits are fractions. That picture was replaced by quantum mechanics ten years later, but the count of states and the symmetry of Section 18.1 survived the change.

What else of Kepler's belongs here. Five more pieces of his work touch the subjects of this paper.

The golden ladder. In his essay on the six-cornered snowflake [69] Kepler observed that the ratios of neighbouring Fibonacci numbers, $5:8$, $8:13$, $13:21$, come ever closer to the golden section. Those ratios are the convergents of $[1; 1, 1, 1, \ldots]$ (Chapter 2): the golden movement of the Suite, three centuries early.

Consonance and the totient. In the first three books of *Harmonices Mundi* Kepler derived the consonances from the regular polygons that can be constructed with ruler and compass. The triangle, square, pentagon, hexagon and octagon give the consonant ratios; the heptagon cannot be constructed and, he argued, that is why 7 gives no consonance [47]. Gauss and Wantzel later settled which polygons can be constructed: the n-gon can exactly when $\varphi(n)$ is a power of 2. That is the totient of Chapter 5, the same function that weights the correlation matrix (Proposition 15.6) and that appeared in the ratio $n/\varphi(n)$ of Proposition 18.5. The polygons Kepler called knowable, with $3, 4, 5, 6, 8, 10, 12, \ldots$ sides, are in modern terms among those with $\varphi(n)$ a power of 2.

Tilings that never repeat. Book II of *Harmonices Mundi* lists the eleven tilings of the plane by regular polygons, and draws a pattern of pentagons and stars that anticipates the non-repeating tilings of Penrose (1974). In those tilings the two kinds of tile occur in the golden ratio, and since 1984 patterns of this kind have been known in real materials, the quasicrystals [118]. The equation of Section 18.4 is the standard model of an electron in a one-dimensional pattern of this kind.

Packing, and π. The snowflake essay also asserts that no packing of equal spheres is denser than the greengrocer's pile, which fills $\pi/\sqrt{18} = 0.7405$ of space. That is how the atoms sit in many metals. The claim was proved by Hales [55].

The first inverse-square law. In his optics of 1604 Kepler stated that the brightness of a light falls off as the square of the distance. The forces of Section 18.1 follow the same law, and it is the reason an orbit in the atom and an orbit in the sky are one problem.

Two tools of his are also at work in the instrument that accompanies this section. Kepler's equation, $M = E - e\sin E$, gives the place of a planet on its ellipse at a given time; the instrument solves it for every frame it draws. And his logarithms of 1624 are the natural measure of a musical interval: the cents used above are a logarithm, and $\log_2(3/2)$ is one of the five numbers of the Suite.

18.6 What is known and what is ours

Still open, and the one item here that could become a theorem: does the ratio of $1/\lambda_{\min}$ to $\lambda_{\max}$ tend to a limit, and if so to what? Propositions 18.2 and 18.4 reduce it to comparing

idea	class	notes
Kepler's law in the atom; the $2n^2$ shells from the Runge–Lenz symmetry	classical	Pauli, Fock. The reading as a repair of the "shared formula" caveat of Chapters 5 and 12 is ours.
Atoms with $n \approx 1009$ and the fourth harmonic	specialist	Stepkin et al.; the arithmetic of the harmonic is our check.
Orbital resonances, Kirkwood gaps, KAM	classical	Standard celestial mechanics.
The big-term test and Table 18.1	new presentation	Continued fractions of period ratios are an old device; the test against random ratios is our computation.
The sign mirror, Proposition 18.2	new, as far as we found	A two-line consequence of Proposition 17.4. May well be known for GCD matrices.
The one-prime chain and its mirror map	classical	The Kac–Murdock–Szegő matrix and the Poisson kernel.
Hofstadter's butterfly, Hall integers, Cantor spectrum	specialist	Hofstadter; Thouless et al.; Avila and Jitomirskaya.
The butterfly read as a number's musical form; the instrument	new presentation	Our computation and instrument.
The harmonic form, Proposition 18.4	new, as far as we found	One line from Proposition 15.6. The harmonic numbers make it worth stating.
The ratio of the two ends, Tables 18.2 and 18.3	new computation	To $m = 3 \times 10^7$. The exact cases of Proposition 18.5 are elementary.
Conjunction figures, Proposition 18.6	new presentation	The cycles are ancient and the three-gap theorem is classical; reading the trigon as a convergent and its drift as the next term is ours.
Kepler's harmonies tested, Table 18.6	new computation	Others have re-examined them; the tests against random ratios, the comparison of 1600 with 2000, and the test on the planets found later are ours.
Constructible polygons and $\varphi(n)$; quasicrystals; sphere packing	classical	Gauss, Wantzel; Penrose, Shechtman; Hales. Only the pointers to this paper are ours.
The second law in Bohr's atom; Sommerfeld's ellipses	classical	Bohr 1913, Sommerfeld 1916.

the Perron eigenvalues of two explicit non-negative matrices of the same kind, both built from the common multiples of pairs of rhythms. The computation of Section 18.3 says the answer for the gcd matrix is close to 2.57 at every size we can reach, that the correlation matrix is converging on the same value from above, and that for any fixed set of primes the answer is 1.

APPENDIX A

Algorithms

This appendix gives the main algorithms in pseudocode. The Python scripts of Appendix C implement each one.

Algorithm 1: Certified continued-fraction terms of a real constant

Input: a procedure that evaluates x to d decimal digits; the number of terms n
Output: the first n terms of x, each guaranteed correct

repeat
  evaluate x at precision d and at precision $d + 200$;
  expand both: $a \leftarrow \lfloor y \rfloor$, $y \leftarrow 1/(y - a)$, repeated;
  keep the common prefix of the two term lists, minus its last term;
  if *fewer than n terms are kept* **then** $d \leftarrow 1.6\,d$;
until *at least n terms are certified*;

Algorithm 2: Convergents (Proposition 2.1)

Input: terms $a_0, a_1, \ldots, a_N$

$p_{-1}, q_{-1}, p_0, q_0 \leftarrow 1, 0, a_0, 1$;
for $n \leftarrow 1$ **to** N **do**
  $p_n \leftarrow a_n p_{n-1} + p_{n-2}$; $q_n \leftarrow a_n q_{n-1} + q_{n-2}$;

Algorithm 3: The rhythm of a number (standard words, Theorem 3.5)

Input: terms $a_1, a_2, \ldots, a_k$ of $\alpha = [0; a_1, a_2, \ldots]$
Output: the level words $s_0, \ldots, s_k$; s_n has q_n beats and p_n hits

$s_{-1} \leftarrow$ `x`; $s_0 \leftarrow$ `.`;
$s_1 \leftarrow s_0^{a_1 - 1}\, s_{-1}$;
for $n \leftarrow 2$ **to** k **do**
  $s_n \leftarrow s_{n-1}^{a_n}\, s_{n-2}$; `// repeat, then close`

Algorithm 4: Bjorklund's algorithm for the Euclidean rhythm $E(k, n)$

Input: $1 \le k < n$

$A \leftarrow k$ groups `[x]`; $B \leftarrow n - k$ groups `[.]`;
while *B has more than one group* **do**
  $m \leftarrow \min(|A|, |B|)$;
  append the ith group of B to the ith group of A, for $i = 1, \dots, m$;
  the unused groups (from A if $|A| > m$, else from B) become the new B;
return *the concatenation of the groups of A and then of B*

Algorithm 5: Lehmer–Powers square search on the continued fraction of $\sqrt{kN}$

Input: odd composite N, multiplier k

$a_0 \leftarrow \left\lfloor \sqrt{kN} \right\rfloor$; $P \leftarrow 0$; $Q \leftarrow 1$; $a \leftarrow a_0$; $A_{-1} \leftarrow 1$; $A_0 \leftarrow a_0 \bmod N$;
for $n \leftarrow 1, 2, \dots$ **do**
  $P \leftarrow aQ - P$; $Q \leftarrow (kN - P^2)/Q$; $a \leftarrow \lfloor (a_0 + P)/Q \rfloor$;
  if *n is even and $Q = s^2$ for an integer s* **then**
    $g \leftarrow \gcd(A_{n-1} - s, N)$;
    if $1 < g < N$ **then return** g;
  $A_n \leftarrow (aA_{n-1} + A_{n-2}) \bmod N$;

Algorithm 6: Counting prime convergents of random reals (Chapter 8)

Input: number of reals M, number of convergents N

for $i \leftarrow 1$ **to** M **do**
  draw a uniform integer $r \in [1, 10^D)$ with $D = \lfloor 1.6N \rfloor + 20$;
  expand $r/10^D$ by exact Euclidean division; keep the first $N + 1$ terms;
  compute p_n, q_n for $n \le N$ (Algorithm 2);
  foreach $n \le N$ **do**
    $E \leftarrow E + \zeta(2)/\ln q_n$; `// heuristic expectation`
    if *q_n has no prime factor below* 3000 *and passes Baillie–PSW* **then**
      count q_n as prime

Algorithm 7: Hausdorff dimension of a continued-fraction Cantor set

Input: allowed terms A; number of Chebyshev nodes m

nodes $x_1, \dots, x_m$ on $[0, 1]$; interpolation matrix V^{-1} for polynomials of degree $< m$;
`rho(`s`)`: matrix $M = \sum_{a \in A} \operatorname{diag}\big((a + x_i)^{-2s}\big)\, V_a V^{-1}$, where V_a evaluates the basis at $1/(a + x_i)$; **return** *spectral radius of M*;
bisect on $s \in [0, 1]$ for $\rho(s) = 1$;

Algorithm 8: Number variance of an unfolded point set

Input: sorted points $u_1 < \cdots < u_N$ with mean spacing 1; window lengths L; trials T

foreach L **do**
    draw T uniform window starts t in $[u_1, u_N - L]$;
    count the points in each window $[t, t + L)$ by binary search;
    $\Sigma^2(L) \leftarrow$ sample variance of the counts;

APPENDIX B

Computational methods

Software. All computations use Python 3 with mpmath 1.3 (arbitrary-precision reals and the zeros of the zeta function), sympy (factorizations, small primality tests), gmpy2 (fast big-integer arithmetic and the Baillie–PSW test), NumPy and SciPy (linear algebra, statistics), and matplotlib (figures). The typesetting uses XeLaTeX with BibTeX.

Continued fractions. The terms of π come from π evaluated to 11,600 decimal digits, expanded by the Gauss map at two working precisions; only terms on which the two expansions agree, less one, are kept (Algorithm 1), which certifies the first 10,000 terms. The constants e, $\sqrt{2}$, φ and $\log_2(3/2)$ are expanded the same way to 4,001 terms, with the precision raised automatically when needed. The pattern of e and the constant terms of $\sqrt{2}$ and φ are asserted, which checks the method.

Rhythms. Rhythms are built with Algorithm 3 and compared, letter for letter, with the floor rule (3.1) evaluated in multiple precision: 140,000 beats for $\pi - 3$ and complete words for 200 random term sequences.

Primality. A number is called prime if it is below 3,000 and in the list of small primes, or if it has no prime factor below 3,000 (one gcd with the primorial) and passes the Baillie–PSW probable-prime test of gmpy2. For the Fibonacci and Pell indices our lists agree with the OEIS [103, 108], and for π with [105, 106].

Long expansions. Chapter 15 uses 12,000,001 terms of π, from a 43-million-bit interval around π expanded by a divide-and-conquer form of Euclid's algorithm that keeps only the terms common to both ends of the interval (Section 15.1); the expansion is repeated at a second precision and checked against the published record terms. Primality tests there use trial division by the primes below 10^5 before Baillie–PSW. Random reals in that chapter are random fractions with 2,600 or more binary digits, expanded exactly. [Added 2026-11-05]

Primality, third procedure. For $20{,}000 < n \le 38{,}571$ for the denominators and $20{,}000 < n \le 42{,}037$ for the numerators (Chapter 17; Teddy Rook's own searches, with the same sieve, reach 50,000 for both) the convergents first pass through the rhythm sieve: the recurrence is run modulo every prime below 10^7, and any q_n or p_n with such a factor is dropped (about

95% of them). Each survivor gets a strong probable-prime test to base 2 and then the strong Baillie–PSW test of gmpy2. The intermediate fractions of Section 17.6 are sieved the same way along their arithmetic progressions. A proof of primality, when one is wanted, needs an elliptic-curve certificate (ECPP), which none of the new primes has yet. [Added 2026-11-10]

Random reals. A random real is represented by $r/10^D$ with r a uniformly random integer of $D = 1{,}620$ digits (Algorithm 6). Its first 1,000 terms are shared by every real in $[r/10^D, (r+1)/10^D)$, as explained in Chapter 8. The generator is Python's `random.Random` with a fixed seed, so the sample is reproducible.

Zeta zeros. The first 5,000 zeros were computed with `mpmath.zetazero` at 20 digits; the script checks that `mpmath.nzeros` counts exactly 5,000 zeros up to the last one, so none is missing, and recomputes a few at higher precision. The explicit-formula figure uses the first 100 zeros.

Random matrices. GUE matrices are $H = (A + A^*)/2$ with A a 1000×1000 matrix of independent standard complex Gaussians, so that $E|H_{ij}|^2 = 1$ off the diagonal; eigenvalues are divided by $\sqrt{1000}$ (semicircle of radius 2) and unfolded with the semicircle distribution function, keeping the central half.

Statistics. Chi-square tests use SciPy's χ^2 survival function; the Kolmogorov–Smirnov distance is the two-sample statistic; Poisson tail probabilities use SciPy. The pair-correlation histogram counts all pairs within 3 mean spacings among the first 6,000 points of each set; the number variance uses 1,500 to 4,000 random windows per length (Algorithm 8).

Structure factors. $S(k)$ is computed directly from (6.1) on a grid of 40,000 wavenumbers, with positions measured from the start of the window; the figures use a Hann taper (weights $2\sin^2(\pi x/\ell)$ across the window of length ℓ) to suppress side lobes, while the peak values quoted in the text are untapered and evaluated exactly at the rational wavenumbers.

Circle map. Rotation numbers are averages over 2,520 iterations (a multiple of every $q \le 10$, so locked orbits give exact fractions) after a transient of 400; step widths are measured on a grid of 40,001 values of Ω.

Dimension. Algorithm 7 with 40 and 60 Chebyshev nodes; the two agree to 13 digits for $E_{\{1,2\}}$, and the 60-node value matches the published $0.53128050627720514\ldots$ [66] to the twelve digits we print.

APPENDIX C

Reproducibility

The full source of this work, including the LaTeX files, the bibliography and every script, is kept with the companion site's source under `docs/papers/rhythm-of-numbers/`. Running the scripts in order regenerates every figure (`paper/figures/`) and every number in the text: each script writes a file of LaTeX macros to `paper/gen/`, and the text uses only those macros for computed values. Scripts stop with an error if any asserted fact fails.

Table C.1. The scripts, in the order `run_all.sh` runs them, with what each computes.

script	computes
`common.py`	shared helpers: certified continued fractions, convergents, standard words, the floor rule, macro output, figure style
`c1_cf.py`	terms of π, e, $\sqrt{2}$, φ, $\log_2(3/2)$; the starting guess; convergent table; Legendre and best-approximation checks; nested-block checks; beats 32,996 and 33,102; three-gap check; Gauss–Kuzmin, Khinchin and Lévy statistics
`c2_music.py`	equal temperaments, comma, thirds; Bjorklund vs. Christoffel; Euclidean table; circle map, tongues, staircase and Farey rule; Farey counts
`c3_coprime.py`	orchard, Euler product, correlation formula and matrix, Balmer, residue bias to 10^7, sunflower and wheels, structure factors, explicit formula
`c4_tools.py`	Lehmer–Powers factoring, necklaces, the prime hunt and its Poisson tails
`zeros.py`	the first 5,000 zeta zeros (about one hour)
`c5_spacing.py`	spacing distributions, pair correlation, number variance, the Fourier picture of the zeros
`c6_fractals.py`, `c6b_gcf.py`	complete quotients, Ford circles, Stern–Brocot, $?(x)$, dimensions, Mandelbrot bulbs, Fibonacci word, six pictures; patterned fractions for π
`c7_primeconv.py`, `c7b_primeconv_figs.py`	prime convergents of five constants and 400 random reals (several minutes); divisibility shares; figures and tables
`c8_suite.py`	the Suite's forms checked against the continued fractions; form figure
`c9_misc_figs.py`, `c10_connections.py`	remaining figures; drum-mode ratios

Table C.1, continued

script	computes
`c11_openq_picf.py` [Added 2026-11-05]	12 million certified terms of π by divide-and-conquer, two precisions, checked against the OEIS records (about two minutes)
`c11_openq_pisearch.py` [Added 2026-11-05]	probable-prime q_n and p_n of π for $n \le 20{,}000$ (resumable; about 45 minutes)
`c11_openq_stats.py` [Added 2026-11-05]	Gauss–Kuzmin, pair, tail and block tests, Khinchin, Lévy and Galambos for 12 million terms, calibrated on 1,000 random reals (about 3 minutes)
`c11_openq_model.py` [Added 2026-11-05]	the exact expectation of Theorem 15.3 and its check by brute force and by 20,000 random reals; the densities for e (about 2 minutes)
`c11_openq_matrix.py` [Added 2026-11-05]	spectra of the correlation matrix to $m = 6{,}000$, the top eigenvalue to $m = 10^6$ (about 2 minutes)
`c11_openq_edo.py`, `c11_openq_primes.py` [Added 2026-11-05]	equal-temperament rankings and tests; tables and figure for π's prime convergents and for Fibonacci and Pell primes
`c12_openq2.py` [Added 2026-11-06]	Chapter 16: the sets E_q and their overlaps for primes below 3,000; e modulo primes; near-copies, spacing ratios and bounds for the correlation matrix; the commas of 171; the Doeblin–Lenstra and serial-correlation tests, calibrated on 300 random reals (about 3 minutes)
`c12_third_pisearch.py` [Added 2026-11-10]	Chapter 17: the search of q_n, $20{,}000 < n \le 25{,}332$, after Teddy Rook's own search (resumable; about an hour)
`c12_third_pi.py` [Added 2026-11-10]	q_{25332} rebuilt from two independent sources of π, checked against the site's copy and tested; the heuristic to $n = 25{,}332$; the digits of Appendix D (about 3 minutes)
`c12_third_sieve.py` [Added 2026-11-10]	the rhythm sieve and the search of q_n ($25{,}332 < n \le 38{,}571$) and p_n ($20{,}000 < n \le 42{,}037$), in several runs (resumable; a few minutes of sieving, then about an hour of tests per run)
`c12_third_newprimes.py` [Added 2026-11-10]	q_{27651} and q_{32394} rebuilt and tested; the counts to $n = 32{,}394$; the odds of a fourteenth; two versions of the pattern tests; the mod-12 motif; the next veins of intermediate fractions; the digits of Appendix D (about 5 minutes)
`c12_third_fourteenth.py` [Added 2026-11-10]	q_{37721} rebuilt and tested; the counts to $n = 40{,}000$; the cluster; the gaps and the odds of a fifteenth; the notes of the prime convergents and their comparisons; the family $(22k+3)/(7k+1)$ and the Bateman–Horn constant; the digits of Appendix D (about 5 minutes)
`c12_third_fifty.py` [Added 2026-11-10]	p_{42037} rebuilt and tested; the counts to $n = 50{,}000$; the tests and odds there; the notes with sixteen numerators; the digits of Appendix D (about 6 minutes)
`c12_third_rate.py` [Added 2026-11-10]	Section 17.7: π's own constant with and without $\zeta(2)$; the rhythm sieve over all $n \le 50{,}000$ and its survivors; the voices test on two million terms, calibrated on 800 random reals (about 40 minutes)
`c12_third_inter.py` [Added 2026-11-10]	prime intermediate fractions at levels 1 to 3,000, with their notes (about a minute) and at level 15,542 (about an hour, resumable)
`c12_third_e.py`, `c12_third_matrix.py` [Added 2026-11-10]	closed forms, forced divisors and periods for e; extreme eigenvalues, bounds and the trace of the inverse of the correlation matrix to $m = 10^6$ (about a minute)

Table C.1, continued

script	computes
`c13_large/verify_ch18.py` [Added 2026-11-10]	Chapter 18: recomputes every number printed in the chapter (250 checks, about two minutes; `--big` adds the 3×10^7 row of Table 18.2, about 4 GB of memory). These are Teddy Rook's own scripts, kept as he wrote them; the chapter's numbers are typed in its text and not inserted by macros
`c13_large/`: `periods.py`, `resonance.py`, `check_numbers.py` [Added 2026-11-10]	the table of orbital periods with its sources; Table 18.1 and the tests against random ratios; the Kirkwood positions and Saturn's ring resonance
`c13_large/`: `mirror.py`, `eig.py`, `ratio.py`, `fast.py`, `tr.c` [Added 2026-11-10]	Section 18.3: the sign mirror, the harmonic form and the two ends of the spectrum to 3×10^7 (the C file is built by the scripts)
`c13_large/`: `butterfly.py`, `kepler.py`, `harmonies.py` [Added 2026-11-10]	Table 18.4; conjunction figures and the second law in the atom; Kepler's harmonies in his figures, in 1600 and in 2000, and for the bodies found later
`c13_large/indep_check.py` [Added 2026-11-10]	a separate spot check, written without the scripts above, of the Bohr numbers, seven period ratios, Laplace's relation, the two conjunction steps, Propositions 18.2 and 18.4 at $m = 80$, the first row of Table 18.2 and three band widths

Building. With Python 3 and the packages of Appendix B installed, run `sh run_all.sh` in `compute/`, then `latexmk -xelatex main.tex` in `paper/`. Random choices use fixed seeds, so a rebuild reproduces the same numbers, except that the GUE samples (seeded too) and timings may differ across library versions in the last digit.

Provenance of non-computed facts. Dates, names and bibliographic details were checked against the sources listed in the bibliography. The musical facts about the Suite come from the companion site's data file `data/music/suite.yaml`, compiled from the engraved score.

APPENDIX D

The digits of the probable primes

Added 2026-11-10.

The eleventh: q_{25332}

These are the 13,017 decimal digits of q_{25332}, the denominator of the 25,332nd convergent of π (Section 17.5), in rows of 80 digits and groups of 10. The SHA-256 hash of the digits written without spaces or line breaks is

`73f98bd668e4c94394d07e7846ca7643d6ecf617317850bcbff60ac260ad7974`.

It is a probable prime: it passed the strong Baillie–PSW test and 25 Miller–Rabin rounds, and a proof of its primality is in progress. The same digits are on the companion site as a plain text file, one line with no spaces [115]. Anyone can rebuild the number from π's continued fraction with the script `c12_third_pi.py`.

```
2759256854 5368344821 6470675444 9619863621 3374710001 5410953258 8150875771 4922843075
6468607451 4718994153 1458542475 4389868197 8938385516 3290758043 9277925476 6223489742
9373949113 8449506040 6579614121 9961078695 0041028984 4076278917 7427057150 0386126115
0217895325 0996733181 7240139065 9037150094 7041729980 9163350607 8869395877 0625000590
1708799103 6112571362 9040756532 1309385501 2452818308 4079421178 3114350677 4002310401
9206798935 7145440912 1328637722 1954188647 8199619166 0808031767 3381687809 9797374850
6442838213 6498005428 6431487795 1852843322 0125178020 5594132541 6029788872 3678811596
6501444739 2386707147 1513935090 1791988211 5034184103 3037057438 0494932063 9534298373
8905476908 6363357487 1004018149 4700171961 7005612928 2542341286 1931132337 7330847163
8762470684 1471236347 8766722915 6863492516 2118468604 6179626560 9286014046 2382389278
9604008836 7736960797 5496445899 6330879438 5042250815 3892106036 0421769039 4850309337
5520690352 7433771935 1690479524 5934433689 3698991169 9064494886 0329436509 8237107348
2173844891 5183469838 7634874291 3999144222 0714657869 9877545633 9740121403 1804694362
1706609658 6720186459 1949834510 8637333357 5175532585 6607747266 9510761471 9756528566
1004749603 9322064679 5160044600 6245525768 2211990616 9047124687 8881922505 5440485936
7944852066 3492352683 4932655969 0560843324 6743702622 9929736120 0896493351 7804042931
3870088938 3445263177 1189105765 2676079868 4428227801 5341081118 7572995746 8710933669
5190428570 4169704817 7128095999 2760603088 9337040918 6934436055 4792403153 9783774746
4959154063 9597032884 2934891583 1731970676 7034148709 4010339499 3974793466 3554558191
1648239820 8579156302 0833443365 8817634601 0796457793 9315257340 7657162007 4801951981
8262780979 4499240873 1406424812 1260901490 2794588741 7918035623 8227568087 7431518294
6255215563 9646983996 6602317316 2165153080 6211092417 9627740721 2193742151 9794003769
7123657851 3400379150 2885978740 7006571330 7281085592 8801282121 3617731982 4446205603
9245276314 5808355460 1529541235 4625695638 8439927457 7290673931 1568232783 6194545295
3191733899 8387670459 9249187700 7106503392 5160499136 4953137413 2241429021 5298216967
```

8656390829 5095915909 2320016761 2974207673 1740435043 4145514244 5673938790 3195678819
7243840343 8084519933 7049616407 0637539295 1860163769 7503193132 7983271430 7922347622
3642963110 1359811050 3625526097 6005879133 6763136401 7977731962 9984302354 9933853297
3287728670 1386181825 0662701691 9355934052 1355461423 6663258076 3878051781 5492739386
4224857519 5540786410 0550404581 2914420021 7446603282 6291761424 7472817074 7143675355
8467165482 9946724079 5690689328 8139522350 8432809518 1982402575 4935413378 5519985385
3142271394 8590866152 4580477227 4883261577 2636359994 0699508870 7172521637 4343090138
8699966999 5059707477 3125366642 2751515104 7762278444 5336462551 0225013819 4900025467
3947968471 6332390718 8870568505 7046186208 8539155360 9157322364 0555584546 3940113974
8731326299 3145087957 1005243455 1420491768 3192028605 5627183409 0595072059 0689881130
1374575055 6474314558 1188825817 1303304661 0862439589 3819546380 4938920324 8948705371
4392954707 4212621629 3187192572 9094769282 7442065837 0754824535 0448740552 2267070063
2013090532 6435816322 4130731643 9568390971 5887023050 0156635116 6805243327 5612179124
2846777181 3222820626 5149005965 6514800888 3882884824 7152352031 5106522935 8530070105
9741923582 4088381163 5609904532 3829185930 9340772908 1536928095 4427872192 8372096137
7035735320 2567429627 2189813877 6722716485 5387672930 3043783259 8495211326 8450841509
5550865807 7259886451 9045768601 2823107086 2521716611 0323880199 8988228012 1962588367
9770981384 0283474621 7315445766 0712467737 9497986579 7855225855 2894108845 6298865167
4519316064 7644768770 0660352377 1564197531 1925337605 6961433967 6128541824 4757266537
6396709730 0555950302 0581893251 9943546589 9014451888 5429359490 3896060984 8612281708
0237560353 8760433606 5717494084 9900737195 5168538556 5853454476 1008988299 6002770681
2195881453 2467605676 1415108131 5778388118 5471134602 9612095608 5678794343 4721285632
2828762502 6727500616 3169579964 7858345733 8595454434 5395658327 6000343615 3466486246
8930135084 9617346543 7677009404 9278797404 6947573638 4140226043 8733994785 7259904442
1705736172 0947948057 2205322278 9712576750 4574253717 0530156302 3689287686 2317272618
4814827192 1430158026 8104718405 9490417080 5713590033 4334288385 1594035967 6812590424
8050778612 9975673520 5439050281 1540351234 6724599391 8577227518 3435944753 2485859265
8517198920 7093703538 3687528222 9123991624 2534475829 2334008067 1353620180 8511281271
4053914835 1437053110 4823532743 5962044276 9035028065 1463700646 0300070581 3352723340
2397527373 0265795357 0150872772 0660141820 1861523319 9219024090 6162303006 0872834193
9672344262 9658774454 6695267985 4645779183 4025160872 8703581074 2337964682 4575200311
1697281237 1758228422 9495811052 4315019346 0748528401 5263648019 7025720628 8632114446
1612351594 6389062742 4080615225 2815481140 3220313518 1054449653 0550036723 2774897837
4663700840 9013009214 7385220371 7982705413 3246792015 6372985203 9780433496 6470066447
3969172847 8415353546 0224987549 8306164384 0833420410 1743176326 0497522435 6393853565
5605480636 4140794750 1407731382 3256572399 0997981282 0359414645 4632760431 9575841852
3710784311 4610255195 4731660382 6439401765 3710027609 7879771504 5699472582 9557789559
4603464890 1962601167 6711369863 8098527680 8539862319 2983362662 9831493025 7349389044
9456246549 4326477815 6769828574 8641426246 7135223050 3390574579 5338768615 4034527517
4231370355 2106110982 2065876556 1233007683 6842062320 9934182891 7018501019 0160135096
3288676016 9703807092 1193880183 2925793671 0671324916 9825700678 1345772892 2763006533
3330859967 4412827400 5237555321 7841983829 6271480286 2932391905 2516808091 6026858698
9801808505 1014026492 7145206850 1625940566 0110523313 9868386526 9386542339 3478991187
5256274905 0971797934 4301086952 5202451033 1269155542 9154016183 4691438440 8059787033
8532148986 1661647605 7745111516 4341121362 7632832936 9271313374 5776466592 7739489487
6919859144 9375893706 3996298644 8405772914 2432945595 4108537916 2374781638 2713622064
0235555145 2839443120 7428885842 9808401353 9277634613 7659750090 3338378792 5995895228
3146493722 4596928221 3741562954 3347043083 7273575310 0878687643 7153122576 0927013674
9738253284 2377096132 6984031460 5016873261 9485018425 5205156813 4426337115 6249956214
1478478473 2190503105 6917450136 7119451723 5848911647 1311281300 5551740838 0008611382
1005417140 3628364822 9442902463 4353510628 2738739032 3888387628 9877955866 9532327253
1276191667 5585977770 0693026992 5997091098 7954822399 9608328685 5012177302 6561616956
7121279022 9557039989 3903292861 4630147373 5146314195 0198279125 0287594348 0123151283
7527176943 5044629534 7670343968 6337246485 8547460041 9841343503 6801483157 7130528285
9743807672 5428699318 2944753632 9459333365 0283949505 6807096523 1848224665 2000816996
3601568841 8722874426 5652354956 5846313123 0720936705 3864981019 4287311028 3481444416
7830337423 7732699571 2032557946 9035282654 5671103232 6328052861 9481993205 8882053181
7540809994 0763797991 0873418888 8925393802 3116891706 8481886462 1809910942 7471521199
2615366034 6326437168 8967125089 0916253398 4513864850 4881543418 4082824655 9892860310
6329946132 3585601616 2627499709 4228934906 8596117156 7889277889 8908995512 0664895680
2513807008 2200398164 8693884104 2922230080 6436663831 0270343116 3933374221 2430457675
6317494995 4441723036 4356170628 4798658861 8904021761 5249454138 1763791457 2420461342
3424256442 6962877053 8218425262 9874527744 8136011163 1784916337 0341510242 5812475682
5376415636 4785686875 0656248771 9752002820 6023095388 0933234502 6720858946 3872179507
1595553565 8594870420 0220632856 4246254540 3281764208 7913944597 6932155154 7465794608
9418216431 1258609970 6343371298 5043926796 5778815127 9417014092 4798097072 4505701391
1812595347 2170035270 0677945911 9780303861 1798188621 7236141849 6225370349 0992488860
5645030619 9720829104 8296733680 5491109370 6967259936 1852037425 0044683471 2197875816

7095231102 9855000954 0176547144 5249173971 3126954790 8287233815 5266495150 7533746229
4358491312 7282432886 4797935465 6433062890 8344113319 1454582236 1876918754 1977025755
9380769199 5447351393 1319095634 9525201933 2996005572 6666987625 6928490798 4356616629
5456513092 3584798816 9398968267 7573635086 3842261127 2052020349 5431306366 3510772647
5516152985 3598564546 3260718753 2849328144 1766925722 4099888419 2549549021 2441910109
5965454768 2420207143 1241583636 7214546205 6634653555 0539891213 1369693173 1671052412
9602691502 0924361603 2526745749 9920411840 8071251466 4774740579 6198925583 4402609083
5397859504 5777353090 5361821231 1569671875 4319007575 0925677306 2090001696 9736481660
5068551571 8735737176 6422442187 1274020073 9265305832 5312821770 9236178019 7305607910
7975033141 2863768179 9111837661 2536337351 1804996519 1291406655 6178519840 4436494810
3593519946 1287439395 6755881043 6921143449 0724933880 2636746641 3295532287 6533875789
7427865021 5171150578 1754873760 8497036684 2471026368 7082417993 3335158102 2979299095
4350510042 9820595916 3412503564 6042541637 4814121234 7374027769 2682432611 5527140703
3042957088 7126290896 2117782788 7267271104 2565497494 9360687836 2795392942 8018141467
8591963515 6616178719 5150839731 0838635542 2755386381 5268411827 1682698535 9649980653
7643306587 4286530265 1810257372 9219741371 2796811664 6593335806 0821883683 5826905358
8115610749 8586475725 9160443204 1387204386 9320620453 3102718524 4767782652 1000020351
9995137616 8427955161 6207012065 4099442721 2425887017 6296318274 0028141046 4653560759
4503027850 4362399285 4106603880 2755300491 1680981149 9342950822 6595736593 6816557978
9330306917 0176795418 8765065341 2610250226 2898193749 1788188037 4038198345 4721594013
3434347974 1267590445 3164517326 2414126261 5634017438 1719236343 3931223822 1799950627
1141817201 3387192843 1232282354 6957720171 3204860663 8052181200 6124530371 9336242457
3100484292 9407844526 6451096958 2125026382 5218093936 3877482174 4599791928 2669110087
4251739320 6531045580 1172573578 2024420985 7977101487 7943275489 4953141688 4703579443
9467624043 4229961565 4321993690 3893422792 0592737408 9271222991 4912603108 0384866082
7422901756 4326833420 0210079863 3083489035 6010773498 1701268557 4714183938 2779686593
5177287732 6590334801 3444313459 8353570705 5807650455 6713621961 3909836113 9178024884
2141785544 1815455555 7882396744 8132026732 1841999314 8339499170 3732314061 2204591097
8969585294 6806755577 2799652548 1506613175 5187238722 4912464266 0584588980 1145980648
1174216950 8090655061 5724661901 9450012043 1701605794 3756483060 0066497894 8650843474
9831172120 4472895875 4376958282 1435872420 4452435522 4847430874 6429782298 3840427426
1465015039 8124707401 6739548786 3645611208 6836667853 2401182145 6049397985 2899520904
3659460131 8039567242 9683741834 6023003246 9693243911 6639207827 5061142360 3668685234
6857186667 3158266104 4653284017 3160340739 8556943376 6400783561 3757055608 1888961042
2071993105 5877164006 0628295906 3474283106 6992486829 5058682286 9346502047 6892748443
1279053929 8375728318 3910265121 7625011475 3423618825 8134110476 8796593362 6033245005
7332434477 1503107565 3708371101 7925139043 4413689002 7786742961 6314309176 4994370566
1914898865 0715953878 8552980223 8795226063 3553140240 9738503052 0423910851 9478251234
8579469687 0652010295 2028586193 5824694907 9190642653 6799658436 8773030116 3843392584
8500291370 5520024743 6296015860 2189941969 9299420387 0783192862 4502766505 3907809291
2656691237 1669449469 6522675868 5791167250 8661882249 1522934980 9599909629 9395696564
9979958608 7718326350 1644152503 1464699164 9637534862 2236556470 3481473692 6434557390
3699363818 9202933253 0249894691 9741987555 3221763661 6958869367 7115935187 2777120206
6277169857 6645530039 2438034985 8479115125 2806713797 4846527205 2974322155 1024151848
4624218883 1218478566 3775189123 4664906617 6122960409 7999351156 7830494514 1448439901
1784046296 6579713921 3426729413 2548624132 7513849872 9966072873 3054904265 8931715540
5842926737 3378549790 4109413954 1606660476 2632649687 3595494280 6773157460 9142314602
0548669001 6986214762 2044446513 9214753383 8694288507 3722090907 3332076045 7349674568
0576119095 8328418231 9520628350 2308184050 4555209566 8402752217 8987854008 6333002457
1685843888 7610218186 0128624819 1231865757 3661427192 8680664912 9817121580 8008835056
7768733271 6785043102 1000639630 9066895833 2929674195 1787660314 1437595533 0783532300
5164899954 4920308802 4086580558 7673265943 0519843337 1821514232 8638234355 5460403705
6101407880 3846289156 7529062339 7188962085 5258400609 9834916276 0336310387 9620658027
2625478249 2651850009 2079811443 6002739273 8314222866 2538550999 6040453109 1764033760
8161009355 2419042870 0902524510 4822656876 6236487740 0638027184 0363614023 0795462156
7359159623 4470000618 2540192737 3978022184 2237824485 2484639084 4134831838 9196545075
8995941426 3354143263 9574440757 0238564911 8185426476 9298350213 4842110250 6015730706
6556510072 9259091059 6579888565 3375855937 9116964079 3033372769 6532808262 5289416221
7892897730 4049374932 6184428912 0836456585 3282968347 4900330186 3768107727 6220077423
7090139627 3990047681 8729704428 3794223217 5032747461 1305963152 2780382774 8500509514
0043577400 0510200310 3759518026 6327587714 1702845634 7893295385 4501234909 8034659316
9686437819 9422633309 8760288512 4312568098 1747076951 1931576906 0654519846 9794500610
6496465494 0120833330 5364394624 8276179666 3531881950 0941154864 0960944432 4708177578
3769442474 7675381661 6135957198 6036657989 8157603467 2923778807 2364101624 5863044260
9089360278 1110515593 7408778924 7667661220 4650801641 7135769939 2179273840 5873878348
8077599135 0356659519 8056720202 4731640765 3122720327 5733258275 3236429505 3181525988
3436893485 2630871468 7187936915 2519715432 6589550878 6089250046 4149697596 1123563181
9272256275 1490612434 0935119856 8476247826 9581711328 1126368438 0728047500 4720007284

```
1297231039 1537741874 1602586728 2739625340 1134127311 8877287050 9515876812 1883823239
6790868102 2927614806 8835468606 9110504103 6014942809 0949703
```

The twelfth: q_{27651}

These are the 14,190 digits of q_{27651}, the denominator of the 27,651st convergent of π, in the same layout. Its SHA-256 hash is

`4373b2822612a676bfbbeec29a0ea3e56aac31467a99903804990469e10b659d`.

It is a probable prime (strong Baillie–PSW and 25 Miller–Rabin rounds); no proof has been started. The script `c12_third_newprimes.py` rebuilds it.

```
3053572350 7832454974 9316705051 0631928841 4639691935 7279903887 1347188395 7654970624
5226242217 3422163009 2195427103 2661728596 1557873442 4406958964 9873874813 5440094954
9124744632 2227781661 9194561734 4308413512 8155416259 0748292673 3134137929 4916564363
7033789130 0215693403 4770485835 5520330816 9414472981 2335528069 7591527488 6161598905
2141371550 9598395316 7191011750 4075834499 6052950516 3909220705 2213568250 4186107731
1976061558 1156371662 6074419195 6985881961 5002609393 3227611181 5945317249 6353280645
6425074115 0870200897 1845036453 6436128028 8543768364 9443619189 8355618173 2098931107
9699597315 6028320336 8628784481 9893571469 1185967898 8303579374 3090746777 3564192236
4908301423 2801731430 6860559535 5123330444 1683789548 8268182959 2707979091 8301344453
5648908396 3223991611 3893936220 1863228089 0025190147 3397306785 1442984981 0068619949
7414295364 3017062764 2023836365 9385912089 6898882236 3688589946 9943932518 9340588119
0036108343 3283572042 4431282680 4077219640 0482542648 7380314904 1057413093 3026771936
5534295919 3347454456 1219216065 9142897941 6102359175 6912473050 1993384613 9560794903
3027566516 2464109840 4559557930 6772399435 0777705955 1099968161 5759762190 5673506081
5273010891 2030548670 5819825708 7327732349 3769300042 0963857524 0841573743 0824197463
1666879143 2520799916 1393849371 5091965664 9452516968 5609862013 2712740864 0896942508
1029693685 9574220230 4007358272 9597080695 8335917832 2558399505 6743010080 0804377955
5969541643 5585257697 5772493696 3173978883 7701933173 3954203215 3137179912 6028599413
6946154561 5671812468 5717529456 2871492416 6269919315 1198769090 9800645555 9640847571
7310104591 8574231194 4726557252 1337527896 6336931117 2482413783 0073478020 5742107450
7274384523 6482993140 8380929798 4263990866 8763446140 1170110089 9894871244 0384824643
9915002615 3481224994 4995715984 9834134403 1777365819 3261869346 7275253533 7091938787
1050734484 3756633566 2343589329 4913838103 4497133105 3102435967 2332844889 7342013483
4066583708 1233897025 8126865095 3806164703 0346746978 0718345335 4983903943 2813489813
0638499713 2350615546 8085891988 5538743965 6668382024 0685627038 0288897974 0804949108
0835946561 7203579113 0840907241 2489940089 7471049376 9911127168 9599114182 5752557168
1992205642 4825283590 4053750136 2811026933 0883612006 1667011671 2421436378 6813719930
9522370224 1976435358 9100554509 4573521763 8429144576 0775399676 1197420109 9003154786
7722051754 7344570199 4746803392 5555375515 0638508090 9351173489 4318457182 6361521270
6360619521 7652841290 6359835735 6819119307 0523382434 2098765198 1993372882 1440323201
5495611238 9840488914 9079594822 8824611254 7209012777 7595078625 8540554991 5267236893
2113038454 2973708482 0930219377 4492314656 0293579464 5623494336 8153239598 5914587145
8587159648 0843167844 9445121214 0565017306 7461643308 8390421804 6906075681 4062390454
7883560133 8137970201 7770533412 5883426138 8205139576 6225998784 5569324718 3021193573
1937372868 2717986984 4022404935 6056216476 8494594224 6657469762 0604623829 4233315683
1202730010 8997947201 3318191437 1453310726 1779073435 5814365609 4072033497 5920117009
3340268422 1091832123 3749773049 2368341797 7951903502 2868448272 1425519876 4867470207
0878128937 9553823480 5988501401 1223305469 2652623046 8763302232 0076650540 0035542041
0817994148 5021391516 7737162615 3498174583 6921626725 4597766112 1923461307 3946558864
8746266965 6822919643 0487032633 3156694939 8441541155 6166886470 9630861596 2977800930
2121704942 5729516913 4477000759 5195110551 9966265763 1462688707 1618521466 7433779947
2979891640 4872553522 3836321042 0416749566 6932371742 8095486934 9368182163 9558621757
5070324752 3499656944 8435806752 8216149636 3375912415 6025786605 4728233166 4576787926
8277088140 4327572407 4787219917 9179166097 2600244754 6372136850 2311865275 0658285177
0192827526 0822218391 6336532712 7904975137 8928126878 8553351462 0315484391 0023479118
1514502142 8429615795 4678486012 0946272564 1889573923 2418494719 8645990661 8359439426
3025990953 9177310474 5804214808 3592970772 5910788799 8249012783 1133645385 8540872426
5401018763 0473688838 1503208831 5436482181 8208458400 9104713684 5871875670 4129536777
3176995530 3410072744 1107791150 8186357316 2670497273 5037524345 4982170301 5126448785
3456210733 5007447670 4183996370 2099195069 9806332345 9616673666 0168869880 7640438289
3145093458 4631997542 5933704660 4463476364 8356601742 2280623808 7206415936 4492076042
```

2545613349 5681126143 0827089021 0309613574 8362619038 8254306680 0495333699 1878670552
4844739250 5754138077 7557140770 4615881964 8118544212 6760432887 0294299242 4497603338
5869072379 8489745061 3154427039 5774404445 6874357900 7777981532 9292900136 6770148867
4812065966 5421717893 4989628605 4498567267 9357978192 6335354449 0967928666 8995807310
3751416436 9564324436 7157519384 9747293717 7616787186 8179625792 1610863880 9598001486
2945521325 4305635312 8422024674 0664014863 4045207493 9975097823 6384996405 5530376425
9171459920 5945958200 8887959965 2515116626 4260657395 1550778682 1413902602 9059175662
9918542274 8102261550 4636060147 9991394216 5450461339 9537709612 3658365650 9361738545
2577039029 9629688533 3795992884 9746961311 8500668078 6737866122 3835203349 8438708374
2299810329 4124034669 5899870426 6352042165 1039791207 5008256538 7148377567 2811587514
4643094476 2860379103 0821631809 5954041562 9430773278 7496173947 3550079482 5992710910
8924117478 0886808217 4213176841 6477129065 5250026222 3632021468 3181342902 6087141003
2979825165 5898039798 5483800491 6168426668 8798728239 3940182924 7852593255 6749940321
5993709445 5647860055 6824240801 1802708276 6699903849 1517664900 0228125068 7850862159
5487206721 7996413809 4180508574 7638492231 1352941507 3668789054 1555946369 9017497918
7704054813 3087040207 2620358649 1772459204 3207854604 7628076579 5021669760 5255338244
1983092676 1444928983 1894315308 1385865695 2706323339 9753413039 9184723016 4471881379
6254682341 6167671838 1968810775 8765244428 7540587525 5955911174 0306897769 7432673846
7696351300 2664978402 1764299138 6821007048 4730486210 1700401760 8922533070 8315058947
8222307534 5620576919 6289255705 6628207583 2161693821 4242214708 2834929535 3313142921
8251855005 8772647397 6336296909 3009007001 1100883889 4500995881 5137467985 5654984180
8572685086 4521136577 3940935383 5214552167 4779492023 2168252536 4137601466 7180934467
8326263064 8579652050 4194572972 6232001198 1707494725 1095912709 8550173284 8845749339
6401212488 2766725221 8852725515 6332422121 8347098285 4413738479 7271673976 2425555002
7205979627 7760241968 2805492622 6126079460 6498507575 3054011430 8910611102 5048522315
3665878647 4028776710 5484134087 1941487153 5270370629 0395317687 1630839766 5203748167
5875098910 3484495920 8564473755 2657510446 9283633773 6194295862 3411132778 6590142466
5239021436 9455612048 2762719591 4073916673 6722008351 0132061016 1504258741 9562376295
5854540527 6612961687 3396045204 0859074464 2457920559 5275171615 0827185121 6984930135
1233665065 0032444915 9319560156 7827537672 2951897208 9634779338 6176407526 4751371186
4382965119 7038837396 8992384685 7598520085 3392473866 6489635719 2694083083 1027791014
4089208089 3382644932 2864156670 8983029213 7339810920 1079697079 0150638005 0039222678
4906099668 0276535850 6924114426 1292341720 1458427264 0527940672 4373580984 4272709944
3852332378 3799245940 5321152816 6195478308 3228650139 6585265404 2961262633 9009732817
9788947488 9251985666 9037789254 5430634359 7405280117 5458378337 4387205547 4042039734
0224628183 0030212022 6991467132 7169778893 0975417864 0258993155 9844253809 4984452841
9889456788 0689275953 7428990891 4221424406 0894969524 5469443768 0004775778 1050939946
1837740462 3082900286 8550817862 6664198809 4890074919 1222338811 7379514712 4554278245
5269308515 3770750377 3532361872 2030519344 7644287903 5739157061 2993027670 0900707325
9152666030 8812468376 3514323337 8490868673 8048900708 0028909986 9677065563 2656331347
5253093147 4750558185 2840362686 3455414733 8335234378 6711647921 6345119805 9637672083
7419793926 2125514374 4361056620 1274632516 0742713334 8911757018 2961933804 6479586602
5174671930 5130509178 2038990044 6796428161 9507616532 7144420679 9952510359 3152626601
1862305881 2565502826 2390532640 9938106274 7442789226 8281544784 3275470074 4460700294
1862837686 2823417645 2016138354 6062836964 9893065171 1843798412 9481077736 6522429904
7264565059 7376287523 1419634491 5957118942 2546321613 2402908506 9675739301 8098452843
7985228009 2689910477 1557286438 3246442977 3995934517 5216753574 7342104588 5277835807
6629828146 0120259400 7973971302 5344183805 1535826096 0780293420 4321325802 3788490154
9898442087 3050971021 9155837325 2939276711 1150471551 2227397343 6011635915 5083327397
1186139359 4473635822 5507641771 1043305790 1026926874 3005631923 9085843869 1419380166
1592730010 8391862289 4785028279 3499785467 8560715767 5249369171 2836565380 7889298846
3111082348 8889154452 2031142349 0388491554 4668150333 2650313493 6677928795 2998699228
5194486150 5330326956 9351162662 7813717562 8984794705 5497318499 7879863424 3803804105
3490199369 1595475689 4999869620 0726381736 5142310806 5610260229 5475316467 9883224987
4648853826 5294052337 5726620873 2904245797 6117142370 9110195193 5603785626 6872413538
1152632506 3186102306 7581214567 9157549427 0786560893 7029973928 3847643541 0973851247
0413172487 4014657205 6425930371 4976674834 2404443158 3409192939 5654103772 2787200712
6773846777 3280744006 8764686061 2680346920 0914662612 0843971774 5396657744 9089773872
6477493643 7070185021 7132119285 2601195091 5620043984 7095372237 9075719437 0100372876
9658870186 2132589159 0069088226 6022718121 7500542004 2441154710 8969111725 5204012219
9735079107 7257939939 7080641095 6115694806 7806403443 3301773995 4220678716 5835157895
0949035016 8461722569 0624094637 0493042364 0968191263 1880329705 0279356873 5707940737
1277286425 6092908798 6413149397 9796896338 1460864802 3705179916 3814455406 2026010287
3453708880 7432609289 4039120052 0207297412 2904600483 5920223610 0893256403 3100002301
5915472740 9283029853 0800771522 7045580623 6073808752 6422466679 8463104617 7309010164
3121852210 5157717695 7473613878 0871151021 0272180630 0169227668 9123957799 8363339663
9592134148 2920679274 2001453998 5899805093 0653989074 3387485037 5343253915 1591932520
8271340809 4315496023 7668849166 4268262185 3655583796 8881649110 4760334776 7323051298

```
8354180797 1288619741 6880632116 8798507783 7398094232 7048386858 6789568630 0424447402
8799773457 6413425090 6521190241 0799201140 5534004159 3310768486 3691057267 4420629246
2822802533 0716102059 7263677610 3051790900 4138008982 8521342182 4827317454 5558200090
8886015161 1611957158 4970971493 5981028624 0425600345 9374604038 5337818642 6677471499
1811565984 7045220460 9758295086 5716852678 1209970333 8601508020 6351010230 4945148004
6247309022 6008597038 1452564177 2270800765 1310464802 2466038566 7373334454 4432917751
3074055699 3043084292 1672605764 2942952732 2712133028 9882812768 1812557393 4625761420
9827329196 6678844628 7626781593 7437112587 3206671957 1601640270 7811466779 3639452456
9955129663 3449591261 6143013920 9394159565 9233712235 0171526123 5889143213 1831218994
5712148492 3289186570 3774028894 2088374617 8879489980 1553299815 6612521904 4409210481
0761989064 1538894014 4639627985 5439825447 9538283059 1211172783 5973860042 3347483488
1322965835 7696857846 2063211240 8038615681 1031311166 0848446850 7690478530 5498710149
1383147131 9855900574 5446874930 5580361525 7247170423 4689081741 0716782185 3666013120
8139397766 9403394410 3839502263 2326478468 7870619663 6379533323 7462765947 7839619414
7501870503 6004449107 6797276136 7763283181 6662134772 3209410445 7820401612 7353275418
8250087131 1166720885 7507661285 3309903411 1789634898 2595796498 8823759202 6613112951
5043749998 8416601147 8346657365 8494766969 7126995862 6990252170 8969850363 8546694385
8658434700 7503606157 6523436949 7059891584 4125439690 6076295010 5562614298 5460623500
2806433177 0157560442 2477619209 9950087500 3004694319 2316268605 2828142466 2272760596
9468635079 4581637418 4373142687 7171035730 2830357324 9105680602 1718422989 4468672768
0002584257 7052512526 3266800972 2305711324 7447111673 5013523777 9785799016 8437217202
2351881271 4171406175 1798122331 0224307789 2365240735 7763700361 6737247408 3545202140
4979366318 7697878851 9598490134 8631138836 3889660938 4812582903 5026312939 6988148549
0473380143 3932884088 0504911864 6523276837 2920522080 2124565781 1689347355 0428811945
9173620331 1009342632 6986261374 4382340092 8973081067 7012455406 8424148009 1154616765
5860521921 7880154173 1514028585 0908158894 0534367070 6932007236 1336328142 0492427310
0918187904 5280991782 3580859594 7460924784 8468410230 2873944852 0432976630 2881900468
3944050557 0915297132 4758517176 5842999377 1123448241 3144878479 0029297853 5480765712
7046181755 7423142422 3850152618 4110709932 9021349191 9064357303 3317519513 0018635786
3680977771 9060251308 0971576574 4488107374 7864589059 4135455138 8239369055 8757446196
1558850327 6108374329 2674194253 4636496535 5845011760 5520432028 0053947085 4578935454
0733711669 0092333042 9578388350 7572082651 0098768910 9643114906 2223218856 3448065104
2970507365 4601747105 4511514115 4299220847 3730012329 9910124589 1676502075 0883850966
7078748120 7223089108 8853844842 9566630273 7465781481 6457358862 0890585907 2960616890
7471846402 7410623528 8056786826 7188047254 0279238759 6191489904 3296636660 7655393954
5101279186 5400886110 9917238990 3035591899 9258537937 2544993166 9661313786 1463435114
5228718528 9205724722 2137344647 9589769330 9068477071 6272333463 7565151884 1754265010
1746411332 5031259051 7696539217 4357566054 8440622398 5490832169 2064685926 9325848816
0899717660 4846867580 7657218411 2336779110 3348871030 5048311103 1267958846 9081854787
6678224306 6738437896 3975853089 8981082276 9006056255 5283273575 6380827012 4064943480
3498267662 2897358179 5003003568 7739060657 7579630607 4609248406 4313589905 9953300041
6877534359 5766998611 3122197027 1044363412 2488760394 2854926706 5911840255 3721623595
0855509141 8012834903 5306168702 8364105026 2675342630 7521564162 8742541489 2512511531
6749209033 7452550127 8968482801 8418215636 7547710408 9096803866 7060935882 8449506192
9296496205 9049040607 5860585983 3060485810 5433561486 3155086218 5806872957 1630220080
1621691818 6517928613 2114891123 5130017121 7678654638 3138745073 8689783240 4156901934
9597299607 6647725592 6694999758 0361863816 2390255378 7228543526 7827527852 1224159691
4476263526 3927922060 5719662798 0685681934 9512616191 8829248490 7882247708 6853951387
6896672463 3035852163 9806473556 4284000739 0413467897 0029125024 6315827197 5056065193
8126105247 4491099169 2694576072 0975544828 0810139277 9418813302 0290124310 1084992732
6575157879 4981923735 7759700965 2716245580 1590021841 3211088476 5820473922 1348520193
4911725641 6375634756 5686914998 0888009344 9497966184 2206207478 3613670736 1828908462
6316150871 8637237314 1355441142 7823924509 2128094611 5670115002 7022740771 5520455239
8699168563 9150231866 7322788916 5506625561 8705666351 0079279613 0480699133 1799025170
8495988705 2347493160 8800831376 6609217526 7899939360 4296300461 2080365783 7909660937
7771660234 1132438196 5589662524 2858157103 5445974057 7912158966 4808174619 6825750762
3696769973 9950647396 0839344819 6634698545 2402830208 9989091549 7520196731 0392119366
9086365024 5093086445 6331297402 9197141597 3532024922 7034276835 1235371919 3930638369
3655084448 4511009313 1433096729
```

The thirteenth: q_{32394}

These are the 16,695 digits of q_{32394}, the denominator of the 32,394th convergent of π, in the same layout. Its SHA-256 hash is

`08bfe3ab0ebbd69b1945ad957a2fbae729aaca5a3e26ef00a80c9650711eb4a6.`

It is a probable prime (strong Baillie–PSW and 25 Miller–Rabin rounds); no proof has been started. The script `c12_third_newprimes.py` rebuilds it.

```
1965707445 0635224428 6366583511 4148142507 4740685207 0213549836 2317105261 8192426904
8220010197 6427732699 7433235264 0250681957 6038118388 7938142333 4235983754 9929437568
7624596481 4564259622 6514857268 5008567529 9803315756 9622870629 6968480852 2229208843
9888150342 1816767900 8885330278 4524614864 6225557032 7089254669 3346444931 1777423895
0034583656 7070101725 6870574648 9877149122 3137504250 8686989190 5684039309 4060885147
2536912330 0380107362 2194332152 7228741269 9542930964 9132005696 1081964786 4328410256
4732073093 0569006181 4529703629 0538415971 0644186404 8229118275 5895361655 1798517196
3482134460 1898066207 5989578285 3099644278 5487728050 0453348297 3488677794 1311038730
6460983654 2149786749 5823503222 8106428316 8910023265 9550612318 2453449276 5808874029
8416409944 5076610033 7571950241 6361542305 3297954791 8742048735 2802814810 2918323183
1018149846 3263851000 9203768169 0559548185 7087398382 6128384744 2361412797 9994221932
2812898067 0149217369 9649039764 7410745765 6124393100 7402943797 5960576619 8811267129
8786288028 4000970942 6924658463 3661662422 7572830273 3743799854 4442723361 9073925476
0382892048 9488251549 3520751843 8679024618 3344383483 9714578140 7492544902 3701887811
5150175473 5485413240 5632949079 5009056091 0224184966 9541625708 0638148444 8831540926
4977454375 5394548189 3254553082 0725286125 6364907388 3587850553 8063793775 7562528685
4067475001 0662501493 4048464662 2106310197 8336374537 3220545676 8815112415 9614759145
8365213305 5616809470 3184817974 2251152079 3730030008 2194703529 6010519643 7543753282
2499311435 2697443311 6469987595 6698129764 0064979364 0760042879 3743687816 4442171653
4958982036 6062015392 0806636460 9732828228 6255863367 9227973895 6562762915 1040206942
8656059374 9987170945 1776137887 8853149335 1983310859 4872146563 4097699836 6484085443
6786005232 4026593356 0935451348 6209779339 3982832411 7067869122 2005206465 8725201470
5164384635 2291059219 8446813476 3527418995 9365006674 0295933241 3160103276 2801053779
2728326182 0086903098 8048178764 7600707031 4118469406 7110725606 5555307752 5635643313
8322636558 8890395823 4584547709 8150901457 1904802697 6606382989 0193507552 3537296654
3033590882 5461711835 3215381117 4276065480 4469965848 0778085396 9507290705 6489602782
2318537652 3082217549 9657932702 3615614930 6326780368 1888948188 3678319569 8027811044
4479153911 8215823926 0675962571 9385714818 0043742072 1498202645 2780702397 2141668932
3442683464 4994020490 5980238733 8344319194 9471020660 4932072436 9138691599 6639650812
3838205432 9855947304 9322049818 7904482637 8092705130 4706328338 9937454011 0106582492
0754661397 8202027756 1434653871 0290651840 5423480423 5505102974 8022444610 1172204515
3232664271 6497407367 3466418275 3554225738 8598964413 2839476417 2261852587 5862563400
5102887593 8848477372 3492134528 4161877712 9784946914 5853401339 2407888976 1166743998
2920505512 2054836655 4372334214 2069300445 9257062246 5348173612 2280994990 0421806633
4678475933 4711170601 1635761682 7621195340 9262423033 7518381162 4221874001 2560969287
7591372652 9326499389 0855271835 8229147722 2579345509 2404727242 2139748351 4239434841
1135865915 1633794200 2582305598 5651914846 5351766421 9554177050 7822952617 2383704912
6608236355 5413561874 3011389853 9774051782 2844974705 6155535726 4440605028 8988405136
1749758913 3690829336 2214252616 5640713279 8952259383 1527764566 0468567901 7096622758
1417314234 8327584342 7427282382 4924257307 4870457222 9792156575 5050677677 3388239302
1084040151 7905314423 1846236428 7141966988 8349951343 6597547292 5905449350 1793045164
4053081020 9387073527 3958226403 3098424770 9663911154 2669021651 6898049632 1462483269
4793232255 5354327723 5608428314 1487862472 7933636269 2047761194 7307665938 9285551014
1528913390 0107193563 5618596449 6292219827 7695218979 6079824678 1039553250 4215426129
0170323097 8231050751 8707133027 7573725326 2500034912 8815152525 4634161543 2167911602
9721149589 0833910600 3778893244 6146771272 3745697642 6520345220 1326814771 7226125567
9858618244 6208763994 1382752266 5910505431 8575433807 2374040878 7450766151 6453933122
2635740991 0674737853 9503010042 3547056984 9848154569 4565456102 1170704093 7076936328
6064639650 2520179871 9014597412 3759475517 4813345142 4409112296 4917497518 1304956735
9144160977 7589050192 6147975962 2613784141 4279585202 6563268576 2844729399 1587851755
4164117071 0303666169 5113044099 1619727653 2173560108 1791064271 5324843766 6278518902
8495532333 3738598609 1740038330 6571190642 0248015944 6439060859 3513746768 6156342047
6113444954 5951053870 8146950775 2305467781 8381752132 2614984806 6001297407 9353320166
2040384019 1841683684 9618802108 5354981334 3773731025 8188110865 3477919949 4693245666
8631304811 3352039013 1918530263 1918773663 4860953491 5978407596 2447822320 3150779284
7564091312 0550654767 7170678385 8552027099 9869125088 9522235173 8964454312 2292328635
8566292504 3836340044 8874398480 0933103434 2018226499 7566842727 9155194213 4331283077
0492497073 9413861934 5188560939 3937109253 0609344153 3632328042 6031050478 1334248478
2378896164 4450729116 5153297351 7526460107 6517169753 4797436690 0766790381 1475722657
9205179449 3897822598 5563107040 6644602191 1055573870 1562343514 1072306956 1503835962
2653556545 6695455370 7654178783 1351351576 2241518535 3166789159 9437459306 4769144019
5835744451 9851340283 5765115368 6143769130 5320269923 8412148129 3105715108 0560153024
```

5724084289 4886862673 5117704734 7969180759 5386827191 3602064135 5245016743 6927518914
7056471907 0028203580 8708865983 1975241065 1579321386 6510905682 0106617351 0685939769
3079925341 4687612056 6177103403 5982325637 8931258990 3903234348 9738401366 1607472577
6671178245 3080492756 3870664259 0858561269 3553091454 2495699922 6241909162 0344564113
5066469352 8385542648 0542207820 2270861552 3410464332 0324636568 5004179507 7901363002
2050216416 8339524314 0357425941 8900541260 3037192149 8498358862 4619630092 2518987370
8076542042 0449519615 2759177945 4077433792 9698662150 5074264972 8795030491 3814839706
5309556866 8895361852 2012361963 4017677582 4996862944 3198689362 8287681767 4962270653
7084961158 9856234494 6322287521 3316835505 2911624880 4151064176 9990171862 9543913477
1325150064 1588446706 3687406173 1438240363 9729950605 5846597856 4513314449 3413963862
1710726954 3127526396 2201015002 1001498459 4271513056 3214016922 5175842216 3858705199
6142492186 7933751819 6505881425 6014312223 3550832401 6739459485 3176278412 8891949915
4637794962 5621764961 1148890236 7717674359 6980064154 8774122055 3096888687 4925792136
7749508296 5495562197 3183610933 6155255661 2551213473 5647693747 1252456987 3682657709
6893149763 4095780734 0056130446 2977094075 0861707761 4721388706 2117182211 4856468784
1032615329 2255945938 8067602415 0994649587 2917743945 8013170512 0630528573 2303238458
2608816703 3596806923 8018562070 5426889626 3030029755 6501713329 0028001767 0907264316
8324026851 8976996750 3695247179 2887299350 0023449036 6616771320 9423589125 3328237784
0223570812 3289345654 5620512119 2565566281 5952825822 8404431785 5981802492 2725493077
2979574460 0115405050 9128503961 8668318784 6673243376 7493291514 0770158261 2170251618
5469541607 1956986518 2867764410 9573444893 6523978369 3821307566 9839061844 5717262908
3228715777 7736584388 5052578278 6261694821 2054829809 2248660739 7858550092 0812955174
8462620121 6786712230 4819442760 7888693124 3097245311 8033980085 3932210130 7425988334
1773742378 1063361852 1305923894 8518345612 2894353089 7315457638 3169442739 3865073846
6402897936 8067939375 0832990555 5204454221 3452401511 4413096288 4737747995 1474584026
7528437655 5613439560 8279935035 5945878097 8941921865 9347027124 0331547505 5100504671
8975139200 3652978356 3186470879 0744006173 8860796697 8260678008 2291290565 8360274506
7951960483 9981866542 8923447487 1604101851 6472847089 7428146810 0228868586 6294453930
1287899258 3913766592 5088718023 5494055489 0849286342 9213765513 2641513753 3780931682
8711789504 6514949644 2930523561 3653676716 5312548627 8752833281 0718280748 3817429587
1078976840 6098367864 9948448044 5164882994 7832689870 1249439375 1030007382 5578719959
7267341169 2015926555 3202392886 1036396133 7804970994 4874359574 1592210074 9785207278
3385345632 9607966230 7576796877 8777764373 8639380463 0065660832 7440986477 2723813876
3314392294 2204016268 1909504793 0708014898 9612377281 3983266376 8180334861 5090766685
5139312850 6848786435 6926768837 4561908346 4022680210 5686004656 0688905176 6834325579
6243472347 4235527021 1643780590 9894237557 6079525386 0487588773 1957869402 1011002929
9206226962 6322811338 9573493850 0259428864 9562263357 8911701072 5123823273 5148106009
1858692695 4751443167 7179570124 0250580888 5811083045 1599363627 7380071975 5326011121
8114472598 1445766298 1247982946 4824258077 0980386061 7689203250 5633146199 3402909456
8246731073 6448347895 2744581607 0986753805 1143297554 0826826609 7652858782 0839144460
9882786307 4487558284 2790620188 1651260455 6417586820 9140495482 6318997990 1781720525
9681844565 3623577831 7110041866 8889040381 0649476215 6210479938 9690261747 3167686337
4223803542 7240449210 1798508655 2331779101 4189873666 4998216188 5684886468 5586686569
4256616842 0099126299 5533037050 4338970996 4205262891 4876866483 6871105129 2421991547
1301948213 6162388707 1309004286 8673720960 3545922807 7133205666 5951416485 5967195556
9313663107 1204031637 0765569202 3376076251 6011913996 9679981597 6633702401 6833797464
9106208580 1746245554 8820735365 9635853499 5575746013 7611764902 9505203082 4633764723
7913992493 1751419753 2155544738 8297568203 4434419082 9114663296 1811826974 3126713417
1171455755 2914017319 5962817413 7699130928 7868967615 9632127651 7090412804 5834619078
2664645531 2789012006 1552535987 6210779053 8911270543 5282115121 6076608220 4066048100
3883130141 3197800355 4276204748 9753862003 3199961284 5693208932 2578197779 5362463969
6935396200 7346172437 8150551473 5589183785 8187410859 3882344177 9599493831 8776705348
2214907313 7186562841 6359837676 4055080051 8346175637 4626580362 9983696758 5780165043
9756212392 3893193659 6537021571 8631086361 0300928496 5209156532 4550679700 1014802779
3024166056 0713875148 9689172575 9702496010 5163896095 3094310591 6991912518 1063175581
5957625286 7075973091 5918220292 2045611884 5931195797 3682690097 7834953412 4951127842
8776140581 8437779298 5345773452 9502156817 8427424900 9589851691 8854101141 1432749367
6915959415 6747783776 1465537966 1129115693 3367946186 6894379029 6727673062 0504872882
7599639223 5867492115 8559700568 3514542683 4684752051 5362006938 6573968641 8899942740
4138763950 4062825900 2303801512 9468645086 9694775332 0104656296 6554756197 5732312204
9866303876 4299318079 3352683506 7122789382 3788116175 7291121090 6512297624 6813769823
1772548433 8233387995 6055367939 1238125184 2586952109 9476308153 5943411619 4726204525
6000382682 0493732330 1118341185 8608865671 0779081436 2201467867 3558136882 1084975559
4042455377 2007580022 7331928905 0360929391 4242990374 1435423231 8924668985 2750537123
5116544493 9373472520 6787750381 1846008424 8151936465 1361669692 2731741438 3283520340
0742603192 3109373520 8789943741 3391157755 8336898597 2160269004 8166761042 8321514396
1857129684 4834119015 6988660472 9060800406 2729198968 8604909516 3823099944 3221661503
5699798171 9870413493 6464790959 6092830222 2319008463 5927240924 9479027051 4623222191

2215511504 3919561566 8254068814 4393331048 7753116835 8980797848 3990112173 9105680701
0649489879 3956812827 6275455474 5081243271 9660439420 0564339352 7851278912 3729231478
7627456727 4215428976 2319278645 3385445368 4281452897 7675214586 1232067135 4292669580
1138017601 6839367336 5331062543 7143568373 9095584558 8803226018 9638045377 6973799120
6217598795 9353457744 9397858075 6867620274 6157874406 2647187847 0915252886 8366384812
3826823740 2778385824 2739551701 9585502639 0733882690 0344749800 6044197093 0821588786
7595955466 8265101999 0280135977 3153332466 8074165584 6720798797 6720353154 5295251623
9197941960 7389603509 2256726487 9350794579 5539382783 1516605080 4360285376 3385972169
7441176433 6373319412 2599524026 7108663225 2396611088 2573351151 1181933864 1081253277
1645636459 6161238024 0789228130 9872361069 5856565262 3739551319 4034795619 5628543861
0380277986 6412955968 2366202529 7511954114 9311403008 3471549650 2890911209 8550266654
3837080612 5795366776 0926622736 8698852026 5620575379 9370416312 2974575092 2669520902
0321024740 0925347081 8610349841 4616493048 1390180773 7071514930 0265793272 3985616804
7631771658 7131282944 1272126645 6862486228 0187030798 9312182965 0638427739 5845691652
3753781436 9478364037 2899096446 4996776975 7972089617 5045638019 2225219928 4157004030
6276678149 5320480583 8467424112 5946261120 5084092056 2492043628 7650906294 0150306111
2380549359 2307970651 4485849305 5114699639 0418899742 8952827870 0810260823 2942819655
3529301192 0791304371 0730518416 0702631748 6850291834 0878649380 2848108740 8374392191
7070852942 3261773306 5266083243 3894635882 4697123548 6617451240 8462371711 2907438074
9029563173 0236464801 7562512519 0635535516 2980493035 4355805818 6873628436 4927353257
1076404136 8766702152 2609360910 5520278182 8333637169 8877928714 5706775708 3805238541
0003723496 9289093499 8740504888 5855105417 7283092256 6238727613 5673841543 5764033564
5845768937 8299006568 3420976776 2305570063 4383850899 4550784947 3185786849 6791927277
5399539982 1080903511 6617712956 9574557927 4449828821 1643906740 4146951046 5375967902
7928470596 6343509089 8456926052 1010481569 5503050163 0786364012 8440618797 0178651636
7112637688 9684620971 9759094207 3172741552 7532319416 7056945277 9113459053 5130569678
1671430453 0878887684 0029835664 0348102439 8406393365 6334562110 1398495126 0158995386
9791752980 0995048366 6333846255 2042773911 0134743583 8690959653 8809840568 2820594954
9118560247 0994118361 8698568928 0235641932 1547077942 8101166693 6017743613 8807789437
7645065082 5968617971 7700787551 7674911208 6709919575 6673559192 6345675699 7817395674
3752421497 0760144341 9800675009 0609816549 6594102231 0830137477 5066401375 7008715556
1015074946 2009737896 8677359294 9430974787 9011701792 4129776299 9604021960 3143258737
5112234423 7397042619 2239113488 2251326272 4447133609 5788939946 3494925092 3578072501
4078092719 3320501473 5993884238 3076192674 6301587271 7573350035 3902591495 9418721720
5747479315 3846640889 0079929186 1614404219 3940675826 1755025565 5071519227 3286835202
2257024914 9968507990 9071170725 0578667376 2449233248 5298467000 9605004143 3550463851
0782307435 1200246087 7162798275 2889603279 9596805585 5298923620 0358565457 0455851569
2778963910 3277204479 8286946741 1268266374 3896363899 5126490612 3106420585 4938154829
7321603062 4469844379 3040747880 9246967025 7226049887 2969166711 7745441155 9883119430
1052230738 6737055193 8844595819 4545843581 7930856574 5735524033 5265132247 8162339840
8143849093 8863084143 9444468469 4983566696 5405907445 2997330272 7499344935 1957973732
9046817943 5854662896 9745255853 0356341913 7364612838 7872001529 5469913519 4106581462
0117823542 3949797258 6167683591 2134979243 2021913674 2719563729 7119924812 2291233853
4038007870 9849563897 8706518520 2861528701 6683268902 7096476074 2901192778 3495573475
6732195133 2444736172 0384962532 7617791710 1511142713 2499561261 6301266471 3673620374
2849168097 4896131732 1702091890 8807701920 0977692827 0113471008 8511842587 6687589282
0860854120 6608913840 4655852806 5325245333 9790837065 3642586877 0015903908 0285386756
2995759210 2931083597 5521622061 3593049170 7153744484 1841391930 6487336493 4841559559
7841544433 3031287446 9700482094 3079545155 6717666621 5999422842 1786370257 2687999888
2370202879 3528023510 0874696384 5040985001 1330730030 8760585416 7023893563 7249081567
4128202441 9408940410 0539291373 6486657526 9559153277 6970574783 3477164996 5587956961
9504077636 8148026872 6895912459 0764584848 7300070316 9855357934 2671235367 3361647429
3444423017 3749189821 9906107315 9176844869 5041419168 5108222858 7726524607 5999497909
8798602574 1587133597 2987933678 7081752913 0407214093 4961960541 6267348589 2960904211
4417926281 7970936539 5194103787 9068532704 2811051483 8038546107 1715651644 4885640064
4022319955 3026404257 9894618815 6552025344 9699685520 2928852369 6701590488 8172750287
6628143552 9068005762 2614831012 5451051683 5982017889 0452937770 7383838693 4300688236
9398834724 0753747244 8871282626 7349898089 1409965175 0293079168 4407392450 5365694427
6032051128 8764309426 7328162767 7654544855 4648384277 8335575938 9240000817 6779381411
0941357329 9524674029 7219973454 1956809046 0058573141 9325962083 9237495163 0540185531
4933066852 6200195904 8736011240 5152196212 8534566845 4684911746 3870269832 5464750987
6718173897 0691418509 8991298206 4360722839 0324585149 5494892690 3895534880 8665137431
5462111467 9561137286 3164033164 3187599257 4178013248 7948644428 5720874000 1358626322
0373050772 5158332883 8856919206 8860323163 5526667156 6146905292 4692695121 6157460554
9833157512 3792667838 8725897085 7309731519 1190109461 4391519839 3033361053 2063081004
8409534897 6421009874 3804775661 5571237468 0564362038 7882730858 5246142280 4335428977
0744710098 1187489580 9722746103 0196654118 6195219910 1952095890 5118827144 1180252711
6565261018 6425875167 6317949850 1359782014 6781920099 8616301862 2162503359 1765525944

```
2832018398 7060737162 4504683584 8267683340 0685542190 2573782891 4481532395 9458206235
8534320891 6219115616 4170980714 3037597346 7915910091 4838602763 6594104879 3670644417
8502819701 0516908548 7867561050 5252158465 7874178495 6621468303 6188681245 1119680245
6970169712 4574709336 6861036002 3050495337 4749755439 9176250442 3842407061 0107359659
9359046421 0918381833 8790826877 1756130429 1417830062 4156965221 4502051786 6841054993
0125928932 9823453650 2611580687 3007637147 3759808090 5642419315 4184723425 7216049998
9633744685 5049228122 9897317952 6963251657 3801307606 2076949116 8354575425 8815756344
6053968498 9359276520 8326496012 0838585385 1401127621 0425189402 6162661755 9941733326
7985805669 1202795071 8601756040 1005826331 0687767554 6789568060 0146478896 3720852900
5875465401 4436317993 8047699697 5027214216 1801770919 5008401187 8783473170 6867472301
0009476079 7287854254 6768588187 6896778091 7109674999 20787
```

The fourteenth: q_{37721}

These are the 19,465 digits of q_{37721}, the denominator of the 37,721st convergent of π, in the same layout. Its SHA-256 hash is

`e400a170de02c0578e23b9d3b2e2ca73f2345ba9a4232cfee009c148d15af77d`.

It is a probable prime (strong Baillie–PSW and 25 Miller–Rabin rounds); no proof has been started. The script `c12_third_fourteenth.py` rebuilds it.

```
1883641595 1238280222 8558781064 9482248099 0366280680 9088537669 0915311848 7002549542
3944183385 4892269040 0123622624 8318951434 3259453671 6276663143 2220747692 7613941913
5800509952 3466966425 6421438197 6038129334 9398700022 3672749189 6862793162 9786296218
0819274118 5390926297 7582123547 8455501643 2357720976 8294045414 8505112341 7987719593
9478935749 2106657359 8046209619 0565811828 2545239803 4251746137 6464722179 1230222738
5945566847 1937871927 2621703314 4416400800 5269263311 0907896575 1500715613 2858078951
4142212554 9400658510 1318721781 3445810603 6203495134 1394118053 6727250357 7877642232
3593065831 4502025100 7733210465 7683224840 4121160000 9946580132 2039122728 2125355191
3613450574 3990853991 0130106601 1333714342 6086069199 5269858138 6700125824 3418817526
1761382087 7745097311 2420091394 6380159850 2700323575 7829036817 9993939457 8366496362
4968563614 1771512291 6639958792 0492792265 4447027419 9092785185 5762559729 0134943766
6290861991 4431754999 8353940686 8861310505 9188824293 4892764951 9066440089 7415677967
6815627365 8001554160 7033110307 5672083283 7074853553 5792174282 2242143435 0720998928
0573329118 7366649217 0192239385 3901414826 1856777465 0195278167 0092726851 5673046825
7570921549 2254725104 7796301166 6886771027 8304206374 2347650711 8139972924 9820618457
8560006617 7311715982 2187000998 2897144208 2316010297 4897740931 2953586401 5611435644
2413439684 6662513165 2353239909 7251583911 1632265710 9696297832 7006130964 0232104745
9244856039 1247120188 9463531301 9420077459 0333228537 8019114724 9072208175 2365438148
4239551707 3535778318 5918684547 5612590544 3732058979 8251767582 7726961775 6798540268
5198606929 2872422320 0596060855 6601495316 2248298021 8573502473 7552429273 6149221611
3571422361 4291013508 7440383081 9191326415 1870376814 5493515327 0196746873 3422478639
2530131802 3457731989 0445767195 5545565116 4816581277 1276188253 0502814090 1059344558
9344015018 1689372707 7585101442 4468326072 4821595948 4784859530 9055485237 9435488160
2835394311 5378046434 0149655625 3052949482 2572508467 5098117172 3110495344 3433903971
1488329882 6254365973 2545784560 5822432511 1401426658 7658552479 4350108172 0921910876
4323998083 5599731199 0875515717 8369762138 9279320938 4915558205 1983349314 6068134536
4349095946 7657782305 1265262521 3408403220 3303950584 1181199632 0625846629 0762834559
7465455178 2570077051 8678905016 9450571154 6142784250 5992543822 1740139790 0255438550
6435966028 5866503397 9897598288 0013055109 3344008271 5877451752 6154351540 6582175469
6532669733 9061465636 0071710546 1585439375 7570893352 6524994025 5669877940 3778696343
5921240236 6054549468 7326568833 3147599457 1265547963 5631235846 2405916619 7997766606
2152243074 5057759556 0591385007 3138214930 4321947452 6763399723 2200130139 6288074072
1410856459 0038583284 8369167199 4919932010 3108979056 5838866608 6633240263 9191386799
3820266797 4506036628 7869293695 7229848126 4120505365 1380293688 4063461403 0837192515
9209118747 2257087350 4889741055 4438862620 6189572599 5436395955 5232487951 9612938631
7957862809 8383284417 8140682732 0159728038 6598316681 5092298557 0726860883 3271311579
7794444171 5875495535 9334326086 0210275662 8537498529 8054807688 9390669958 9365925138
2974751462 0453103797 3246355899 0087430840 3322502967 1705032499 4702867067 1512448078
6417351506 8483505049 9413869925 0227866562 4938105223 0637193757 8675706435 7816184047
8311131533 8878341592 9814180755 5569878779 6482781325 5227598181 7303512292 2769346602
2342875566 1460887533 5962531192 8194300625 2305466660 8466115689 2364083793 0159500378
5623457787 5610944437 0900237888 6614729223 1482591360 6665050003 2655266707 1608505446
```

1778133268 3845665438 4655343366 6648591274 2208148691 6314507406 1340532873 9668207373
4133822317 5900859015 5181053919 8158026847 2165210686 8639469293 7888774165 9250023976
9708089768 2104501678 2905370414 0646950401 9938914328 1685556595 6880033477 0320244094
9940397746 1524566946 6221219433 5443818962 4917050282 8518562290 8659473391 2929968112
8246671611 1105956609 0649946966 2502741729 5563840737 3295231942 1836355756 9310832297
2482176797 8456175808 9995685444 7277939836 0720696463 0355398401 5591741703 6239121658
0339745557 7048836109 9337498573 0793166464 2369646147 4726239389 4454594079 8497856670
0472393877 9714472838 4575867914 8995336571 1536329203 5353469740 5349685114 7812607312
7065037839 9317277629 7034652347 1185115876 1183998942 4020286684 2307398365 2149108814
5455031828 5690285744 5409105410 8773903279 9024718231 9901947349 9503037181 3063794562
7086536283 1325528564 7889003544 0852022742 9538735902 9829368750 5075211917 5724750070
6735291889 6319156679 4581928601 7668936709 1238790925 4214629138 4924513868 0970338816
8479280989 9283376677 6692727419 4129846286 8248401462 8814643039 2051705066 5039692720
8682513110 5853848733 0218569438 2386742143 5661677898 8796885003 8177425619 5475943158
2329108535 0298808989 9466760114 7705200761 9962336368 5393738846 8074240251 7368134035
8356315845 3504544703 4711623968 7021899835 3480037155 0434685667 5751804842 1441264368
3380427787 7306707806 3007633798 3001974656 1814848508 6153519170 0802371497 7959464453
3949059706 6380914203 2270645849 6135782940 3093356315 9366432590 9794236907 1186530344
0076141937 9479930704 9548370999 3009871378 6852588058 4346955201 2684462568 3946592703
9899728426 0510046212 3391294433 3644255869 7271170959 5667312916 4551203348 4628209368
6573074944 9211401047 4992048566 3340151192 1267838341 1066189858 9070077315 1349462321
8583764337 9390601892 2131992163 0331348338 6963195719 7506734033 7393452263 5920714360
3379599686 1699854148 7895611003 5409961599 8317717742 9808094387 0863119084 8919496870
7104029122 5315900797 3531544035 8547821020 0003231177 3529092904 7159356941 7276271669
5218932076 8613947932 7088467758 4716561309 4167311639 7534640384 2430759351 8826183422
4569193201 1518131504 0457127270 5738867944 2899210828 1761807320 6285401833 4455192874
7998566072 4048645715 4508497491 6213104732 7340061887 1945095853 9276903218 9652487780
4226372813 9483118749 2445226970 9583775193 4900304933 2664311978 5768853605 0229399526
5224836921 7285415028 9254515345 4693422707 6652285441 1537407042 3056062876 5614372213
7069876975 6578968813 0879113004 2364023026 0287441363 4485048279 6750463842 0088104477
1997332415 2238135575 0717468704 1989452173 5395585059 0772068065 0547165476 9583629847
5781307383 2594046493 1588217607 9703185390 2710228320 6887262089 3079268447 9750694742
6778771795 8652811432 9154817832 3960732637 8355628585 7374016093 7896910477 0755645717
2528358057 8241918350 4950931065 4464492622 3693861477 0338670254 8085566560 0866170300
1678084301 7233718160 9903925945 0498208860 2843333709 3264010003 7135572435 9240712296
6259616784 2389067219 3779245895 6447223133 1665381845 9445626533 1396702451 7517885881
1751183079 4025111848 5877089992 3489241673 3889710355 2637725813 3838719344 9779324896
7943358813 4392899946 1103170925 2172062363 3964340412 8984656851 1383083484 3907950134
7791976814 7486926384 9234276319 1699626581 2472175255 6539591520 2843261026 3039090268
9523802486 2622942884 2692373382 1729839736 9461767441 8590074119 3995995340 0550987358
7773021788 4583115503 8662912704 8713700975 9834962325 8639152309 4228151777 9356451528
7414939453 7024428910 1297547849 6549471821 1559283256 5346621978 8486759722 2120536489
3721630929 8721650273 8953030022 7475803020 8279461286 1967477227 7134628812 2539927298
6660456903 5773225432 3707787506 0651522310 7443247873 0944159830 7708169212 3569752062
2572870723 2985210320 5218351024 4103956503 9201222188 1400379752 6889207486 0796993539
7317581949 4204164459 5216904058 6977852935 1750615860 0955056478 2259547713 3836452910
8478467900 7513820932 4964254504 7460297088 4151621328 9639258682 8300759710 7702545084
2312468211 8087265580 6112371310 3736368826 5291063039 0593814996 7691895796 7176682979
2693469807 2432105167 9334205663 2233537958 1031530809 7955387566 5071758530 8341420027
0422227678 3456771130 8170157602 7043633344 0744860965 2422712059 9936674842 4602314996
6268882217 3216860456 5233961167 9760934574 5081364735 0774346718 0386581982 0949580690
3050323058 5178829968 3233450523 5661305682 0973445479 9540983304 8110912997 7178293498
4934089292 3975936293 7497604871 7974697068 4959685584 5723108295 0428356570 4770211477
8246379422 3743280017 2174721538 5995417913 1800171297 9563643195 0099582842 7620751196
7887785292 4795406595 9011587890 6182005734 1363121155 4901965273 6175956971 5487498817
5027826629 6328718128 8231403943 3935459764 5529019859 3872626105 1493450479 3090325529
5596275662 1567714606 9709472074 0464291375 9829130327 4413216391 4685081410 2122727184
4204563869 9584587642 4117560143 5404546260 6053584169 8594359184 8147653597 1109569402
7528095604 5089944453 6975397836 5053401590 9087266364 1547204933 1147850385 0495030604
7071075079 7289730134 8354758784 6820129681 7435711339 6096492955 5218648776 7099620262
4997343533 8934090315 0945283731 9182600083 1445978991 3198266294 9225999866 9892871635
2799811973 1048830010 1106989425 0835106901 0992532697 5351715034 2013688907 2557865590
8428860591 7160138620 9688746695 9961136580 6182289419 2585857925 3089019312 5263264914
6702401505 6887384418 4495553470 0027122655 9250066169 7466887371 0338130200 5784013901
2146179985 7925842880 8635082381 1644198124 3326917136 2840107314 4346504556 5754139286
1118482624 8141515777 3739178664 7424534444 3466839460 1741196576 8706760121 6756267387
9462434980 4300209103 9452321450 8143243461 5879567198 0335745453 4581345016 6244371826
7331608215 8926809998 1858621340 2244345511 3181555637 8520552683 8292542330 0580342260

4507044680 6479693139 8141946092 0854550118 5929601515 5785096681 7229901855 5931584715
6341789193 0074148755 2112958689 3945087524 8049106797 0968448442 0641142455 9462638659
8220398285 0757617323 9545321308 2309161948 6113195495 3380559120 0527958570 2613213382
6360781273 0943057525 2547793944 6395493560 5021270886 1233666378 9642318594 0016615626
5316222282 3797906563 2033792393 2947595087 5673398243 9882697440 4490115493 1772977437
2040607018 3434996718 5209936194 2604723659 0427353159 1987888675 1754017925 2825080158
5197144763 8157015631 7007364689 2590956334 2969206103 3144763481 0776017581 3298588241
5072194892 0148414782 3433666700 8831967480 9039169653 9318535389 8807363196 3593256969
6519107187 6816910839 1397487998 1442372102 6491142653 0188525447 9378078019 0965116105
4120365732 2413245048 6547130450 8494714029 5552254699 9260500435 3072069665 3212465907
5878408921 7506866314 4958720862 8827408524 9917204344 7894135596 9343584770 0238532141
5754664687 1992103282 5823660684 6739771641 7610913242 7623602095 4866613182 3354202335
9783539197 1124881564 3161223298 8709995795 2376564849 0670231182 7801462973 2172805736
7593149059 9912382098 8639425972 1421990768 9058874861 6914767316 7219558434 5762623108
0257884558 4788029219 8495486617 5049928174 5951770309 2189278183 7902747365 7123579152
4484560607 8904014041 5912886035 6684014082 4063495802 1872726990 0312830353 9880859152
9714558107 4229522808 0620615008 1613187788 2252924884 6640863179 0080910778 4632365184
3978329860 9742664772 8458163581 9147367218 3474996382 3524015353 3863702350 8052667113
9854879886 2794037698 6148285330 2204076034 0789752875 6598466667 0078607741 2989845999
4466426202 9033729091 4790321590 0899209789 6249079322 0863927139 9816219059 3241423872
9868076988 1075212498 3874687571 6832860959 2940165882 3146680490 3226192807 7682493673
4529230842 6692441180 6239115602 9546183334 8205853054 6835339148 7125882694 3202618092
4564904723 1850414296 1684868620 3484956021 0004523563 8534284931 2935149138 8447387346
7962441446 4794528136 4309548436 1043204197 3462392442 0321162828 3052985432 9125236836
2136249139 4228652243 7153395078 9012926868 7830374105 2956095155 2712207450 7267982969
7459417307 0213441783 9855133857 6437993510 6261721644 6713193501 4255170883 2870747441
8876095526 7740815861 6670989607 5081851513 3897410389 6634152632 4222594358 1388191435
4499434805 8043065869 6225404759 5371982485 8264865851 1500280546 2563555972 8935866933
3657396126 9027162015 1491652715 1285333863 9636506232 3199553247 1082185461 3234914714
4245402771 7574071794 4907750009 5234207956 5375858864 1154945114 7418161009 6498653520
5790059542 9804168110 2673291428 0188988431 1349548687 6285743244 8852501385 5521015812
2113748871 7312085496 1555748076 6514455260 2114754719 9552699489 8568742282 7515371740
0172322232 4339493015 7655139628 3186716099 9711858259 2606598668 6032308478 5950764428
0615220755 0894111798 0148286500 5143038420 2727684784 1376525689 4568868729 7816418531
7674975000 9816852037 7154303152 1600952951 8187764185 8014660727 4438109976 3084478230
1268493913 7925221350 9004027439 0312000328 9378945296 6013127485 1594282837 1032340201
2176331726 6079990283 7152087493 6335056738 1036481383 3051581097 8364677566 8027675154
8973745324 4169305923 3741486798 6181825932 3454707664 6626304641 0327893328 2812842732
6575906501 1349048462 7822786292 7651200444 9726622068 3806467064 8107168334 7966589661
8846962447 8963896364 0421254066 1945572817 8684757514 6221152012 0705391350 2875036601
6451952238 3331868703 4495965057 4007714520 0464990942 8699197756 6847194792 2766862401
8232174174 3118139919 5856597800 6709705117 1180094667 3307163112 3764196216 7633396060
8133061850 7748334514 7938399948 7235989348 7094044426 8099700013 3530029865 7788027018
9990454197 9229597521 3749968520 2868205857 6330467331 3090439862 4515850049 0577815020
1116007188 4367566372 0133830829 6683296073 4239806023 2625498106 8770125374 1582468731
4182820365 3904796244 1750776374 0986563892 0291954181 5098586653 4536574736 1254780780
9507879163 7742317811 3881371067 2154611691 7550985352 9286613121 2937249709 7445940819
7711911746 1551860140 0331190793 2261764881 1391840133 5867838919 5552306453 3290118167
6513709712 1327913051 0878127397 0599438693 6822663166 4836562454 3542561915 0013536370
8201291165 0244761889 4587595117 0733847914 7577257282 2185865516 0616332820 6030137390
8502562500 5664742647 6680592787 7196974752 4611185996 1743269589 9125444812 0432039414
2619611148 8160564182 3931434553 7345564150 8859690674 1604505198 5071697216 2182418128
0757050978 7833160921 4727253123 0352871411 4730423146 6448021897 3405056997 1610024798
0293306473 6246427682 9579305898 3467415916 8814925775 1571782383 0340538207 5261922781
0001601579 9907430813 8851522655 2855399789 9854135363 8717795730 8710717077 4476291481
0873254550 6549065690 8548349718 0688548624 3751138517 6610139041 0651980103 7347999872
1978438082 6508093373 4750477308 2282703118 2968104394 8327291722 3581505053 9288744854
9557083450 6731676308 4388036766 8212669431 6382915432 0237912533 0034037791 3162473132
3429782168 5181354636 6457422034 6293661959 0105114383 8526671583 1004950425 0168670164
4723718078 2112638739 8312662986 4325817527 0199652599 3585040698 5173271334 6375294296
6771319276 9751442620 2080814937 9394978703 8142822233 4056673312 3415200341 7027725006
8549236187 4494374537 3475814636 0905374535 8655922646 9778141285 0696515612 3218419315
9869025652 7190725960 1911082531 1570834417 0539922353 6246042124 3404786087 0160169960
9242102162 8071841480 8081584033 4475905841 9657342543 2580245881 7386744469 6306998368
2210480943 7849916266 2021067028 9704412722 9411324400 9876615472 2744707975 4004624107
5021175333 4332888892 8516277480 6992571182 4171415845 8944674521 1116677588 7657574411
2700827172 1432338951 3661362105 8689083659 5995552734 7579599407 6188700581 2949379121
3554377255 9337936945 4092076334 5472187131 7913421587 3789560826 0729337574 3044459504

7076369485 1004221316 0502898258 1606488431 3199725703 5862689437 8301643325 8105519487
3104332903 1980079279 6382691737 2550058115 6773555343 7523385871 9930017965 9459110185
0937538820 8887611865 0955816160 9482115326 3487271398 9996799061 4354500434 4482939176
3964236983 6033695888 5876589957 6822154047 2216776397 9837446145 5530392286 9537422252
1030282094 8737278352 9004864618 4039990879 5590351807 7765500282 2264917568 5880378816
9772240891 7293111909 5483457094 5449152285 1723984112 6128314361 4422761617 4290574118
0836389442 7310308204 0713582556 1152040498 2035135375 3867646334 1123218156 4930409818
3238662441 1978388479 5612398952 6036146676 9792085658 8775948957 4914676560 2140493460
4616331050 0845004031 1433783401 4050685364 8133297107 5841663482 8494741235 0932391781
6341211164 2270403400 5861766095 3182847635 6470379718 0114042163 1261897379 1841158290
0157045342 6951458199 3363532390 2390693384 2588595075 5272183184 7452659722 9153650643
1275912416 4882749814 6366254855 3676473995 2971915924 7894297004 2139289185 7352506996
4259613930 8193620407 8762923443 5441907353 5524139588 2261102600 5580408784 3076485427
7871525758 7196968448 5334168325 9425453708 5778195920 2021739943 5193491694 7897248888
7955985198 2766602002 1684700141 6899728052 9565197947 7870569591 6719945524 5960686660
4745901637 3967138224 2465789155 2324944315 3346497203 8367557508 2177320583 0782799428
4166616097 3654811674 3733760964 7988973750 5165656908 5135016658 1772381629 6474658857
6446851125 9737784374 2109965935 8089526478 5456162906 1213027398 6445819033 2929302562
1055925338 4291385743 0363731047 0296218025 5509159376 8705772034 5923548780 0627893135
9242769736 8512567275 8579167587 6581288583 2305550021 2809750068 6975005570 4329081665
9615951310 1512895833 9337487139 4753682370 6284454014 2953434691 8010066862 1672104674
5828393348 7642587662 6738266545 3095432904 8240143648 1006970559 4617162062 7095264699
5786892585 8351828279 5750474965 7852210583 4199554799 5653413050 0895224846 6443520429
4338633283 0986251207 6362185626 5743855562 0933109234 3406834735 9570746259 0195307998
8292848046 6548963798 3854138099 4242695764 8680709248 7238049035 8138660585 2778677606
8913362002 7463010906 2262012872 2962773659 4362314998 9457966465 3067893844 5836825667
9854192050 4018618968 0432932501 3014031944 9645132545 3971241354 1870412439 2352898583
5892396124 3292300458 6624655837 5520454459 5460576325 4238262975 9331878340 6047694022
9777667632 6786563312 0266802731 8251527744 9848392257 1305761027 6658627896 6801265518
1448917427 5326996785 6698093884 2831360265 5119482238 3117528517 3814558214 2961373025
3250749657 0763564201 3679414106 9216676974 7615852620 6421770471 3921484602 0489958871
3688821653 4481804333 5467367328 1402297134 9437575705 3984418606 6730226120 2339225090
1775073870 6476302773 8645538580 1198829864 7361514068 8265480407 5006470275 7493412960
6525781846 2993756529 8608536137 8755922095 9176948146 6949698884 5771188376 5579910468
1357225864 5209142154 4902110704 8679535218 6937386944 0692228993 4587118557 2417665200
8792327946 5345146577 2303203272 5804002016 6487770275 4224418197 5464340358 7973486269
2508292092 2561658168 3116684196 7289200015 2219420875 7572856752 4360418041 5094949876
1998136711 7147999655 9845081759 8624591019 7136820245 7693632876 0869568837 8900506155
2668619899 8822324892 9855617285 0699163543 7222915846 0673583732 9062165566 1788102326
8291391567 3888964834 8496693017 5105950965 3938050845 8124525350 3042631404 7905316196
6478645216 3055099261 0651722097 7634258041 0840523721 9436607119 8134058733 3739968662
1429173768 7034233148 7568968761 2218039567 4348531534 6842370028 2969062383 6525523616
7426756817 5514441674 0132846570 0279697260 3163223831 3415862499 1944458153 1323080073
4970175364 4534806706 6107729222 2192562305 9836877825 3270978575 9704389946 9742351070
6730071934 9999442180 9282750939 6569757353 1574151541 4817568017 9780789504 1051384574
6338728800 8618286776 4515524820 2336744461 8488546257 9518721810 6769790524 6748007929
3991046302 6804086669 1926720759 2056935400 5457280950 2055657866 8811658948 7264905195
1729871518 1948022205 5525888169 9929463400 6377556069 7893817146 3110162713 3832739779
5756723399 1606552861 7726671426 6510307797 1031463903 2808094291 6283901858 0724128319
5046314994 5169727032 6986923842 5569388441 7687545148 9554560819 4956791053 1234161257
0143421696 1677139846 1438821682 6462889425 6954598768 6650938652 5304188049 1240128497
3550975645 8658079670 9740215044 9145792884 3032372501 3089732311 2359833114 7230702131
7117534141 3458078052 4580323388 9546095323 5430674782 8717576241 9086118840 6212934917
7703336859 3638824463 3824884917 4540869080 0526667056 9527418464 6997787478 2441719286
3530078402 3549582497 3800548610 8192514297 4281252747 1015662224 2508129611 2248557914
6799547959 2010940286 4728750787 4814519335 0336922618 7803521215 3045159320 2738096594
9272882105 8108746578 5385463921 6039964680 6280469722 5696478424 7294616543 7964246020
8827375133 3780719577 8968416071 9665338968 9105082341 5063246562 9711925078 2739562677
0197433244 8722596400 5653688907 1792437286 5276724944 4358169493 7787352661 3271866862
7432769745 1293159185 5624855400 3405901146 3418768610 3349267575 0306950063 8511473631
3123931042 8348537565 1064287668 4259872519 0408580729 9652241214 3184492566 6243545700
9373682680 9411746846 4848377921 9123467678 5683848178 7771711756 4503905306 3309442647
2376609925 5541628466 9307166723 5561478066 6185861462 5861942691 6613364668 1998291459
8448227401 0512127152 9679039155 4779406915 2473486576 4481814821 5877907221 9248808583
0922784832 1462270344 0422351595 8875978313 0392091036 0746024218 0257007050 1771491325
3006430753 8061988390 61119

The sixteenth numerator: p_{42037}

These are the 21,667 digits of p_{42037}, the *numerator* of the 42,037th convergent of π, in the same layout. Its SHA-256 hash is

`ca837a3355299394dacfb035d2b07ad092625da952405d18ef3a1b2389510098`.

It is a probable prime (strong Baillie–PSW and 25 Miller–Rabin rounds); no proof has been started. The script `c12_third_fifty.py` rebuilds it.

```
4192028622 9331659941 9568810055 6743296102 5686683687 3732680900 4362136858 5956099983
1477699698 4269066073 3608329734 5784569267 7168684778 7575722945 9629538591 7085619245
9880461595 6430895778 6563734922 7070110720 0433462747 2747899069 4721328255 1712958488
8584468069 2237946118 4368230798 0834195498 2749397653 1945275040 5883739885 0554686005
4925063847 8008841357 4000172862 6900303515 1960442210 8328057259 3219736748 5292896417
7289576120 2461724207 3037787396 3075141774 0320830332 3936089399 7714081665 2625567546
6256021945 8457037805 9737500694 8908614073 7077216748 5636643960 6910407569 1547265785
7456394237 1220019502 1420059983 9914151786 6697666158 4877988925 6814634338 4010193961
5397714246 6243372352 8471370410 4541456320 8008110224 0829491247 5939285889 2219691249
8882392834 1633154595 4197833937 7341372558 9821216620 8830323456 5434185276 5689820591
6130269526 1440719955 0217731892 0329740986 7282880569 6002999552 4087896446 3888857855
7729416900 9784677741 9977519787 0535387506 3194443963 7244980494 9619006204 0924015432
6460546333 2326996663 5886513785 0492550525 2305763548 5287419345 5220441489 4269275299
7887016495 8844642531 0455327106 2217194226 9605810425 8968834092 9627383956 3143227144
0500645598 0042763265 9945021426 6592183162 0917249375 2807738160 9220037364 9396015036
6944066246 9385204941 4513311329 8045291335 0027389461 2339620522 9847951158 6295828248
2389334862 7990100831 5846403447 2525921822 9437875079 7590492719 2882023591 9343883107
3655171840 5200723866 5092815181 3713779928 7686045460 9947348915 6302657542 6801488697
4367356537 0178315229 0426073275 3594035612 9964393510 7043117667 8984841926 3467513054
7328031850 5438234265 7575537866 4716560472 4268503484 9228337182 2184919435 8156481540
9612418779 7778344579 7406918616 9559875369 6206406586 9987085678 1406852082 6189628200
0445344812 6620943212 5293107936 0630995936 3784526880 7579768267 7492661033 4557751681
9533055146 6149186074 5910121409 1675395134 8152263638 0443478698 8590383125 3085659421
8400664014 4696696674 9523980836 5573704836 9448492508 2048362388 7688332524 0380393764
5732890515 7635429301 5855049596 8027490879 7870105771 7282467781 6174541010 9249596211
9402770382 0580502446 6127003017 4314519416 0696113570 9195810212 6954045643 3343452890
4325190843 9083203926 6300391153 5404439684 9443965306 3834086216 3445407186 1381150549
9761426710 3373941077 9179604096 4956833390 5255430207 8902990968 3304519478 0584475362
7777365684 1020138103 5158911693 2285200579 2395026054 6939478457 7326827395 0511591850
5546474830 6005795022 9329031482 5479276430 2677510784 5069202821 5567239956 0797637821
3212343203 6097432819 3100679273 8914809961 6129337355 4460773005 7343232694 8177524418
2939281313 9535913462 0540252911 8622669439 7738881425 4654625210 9350773791 0523187487
5774586616 6946195019 2756800301 7174793050 5299478886 9232835581 6489605245 7093608288
9741976870 4866737819 8957150425 6452946587 2007706069 9356220487 1115901314 2298140254
6595049182 7297086977 9323409184 2303244799 9059423204 4258757550 4499416632 7608861640
4071467364 3693463188 4862485082 1867083011 2101640333 2343230715 4931740838 2372755124
6513712331 6285204216 8786635303 9599796721 6742578404 1230061349 4656760298 1240400534
3639934178 4224422658 4665904314 1316367185 4068537617 5662036454 2547025015 2211262560
9738800988 0568958943 8025658343 2834104527 1058314469 4190249510 5951293873 6760744448
4045754391 0264757022 7707397829 2298680929 6376754402 2517281216 1874773843 5798018049
5047212509 8801365211 1217060727 6510609201 4465349925 2639008152 3943635879 9689313635
6086660774 4530218871 8403630402 5135670512 1321146707 9241399602 6146879895 0179452554
8525842841 5480384317 5852366346 6113458344 4839574265 1922939290 1825146982 8086861759
7180294459 1638248560 8957601849 1013565207 0102424562 6375049308 6137961070 6253134809
2394621232 5513935837 0923012226 3616243652 3416604197 7156468859 4282222027 9071832522
1457896072 2830935485 4384875558 6241532418 5854139404 3550585172 8879572650 6334017228
4727443201 2724723118 5009997793 1786932210 5709446445 2884922991 8092012553 7888161191
2628909961 5226538173 9347365674 3844815147 9654536434 7085167155 1787054970 7063373861
7405829457 7399940588 1058126072 6601474773 5851323720 1751713386 6415619060 4419377633
9379051835 8565820108 8219644743 1486458571 6211183732 3708117966 1500356480 6655109691
0661089896 0093292694 1933745401 4036671140 9980685048 9087660653 0184022780 9226693481
2189264736 2075897929 3302598724 0454532941 3348163193 0825259055 9865672388 0054507517
7820956956 9553717284 0273081210 0668160639 0518771479 6524918968 9605890386 2529742535
8812478346 2124314250 4825474683 1658877347 1082497641 3318427684 2636001463 2095756036
7160624391 8183250879 3340509090 3126555338 5513271086 0855291895 5453117440 3206627426
1668588175 9966164806 0061936362 5361217464 9666000389 0742566694 7261679620 6443294397
```

0031392275 9952566180 2585720202 9700977120 6532133030 0878302212 2261506304 8987586785
0583517849 8825184359 5600523368 4559517561 2450958937 0399415857 5969181844 0133509190
5894901391 6369656335 3498735136 8526378068 1786950386 1898324218 2760251688 9544775555
8882767195 4639302762 7482199972 7093698565 6528123716 0449744569 6785958539 1929071152
9159115020 6156806055 2278015610 5641394347 3144735488 7367468062 4076916246 7224678528
5620043147 3541564704 7806527074 6705034402 3350525232 7525887530 9877015230 9279443779
0750391281 4932305164 0922974059 6767357508 3443479575 5290396180 3731389306 4136442550
5135019497 9532092075 1444526475 4462099068 6547326164 5208598646 7923157738 3963101566
3615302649 6976488303 8718282716 0934266585 2885753049 3089347843 6596586858 8472925067
4029885171 7318121747 0417098729 5595476157 8934710004 7822642301 6953852356 2676388592
4647571019 9258165050 9279195920 3245896566 4352824501 5401224489 6063719622 2814682029
7338848534 0261966924 6322752880 4513274894 0576755182 1765586350 7131327105 8214153281
0563149508 6058316636 3680303689 0391969808 9881631331 8481529909 2332892238 7923201842
8719387612 1800772123 0191433267 4695046882 8805987147 7767882477 7639147960 2795376910
8611039594 4591613143 6389768921 9987764365 4945965501 7957156697 0957706926 4447226583
0512165028 0604798349 9641591061 5843289650 9670540395 4217626129 0765295375 2524310436
2994102758 9519622396 8154547718 9453654620 4646802186 5681940950 4774327116 9621541875
3818076681 3410995046 2132279970 6587337203 1756767783 0851647913 0162497091 2127160870
9393265671 2334875367 6754789475 8041211178 3940238562 9098969490 3722454214 4715386611
4946181174 8691310211 7564671794 8524800646 0935309720 0023367979 7359491490 7969801296
8375078804 8873948317 8280373966 2194663454 9423202012 5593808715 3948448123 5284726805
0328060961 0367640868 8041301174 9290271645 4099254985 8199999832 4946811396 1714143954
7168646751 4092499983 3004116174 0529089394 2094649794 7207354983 6962897083 0798283359
2322711465 4167073276 5333756123 7566255019 1106981419 2401836027 4789078350 7025408904
2368913629 6572627725 1237451735 2003299411 8298248434 5834434850 5266601208 4691303055
9633951001 0802915293 0282945906 7843573738 2005146825 8134790429 7908390582 4972075409
6635774133 3293336690 3416049684 0517888087 9974711587 3000566670 3165806736 2146832260
7686662398 8841882410 3165842157 9707158809 3955760331 7909154122 1402948720 1775556748
0137187697 9675179031 3746662906 9529594815 9472985171 2875965857 3545570241 5668995875
5109646524 5722583238 9684151266 7705209154 5711708189 8887688729 5915665909 1660769242
6313127711 7088955817 3758776229 3533811981 8540543204 1031550062 9425640425 0216356777
6632873187 1532313541 1707513381 5711879776 7789694218 4040015941 3539540898 0741887346
3404008981 9246892890 5078042255 7038487996 1970268830 2835829764 9416041120 9871022623
4561931199 4742660270 5528445712 5562184592 7812206264 0090000136 2746540684 3378792049
5756283002 9452705705 8918593617 8194498983 4576311690 6983826049 3829541228 7495126379
9196813798 8936010743 2973668520 4984853145 8370769663 4174908240 1714092072 2217237281
0377868823 5904474604 7550168390 2477454956 9879561814 8937006222 5668494695 4695859302
7511634355 9412259174 8384833927 1866084205 4232589320 2145141815 5124208692 9764202652
2172504519 4566776839 1529909264 8458947332 0609872171 0377099701 9397977658 3605932942
7339200135 7338075325 4937500913 3239786426 8719727146 6779848971 1566231160 7616193976
4195131446 2986655518 3474475186 2807591337 2558805425 4233747828 3003455999 8187285186
2056725298 7195114137 7275769294 6117448213 4347842739 7937703902 4614860668 1659065255
9409521644 6227797028 6789056108 7762828597 7953504553 9790963434 9602186613 0609193763
2545437041 7929322255 5441282506 0609105891 3797349620 5740008844 4582869378 5558736051
8553365982 7357375053 0200065962 0182395171 5043371240 6812136885 1116448822 6563688139
8371086096 6229211653 0351304715 9488128867 3294545059 8883702385 8073731695 5642530895
2226064497 4656228543 3086465020 1262576801 8934244973 5974458522 5943465953 9435768566
0218865549 6782352421 9753184902 5571390446 8578691427 3683522105 8335177400 2875434149
9028440340 6646373495 0793336244 9396387375 2856191725 7308723165 4431827657 6956489376
7880160992 4945735387 7626269707 5429549633 5224759304 3343832737 5331983598 6757563012
1694892272 5085192769 1170408111 8071616095 3577855799 6509406511 7011396950 2160830596
5512594966 6061047901 4409907366 5621999117 4975813549 6510923970 1588723893 1981442009
1086460139 8850272450 3086311704 4054600616 1416386812 6439098702 5999670169 7224199808
5870590881 6964887278 4541728063 1371284588 4356661171 0238890356 0377054087 0742507754
3756981098 5189597914 7382305882 0093339298 3000669915 7645408404 1330321530 5518514340
6122768026 8287987796 3486885719 3736716419 6365214121 8289787616 7484569789 1088745139
8027462405 8133043478 2318543199 6426515980 7493410733 0756237855 1148823387 0898757896
0865186157 8249199044 6599354829 4047928553 8405213647 5980919073 0419564180 6842204547
4636414354 1939299267 4622029963 3102772777 5841459039 6508330798 2421991631 9793011470
6313267159 8656047798 4063936510 2780445719 6965365016 8488357452 7241461191 6864686816
2628211911 2012285348 4458650423 6939320936 3567757644 2447748594 9778260249 3346468473
1598355831 9731905939 9085623231 6509506121 7639164869 5587748966 8660273372 6219464695
4002053876 4722989836 4534066562 5881554935 6481518497 5265573788 9982940376 1490157230
7205189311 5592844839 3285755368 2796020119 0519113460 9796202568 5761129829 7895037710
8860028886 6459330296 0164736679 3808327533 6147474002 5856886468 0762711584 2318940756
9004735488 1340044889 7981086886 9295533919 4032557836 6803517566 7006739826 0708961975
7553723666 9981589670 9858735443 7052857614 9455684156 1935709569 8414753374 7614119951
0440443347 9963887755 9469350668 2536036099 6039131406 6671723022 2355057409 8264869351

9276628821 1991943796 3442199818 1728448280 6536162474 5609774026 1249623120 7104992932
1136933209 1511306994 7664259760 1328798640 8657768574 6756156023 5209524847 9242716112
3126839647 7089419389 2566111531 8793051203 1921315464 0398309737 5199466966 7359000042
8060985455 8101306810 7818267581 7871862685 5382259887 5972054329 3317517076 1687288002
0817855912 9184513762 8583567960 3837852985 0642446502 0658991843 9437541513 5445283515
3215298328 5884588041 0961042911 9559759999 3801829358 6142226578 7324458252 6049925076
2927133875 2566175741 2230083774 6232673476 9731164785 1968679882 4564173210 0720047454
3938222036 0053411944 3927568303 8648660411 5518895910 3299250046 8250487970 3338273982
9160226032 7492311403 6549975496 3426282838 9051023224 1262138893 5813573870 3758931736
7548874692 0294815900 3835522891 7019162731 5684614605 8974431382 5401114409 9113579151
5074952588 5125520555 1902146612 8526896868 2097009154 1491492407 1806261847 6918728269
4439092012 8059345316 9756322734 0789472990 9867853240 2722202453 7630857903 2639704510
7846532161 3762147624 6178664855 1960481929 4204525118 7675274461 9637055197 1200676410
2879794777 4397918507 0178054998 7072478659 0524516733 7567063446 7395577041 5959652768
1446139751 9087051484 1280406305 5903942513 6985239299 9615487743 8360984749 7583427950
2466064736 3398180823 5981747301 9161805556 2866711545 5036504850 6021587193 1509445576
8293368827 1682833257 5838559951 0012415912 8021840081 4719472894 9414892600 6242664008
2376011635 6043397062 3049686538 1270900607 3318652968 0378361654 5946390654 8148399294
3461576394 8863521621 2254589486 8538185358 6235707844 2758242325 3290349083 9909818244
6170900763 6012950631 8825228416 1327892785 9899824708 5724486388 9063511490 0898557189
2948800862 2086895945 5774545462 5565226803 1739710326 8541994693 2168758462 5392696071
8599304374 0873260263 9421880816 9792800002 4399430878 7367144361 2544414869 5354036845
4822260955 3623135059 2892266754 6669997095 9556329184 2818093834 7047482455 4103653985
4230056164 4040647725 1982821676 8429373580 3780104531 5825968524 0161053299 7584754294
7968500793 4626781453 6593465750 7018285678 7638285136 7324149278 3737407205 9807406177
0064083200 9244196521 2581750512 2535275664 2399417260 0377604416 6119692838 1332209497
8772572021 5667899059 2891436396 8298909346 9241833509 9849612251 2341607199 6280157426
6125851338 5665601241 8284506172 2533355077 0249370404 2459842221 6404233547 4041251037
9660039848 1064766357 5889964729 1992327495 9634125356 9369044458 1221257713 8624005450
3046980535 7134253283 7276344842 8530924705 2721085680 5663679544 5724763564 9254792535
9530292299 2669737376 3580169049 8991808208 8684884562 1262193463 4599348623 2038966770
5505930008 0174577592 3387668771 8502000635 8981401688 8549368084 8897607139 6652598032
3966271630 1788812470 3224137285 3214375446 0820523579 3813521549 3735622819 4947034799
7104611998 6873249904 0505365554 0866764700 2045858427 1632335871 9824360700 6383800820
0600173075 4111842500 6811431378 2974272787 1236152591 2839098169 1480041508 5129051166
2218332897 6756688363 0856930936 1724137412 7618858245 4513893557 5353356384 2151156528
0957907172 1570215593 2666409489 5138865632 6459789814 1102312980 1067031920 9374904353
4600139009 0990333140 6145507272 3860897151 4637414040 3826052634 7265319401 5127053053
2666859960 8612201473 7685074385 2083988859 5768116048 1844156401 9337700719 1277208503
5453974725 8708900255 4485874519 5045780157 8677828288 5602729982 8956976662 7130267956
9146993919 0278001001 9804429724 6223727490 4018665163 0796542028 0791782074 6838028660
3642657530 8653534216 0114858015 6028977516 0075559515 2690713070 1137814729 2072426674
1454554824 7131551862 6321261763 6278291934 4980348504 8632008145 3749911923 0278132632
4970929811 7412196193 5836251828 6151850714 5218817565 1333581567 8689231555 3596344701
2337895806 0030114018 6612932322 1643076038 1628699722 5044388037 0002458440 1289387365
2535801078 7113773013 7812293941 6526347463 0840163959 3710466607 4869900014 0973436509
2685522102 6439159661 4904667858 7228148715 3736073538 7218544982 3436877984 8486914810
1691148008 2909039768 7775354020 7360993653 8015088398 8617658311 2497113413 6212308331
6011914569 8603769466 0034785467 4028053717 8190683313 3791887502 8739473156 0603312396
6916858565 9849804203 8563495586 4180344717 6986265177 5872628809 5921593087 1237808642
6522704153 0438583023 5697570625 1176516105 7851461416 5334296994 6851452818 5824774203
2034510095 7112099391 4872020562 6850173975 6040741259 7309222826 8328467593 9597715982
3750235840 0468243290 8528621473 9739207266 8971970989 1458346648 2195097671 7914657167
5180442388 5928455735 7360488160 6770663811 7853139367 1828808898 0911001412 4902242614
2470916800 0734023527 7020003084 0906102867 5640339141 9420374848 9852723924 7015408829
6860929432 2186299291 3439509159 7654674543 9541844203 0832686073 5287008830 4549447557
2216605553 8531648166 3399375834 8909910770 3464512049 2715271540 8193447247 7875337477
2593257424 9019458271 9305826667 8941170533 6186756066 2198050159 2728491371 3635485213
6666880716 5978220070 3206642401 6927232844 7900322932 6136341464 2591248171 0792542423
0158393220 6660907688 6343046629 0416041940 3073073014 3991800955 5300281791 3658305653
3150871439 8533978129 4122041310 3686789186 1284765891 5158048134 5978794617 0293138962
6013122806 6633780232 2254236671 5697232347 6764597766 4075999785 6379848898 3453365498
9782693393 5980637547 1164035861 8280153162 6429533172 1365114291 4073977996 2126388166
3911135259 1795033018 9374010506 7621092020 6132760812 3495359505 7908875699 7156109506
5857866688 3712314817 5804718611 0740385482 0489182117 6472887776 9974259955 1931890382
3491548672 4173555520 6286571338 0923490140 3871442912 2677556410 8180577795 9095528485
6488402028 1794954361 5712147651 2167948960 2467305823 4671448059 8064517416 2856905279
0765837354 0771347529 4267912453 2380437768 6850937212 3579730511 6836604403 3836519435

2651873028 1362754604 8298753878 0571606725 1592087343 8994552832 4867864412 2833090096
9865201544 9633822663 7685991526 4026225514 0441219710 1399076771 8575258814 9668741099
5116669900 5391827333 5321288387 8093945140 1352482046 3222131364 7589711118 5480155706
2679252415 6898736810 0751155418 8882492654 0070995773 5877221512 4242776895 6181318043
6428930556 2589728639 7459436390 2139748386 9102180889 7983666096 4406149399 9565062560
0465271585 8912277798 7721553226 5503525231 4241971697 1419917235 8431093982 9287776011
2471063789 9015962807 7805625303 4159338561 0912780647 4912088564 4590361261 9082306694
2334919154 5052896392 1731505928 1555850962 6835033599 1302589106 7356854981 7225756902
7780697885 4368820534 3362103603 6444049961 0099205441 8089834848 4064111127 8678635011
9052580932 0849729984 2142079204 9223310616 7835642636 4605751446 5917811736 5228766074
4151263522 5506435138 2463076291 6435322457 0861263550 1029694550 6749904481 2054022796
9380632282 8507517922 8482115734 7530158681 8015741216 8535987736 0321795533 6267121865
9074252357 9844602889 1662043107 4406580200 4425201330 7622977436 6104008664 9902754422
5298522578 0733505835 7470165112 7670661292 7187138858 0594160177 4543629362 6490254515
5029060108 3555453550 9192506001 1937483998 9533790098 7878460589 9991342557 9693032736
7596580364 6132169946 6885360641 1652428418 9590014893 3214106952 9946913237 9310437154
4546678919 6150710309 1894725838 4523538372 6428339219 5797688981 3205242451 1338500136
7892064676 5599880040 3332888954 3557102630 8935349189 8158365251 9594094331 9628417690
2101365726 0463014599 7473500699 4558395941 5013749402 4176732668 1049294706 7455092868
6473499278 0523148273 4309747595 1169587842 3939953030 6885087752 5008718112 2299122007
3199076899 7004946577 8950354711 3485226058 5654967044 9500264630 7087401851 7867513815
3976551518 5639523351 4866248905 3241669803 9279827549 3250591702 5816700449 8289450182
0502144717 1687108654 0833023217 2508933249 1983578996 5112070942 9228351452 3898464472
6046761975 6962450686 1617911403 8650701736 5353776637 0872750514 8027858997 8833721457
1655934142 0426032429 8690768263 2840274848 4037045522 3973327246 7673914710 5069929625
5874783177 8952064549 5330671162 2634257441 1228321463 5089313963 8532101581 5795442268
8900985643 4087838787 8466779558 0981831847 8094954329 6077585441 4261464820 4827907613
8737412928 2806326244 1155541668 3473210184 9909788010 8653987549 1116894713 4225185897
2345497191 5058356800 6225323898 7379780328 8597530122 1664283646 7899466014 3865844187
7283979985 7916888831 0708863938 3923295056 1194715683 8696302443 6042570423 9595212784
2256040782 0080635713 9512138634 3981043403 2848120997 9122315765 9014408628 2983465335
1433411656 7426240532 9166810453 6111893776 6698212449 8297246996 8198685211 0315701159
4176108678 6517473928 8523718020 2620249650 0550401480 6665845841 0619733599 1878966959
5194743992 1731875292 8134172642 9344485593 4325400171 0588724129 7139644161 1038982881
9452379473 7235819790 2850248583 0683666413 2485851289 7650984026 9980050516 9310477225
8508734788 0968457767 4726351982 1954660828 6246040654 3580764525 7571888691 6098753168
6065895975 2701564812 5642281986 9762411566 3060397471 3675899368 2411841435 3205157585
0857886968 9300453037 3465820017 1919095187 9323547783 4898431429 6805014372 4048168173
5643450061 5548278727 8308367252 2990293024 2784400841 1693394666 1927451366 4998678747
5893527984 7550653728 1348533001 2132443923 3636908414 3860886489 8133648349 6611656377
1431107694 9810237320 8375085877 9614499938 6288740615 2759128020 4113937693 5788585548
6984461000 5588765676 3805305150 7385631345 3847424655 6875214620 5883516078 9274031937
9081376673 3310979716 4967814956 2558456300 7533086819 5718923294 0069415750 7022547179
0496225885 3992524035 5635912007 6592519833 8428333601 3651740122 7879459788 4848758769
2790220167 4661680001 6492332409 4962686157 7778133474 8738586334 8875664939 4016159027
5077160199 4441779616 0136356993 5353708556 9297035151 9594003361 2430433178 3620692297
7846867756 0730372421 6626391128 0577245779 5229185689 6286528470 8334893264 2346834952
8435554707 6788434983 8302910749 4371620725 3320273093 6756669420 2362111391 2009449131
1763745459 7601166417 0378936642 9569538306 1347341528 9953939725 5951808448 6409175665
4934369731 4952846519 4257673088 6498024329 2925869843 6714009588 3983534821 4989521218
7094246399 2710304572 9927181947 7785765442 4853530964 8883304295 8922847237 1688966071
5924034981 8439967994 3630465732 0285838186 9998704978 1613699627 4519556923 5890969083
8300257488 4050192965 0625429284 7788925761 9263079555 9936489607 7087671734 3915109349
7088376000 0609685497 3082016324 7419173804 9074384297 8234542459 7866603307 2681090694
4665083592 3604782887 7933316231 1359690343 3795993202 3331239136 5986785437 2873082366
6664345441 1317472991 3318742075 5037390265 3753836002 7387566463 9346054810 9014994313
2677446793 0920776238 9473118764 9031776156 7411714638 7239156030 0293875522 5523472022
1022345480 9769956353 5831932803 5327635928 0497175843 2493295738 8604587256 9768323770
3144168859 2905467973 5600226198 8110980913 3040090397 4924646150 4017595514 4287560214
4613390919 0965240117 0996777798 1004757367 4591650742 2229055934 7659785277 3597814138
7635992188 4774032857 6595025579 5330967970 4847848399 2796480251 9110277567 4962170006
8056223944 0832955138 9492948174 9012433642 7233850384 1987400700 9424038880 7052970431
1168742447 4441026050 6869169300 9878846418 3403942629 4458411562 6083579648 4455218636
4830360983 7217500514 0208632406 7393974025 7674418994 3927573771 4272713912 9313076336
8286311559 9404165276 5469771736 2955980323 2135808936 6077507569 4672661243 3583281238
0406123119 9355768178 9421925792 5378662533 9721855302 0729747220 0142913830 8020665755
4223216205 1348702682 1244459257 0727848723 3804877988 0169051871 6966136717 8183165745
3382483826 0561633186 6362681392 1792139329 3275661402 3538480875 7620029738 4567729230

6088144674 8244500579 5624473443 6339531934 2921833762 7106343881 7405234462 7539428124
0382249095 2440492275 1868669536 8259457833 2228695598 1771157786 7431713841 6710385542
8938163156 7135003649 2967046607 9131738896 3612268676 7639686564 9306202876 7973848660
6326768837 1448943851 4734999520 7913900766 1504778991 1880283884 4457788536 2833689157
4989451527 6668768066 8744511425 5740655253 0149784507 2167048812 8705387795 2285624122
6782002501 8355280459 9037266481 0212023736 8413122513 8745308411 0197750209 0499732015
1778001074 5388254882 3755419263 5819945858 4307138369 7600181824 0042658922 0476243775
4754628395 7550573857 8740309173 7278594571 3599265741 8376898345 2730463373 0732407270
3757482497 4249085880 2516060925 9619605350 5786256374 8177616179 4706025045 3597973371
8077121254 9713676390 7446682431 5333648940 4353498040 1319463684 9778484536 0916360054
6729406809 6639786209 3128298309 5376765387 8256921258 0126280918 0879521

Bibliography

[1] Eric Agol et al. Refining the transit-timing and photometric analysis of TRAPPIST-1: masses, radii, densities, dynamics, and ephemerides. *The Planetary Science Journal*, 2 (1):1, 2021.

[2] Christoph Aistleitner, István Berkes, and Kristian Seip. GCD sums from Poisson integrals and systems of dilated functions. *Journal of the European Mathematical Society*, 17(6): 1517–1546, 2015.

[3] Pascal Alessandri and Valérie Berthé. Three distance theorems and combinatorics on words. *L'Enseignement Mathématique*, 44:103–132, 1998.

[4] Jean-Paul Allouche and Jeffrey Shallit. *Automatic Sequences: Theory, Applications, Generalizations*. Cambridge University Press, 2003.

[5] Y. Y. Atas, E. Bogomolny, O. Giraud, and G. Roux. Distribution of the ratio of consecutive level spacings in random matrix ensembles. *Physical Review Letters*, 110:084101, 2013.

[6] A. O. L. Atkin and François Morain. Elliptic curves and primality proving. *Mathematics of Computation*, 61(203):29–68, 1993.

[7] Artur Avila and Svetlana Jitomirskaya. The Ten Martini Problem. *Annals of Mathematics*, 170(1):303–342, 2009.

[8] Michael Baake and Uwe Grimm. *Aperiodic Order. Volume 1: A Mathematical Invitation*, volume 149 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2013.

[9] J. J. Balmer. Notiz über die Spectrallinien des Wasserstoffs. *Annalen der Physik und Chemie*, 25:80–87, 1885. Third series.

[10] Paul T. Bateman and Roger A. Horn. A heuristic asymptotic formula concerning the distribution of prime numbers. *Mathematics of Computation*, 16(79):363–367, 1962.

[11] Matthew Bennett, Michael F. Schatz, Heidi Rockwood, and Kurt Wiesenfeld. Huygens's clocks. *Proceedings of the Royal Society A*, 458:563–579, 2002.

[12] Dave Benson. *Music: A Mathematical Offering*. Cambridge University Press, 2006.

[13] Lennart Berggren, Jonathan Borwein, and Peter Borwein. *Pi: A Source Book*. Springer, New York, 1997.

[14] M. V. Berry. Semiclassical formula for the number variance of the Riemann zeros. *Nonlinearity*, 1(3):399–407, 1988.

[15] Jean Berstel, Aaron Lauve, Christophe Reutenauer, and Franco V. Saliola. *Combinatorics on Words: Christoffel Words and Repetitions in Words*, volume 27 of *CRM Monograph Series*. American Mathematical Society, 2008.

[16] E. Bogomolny, O. Bohigas, P. Leboeuf, and A. G. Monastra. On the spacing distribution of the Riemann zeros: corrections to the asymptotic result. *Journal of Physics A: Mathematical and General*, 39(34), 2006. doi:10.1088/0305-4470/39/34/010.

[17] Andriy Bondarenko and Kristian Seip. GCD sums and complete sets of square-free numbers. *Bulletin of the London Mathematical Society*, 47(1):29–41, 2015.

[18] Andriy Bondarenko and Kristian Seip. Large greatest common divisor sums and extreme values of the Riemann zeta function. *Duke Mathematical Journal*, 166(9):1685–1701, 2017.

[19] Bence Borda. Equidistribution of continued fraction convergents in $\mathrm{sl}(2,\mathbb{Z}_m)$ with an application to local discrepancy. *Journal of Modern Dynamics*, 2025. doi:10.3934/jmd.2025005; arXiv:2303.08504.

[20] W. Bosma, H. Jager, and F. Wiedijk. Some metrical observations on the approximation by continued fractions. *Nederlandse Akademie van Wetenschappen. Indagationes Mathematicae*, 45(3):281–299, 1983.

[21] Achille Brocot. Calcul des rouages par approximation, nouvelle méthode. *Revue Chronométrique*, 3:186–194, 1861. Volume and pages as cited in "On Euclid's algorithm and elementary number theory", arXiv:1506.05981; we have not seen the original.

[22] T. C. Brown. Descriptions of the characteristic sequence of an irrational. *Canadian Mathematical Bulletin*, 36(1):15–21, 1993.

[23] Clifton Callender. Sturmian canons. In J. Yust, J. Wild, and J. A. Burgoyne, editors, *Mathematics and Computation in Music (MCM 2013)*, volume 7937 of *Lecture Notes in Computer Science*. Springer, 2013. doi:10.1007/978-3-642-39357-0_5.

[24] Clifton Callender. Infinite rhythmic tiling canons. In *Proceedings of Bridges 2015: Mathematics, Music, Art, Architecture, Culture*, pages 399–402, 2015.

[25] Norman Carey and David Clampitt. Aspects of well-formed scales. *Music Theory Spectrum*, 11(2):187–206, 1989.

[26] Norman Carey and David Clampitt. Self-similar pitch structures, their duals, and rhythmic analogues. *Perspectives of New Music*, 34(2), 1996.

[27] Ernst Florens Friedrich Chladni. Entdeckungen über die Theorie des Klanges. Leipzig, 1787.

[28] David Clampitt and Thomas Noll. Modes, the height-width duality, and Handschin's tone character. *Music Theory Online*, 17(1), 2011.

[29] John Clough and Jack Douthett. Maximally even sets. *Journal of Music Theory*, 35: 93–173, 1991.

[30] Henry Cohn. A short proof of the simple continued fraction expansion of e. *American Mathematical Monthly*, 113(1):57–62, 2006.

[31] C. R. Dean et al. Hofstadter's butterfly and the fractal quantum Hall effect in moiré superlattices. *Nature*, 497:598–602, 2013.

[32] Arnaud Denjoy. Sur une fonction réelle de Minkowski. *Journal de Mathématiques Pures et Appliquées*, 17:105–151, 1938.

[33] Robert L. Devaney. The Mandelbrot set, the Farey tree, and the Fibonacci sequence. *American Mathematical Monthly*, 106(4):289–302, 1999.

[34] Manuel Domínguez, David Clampitt, and Thomas Noll. WF scales, ME sets, and Christoffel words. In *Mathematics and Computation in Music*. Springer, Berlin, Heidelberg, 2009. doi:10.1007/978-3-642-04579-0_48.

[35] Freeman Dyson. Birds and frogs. *Notices of the American Mathematical Society*, 56(2): 212–223, 2009.

[36] Freeman J. Dyson. Statistical theory of the energy levels of complex systems. I. *Journal of Mathematical Physics*, 3(1):140–156, 1962.

[37] Andreas Enge. FastECPP over MPI. arXiv:2404.05506, 2024. Software: CM, `https://www.multiprecision.org/cm/`.

[38] Paul Erdős. On the distribution of the convergents of almost all real numbers. *Journal of Number Theory*, 2:425–441, 1970.

[39] Paul Erlich. A middle path between just intonation and the equal temperaments, part 1. *Xenharmonikôn*, 18, 2006.

[40] Leonhard Euler. De summis serierum reciprocarum. *Commentarii academiae scientiarum Petropolitanae*, 7:123–134, 1740. Eneström index E41.

[41] Leonhard Euler. Variae observationes circa series infinitas. *Commentarii academiae scientiarum Petropolitanae*, 9:160–188, 1744. Eneström index E72.

[42] Philippe Flajolet and Bruno Salvy. Euler sums and contour integral representations. *Experimental Mathematics*, 7(1):15–35, 1998.

[43] V. Fock. Zur Theorie des Wasserstoffatoms. *Zeitschrift für Physik*, 98:145–154, 1935.

[44] L. R. Ford. Fractions. *American Mathematical Monthly*, 45(9):586–601, 1938.

[45] A. S. Fraenkel, M. Mushkin, and U. Tassa. Determination of $[n\theta]$ by its sequence of differences. *Canadian Mathematical Bulletin*, 21(4):441–446, 1978.

[46] J. Galambos. The distribution of the largest coefficient in continued fraction expansions. *Quarterly Journal of Mathematics, Oxford Series*, 23:147–151, 1972.

[47] Owen Gingerich. Kepler, johannes. In *Dictionary of Scientific Biography*, volume 7, pages 289–312. New York, 1981. Volumes 7 and 8 bound together.

[48] Solomon W. Golomb. Combinatorial proof of Fermat's "little" theorem. *American Mathematical Monthly*, 63(10):718, 1956.

[49] Francisco Gómez-Martín, Perouz Taslakian, and Godfried T. Toussaint. Structural properties of Euclidean rhythms. *Journal of Mathematics and Music*, 2009. doi:10.1080/17459730902819566.

[50] I. J. Good. The fractional dimensional theory of continued fractions. *Proceedings of the Cambridge Philosophical Society*, 37:199–228, 1941.

[51] Ronald L. Graham, Donald E. Knuth, and Oren Patashnik. *Concrete Mathematics: A Foundation for Computer Science*. Addison-Wesley, second edition, 1994.

[52] Jon Grantham and Andrew Granville. Fibonacci primes, primes of the form $2^n - k$ and beyond. *Journal of Number Theory*, 2024. doi:10.1016/j.jnt.2024.02.002; arXiv:2307.07894.

[53] Jennifer Mary Knightley Gray. *A Study of Babylonian Goal-Year Planetary Astronomy*. PhD thesis, Durham University, 2009. https://etheses.durham.ac.uk/id/eprint/101/.

[54] Great Internet Mersenne Prime Search. GIMPS discovers largest known prime number: $2^{136,279,841} - 1$. Press release, https://mersenne.org/primes/press/M136279841.html, 2024.

[55] Thomas C. Hales. A proof of the Kepler conjecture. *Annals of Mathematics*, 162:1065–1185, 2005.

[56] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Oxford University Press, sixth edition, 2008. Revised by D. R. Heath-Brown and J. H. Silverman.

[57] Glyn Harman. *Metric Number Theory*, volume 18 of *London Mathematical Society Monographs, New Series*. Clarendon Press, Oxford, 1998.

[58] Charles Hermite. Sur la fonction exponentielle. *Comptes rendus de l'Académie des sciences*, 77:18–24, 74–79, 226–233, 285–293, 1873.

[59] Titus Hilberdink. An arithmetical mapping and applications to Ω-results for the Riemann zeta function. *Acta Arithmetica*, 139(4):341–367, 2009.

[60] A. J. Hoffman and H. W. Wielandt. The variation of the spectrum of a normal matrix. *Duke Mathematical Journal*, 20(1):37–39, 1953.

[61] Douglas R. Hofstadter. Energy levels and wave functions of Bloch electrons in rational and irrational magnetic fields. *Physical Review B*, 14:2239–2249, 1976.

[62] Shaofang Hong and Raphael Loewy. Asymptotic behavior of eigenvalues of greatest common divisor matrices. *Glasgow Mathematical Journal*, 46(3):551–569, 2004.

[63] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, 2004.

[64] H. Jager and P. Liardet. Distributions arithmétiques des dénominateurs de convergents de fractions continues. *Indagationes Mathematicae (Proceedings)*, 50:181–197, 1988.

[65] Oliver Jenkinson and Mark Pollicott. Computing the dimension of dynamically defined sets: E_2 and bounded continued fractions. *Ergodic Theory and Dynamical Systems*, 21 (5):1429–1445, 2001.

[66] Oliver Jenkinson and Mark Pollicott. Rigorous effective bounds on the Hausdorff dimension of continued fraction Cantor sets: a hundred decimal digits for the dimension of E_2. *Advances in Mathematics*, 325:87–115, 2018.

[67] M. H. Jensen, P. Bak, and T. Bohr. Complete devil's staircase, fractal dimension, and universality of mode-locking structure in the circle map. *Physical Review Letters*, 50: 1637, 1983.

[68] Johannes Kepler. Mysterium cosmographicum. Tübingen, 1596.

[69] Johannes Kepler. Strena seu de nive sexangula, 1611. English translation by C. Hardie, *The Six-Cornered Snowflake*, Oxford, 1966.

[70] Johannes Kepler. Harmonices mundi. Linz, 1619. English translation by E. J. Aiton, A. M. Duncan and J. V. Field, *The Harmony of the World*, American Philosophical Society, 1997.

[71] A. Ya. Khinchin. *Continued Fractions*. Dover Publications, Mineola, NY, 1997. Translation of the third Russian edition; first English edition University of Chicago Press, 1964.

[72] A. Khintchine. Metrische Kettenbruchprobleme. *Compositio Mathematica*, 1:361–382, 1935.

[73] Sergey Khrushchev. A recovery of Brouncker's proof for the quadrature continued fraction. *Publicacions Matemàtiques*, 50(1):3–42, 2006.

[74] Dimitris Koukoulopoulos and James Maynard. On the Duffin–Schaeffer conjecture. *Annals of Mathematics*, 192(1):251–307, 2020.

[75] Dimitris Koukoulopoulos, James Maynard, and Daodao Yang. An almost sharp quantitative version of the Duffin–Schaeffer conjecture. arXiv:2404.14628, 2024.

[76] L. Kuipers and H. Niederreiter. *Uniform Distribution of Sequences*. Wiley, 1974.

[77] R. O. Kuzmin. On a problem of Gauss. *Doklady Akademii Nauk SSSR*, pages 375–380, 1928.

[78] Laurent Laloux, Pierre Cizeau, Jean-Philippe Bouchaud, and Marc Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83:1467, 1999.

[79] L. J. Lange. An elegant continued fraction for π. *American Mathematical Monthly*, 106 (5):456–458, 1999.

[80] J. Laskar, A. Fienga, M. Gastineau, and H. Manche. La2010: a new orbital solution for the long-term motion of the Earth. *Astronomy & Astrophysics*, 532:A89, 2011.

[81] D. H. Lehmer and R. E. Powers. On factoring large numbers. *Bulletin of the American Mathematical Society*, 37(10):770–776, 1931.

[82] Robert J. Lemke Oliver and Kannan Soundararajan. Unexpected biases in the distribution of consecutive primes. *Proceedings of the National Academy of Sciences*, 113(31): E4446–E4454, 2016.

[83] Ernő Lendvai. *Béla Bartók: An Analysis of his Music*. Kahn & Averill, London, 2000.

[84] Paul Lévy. Sur les lois de probabilité dont dépendent les quotients complets et incomplets d'une fraction continue. *Bulletin de la Société Mathématique de France*, 57:178–194, 1929.

[85] Paul Lévy. Sur le développement en fraction continue d'un nombre choisi au hasard. *Compositio Mathematica*, 3:286–303, 1936.

[86] Linda Hall Library. Scientist of the day: John R. Newlands. https://www.lindahall.org/about/news/scientist-of-the-day/john-r-newlands/, 2026. Accessed November 2026.

[87] Peter Lindqvist and Kristian Seip. Note on some greatest common divisor matrices. *Acta Arithmetica*, 84(2):149–154, 1998.

[88] M. Lothaire. *Algebraic Combinatorics on Words*, volume 90 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2002.

[89] R. Luque et al. A resonant sextuplet of sub-Neptunes transiting the bright star HD 110067. *Nature*, 623:932–937, 2023.

[90] V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.

[91] Jean-Claude Martzloff. *A History of Chinese Mathematics*. Springer, 2006.

[92] Madan Lal Mehta. *Random Matrices*, volume 142 of *Pure and Applied Mathematics*. Elsevier/Academic Press, third edition, 2004.

[93] Hermann Minkowski. Zur Geometrie der Zahlen. In *Verhandlungen des III. Internationalen Mathematiker-Kongresses in Heidelberg*, pages 164–173, 1904.

[94] Richard Moeckel. Geodesics on modular surfaces and continued fractions. *Ergodic Theory and Dynamical Systems*, 2(1):69–83, 1982.

[95] Alexis Monnerot-Dumaine. The Fibonacci word fractal. HAL preprint hal-00367972, 2009.

[96] H. L. Montgomery. The pair correlation of zeros of the zeta function. In *Analytic Number Theory*, volume 24 of *Proceedings of Symposia in Pure Mathematics*, pages 181–193. American Mathematical Society, 1973.

[97] H. L. Montgomery and R. C. Vaughan. The large sieve. *Mathematika*, 20:119–134, 1973.

[98] Michael A. Morrison and John Brillhart. A method of factoring and the factorization of F_7. *Mathematics of Computation*, 29(129):183–205, 1975.

[99] NASA Space Science Data Coordinated Archive. Planetary fact sheets. https://nssdc.gsfc.nasa.gov/planetary/factsheet/, 2026. Accessed November 2026.

[100] A. M. Odlyzko. On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation*, 48(177):273–308, 1987.

[101] OEIS Foundation. Sequence A001203: Simple continued fraction expansion of Pi. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A001203, 2026. Accessed November 2026.

[102] OEIS Foundation. Sequence A001517: Bessel polynomials $y_n(x)$ evaluated at 2. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A001517, 2026. Accessed November 2026.

[103] OEIS Foundation. Sequence A001605: Indices of prime Fibonacci numbers. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A001605, 2026. Accessed November 2026.

[104] OEIS Foundation. Sequences A033089 and A033090: Incrementally largest terms in the continued fraction for Pi, and their positions. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A033089, 2026. Accessed November 2026.

[105] OEIS Foundation. Sequence A086785: Primes found among the numerators of the continued fraction rational approximations to Pi. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A086785, 2026. Accessed November 2026.

[106] OEIS Foundation. Sequence A086788: Primes found among the denominators of the continued fraction rational approximations to Pi. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A086788, 2026. Accessed November 2026.

[107] OEIS Foundation. Sequence A094007: Numbers k such that the denominator of the k-th convergent of the continued fraction expansion of e is prime. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A094007, 2026. Accessed November 2026; records $a(11) > 50000$.

[108] OEIS Foundation. Sequence A096650: Indices of prime Pell numbers. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A096650, 2026. Accessed November 2026.

[109] OEIS Foundation. Sequence A117536: Nearest integer to locations of increasingly large peaks of $|\zeta(1/2 + 2\pi it/\log 2)|$. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A117536, 2026. Accessed November 2026.

[110] OEIS Foundation. Sequence A259490: Numbers k such that the denominator of the k-th convergent of the continued fraction expansion of Pi is prime. The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A259490, 2026. Accessed November 2026.

[111] W. Pauli. Über das Wasserstoffspektrum vom Standpunkt der neuen Quantenmechanik. *Zeitschrift für Physik*, 36:336–363, 1926.

[112] Vasiliki Plerou, Parameswaran Gopikrishnan, Bernd Rosenow, Luís A. Nunes Amaral, and H. Eugene Stanley. Universal and nonuniversal properties of cross correlations in financial time series. *Physical Review Letters*, 83:1471, 1999.

[113] Bernhard Riemann. Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse. *Monatsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*, November 1859.

[114] Teddy Rook. Where music, patterns, math and chemistry meet. Teddy Rook Book, `https://teddyrookbook.com/connections/`, 2026.

[115] Teddy Rook. The rhythm of numbers. Teddy Rook Book, `https://teddyrookbook.com/connections/rhythm-of-numbers/`, 2026.

[116] Teddy Rook. The Continued-Fraction Suite: six pieces for piano, one number each. Score, Teddy Rook Book, `https://teddyrookbook.com/music/continued-fraction-suite/`, 2026.

[117] Raphaël Salem. On some singular monotonic functions which are strictly increasing. *Transactions of the American Mathematical Society*, 53(3):427–439, 1943.

[118] D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn. Metallic phase with long-range orientational order and no translational symmetry. *Physical Review Letters*, 53:1951, 1984.

[119] H. J. S. Smith. On the value of a certain arithmetical determinant. *Proceedings of the London Mathematical Society*, 7:208–212, 1875.

[120] Arnold Sommerfeld. Zur Quantentheorie der Spektrallinien. *Annalen der Physik*, 356 (17):1–94, 1916.

[121] V. T. Sós. On the distribution mod 1 of the sequence $n\alpha$. *Annales Universitatis Scientiarum Budapestinensis de Rolando Eötvös Nominatae, Sectio Mathematica*, 1:127–134, 1958.

[122] K. Soundararajan. Extreme values of zeta and L-functions. *Mathematische Annalen*, 342 (2):467–486, 2008.

[123] E. M. Standish and J. G. Williams. Approximate positions of the planets. JPL Solar System Dynamics, `https://ssd.jpl.nasa.gov/planets/approx_pos.html`, 1992. Accessed November 2026.

[124] S. V. Stepkin, A. A. Konovalenko, N. G. Kantharia, and N. Udaya Shankar. Radio recombination lines from the largest bound atoms in space. *Monthly Notices of the Royal Astronomical Society*, 374:852, 2007. arXiv:astro-ph/0610407.

[125] M. A. Stern. Ueber eine zahlentheoretische Funktion. *Journal für die reine und angewandte Mathematik*, 55:193–220, 1858.

[126] S. Świerczkowski. On successive settings of an arc on the circumference of a circle. *Fundamenta Mathematicae*, 46:187–189, 1958.

[127] P. Szüsz. Verallgemeinerung und Anwendungen eines Kusminschen Satzes. *Acta Arithmetica*, 7:149–160, 1962.

[128] D. J. Thouless. Bandwidths for a quasiperiodic tight-binding model. *Physical Review B*, 28:4272–4276, 1983.

[129] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs. Quantized Hall conductance in a two-dimensional periodic potential. *Physical Review Letters*, 49:405–408, 1982.

[130] Salvatore Torquato and Frank H. Stillinger. Local density fluctuations, hyperuniformity, and order metrics. *Physical Review E*, 68:041113, 2003.

[131] Salvatore Torquato, Ge Zhang, and Matthew de Courcy-Ireland. Uncovering multiscale order in the prime numbers via scattering. *Journal of Statistical Mechanics: Theory and Experiment*, page 093401, 2018. doi:10.1088/1742-5468/aad6be.

[132] Godfried T. Toussaint. The Euclidean algorithm generates traditional musical rhythms. In *Renaissance Banff: Mathematics, Music, Art, Culture (Proceedings of Bridges 2005)*, pages 47–56, 2005.

[133] I. M. Vinogradov. *The Method of Trigonometrical Sums in the Theory of Numbers*. Interscience, London and New York, 1954. Translated, revised and annotated by K. F. Roth and Anne Davenport.

[134] H. von Mangoldt. Zu Riemanns Abhandlung "Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse". *Journal für die reine und angewandte Mathematik*, 114:255–305, 1895.

[135] André Weil. On some exponential sums. *Proceedings of the National Academy of Sciences of the USA*, 34:204–207, 1948.

[136] Eric W. Weisstein. Pi continued fraction. MathWorld, a Wolfram Web Resource, `https://mathworld.wolfram.com/PiContinuedFraction.html`, 2026. Accessed November 2026; lists the computations of Gosper (1977), Havermann (1999, 2002), Bickford (2010), Weisstein (2010–2013), Fahad (2021) and Frank (2025).

[137] Xenharmonic Wiki. 171edo. `https://en.xen.wiki/w/171edo`, 2026. Accessed November 2026.

[138] Xenharmonic Wiki. Ennealimmal. `https://en.xen.wiki/w/Ennealimmal`, 2026. Accessed November 2026.